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A Boundary Element Method for the Elastodynamic Analysis of Underground Structures Subjected to Plane and Explosive Waves

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By

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ABSTRACT

The Boundary Element Method (BEM) has been successfully used in Engineering analysis in recent years and proved to be particularly suitable for Geomechanics applications.

In this thesis the direct BEM is applied for solving transient elastodynamic problems associated with underground structures subjected to plane and explosive waves, under plane strain conditions.

The two dimensional analysis can be performed for any underground structure, involving complicated geometry and boundary conditions. The structures may be unlined or lined openings of arbitrary shape and the ground medium is infinite or semi-infinite.

The solution is applicable either in Laplace transform domain or in time domain. In the first case the solution is accomplished by applying the Laplace transform to the equation of motion, by implementing the direct BEM for the numerical solution of the problem and by numerically inverting the transformed solution using Fast Fourier Transform. In the time domain case, the BEM is applied by numerically integrating in the real time domain.

The available forcing function can be a plane or an explosive P or SV shock stress wave.

Most of the work associated with this thesis was directed in the development and implementation of a computer package capable of dealing with elastodynamic problems, related to underground structures. The computer package of programs that has been developed has pre-processing, processing and post-processing capabilities. In addition, automatic mesh generation facilities have been included to facilitate the modelling of specific problems.

The most important contribution of this work, in engineering problems associated with underground openings, is the use of BEM to obtain accurate solutions for the dynamic loading associated with blasting and explosives from military weapons.

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CHAPTER 1

INTRODUCTION

1.1 GENERAL INTRODUCTION

Analytical methods in elastodynamic problems, were first used in early 1960s for the calculation of stress and displacement fields around two dimensional idealization of underground openings, subjected to harmonic or transient dynamic loading. Such openings were characterized by a circular, elliptical, or parabolic cross section, surrounded by elastic or viscoelastic medium extending to infinity. The text book by Pao and Mow (1965) is an established reference for this type of work.

The development in computer technology after 1960s, enabled the numerical methods to advance rapidly. The Finite Element Method (FEM) was widely used and as the most popular numerical method for this kind of work, is a very effective technique especially for complex structure geometries, inhomogeneities in the material and complicated constitutive behaviour. However the FEM is limited in that the infinitely extended medium must be modelled as finite, and transmitting boundaries must be used at the sides of the mesh. Normally such boundaries are placed far enough, to allow for energy radiation, however the resulting solution still suffers from an element of inaccuracy. The computational effort needed to model infinite continua is considerable, even with sophisticated packages.

In the last fifteen years the Boundary Element Method (BEM), has emerged as a powerful alternative to the above methods, (Brebbia et al.,1984; Banerjee and Butterfield,1981). The BEM is ideally suited for modelling infinitely extended media in elastodynamic problems, because the radiation and regularity conditions are automatically fulfilled, thus eliminating wave reflections from infinity (Eringen and Suhubi,1975). Furthermore, for linear problems, only the surface on the boundaries of the surface needs to be discretized resulting in reduction of the computational effort and a simpler mesh.

The BEM method in conjunction with the Laplace transformation has been used successfully in the past for solving several problems. Cruse (1968) and Cruse and Rizzo (1968) used the direct BEM with Papoulis' inversion algorithm of the Laplace transform, to solve a half plane propagation problem. In addition Manolis (1980) and Manolis and Beskos (1981) used the BEM with the inversion algorithm introduced by Durbin (1974), to study two dimensional problems. The BEM, in conjunction with Laplace transform, has also been used for solving problems of explosions inside a cylindrical or spherical cavity. Kontoni et al. (1987), solved the problem of suddenly applied and maintained pressure due to an

explosion inside a cylindrical cavity. Manolis (1984) and Ahmad and Manolis (1987), addressed the same problem but for a spherical cavity.

Solutions of elastodynamic problems by BEM in the time domain have been presented by many authors in the published literature. Two-dimensional plane strain problems were examined by Niwa et. al. (1980) for two-dimensional cases and Mansur and Brebbia (1982), for the scalar wave equation. Niwa et al. (1980) and Manolis (1983) employed a three-dimensional methodology by BEM to investigate the scattering of elastic waves around cylindrical openings under conditions of plane strain. Manolis et al. (1986), Banerjee et al.(1987), Karabalis and Beskos (1984) employed BEM to solve three dimensional problems.

Manolis and Beskos (1988) presented a comprehensive account of the formulation and numerical implementation of the BEM for elastodynamic problems.

1.2 NEED FOR CURRENT RESEARCH

Underground structures are of great importance in engineering. For civil purposes tunnels are usually used for transportation (traffic, water supplies). Moreover, many plants are constructed underground for environmental and safety reasons. Underground reservoirs are usually used for fuel and gas storage. Nuclear waste disposal facilities are constructed always underground.

For military purposes, underground structures are extremely important. Protective shelters, headquarters, weapon magazines, fuel tanks and many other installations are usually constructed underground.

The calculation of underground structures under dynamic loading is an extremely difficult task. Because of the uncertainties concerning the properties of the ground and the behaviour of the structure under dynamic loading, over-simplified assumptions are usually introduced referring to the prediction of the loading and the behaviour of the structure. These calculations lack accuracy and often are performed in an empirical manner rather than a scientific accurate one.

This work, is concerned with the development of an accurate and systematic way of solving underground structures, modelled under two dimensional plane strain conditions and transient elastodynamic loading. The BEM methodology is used being a particularly suitable numerical tool for problems associated with infinite and semi-infinite media.

In order to analyze a variety of structures of complex geometry and loading, a computer package of programs has been developed with special attention to the simplification of data introduction. Furthermore, the modular form of the structure of the program, offers the

ability to extent the capabilities of the package without considerable effort. The accuracy of the program is demonstrated by comparing results with solutions available in the literature.

1.3 LAYOUT OF THE THESIS

This thesis presents the work associated with the complete numerical implementation of the BEM method used for transient two-dimensional elastodynamic problems of openings of arbitrary shape for isotropic piecewise homogeneous solids.

Chapter two deals with the Laplace transform domain BEM. The Laplace transformation of the loading, the basic formulation and the inverse Laplace transformation are discussed first followed by the presentation of the BEM formulation and finally a number of applications are examined.

In Chapter three, the BEM formulation for time domain BEM is presented in detail and several relevant applications are also examined.

Chapter four is devoted to the examination of the fundamental topics used throughout the present work. More specifically, reference is made to the formulae used for evaluating the pressure due to an explosion at a point, the numerical integration techniques for calculating singular and non-singular cases, the isoparametric representation of line elements and the method of numerical calculation of strains, the numerical evaluation of the local to the elements co-ordinate system and finally the stress and strain tensors, traction vector and transformation of stresses are presented.

The BEM formulation for problems with zones characterised by different properties to those of the surrounding material is implemented in Chapter five. Further applications are presented, with special emphasis to lined openings.

In Chapter six, problems relative to wave propagation are presented in a general form suitable for programming. The reflection conditions on the ground surface and reflection coefficients for the case of harmonic waves are derived. Problems related to the incident and reflected waves are examined. A number of applications for semi-infinite media are also presented.

The description of the program that has been developed in the context of this work and the description of the subroutines of the package, are presented in Chapter seven.

CHAPTER 2

LAPLACE DOMAIN BEM

2.1 INTRODUCTION

In this chapter the implementation of the BEM for transient two dimensional elastodynamic problems in Laplace transform domain, is presented. The solution is accomplished by applying the Laplace transformation to the equation of motion, by implementing the direct BEM for the numerical solution of the problem and by numerically inverting the transformed solution using Fast Fourier Transform (FFT). The available forcing function can be a plane or an explosive P and SV shock stress wave.

A number of applications are finally presented to demonstrate the accuracy and applicability of the present implementation.

2.2 FORMULATION OF THE PROBLEM

For the solution of a transient elastodynamic problem, the following steps have to be applied:

a. Direct Laplace transformation of the time function of loading by analytical or numerical methods.

b. Solution of the problem in the transformed steady state domain, by applying the BEM.

c. Numerical inversion of the solution by the FFT algorithm, in order to obtain the solution in the time domain.

2.2.1 Laplace Transformation of the Loading

The direct Laplace transformation of a time function f(t) is (Spiegel, 1965)

$$\mathcal{L}[f(t)] = \bar{f}(s) = \int_0^\infty f(t)\, e^{-st}\, dt \qquad (2.1)$$

where s is the Laplace transform variable and $\mathcal{L}$ is the Laplace transform operator.

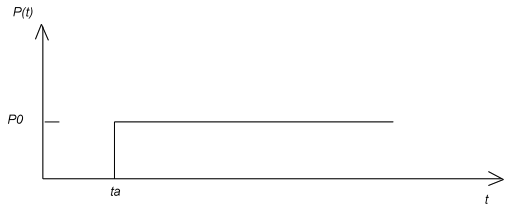
This transformation for the time function of loading can be performed analytically for relatively simple cases of loading or numerically for complicated histories of loading (e.g. accelerograms).

2.2.1.1 Analytical Laplace Transformation

a. Plane Wave

This is an incident stress wave that is applied as a time varying function. In this work the following two forms of loading are examined :

(i). As a suddenly applied pressure P_0 and maintained constant in time. The time history of the tractions may be expressed as the product of P_0 with the Heaviside unit step function [H(t-t_a)], where t_a is the time of arrival of wavefront at the node. The Heaviside function satisfies the following conditions

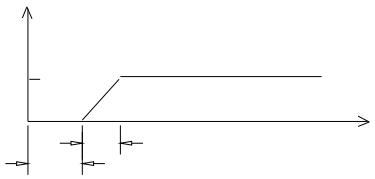


$$[H(t-t_a)] = 0 \quad for \quad 0<t<t_\alpha$$
$$[H(t-t_a)] = 1 \quad for \quad t_\alpha>t>\infty$$

The Laplace transform of this product according to Spiegel (1965), is

$$\mathcal{L}\,[\,P_0\,H(t-t_a)\,] = P_0\,\frac{e^{-t_a s}}{s} \qquad (2.2)$$

(ii). As a ramp-like function, varying linearly up to a value and then maintained constant. The Laplace transform in this case is



b. Wave generated by an Explosion Point

$$\mathcal{L}\,[\,P(t-t_a)\,] = P_0\left(\frac{1-e^{-t_c s}}{t_c s}\right)\left(\frac{e^{-t_a s}}{s}\right) \qquad (2.3)$$

The pressures generated by an explosion at a point will be presented in Chapter 4, section 4.2.

Here the Laplace transform of the forcing function is examined with or without decay in time. Using equation (4.1) of Chapter 4, the Laplace transform of the pressure function without decay in time, may be expressed as: and with decay in time using equation (4.8) of Chapter 4.

where t is the elapsed time counted from the time t_a of arriving of wavefront at node i, while $t_a = R_i/c$, R_i is the

$$\mathcal{L}\,[\,P_{0i}H(t-t_a)\,] = P_{0i}\frac{e^{-t_a s}}{s} \qquad (2.4)$$

$$\mathcal{L}\left[\,P_0 e^{-\frac{t}{t_a}}\,\right] = P_0\left(\frac{t_a}{1+t_a s}\right)e^{-t_a s} \qquad (2.5)$$

distance of node i from the wavefront line at time t=0 and c is the seismic velocity.

2.2.1.2 Numerical Laplace Transformation

The analytical direct Laplace transformation is not always possible for the given function of loading, as in the case of complicated analytical or discrete loading histories, e.g. accelerograms. For the cases where numerical transformation is necessary, this may be accomplished by implemeneting the following ways (Narayanan and Beskos, 1982):

a. Using a modification of Durbin's method of numerical inversion, as will be examined in the forthcoming, the discrete direct Laplace transform for a given spectrum of values of the function f(t) at N equidistant stations (time steps), the direct Laplace transform of these values at the same stations are given by the formula:

$$\bar{f}(s_k) = -\frac{1}{2}f(0) + \sum_{n=1}^{N}(A(n) + iB(n))W^{kn} \qquad (2.6)$$

where

$$s_k = \gamma + i\left(k\frac{2\pi}{T}\right), \qquad W = e^{-i\left(\frac{2\pi}{N}\right)}$$

$$A(n) = \sum_{l=0}^{L}\left[f((n+lN)\Delta t)\, e^{-\gamma(n+lN)\Delta t}\right]\Delta t$$

$$B(n) = 0$$

$$T = NL\Delta t \qquad n = 1,2,\ldots,N$$

$$k = 0,1,2,\ldots,N-1 \qquad l = 0,1,2,\ldots,L \qquad (2.7)$$

In this work L is taken as 1.

b. For loading functions that vary linearly with time, the discrete values of the direct Laplace transform at N equidistant stations, is given by

$$\bar{f}(s) = \sum_{n=1}^{N-1}\frac{1}{s^2\Delta t}\left\{s\Delta t\left(f(t_n)\, e^{-st_n} - f(t_{n+1})\, e^{-st_{n+1}}\right) + \Delta f\left(e^{-st_n} - e^{-st_{n+1}}\right)\right\} \qquad (2.8)$$

this approach is very useful for accelerograms associated with blasting vibrations or earthquakes.

2.2.2 Solution by the Boundary Element Method

2.2.2.1 Formulation

For a homogeneous, isotropic, linear elastic material and small displacements, the motion is governed by the well known Navier-Cauchy equation, which may be expressed according to Banerjee et al. (1987) as:

$$(c_1^2 - c_2^2)\, u_{i,ij} + c_2^2 u_{j,ii} + b_j = \ddot{u}_j \qquad (2.9)$$

where u denotes displacements and b signifies the body force per unit mass and i and j vary from 1 to 2. Furthermore,

$$c_1 = \sqrt{\frac{\lambda+2\mu}{\rho}}, \qquad c_2 = \sqrt{\frac{\mu}{\rho}}$$

$$\mu = \frac{E}{2(1+\nu)}, \qquad \lambda = \frac{\nu E}{(1+\nu)(1-2\nu)} \qquad (2.10)$$

where λ, μ are the Lame's constants, E is the Young's modulus, ρ is the mass density and ν is the Poisson's ratio. The constitutive equations are

$$\begin{aligned}\sigma_{ij} &= \lambda\varepsilon_{kk}\ \delta_{ij} + 2\mu\varepsilon_{ij} \\ &= \lambda u_{k,k}\ \delta_{ij} + \mu(u_{i,j}+u_{j,i}) \\ &= \rho(c_1^2 - 2c_2^2)\,u_{k,k}\ \delta_{ij} + c_2^2(u_{i,j} + u_{j,i})\end{aligned} \qquad \textbf{(2.11)}$$

where δ_{ij} is the Kronecker delta and σ_{ij} and ε_{ij} are the stress and strain tensors respectively and $\varepsilon_{kk}=\varepsilon_{11}+\varepsilon_{22}+\varepsilon_{33}$ is the first strain invariant. Furthermore, the strain-displacement relation is

$$\varepsilon_{ij} = \frac{1}{2}(u_{i,j}+u_{j,i}) \qquad \textbf{(2.12)}$$

The boundary and initial conditions are:

$$\begin{aligned}&t_i = \sigma_{ij}\ n_j \\ &u_i(P,t=0) = U_{i0}\ , \quad \dot{u}_i(P,t=0) = \dot{U}_{i0}\end{aligned} \qquad \textbf{(2.13)}$$

where P is the source node (field point), $u_i(P,t)$ and $t_i(P,t)$ is the displacement and traction vector at the point P(x,y) and at time t, and n_j is the outward normal to the surface unit vector. Using the dynamic Betti's reciprocal theorem on the surface A of the mesh, for the tractions and displacements matrices of the source node and the fundamental singular solutions of tractions and displacements, the following integral equation is derived:

$$c_{ij}(Q)\,u_i(Q,t) = \int_A [\,G_{ij}(P,Q,t) * t_i(P,t) - F_{ij}(P,Q,t) * u_i(P,t)\,]\,dA(P) \qquad \textbf{(2.14)}$$

where Q is the receiver node and G,F are the fundamental singular solutions. The discontinuity term c_{ij} takes the values: 1 for Q located in the interior of the region, 0.5 for Q on the surface of the smooth boundary and 0 for Q located in the exterior of the region. The * indicates convolution defined for time T as

$$G_{ij}\ (P,Q,T) * t_i(P,T) = \int_0^T G_{ij}\ (P,Q,T-\tau)\ t_i(P,\tau)\,d\tau \qquad \textbf{(2.15)}$$

Applying the Laplace transform with zero initial conditions and zero body forces, the following equations in the Laplace transformed domain are obtained:

$$\begin{aligned}&(c_1^2-c_2^2)\,\overline{u}_{i,ij} + c_2^2\ \overline{u}_{j,ii} = s^2\overline{u}_j \\ &\overline{\sigma}_{ij} = \lambda\overline{u}_{k,k}\ \delta_{ij} + \mu(\overline{u}_{i,j}+\overline{u}_{j,i})\ , \quad \overline{\tau}_i = \overline{\sigma}_{ij}\ n_j\end{aligned} \qquad \textbf{(2.16)}$$

the integral transformed equation is

$$c_{ij}(Q)\,\overline{u}_i(Q,s)=\int_A\left\{\overline{G}_{ij}(P,Q,s)*\overline{t}_i(P,s)-\overline{F}_{ij}(P,Q,s)*\overline{u}_i(P,s)\right\}dA(P) \qquad (2.17)$$

where the overline indicates the transformed value of a variable. This can be written for convenience as:

$$c_{ij}\overline{u}_i=\int_A\left\{\overline{G}_{ij}*\overline{t}_i-\overline{F}_{ij}*\overline{u}_i\right\}dA \qquad (2.18)$$

or equevalently (Banerjee et al., 1987)

$$c_{ij}\overline{u}_i=\int_A\left\{\overline{G}_{ij}\,\overline{t}_i-\overline{F}_{ij}\,\overline{u}_i\right\}dA \qquad (2.19)$$

In order to solve this integral equation, the Cartesian co-ordinates, the displacements and tractions at every node of the mesh, are expressed in terms of shape functions and the nodal values of these quantities, as will be discussed in Chapter 4, section 4.4.

$$x_i(\xi)=M_\alpha(\xi)X_{i\alpha}\ ,\qquad \overline{u}_i(\xi)=M_\alpha(\xi)\overline{U}_{i\alpha}\ ,\qquad \overline{t}_i(\xi)=M_\alpha(\xi)\overline{T}_{i\alpha} \qquad (2.20)$$

`where $X_{i\alpha}$, $U_{i\alpha}$ and $T_{i\alpha}$ are the nodal co-ordinates, displacements and tractions respectively of the examining element, the index i=1,2 corresponds to the two co-ordinates, α=1,2,3 refers to the three nodes of the quadratic element and ξ is the curvilinear co-ordinate of the node under consideration that takes values in the interval [-1,1]. Combining equations (2.19) and (2.20) according to Manolis and Beskos (1988), results in:

$$c_{ij}\overline{u}_i=\sum_{m=1}^{L}\left\{\int_A\overline{G}_{ij}[M_\alpha\overline{T}_{i\alpha}]\,dA-\int_A\overline{F}_{ij}[M_\alpha\overline{U}_{i\alpha}]\,dA\right\} \qquad (2.21)$$

where L is the number of the total elements of the mesh, s is the length of the m element.

Employing equation (2.21) to all the mesh nodes and applying summation results in

$$\sum_{l=1}^{N}c_{ij}\overline{u}_i=\sum_{l=1}^{N}\sum_{m=1}^{L}\left\{\int_A G_{ij}[M_\alpha\overline{T}_{i\alpha}]\,dA-\int_A F_{ij}[M_\alpha\overline{U}_{i\alpha}]\,dA\right\} \qquad (2.22)$$

where N is the total number of the nodes of the mesh.

Also (Chapter 4, section 4.4)

$$ds=|J(\xi)|\,d\xi \qquad (2.23)$$

and equation (2.21) becomes

$$\sum_{l=1}^{N} c_{ij}\overline{u}_{i} = \sum_{l=1}^{N}\sum_{m=1}^{L}\left\{\int_{-1}^{1} G_{ij}\,[M_{\alpha}\overline{T}_{i\alpha}]\,|J|\,d\xi - \int_{-1}^{1} F_{ij}\,[M_{\alpha}\overline{U}_{i\alpha}]\,|J|\,d\xi\right\} \qquad (2.24)$$

Extracting out of the integral the nodal values of the tractions T_{ik} and displacements U_{ik}, yields

$$\sum_{l=1}^{N} c_{ij}\overline{u}_{i} = \sum_{l=1}^{N}\sum_{m=1}^{L}\left\{\overline{T}_{i\alpha}\int_{-1}^{1} G_{ij}M_{\alpha}\,|J|\,d\xi - \overline{U}_{i\alpha}\int_{-1}^{1} F_{ij}M_{\alpha}\,|J|\,d\xi\right\} \qquad (2.25)$$

that may be expressed for clarity

$$\sum_{l=1}^{N} c_{ij}\overline{u}_{i} = \sum_{l=1}^{N}\sum_{m=1}^{L}\left\{\sum_{\alpha=1}^{3}\overline{T}_{i\alpha}\int_{-1}^{1} G_{ij}M_{\alpha}\,|J|\,d\xi - \sum_{\alpha=1}^{3}\overline{U}_{i\alpha}\int_{-1}^{1} F_{ij}M_{\alpha}\,|J|\,d\xi\right\} \qquad (2.26)$$

For the formation of the system of equations (2.26), the source (field) point is allowed to coincide sequentially with all the nodes of the mesh (inner summation). Further for every position of the source point, the receiver point is allowed to coincide sequentially with all the elements of the mesh (outer summation).

Finally, for every node P of the element (so urce node), the fundamental solutions G_{ij} of displacements and F_{ij} of tractions (Appendix A) at every node Q of all the elements (receiver node) are calculated and multiplied by the Jacobian of transformation and the shape functions (Chapter 4, section 4.4). For the fundamental solutions at the receiver node Q, F_{ij} is the traction at Q in i direction, due to a concentrated unit pulse at point P in j direction. Similarly G_{ij} is the displacement at Q in i direction due to a unit pulse at point P in j direction. The numerical integration is performed by Gaussian Quadrature formulae and indirect schemes as will be discussed in Chapter 4, section 4.3.

2.2.2.2 Solution of Equations

The integrated fundamental solutions are collected in arrays and the system of equations in matrix form resulting from equations (2.26) is
where n is the examining time step and $\{t\}^{(n)}$, $\{u\}^{(n)}$ are vectors of tractions and displacements respectively at all

$$[F]\,\{\overline{u}\}^{(n)} = [G]\,\{\overline{t}\}^{(n)} \qquad (2.27)$$

the nodal points. [G], [F] are matrices containing the summations of the contributions of the kernels G_{ij} and F_{ij} respectively. The contribution of the jump term c_{ij} has been absorbed in the matrix [F].

By knowing the boundary conditions {t}, {u} or mixed, the above equation can be solved for the rest of the unknowns.

For instance, if {t} is known then

$$\{\bar{u}\}^{(n)} = [F]^{-1}[G]\{\bar{t}\}^{(n)} \qquad (2.28)$$

This procedure is applied to all the examining time steps as they are determined in section 2.2.3.

2.2.2.3 Stresses on the Boundary

After solving the equations for tractions and displacements at the nodes of the mesh in the transformed domain, the stresses on the boundary may be obtained, with one of the following ways:

a. Employing the constitutive equations and the boundary conditions, for every node, the following system of equations is derived:

$$\begin{aligned} \bar{\sigma}_{ij} &= \lambda \bar{u}_{k,k}\delta_{ij} + \mu(\bar{u}_{i,j}+\bar{u}_{j,i}) \\ \bar{t}_i &= \bar{\sigma}_{ij}n_j \\ \bar{u}_{i,s} &= \bar{u}_{i,j}q_j \end{aligned} \qquad (2.29)$$

Rearranging these equations (according to Manolis and Beskos 1981) the following system in matrix form is obtained:

$$\begin{bmatrix} -1 & 0 & 0 & \lambda+2\mu & 0 & 0 & \lambda \\ 0 & -1 & 0 & \lambda & 0 & 0 & \lambda+2\mu \\ 0 & 0 & -1 & 0 & \mu & \mu & 0 \\ n_1 & 0 & n_2 & 0 & 0 & 0 & 0 \\ 0 & n_2 & n_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & q_1 & 0 & q_2 & 0 \\ 0 & 0 & 0 & 0 & q_1 & 0 & q_2 \end{bmatrix} \begin{bmatrix} \bar{\sigma}_{11} \\ \bar{\sigma}_{22} \\ \bar{\sigma}_{12} \\ \bar{u}_{1,1} \\ \bar{u}_{2,1} \\ \bar{u}_{1,2} \\ \bar{u}_{2,2} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \bar{t}_1 \\ \bar{t}_2 \\ \bar{u}_{1,s} \\ \bar{u}_{2,s} \end{bmatrix} \qquad (2.30)$$

where $u_{1,s}$ is the strain in the axial (tangential) direction of the element, at the examining node, n_1,n_2 are the direction cosines of the unit normal vector with x,y global axis respectively. Similarly q_1,q_2 are the direction cosines of the unit tangential vector with the x,y global axis as depicted in Figure (4.5) and will be discussed in Chapter 4, section 4.5.

The system of equations (2.30) may be expressed in matrix form, according to:

$$[A]\{X\} = \{B\} \tag{2.31}$$

and may be solved for the unknowns by inverting the matrix A:

$$\{X\} = [A]^{-1}\{B\} \tag{2.32}$$

The stresses on the boundary are obtained by resolving them in the radial and tangential direction for every node of the mesh. All the tractions, strains, displacements and direction cosines of the unit normal and tangential vectors, that are required in the above equations, are obtained via shape functions for every element and node of the mesh as in Chapter 4, section 4.4 will be presented.

b. Another technique consists in applying Hooke's law as follows:

The tractions and strains are determined first, as previously, in the global co-ordinate system. In turn they are transformed to the local co-ordinate system, with x axis the normal to the surface of the mesh at the examining node, and n axis tangential to the surface (Figure 4.5).
Applying Hooke's law in the local system results in:

$$\begin{aligned} \sigma_{qn} &= \sigma_{nq} = t_s \\ \sigma_{nn} &= t_n \\ \sigma_{ss} &= \frac{1}{1-\nu}(\nu\sigma_{nn} + 2\mu\varepsilon_{ss}) \end{aligned} \tag{2.33}$$

Finally in both cases we have to add the hoop stresses of the incident wave.

2.2.2.4 Interior Points

Once the boundary solution is obtained, the displacements at interior points can be calculated from equation (2.19)

$$c_{ij}\overline{u_i} = \sum_{\alpha=1}^{k} \int_{\tau_{\alpha-1}}^{\tau_\alpha} \left\{ \int_A \left[G_{ij}\overline{t_i} - F_{ij}\overline{u_i} \right] dA \right\} d\tau \tag{2.34}$$

where k is the examining time step.

The displacements and tractions kernels G_{ij} and F_{ij} are calculated by taking $c_{ij}=\delta_{ij}$ and as t_i and u_i the known tractions and displacements on the boundary. In this case as source point is accepted the examining internal point and as destination the nodes on the boundaries. The displacements at the time step k can be found by using equation (2.34) as

$$\{\overline{u}\}^k = \sum_{\alpha=1}^{k} \left\{ \sum_{l=1}^{N} \left([G_l]^k \{\overline{t_l}\}^\alpha - [F_l]^k \{\overline{u_l}\}^\alpha \right) \right\} \qquad (2.35)$$

where N are the total nodes of the mesh.

The interior stresses can be calculated either numerically or analytically.

(a). In the numerical case, the displacements at the nodes of a neighbouring triangular cell are calculated and then the stresses are derived by using interpolation functions (Mansur, 1983) and taking the derivatives of the displacements.

(b). In the analytical case, interior kernels are used, which are calculated by using the constitutive equations (Eringen and Sihubi, 1975). The stresses are then given by

$$c_{ij}\overline{u_i} = \int_A \left\{ \int_0^\tau \left[G_{ij}^\alpha \overline{t_i} - F_{ij}^\alpha \overline{u_i} \right] d\tau \right\} dA \qquad (2.36)$$

where t_i and u_i are the known boundary tractions and displacements and G_{ij}^α and F_{ij}^α the interior kernels with the interior node as the source point.

2.2.3 Numerical Inversion of the Laplace Transform

Following the solution of the problem in the Laplace transform domain, the solution in time domain has been obtained by inversion of the Laplace transform. This is performed by the Durbin's inversion algorithm (Durbin, 1974), with the use of FFT, according to Manolis and Beskos (1981). This algorithm is an improvement of the Dubner and Abate method and is based on the Fourier cosine and sine transforms. Knowing the values of the transform Laplace function $\bar{f}(s_n)$ at N equidistant stations (time steps), their inverse transformation is given by the formula:

$$f(t_k) = 2\frac{e^{\gamma k \Delta t}}{T} \left\{ -\frac{1}{2} Re[\bar{f}(\gamma)] + Re\left[\sum_{n=0}^{N-1} (A(n) + iB(n)) W^{kn} \right] \right. \qquad (2.37)$$

where

$$s_n = \gamma + i\left(n\frac{2\pi}{T}\right) , \qquad W = e^{i\left(\frac{2\pi}{N}\right)}$$

$$A(n) = \sum_{l=0}^{L} Re\left[\overline{f}\left(\gamma + i(n+lN)\frac{2\pi}{T}\right)\right]$$

$$B(n) = \sum_{l=0}^{L} Im\left[\overline{f}\left(\gamma + i(n+lN)\frac{2\pi}{T}\right)\right]$$

$$T = NL\Delta t \qquad n = 1,2,\ldots,N$$

$$k = 0,1,2,\ldots,N-1 \qquad l = 0,1,2,\ldots,L \qquad \textbf{(2.38)}$$

and $t_k = k\Delta t = k(T/N)$, T is the total examined time interval. It is suggested $\gamma T = 5$ to 10 and LxN = 50 to 500. In this work γ is taken 6 and L=1.

This algorithm operates with complex data and the inversion may be achieved for the displacements, tractions and stresses on the boundary for all or the most important mesh nodes.

2.2.4 Linear Viscoelastic Behaviour

The linear viscoelastic behaviour may be introduced by replacing Lame's constants λ and μ by the corresponding complex values (Manolis and Beskos 1988). For instance for the Kelvin material:

$$\mu_s = \mu + \eta s , \qquad \lambda_s = \lambda - \frac{2}{3}\eta s \qquad \textbf{(2.39)}$$

and for the Maxwell material:

$$\mu_s = \frac{\mu s}{s + \frac{\mu}{\eta}} , \qquad \lambda_s = \frac{\lambda s + \frac{\mu}{\eta}\left(\lambda + \frac{2}{3}\mu\right)}{s + \frac{\mu}{\eta}} \qquad \textbf{(2.40)}$$

where η is the coefficient of viscosity and the transform variable s takes complex values for the Durbin's inversion algorithm of Laplace transformation.

The Lame's constants must be replaced by their complex counterparts. The seismic velocities will be replaced by complex counterparts :

$$c_{1s} = \sqrt{\frac{\lambda_s + 2\mu_s}{\rho}} , \qquad c_{2s} = \sqrt{\frac{\mu_s}{\rho}} \qquad \textbf{(2.41)}$$

which are used in calculating the delay times of the wavefront arriving at a node. Moreover, the Poisson's ratio has to be replaced by:

$$\nu_s = \frac{\lambda_s}{2(\lambda_s+\mu_s)} \qquad (2.42)$$

In the case of waves reflection on a free surface, the complex values of Lame's constants must be also introduced, in the stress and amplitude reflection coefficients equations (Chapter 6, section 6.2.2 and 6.2.3). All the constants that are extracted from these are also replaced by the complex values as above.

2.3. APPLICATIONS

In the following applications the material properties used are: Young's modulus E=20,000 MPa, Poisson's ratio ν=0.25 and density ρ=2.55 Mg/m^3. Furthermore, in all applications the total time interval is 0.015 s implemented in 20 time steps.

2.3.1 Plane Wave

2.3.1.1 Circular Cross Section Cavity in an Infinite Medium Swept by a Plane P Wave

This problem has been previously solved, by a combination of analytical and numerical approach (Pao and Mao, 1973), and by direct BEM (Manolis, 1980; Manolis and Beskos, 1981). The radius of the cavity is 5.0 m. The incident wave is a shock compressional plane wave of -10.0 MPa magnitude that is suddenly applied and maintained constant in time. The angle of incidence of the incoming wave is 0°, measured from the global horizontal x axis. Eight quadratic elements are used to represent the surface of the mesh, corresponding to 16 nodes (Figure 2.1). The position of the plane wavefront at t=0, is considered as tangent to the circle, at node 1.

The variation of these stresses in time, for the nodes 1,9,11,13,15 which are at angles 0°, 180°, 135°, 90° and 45° respectively from the x axis, is shown in Figure (2.1). The most compressive hoop stress appears at node 13 at the 12th step and is maintained approximately constant, at -29 MPa, for the subsequent steps, which is in good agreement with the results of Manolis (1980) and Pao and Mao (1973).

The tangential stresses on the periphery of the opening for the time step 14, for all the nodes, is shown in Figure (2.2) while the deformed shape of the mesh at time step 14 are shown in Figure (2.3)

Furthermore, the displacements in x direction in mm versus time steps is shown in Figure (2.4).

It can be seen that the stresses and displacements in late times, tend to be equal for the nodes symmetric to the y axis (nodes 1 and 9, 15 and 11). The cavity behaves similar to the elastostatic case loading and the dynamic solution approaches the static solution.

2.3.1.2 Square Cross Section Cavity in an Infinite Medium Swept by a Plane P Wave

The problem has been solved by direct BEM in conjunction with Laplace transform by Manolis (1980), for a wave striking perpendicularly to one of the sides of the square. The same problem is solved here in order to investigate the accuracy of the developed program. The side length of the square is 5.0m. The incident wave is

a shock compression plane wave of magnitude -10.0 MPa, that is suddenly applied and maintained constant in time, impinging at time t=0 at 0° angle with the horizontal x global axis, at node 1. Two cases of discretization of the free surface are examined here of equal elements or unequal ones.

a. Equal Elements

Forty equal quadratic elements are used to model the opening,with total of 84 nodes (Figure 2.5). The corner nodes are duplicated in order to take account of the discontinuity of tractions at the corners.

The variation of the tangential stresses in time is presented in Figure (2.5). The most compression stress is -50.3 MPa and appears at the corner node 81 which forms an angle of 135° with the x axis.

In Figure (2.6) the hoop stresses at the 7th time step for all the nodes are presented. The two sides of the square parallel to the wavefront, as was expected, are subjected to compressional stresses and the middle part of the two vertical sides to tensional stresses.

The deformed shape of the mesh at the 7th time step and the x displacements versus time steps are presented in Figures (2.7) and (2.8) respectively.

b. Unequal Elements

Forty unequal quadratic elements are used to model the opening,with total of 84 nodes (Figure 2.9). The size of the elements progressively decreases towards the ends of the sides, as the corners of the opening are approached. In this example, the element size reduction ratio is specified as 3.0 for both sides. Application of this ratio results in the introduction of small elements very close to the corners, as shown in Figure (2.9).

In Figure (2.10) the tangential stresses on the boundary of the opening for the time step 7, for all the nodes are depicted. In this Figure, two values are displayed at each corner, as a result of the existing discontinuity of stresses. The deformed shape of the mesh at the 7th time step is shown in Figure (2.11).

Figure (2.12) shows the variation of the hoop stress versus time steps, for nodes 51,81 and 84,71 that are near to the upper left and right corner respectively of the square and at the nodes 1,41 and 61 that are at the middle of the sides of the square. In Figure (2.13) the variation of the hoop stresses for nodes 49,50,51,52,53 and 81 near to the upper left corner of the square and in Figure (2.14) the respective stresses 69,70,71,72,73 and 84 near to the upper right corner are presented. Examination of these diagrams, indicates that the corner nodes are characterised by high stress concentrations of about -182.0 MPa, that are reduced rapidly at a small distance from the corners.

Figure (2.15) shows the variation of the x displacement with time steps, for the nodes 51,81,84,71,1,41 and 61.

It can be seen that the displacements at the duplicated corner nodes are equal and the continuity of displacements is preserved (nodes 71,84 and 51,81).

It is also clear here that at late times the cavity behaves similar to the elastostatic case loading .

2.3.1.3 Rectangular Cross Section Cavity in an Infinite Medium Swept by a Plane P Wave

A rectangular opening with sides along the x and y directions of 6.0m and 4.0m respectively is examined here

to demonstrate the optimisation technique of the program. Forty unequal quadratic elements are used to model the opening, 11 along the x side and 9 along the y side (Figure 2.16). The size of these elements progressively decreases towards the ends of the sides, as the corners of the opening are approached. In this example, the element size reduction ratio for the x sides is specified as 3.0 and for optimising the results at the corners, the program automatically calculates the ratio of 2.54 for the y sides. Application of these ratios result in the introduction of small elements very close to the corners, as shown in Figure (2.16). Because of the optimisation process, the corner elements with nodes (70,71,72) and (84,73,74) have equal length.

The distribution of the tangential stresses on the boundary of the opening for the 7th time step is shown in Figure (2.17), where two values are displayed at each corner as a result of the existing stress discontinuity. Figure (2.18) shows the variation of the hoop stress versus time steps, for the corner nodes 50,81,72 and 84 and for the middle to the sides nodes 1,41 and 81. These corner nodes are characterised as previously by high stress concentrations. The variation of the x displacements with time steps, for the same nodes are presented in Figure (2.19), where one may notice that the displacements at the duplicated corner nodes are equal (nodes 72,84 and 50,81).

The cavity behaves similar to the elastostatic case loading in late times, as was confirmed previously.

Consideration of the two previous examples, indicates that it is possible to specify the required position near a corner where stresses will be calculated. Furthermore, it is evident that even in the case of sharp corners, the stresses can be approximated with the highest possible accuracy without any increase in computational effort.

2.3.1.4 Circular Cross Section Cavity in an Infinite Medium Swept by a Plane SV Wave

In Figure (2.20), the same cavity as in Application 2.3.1.1 (radius 5.0m discretized into 8 elements) is examined, loaded by an incident SV plane wave that carries a shear stress of -10.0 MPa, at 0° angle of incidence.

The variation of the tangential stresses in time, for the nodes 1,9,11,13 and 15 which are at angles 0°, 180°, 135°, 90° and 45° respectively from the x axis, is shown in Figure (2.20).

The tangential stresses on the boundary of the cavity, for the 14th time step are depicted in Figure (2.21). In this case the most compressive stress -42.2 MPa appears at node 11 and -38.9 MPa at node 3, that form an angle of 135° and 315° respectively with x axis. Also significant tensile stresses, of the same magnitude as previously, appear at nodes 7 and 15 that form angles of 225° and 45° respectively with x axis.

The x and y displacements versus time steps are presented in Figures (2.22) and (2.23) respectively. It is also evident here that the behaviour of the cavity at late times is similar to the elastostatical case loading.

2.3.1.5 Circular Cross Section Cavity in an Infinite Medium Swept by a Plane P+SV Wave

In Figure (2.24), the same cavity as in Application of section 2.3.1.1 cavity (radius 5.0m discretized into 8 elements) is examined when loaded by a combination of an incident P of -10.0 MPa and an SV of -10.0 MPa shear plane wave, at 0° angle of incidence. The discretization is similar to the application of section 2.3.1.1.

In Figure (2.24) the variation of the hoop stresses in time at nodes 1,9,11,13 and 15 is presented.

The tangential stresses on the boundary of the opening for the 14th time step are presented in Figure (2.25). The most compressive stress -117.3 MPa appears at the 14th step at the node 11 and -110.8 MPa at node 4, that form an angle of 135° and 292.5^o respectively with x axis. Also significant tensile stresses of 55.9 and 52.4 MPa

appear at nodes 8 and 16 that form angles of 202.5° and 22.5° respectively with x axis.

The x and y displacements versus time steps are depicted in Figures (2.26) and (2.27) respectively. Once more, the behaviour of the cavity for late times is similar to the elastostatical case loading.

2.3.2 Explosion at a Point

2.3.2.1 Circular Cross Section Cavity in a Full Space Loaded by an Explosion Wave

A circular cross section cavity of radius 2.50 m, located in the same material used in all previous applications is examined under the loading of an explosive stress wave generated by an underground explosion at a point. The co-ordinates of the explosion point with respect to a general co-ordinate system with origin at the centre of the cavity, are (8.0, 0.0) m. The three types of formulas as will be presented in Chapter 4, section 4.2 are used here, with or without decay of the wave in time. The equivalent charge weight of a GP 1000 lb bomb (U.S.Army, 1984) which is 188.7 kg of explosive weight is used here. The attenuation coefficient is taken n=2.75, and the coupling factor f=1.0. The total time interval examined here is 0.015 s separated into 20 time steps.

a. Cubic Root Formula Without Decay in Time

The stress wave generated by the explosive, is calculated using the cubic root formula for every node and is maintained constant without decay in time. The direct compressive stress derived from this formula at node 1 has been calculated analytically and is -33.7 MPa. In Figures (2.28) and (2.29), the tangential stresses on the periphery of the opening and the deformed shape of the mesh at the 6th time step are presented. The change of these hoop stresses in MPa and displacements in mm versus the time steps at the nodes 1,15,13,11 and 9 is demonstrated in Figure (2.30). The maximum tangential compressional stress of -43.3 MPa is taking place at node 15 at the 5th time step, that forms an angle of 45° with the x global axis. The maximum tensional stress is 10.3 MPa and appears at node 1 at the 6th time step. The maximum stresses are decreasing slightly in the subsequent time steps. The peak values of displacements appear at the 9th time step with values -7 mm at node 1 which is increasing slightly in time, and +1.5 mm at node 9 (Figure 2.31). At late times, all the deformation curves of the nodes assume a parallel orientation indicating that the cavity undergoes a rigid body translation.

b. Cubic Root Formula With Decay in Time

The direct compression stress computed from this formula at node 1 has been calculated analytically and is 33.7 MPa. The plots of the preprocessing phase of the program, are illustrated in Figure (2.32), where the mesh and the rays of the incident wave are shown. Figure (2.33) shows the tangential stresses on the periphery of the opening and the deformed shape of the mesh at the 5th time step. The change of the tangential stresses and displacements versus the time steps at nodes 1, 15,13,11 and 9 is demonstrated in Figure (2.34). The maximum tangential compressional stress is -27.5 MPa and appears at node 14 at the 5th time step, which forms an angle of 67.5° with the x global axis and the maximum tensional one is 5.6 MPa and appears again at node 1 at the 5th time step. The stresses rapidly increase to the peak value and consequently drop to zero at the 16th time step. The peak values of x-displacements appear at the 6th time step for node 1, and at 9th time step for node 9 (Figure 2.35).

At the 16th time step, the stresses and displacements are reduced to approximately zero and the cavity returns to its original undeformed shape, following the passage of the shock wave.

c. Square Root Formula With Decay in Time

The attenuation coefficient n and the empirical coefficient B are taken 1.57 and 11.65 respectively. In using these coefficients in the square root formula, results in direct compressive stress about -33.7 MPa at node 1. This is the same stress value as the one calculated in the previous case of the cubic root formula indicating that the adoption of these values for n and B makes the comparison between the two cases easier. Figures (2.36) and (2.37) present for the 6th time step, the tangential stresses on the periphery of the opening and the deformed shape of the mesh respectively. The change of these hoop stresses and the displacements versus the time steps for the nodes 1,15,13,11 and 9 is demonstrated in Figures (2.38) and (2.39) respectively. The maximum tangential compressional stress is -32.5 MPa and appears at node 15 at the 6th time step, that forms an angle of 77.5° with the x global axis. The maximum tensional stress is 2.6 MPa and appears at node 1 also at the 6th time step. The maximum stresses are decreasing rapidly at the subsequent time steps. Similarly the peak values of displacements appear at the 6th time step with value -3.5 mm and is maintained about constant in time for node 1, and 1.7 mm for node 9 (Figure 2.39).

Here is also observed that at time step 16 stresses and displacements are reduced to approximately zero and the cavity returns to its original undeformed shape, following the passage of the shock wave.

2.3.2.2 Rectangular Cross Section Cavity in an Infinite Space Loaded by an Explosion at a Point P Wave

A rectangular cross section cavity of 6.0m width by 4.0m height, is subjected to the loading resulting from an explosion at a point. The equivalent charge weight of a GP 1000 lb bomb (U.S.Army, 1984), corresponding to 188.7 kg of explosive weight is used, while the attenuation coefficient is taken as n=2.75 and the coupling factor is set to f=1.0. The charge is located at a point with co-ordinates (8.0, 0.0) m to a general co-ordinate system with origin at the centre of the opening. The cubic root formula with decay in time is used for the estimation of the incident pressures at the nodes of the mesh.

In a similar manner to the application of section 2.3.1.3 (Figure 2.16), forty unequal quadratic elements, as in of section 2.3.1.3, are used to model the opening, 11 along the x side and 9 along the y side. The size of these elements progressively decreases towards the ends of the sides, as the corners of the opening are approached. In this example, the element size reduction ratio for the x side is specified as 3.0 and the program automatically calculates the ratio of 2.54 for the y side. The distribution of the tangential stresses on the boundary of the opening is shown in Figure (2.40), where two values are displayed at each corner as a result of the existing stress discontinuity. The deformed shape of the mesh for the 6th time step is shown in Figure (2.41).

Figure (2.42) shows the variation of the hoop stress versus time steps, for nodes 68,70,71,72,84,73 and 74 that are near the upper right corner of the rectangle. The concentration of high stresses is rapidly reduced at a small distance from the corners.

Figure (2.43) shows the variation of the x displacement in time, for the same nodes.

At late times stresses and displacements are reduced to nearly zero values and following the passage of the shock wave, the cavity returns to its original undeformed shape.

It seems that it is possible to specify the required position near a corner where stresses will be calculated as it has been mentioned in application 2.3.1.3. Moreover, it is evident that even in the case of sharp corners, the stresses can be approximated with the highest accuracy without any increase in computational effort.

2.3.3 Explosion Inside a Cylindrical Cavity

The problem of a cylindrical cavity in infinite medium, subjected to a suddenly applied internal explosion has been examined by the direct BEM by Kontoni et al. (1987) in Laplace transform domain. The same problem is solved here in the same domain.

The radius of the cavity is 6.35 m, the modulus of elasticity is 207,000 MPa, Poisson's ratio is 0.33 and the mass density is 8.48 Mg/m^3. The internal radial pressure of 10.0 MPa is applied suddenly and maintained constant.

Figure (2.44) shows the variation of the hoop stress with time for the node of the cavity that is at 0° angle with x axis. The respective results of Kontoni et al. are also presented. It can be seen that there is a good agreement between the two solutions.

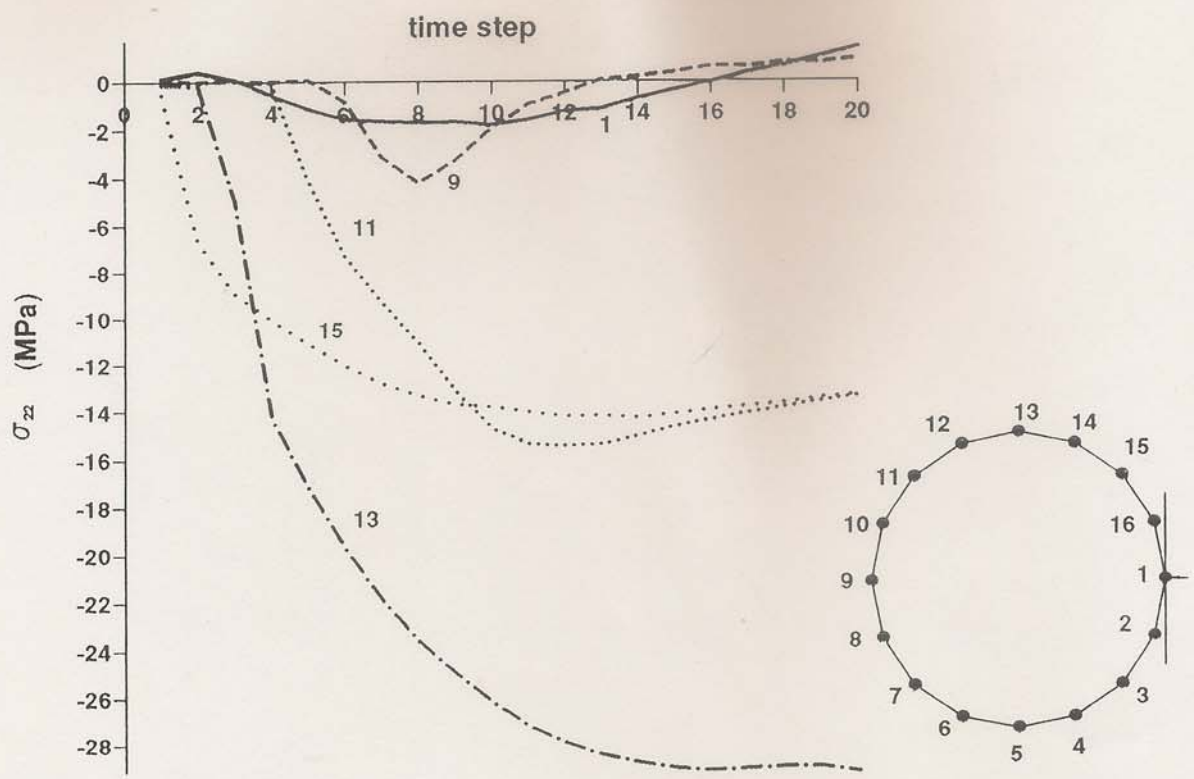


Figure 2.1 Tangential (hoop) stresses versus time steps

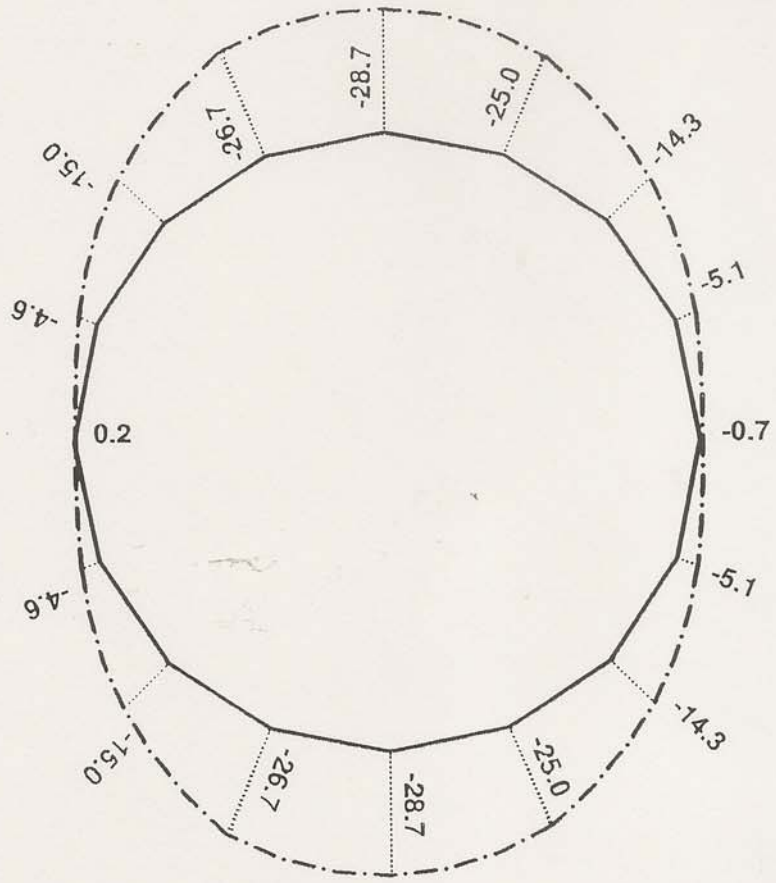


Figure 2.2 Tangential (hoop) stresses at 14th time step

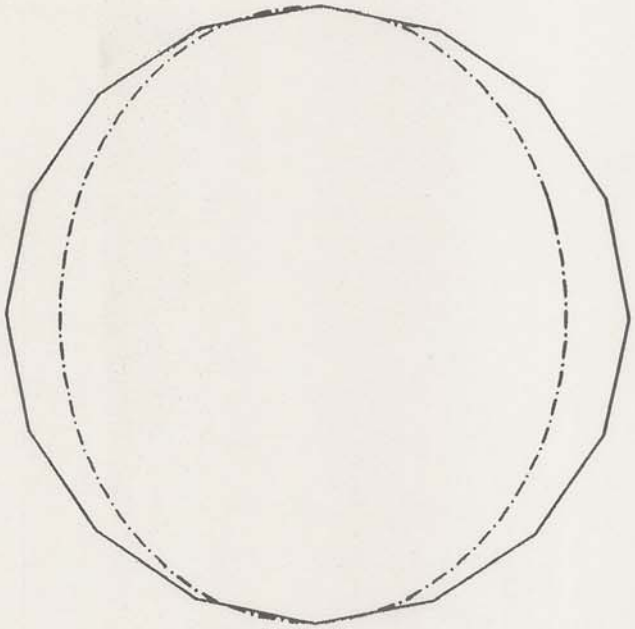


Figure 2.3 Deformed shape at 14th time step

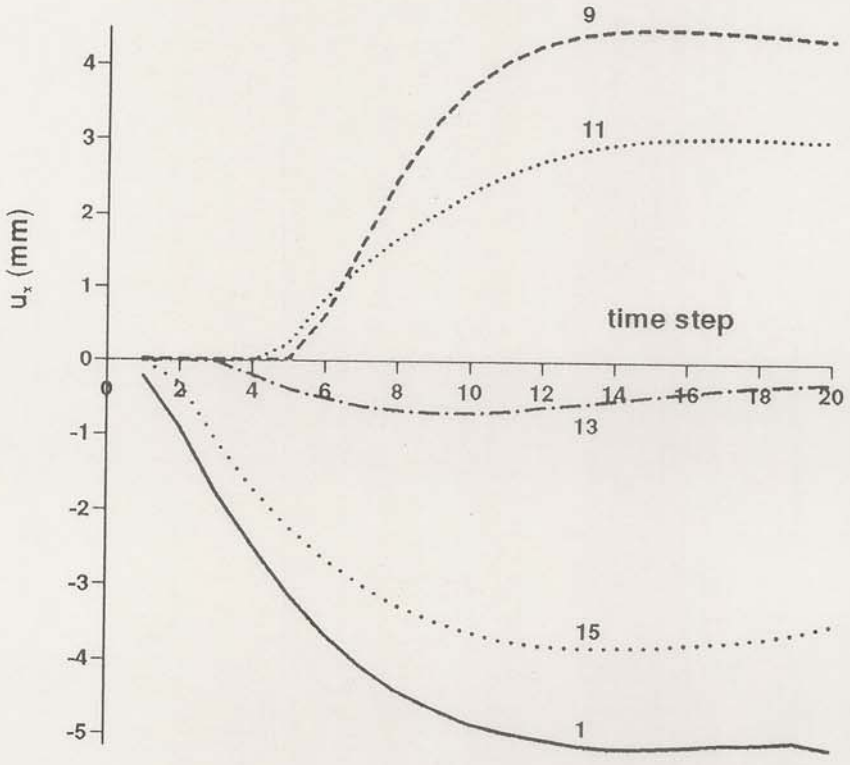


Figure 2.4 x displacements versus time steps

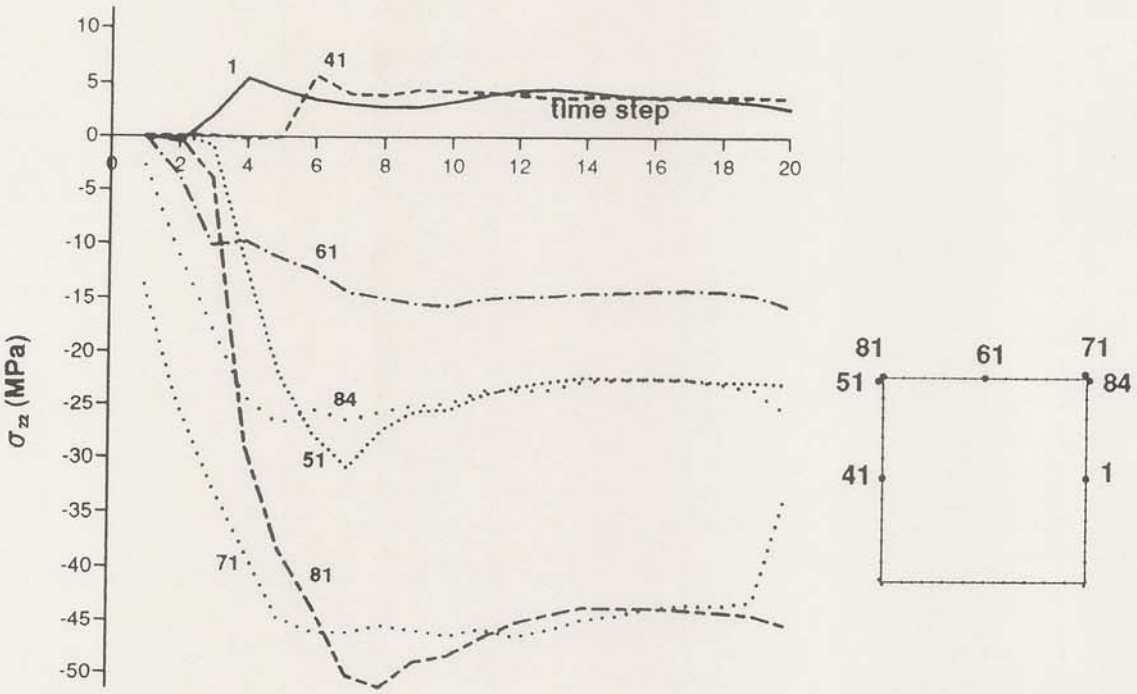


Figure 2.5 Tangential stresses versus time steps

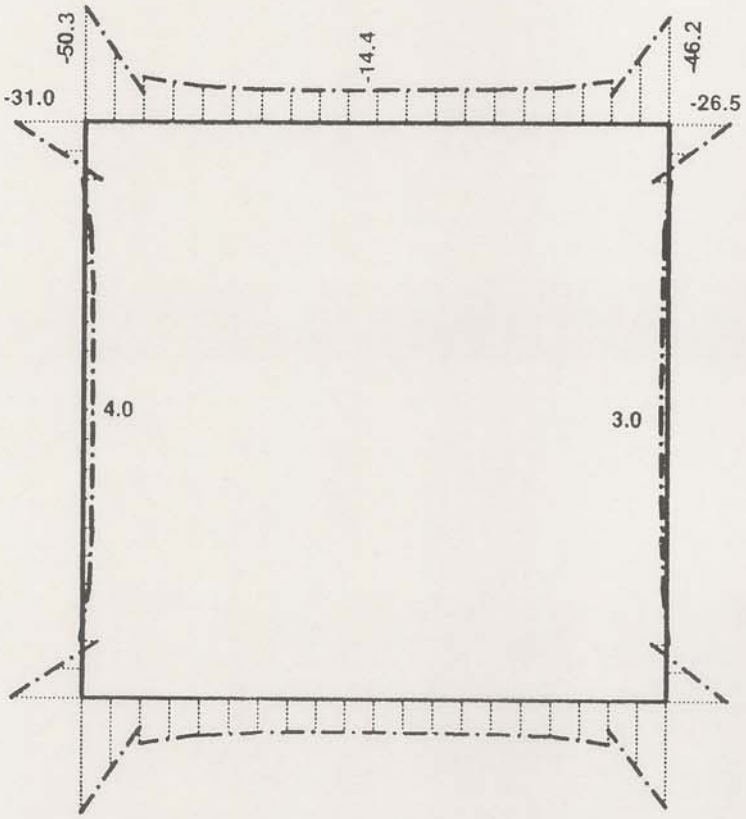


Figure 2.6 Tangential stresses at 7th time step

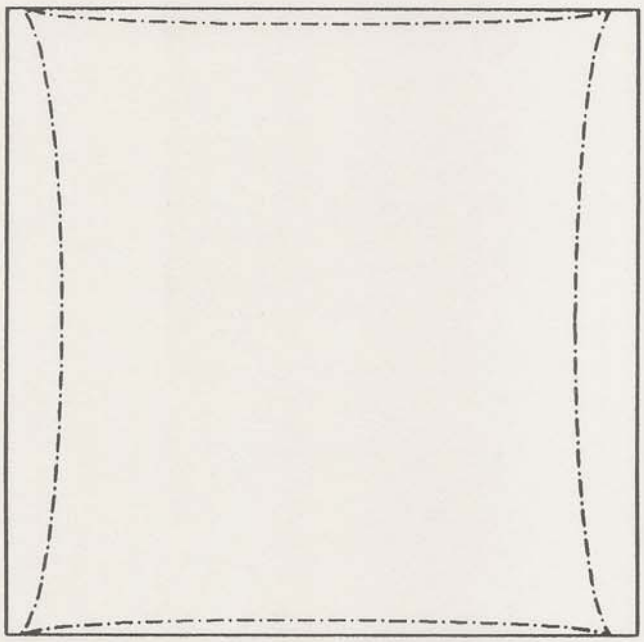


Figure 2.7 Deformed shape at 7th time step

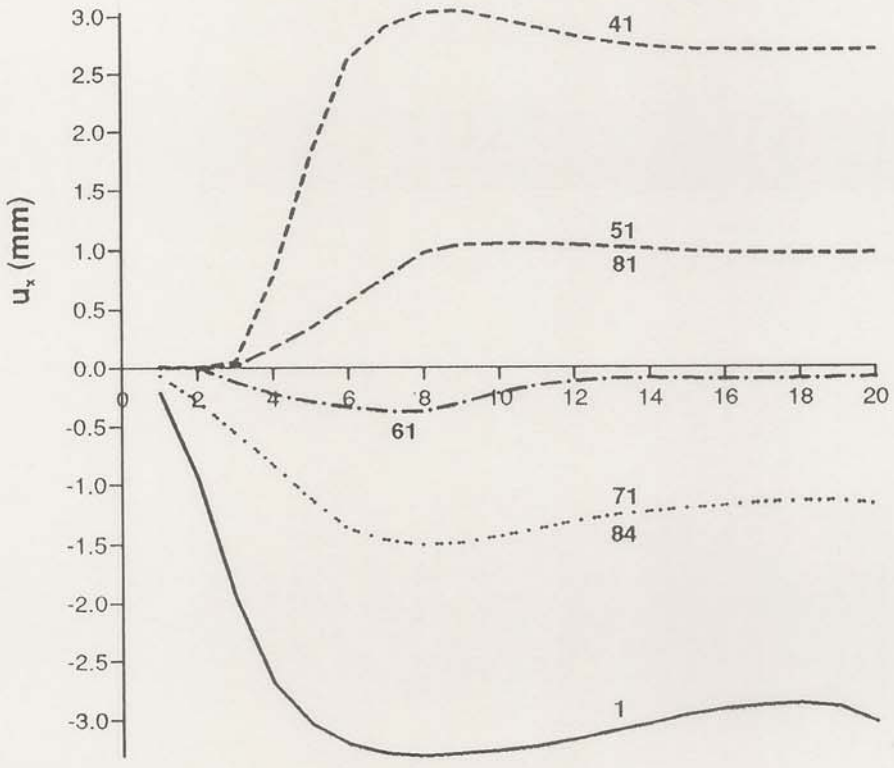
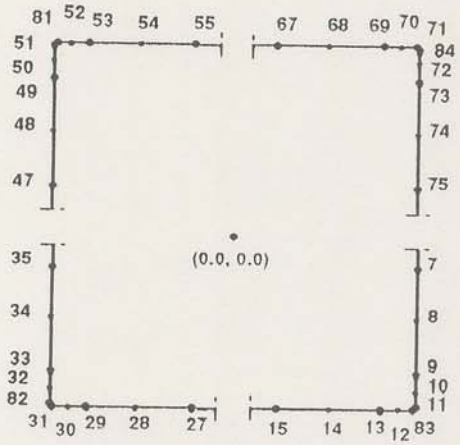


Figure 2.8 x displacements versus time steps



Node co-ordinates

7 (2.500,-2.417)
8 (2.500, -2.448)
9 (2.500, -2.479)
10 (2.500, -2.490)
11 (2.500, -2.500)
12 (2.490, -2.500)
13 (2.479, -2.500)
14 (2.448, -2.500)
15 (2.417, -2.500)
83 (2.500, -2.500)

Figure 2.9 Square discretized into 40 unequal elements

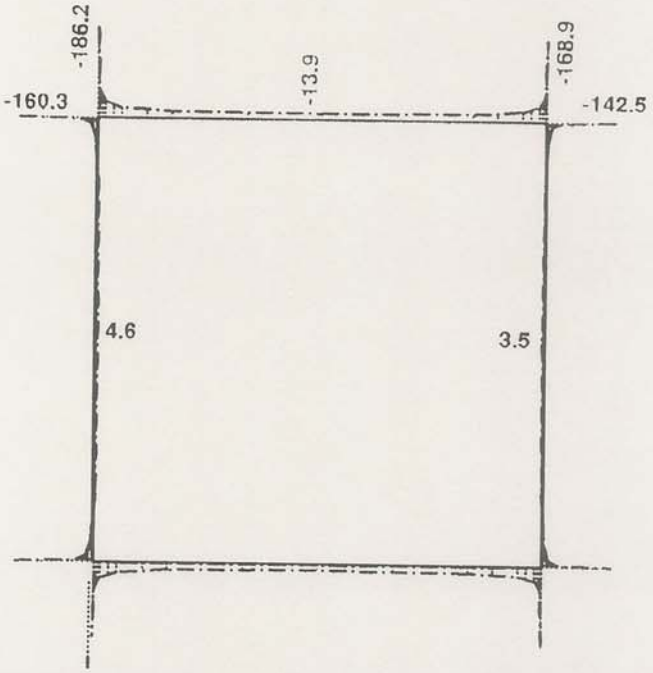


Figure 2.10 Tangential stresses at 7th time step

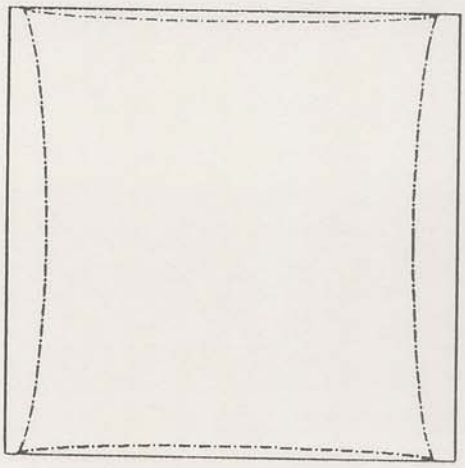


Figure 2.11 Deformed shape at 7th time step

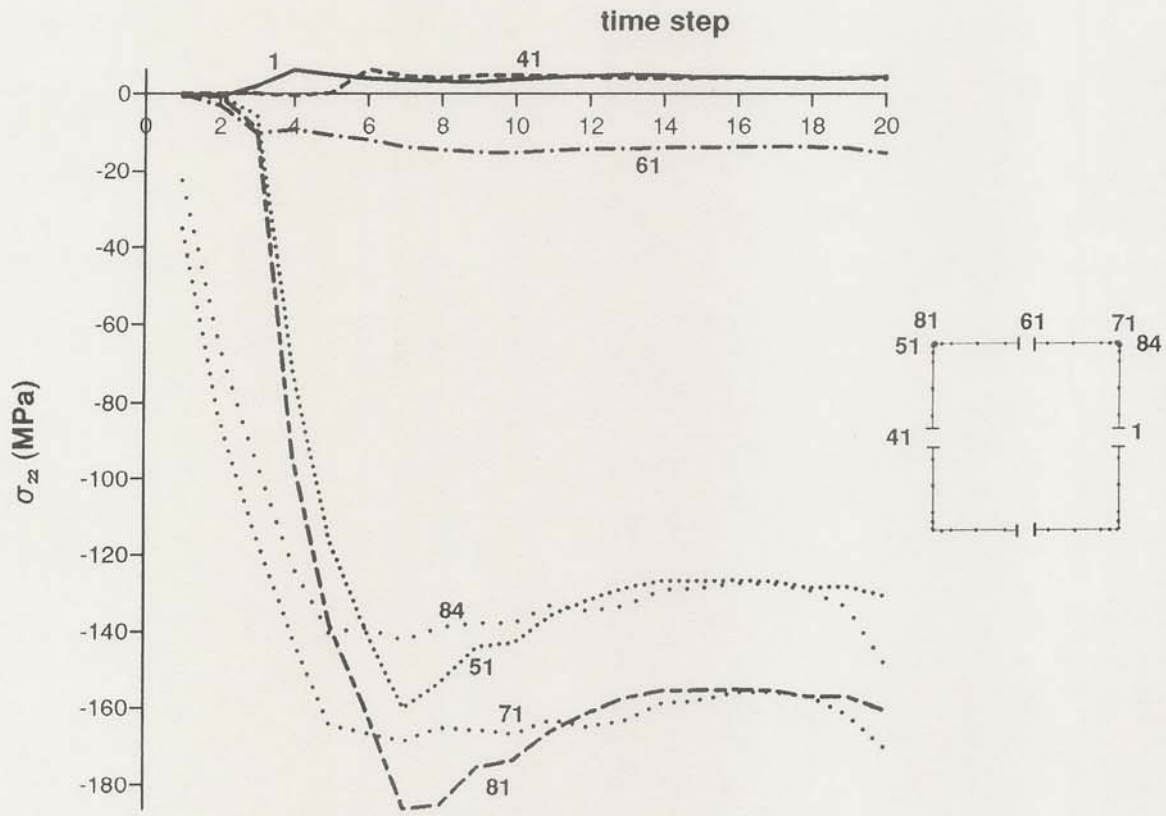


Figure 2.12 Tangential stresses versus time steps

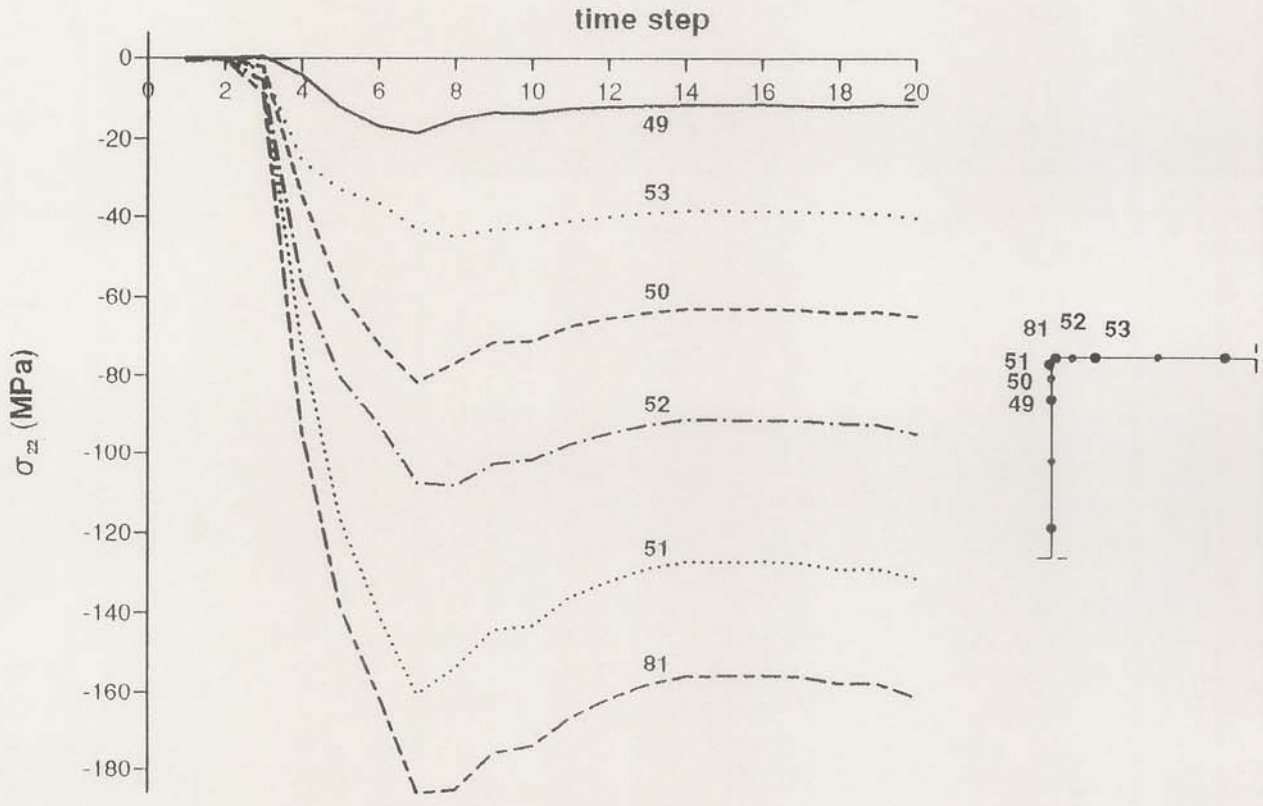


Figure 2.13 Tangential stresses versus time steps

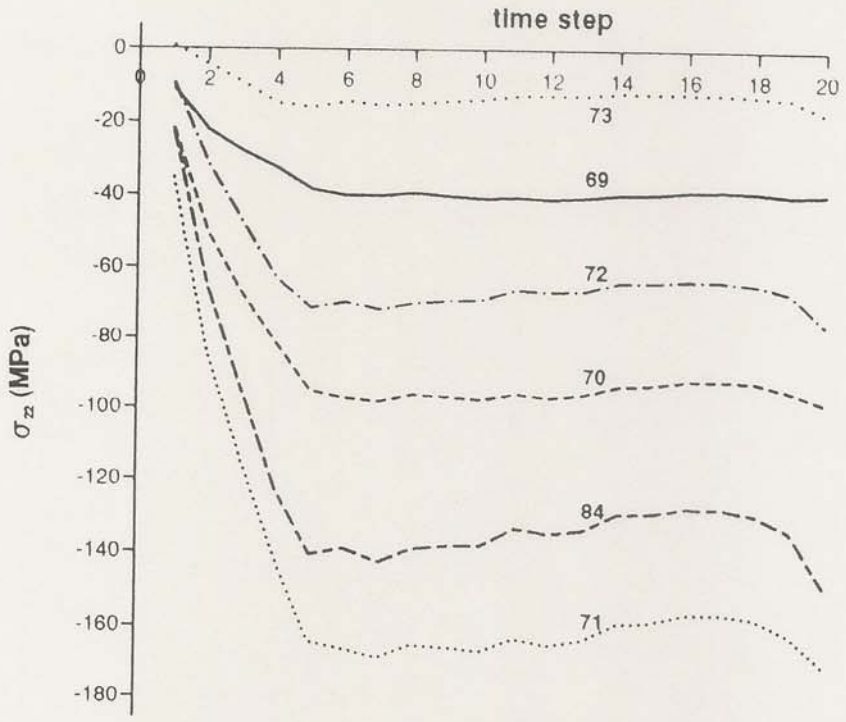


Figure 2.14 Tangential stresses versus time steps

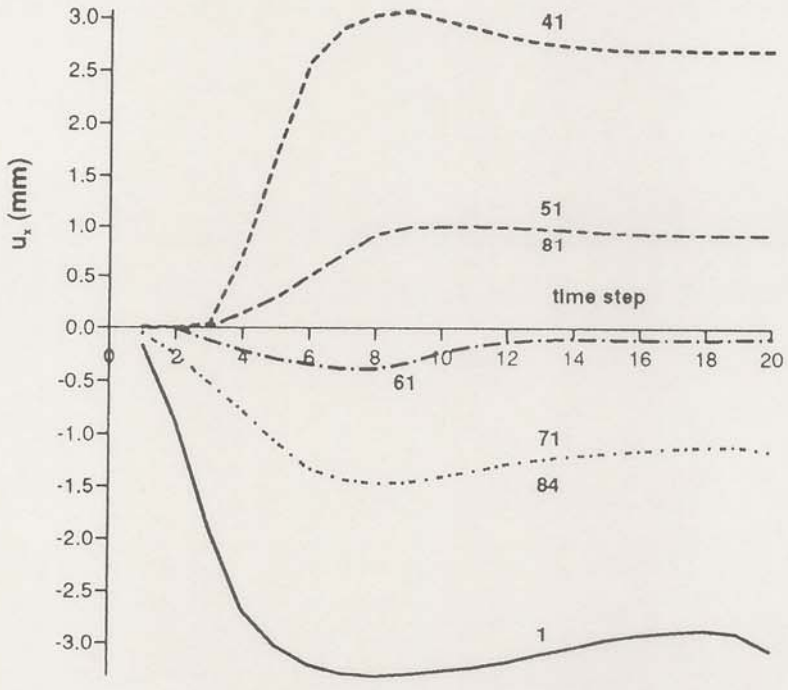
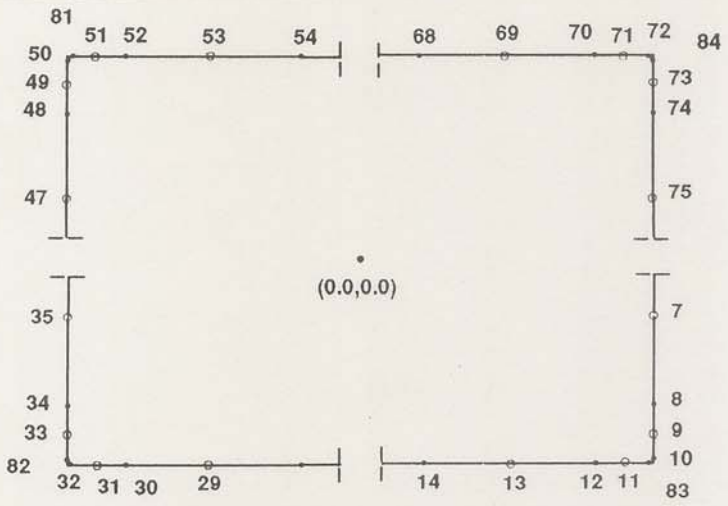


Figure 2.15 x displacements versus time steps



Node co-ordinates for progressively decreasing rates r_x / r_y

2.54 / 3.0	3.0 / 3.0
68 (2.912, 2.000)	68 (2.951, 2.000)
69 (2.944, 2.000)	69 (2.969, 2.000)
70 (2.975, 2.000)	70 (2.988, 2.000)
71 (2.988, 2.000)	71 (2.994, 2.000)
72 (3.000, 2.000)	72 (3.000, 2.000)
73 (3.000, 1.988)	73 (3.000, 1.988)
74 (3.000, 1.975)	74 (3.000, 1.975)
75 (3.000, 1.938)	75 (3.000, 1.938)
84 (3.000, 2.000)	76 (3.000, 1.901)

Figure 2.16 Geometry of the mesh with unequal elements

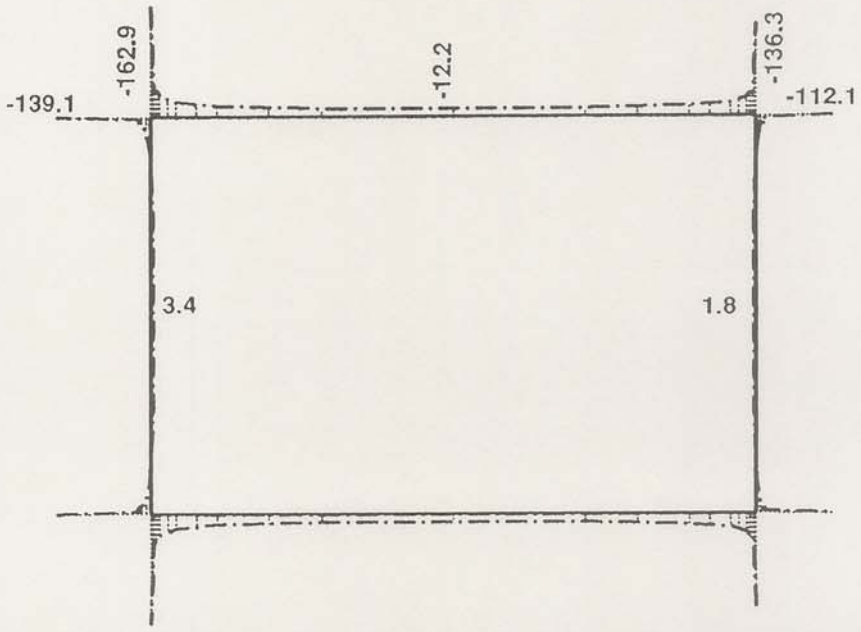


Figure 2.17 Tangential stresses at 7th time step

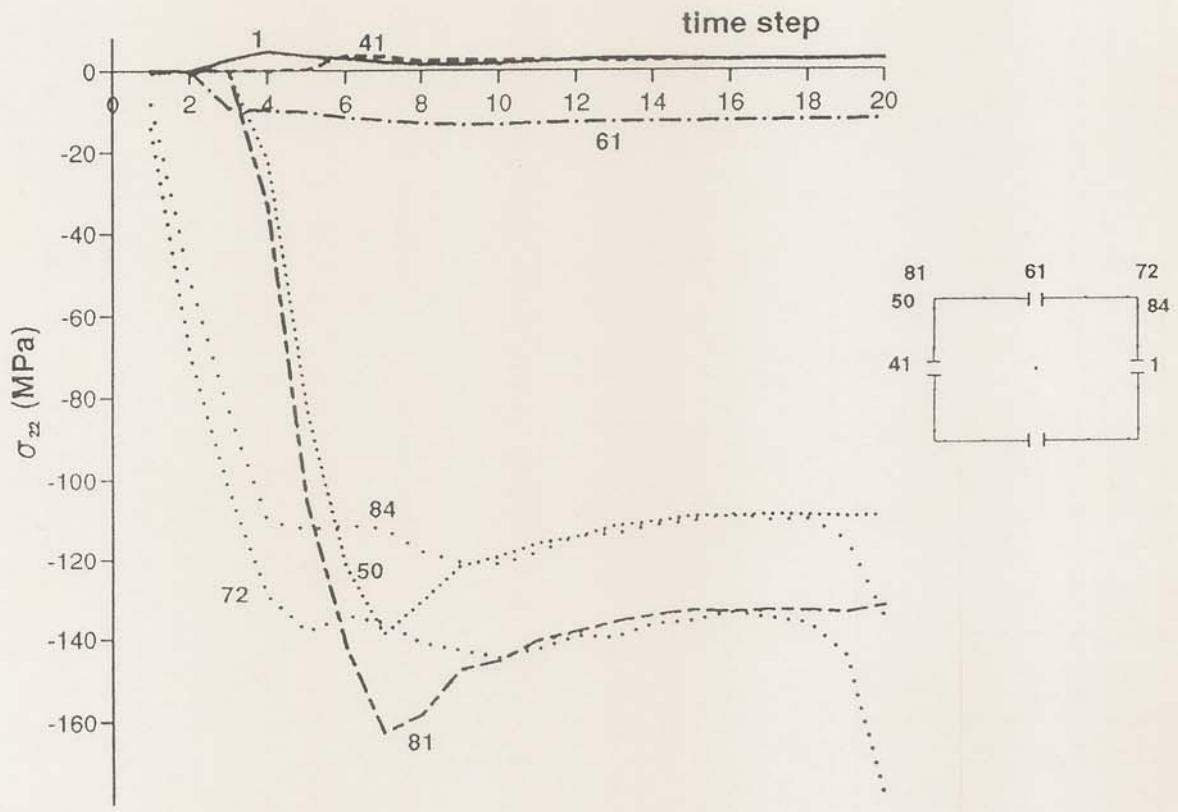


Figure 2.18 Tangential stresses versus time steps

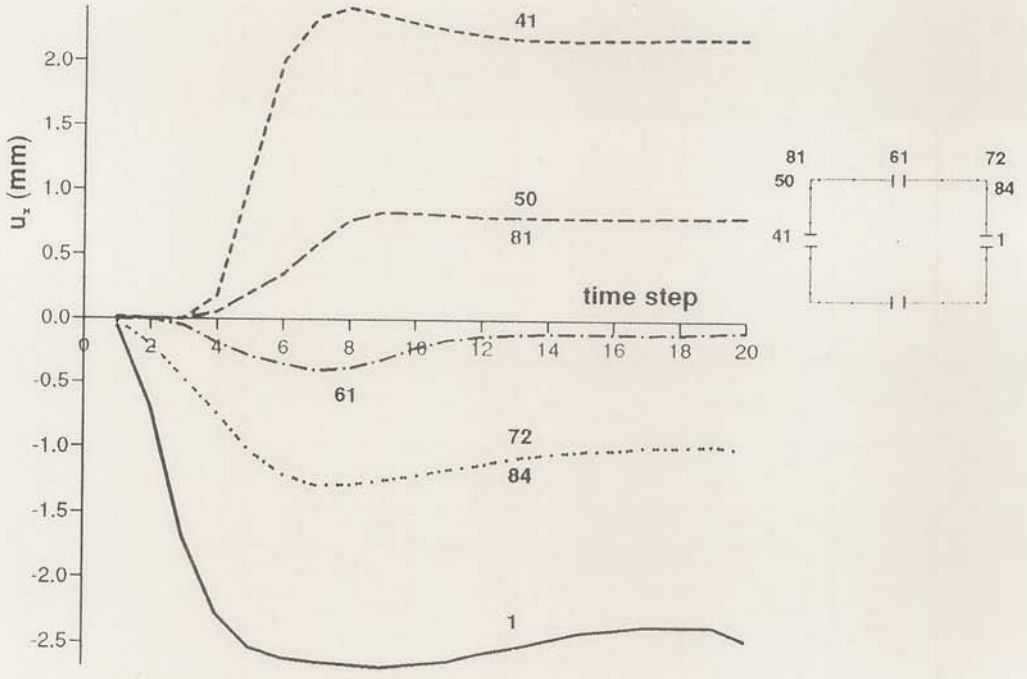


Figure 2.19 x displacements versus time steps

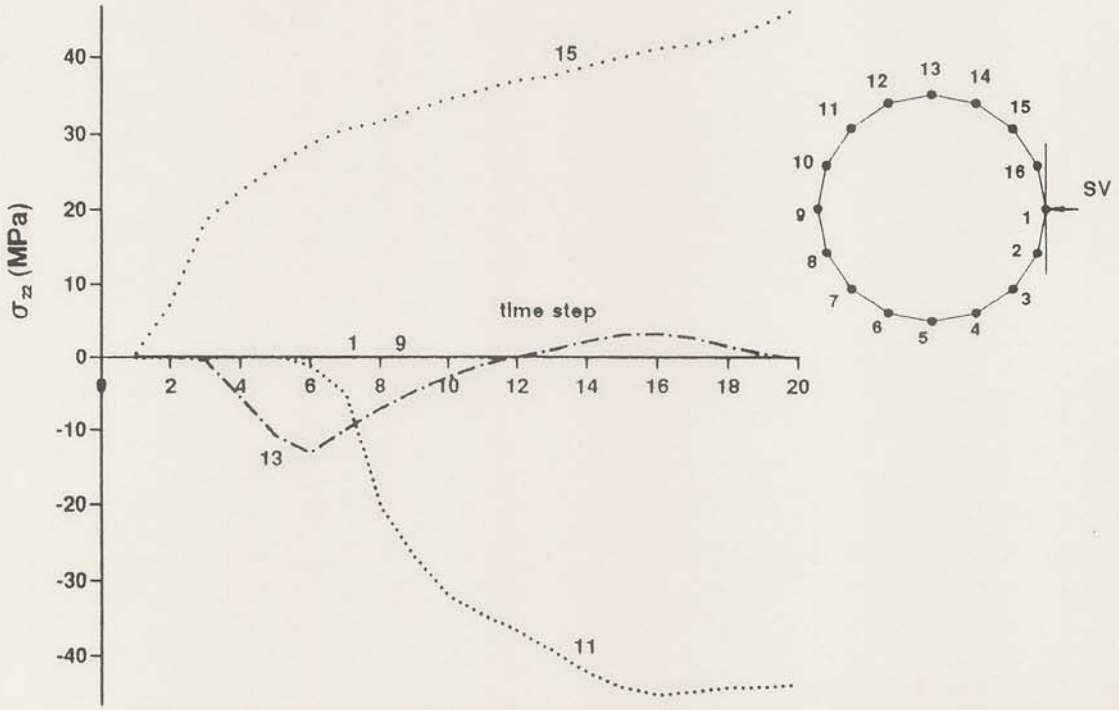


Figure 2.20 Tangential stresses versus time steps

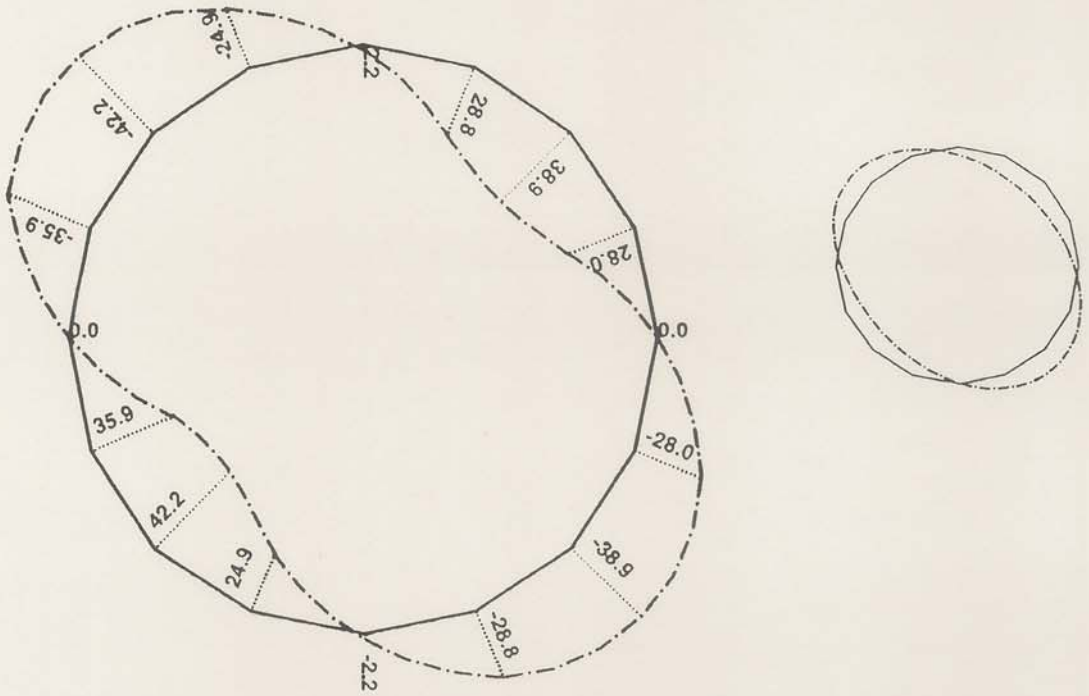


Figure 2.21 Tangential stresses and deformed shape at 14th time step

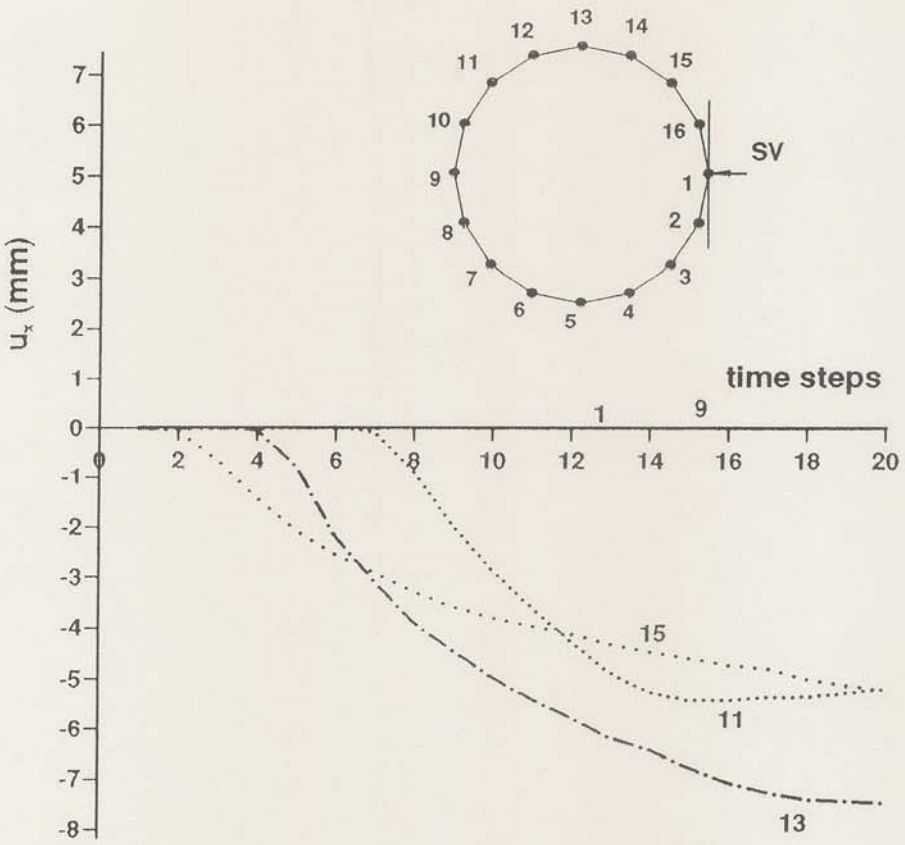


Figure 2.22 x displacements versus time steps

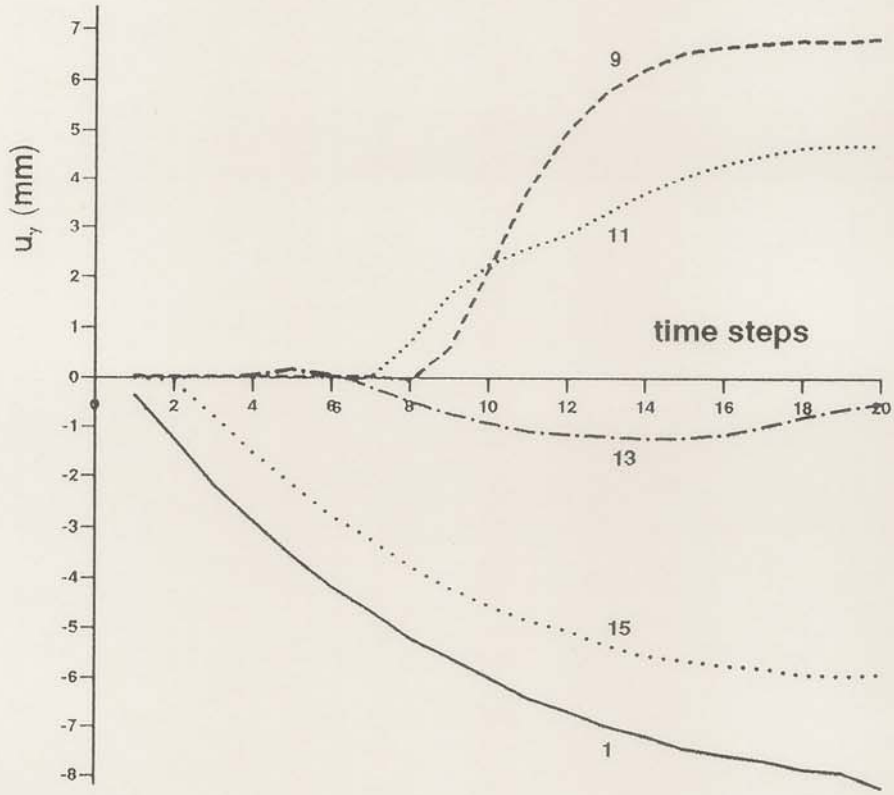


Figure 2.23 y displacements versus time steps

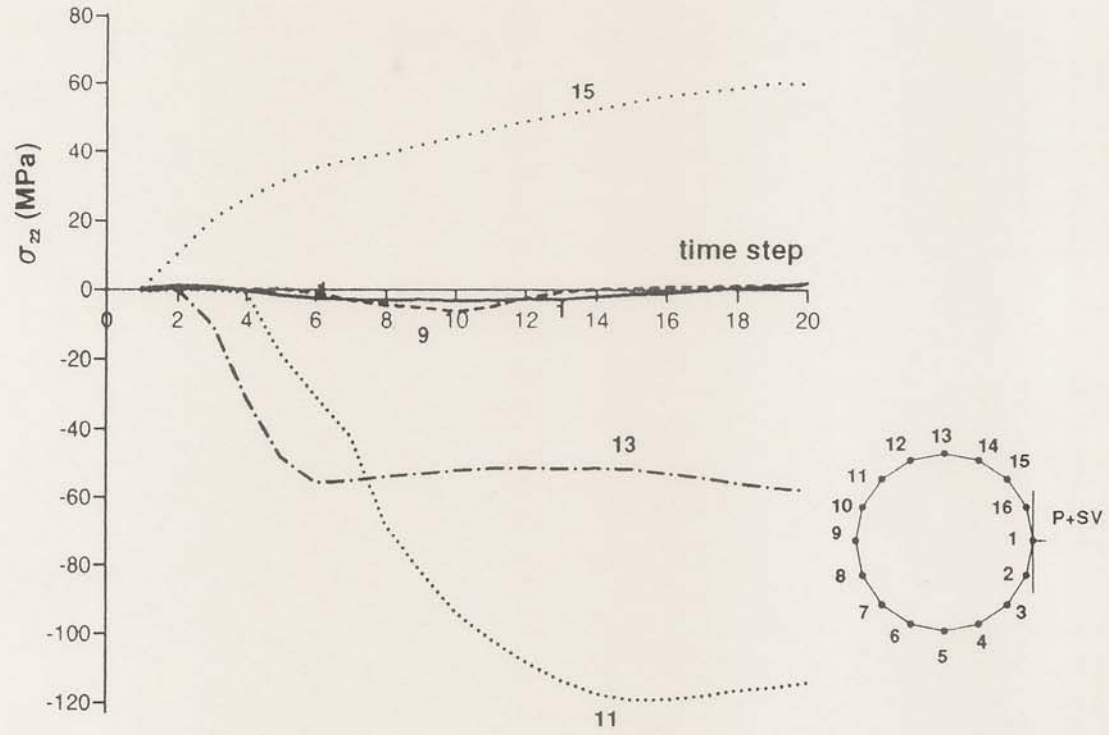


Figure 2.24 Tangential stresses versus time steps

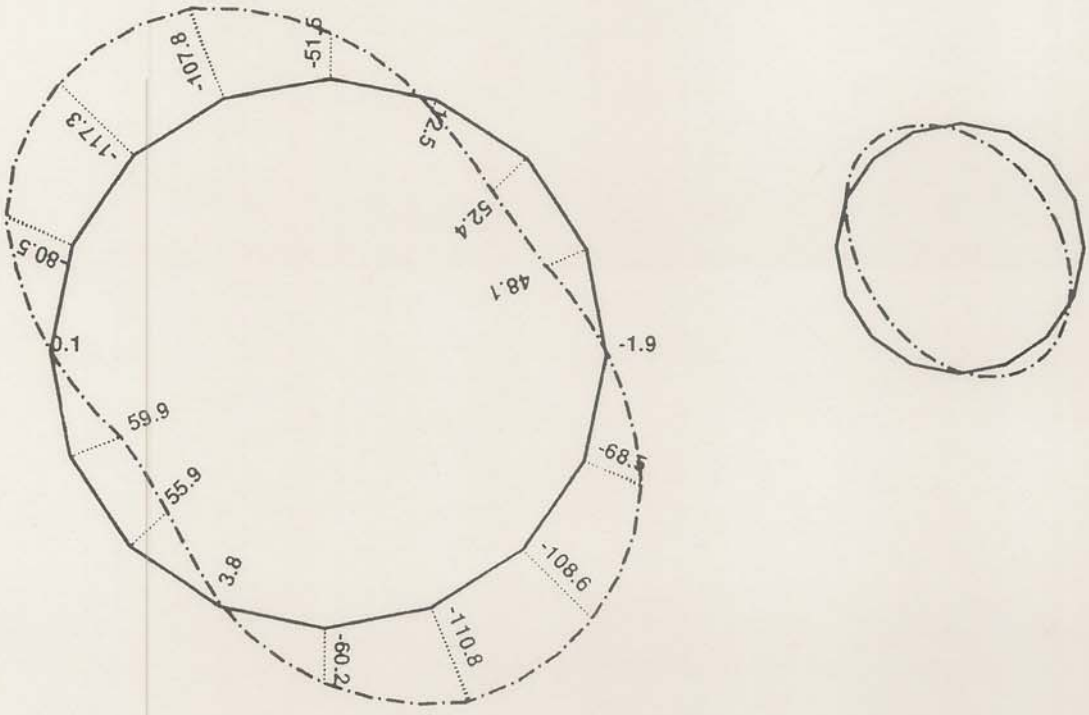


Figure 2.25 Tangential stresses and deformed shape at 14th time step

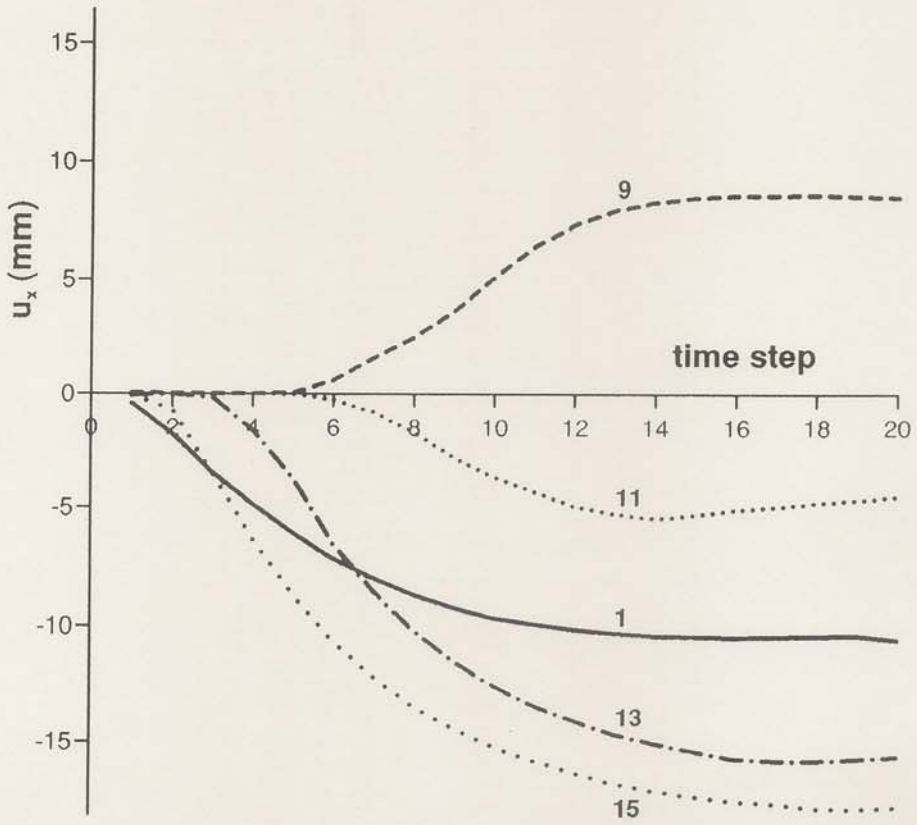


Figure 2.26 x displacements versus time steps

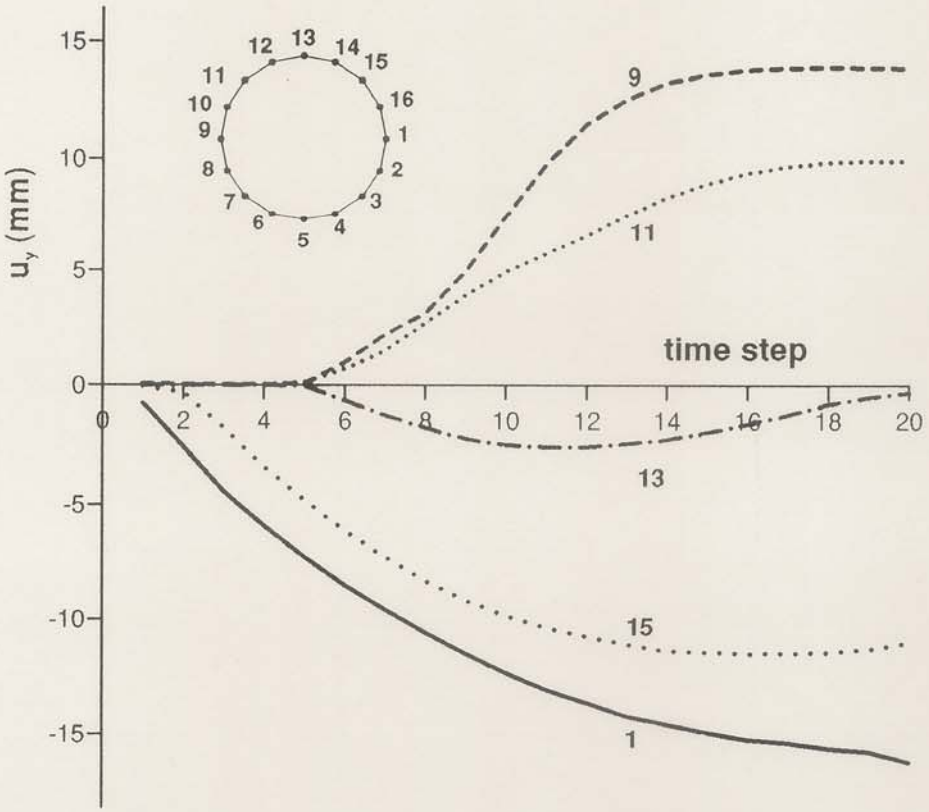


Figure 2.27 y displacements versus time steps

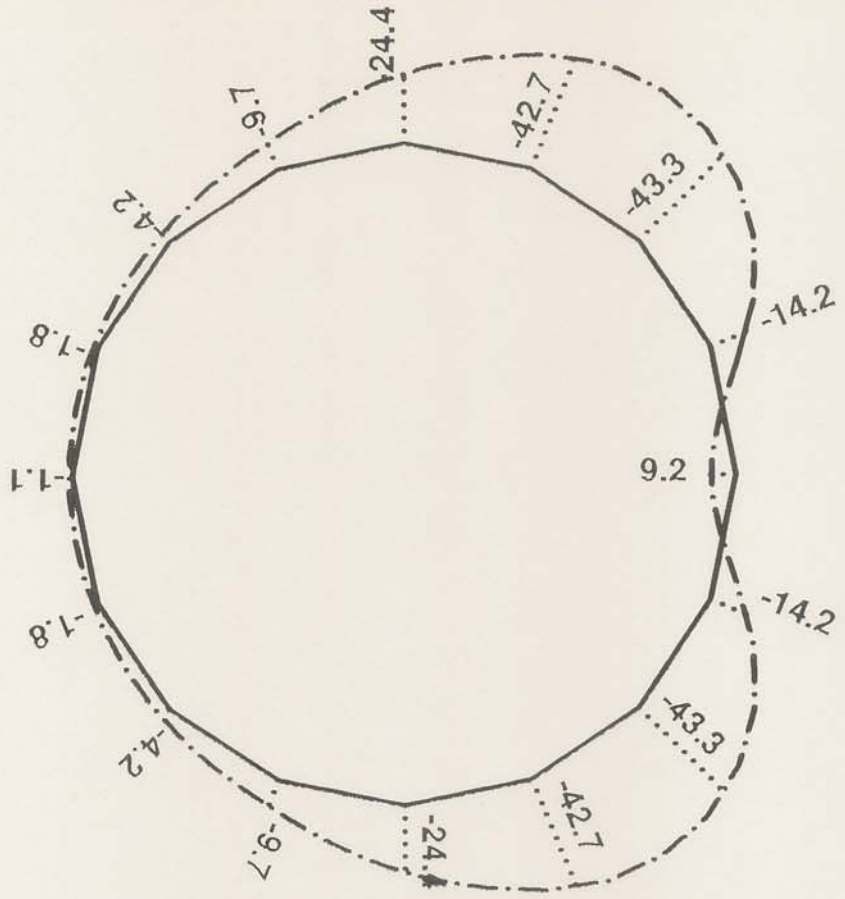


Figure 2.28 Tangential stresses at 6th time step

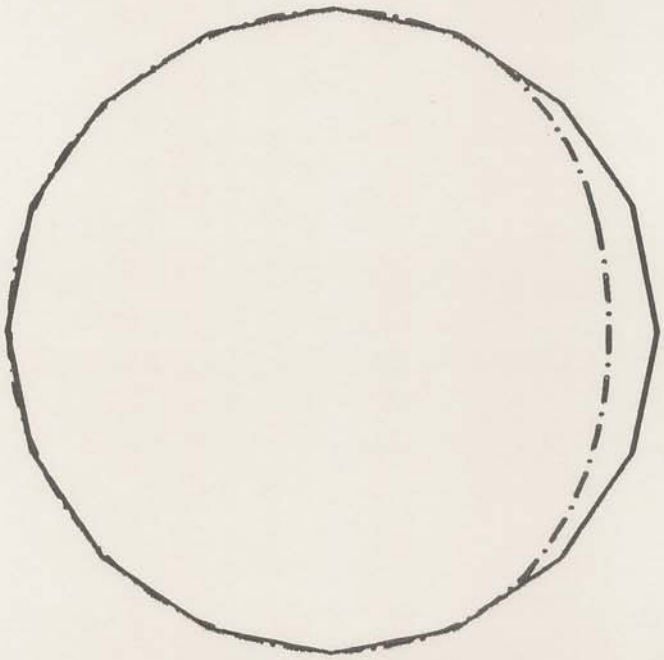


Figure 2.29 Deformed shape at 6th time step

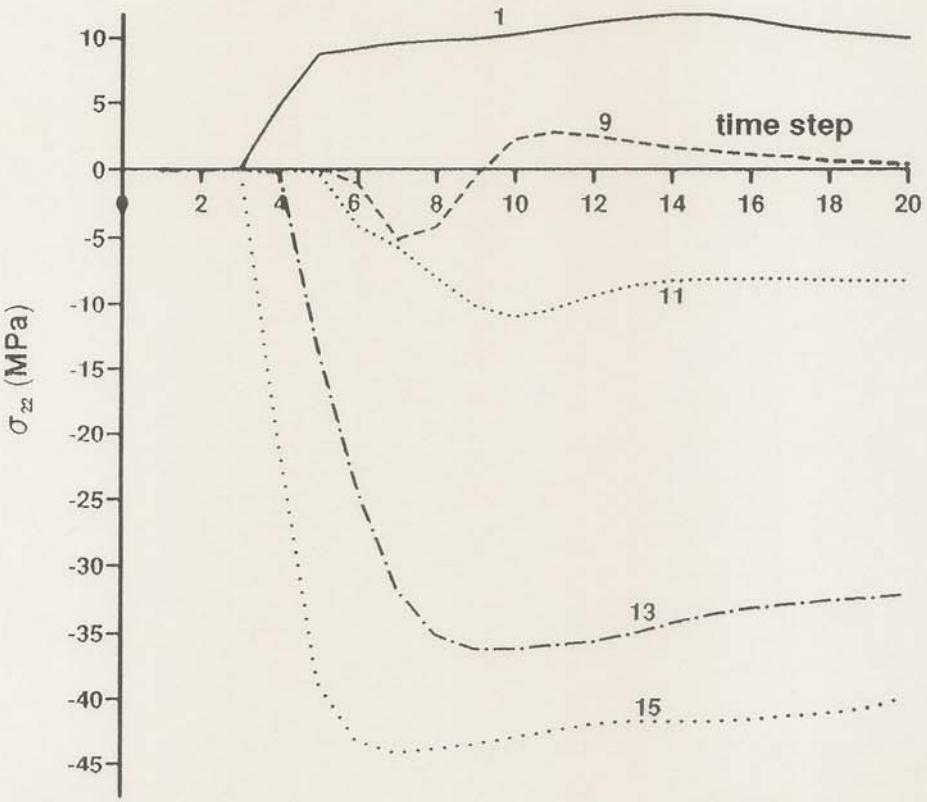


Figure 2.30 Tangential stresses versus time steps

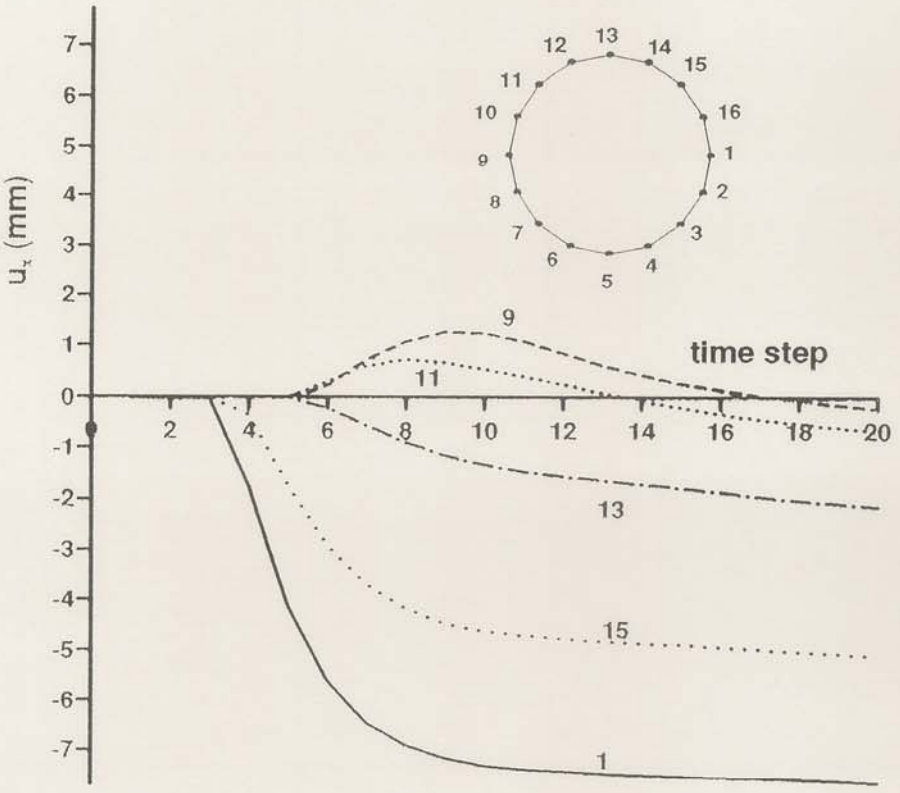


Figure 2.31 x displacements versus time steps

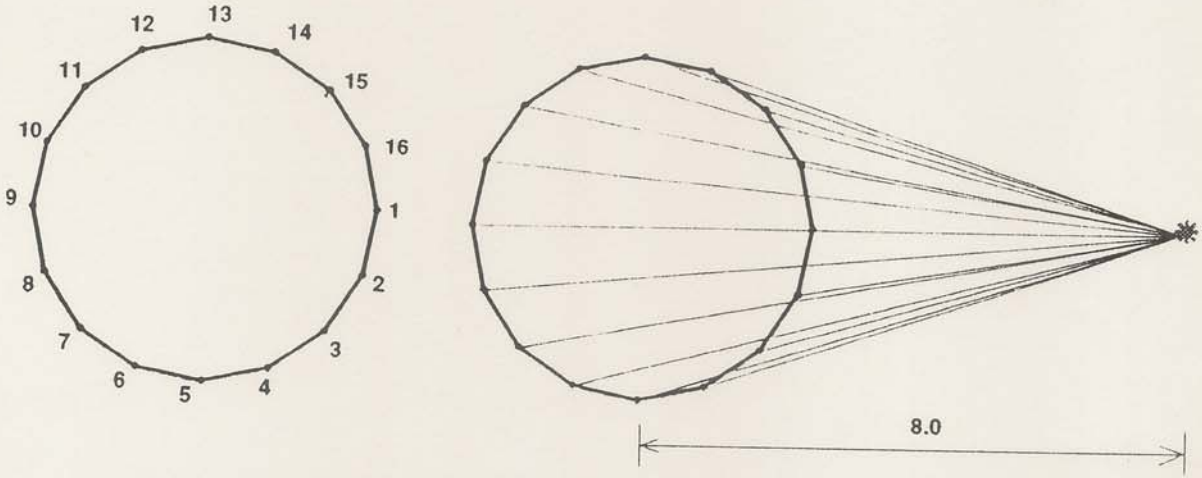


Figure 2.32 Geometry of the problem

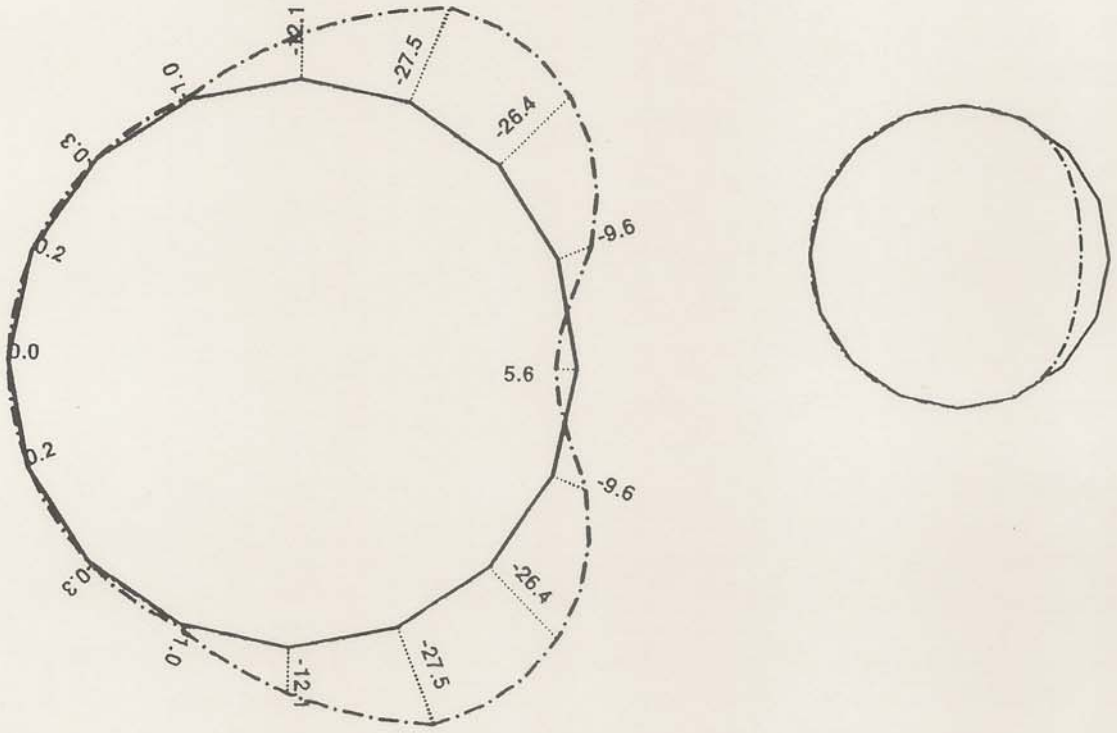


Figure 2.33 Tangential stresses and deformed shape at 5th time step

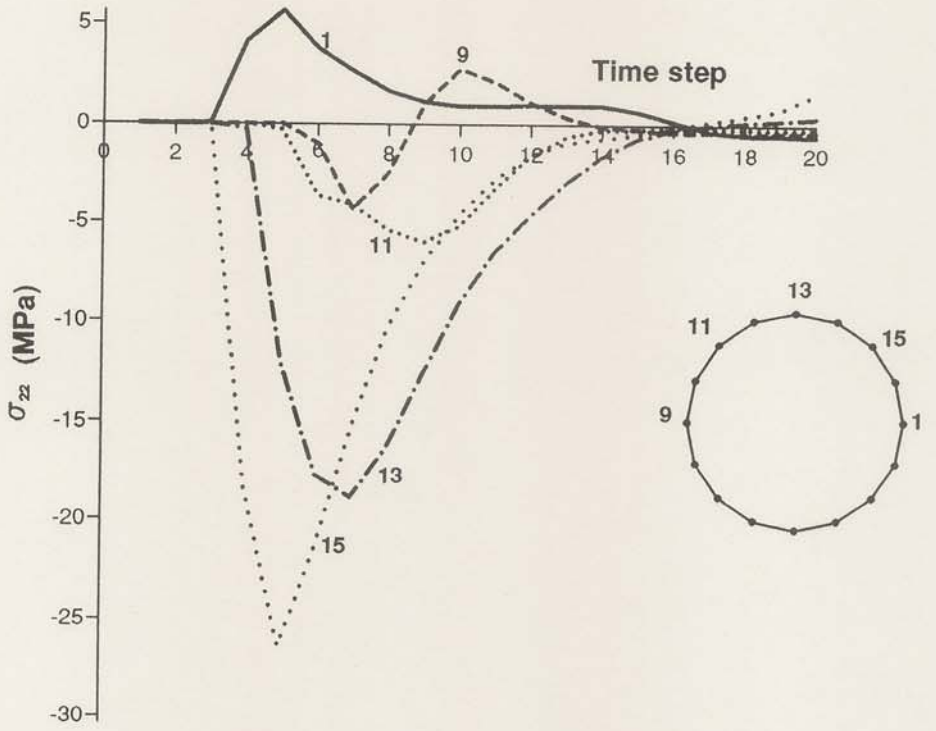


Figure 2.34 Tangential (hoop) stresses versus time step

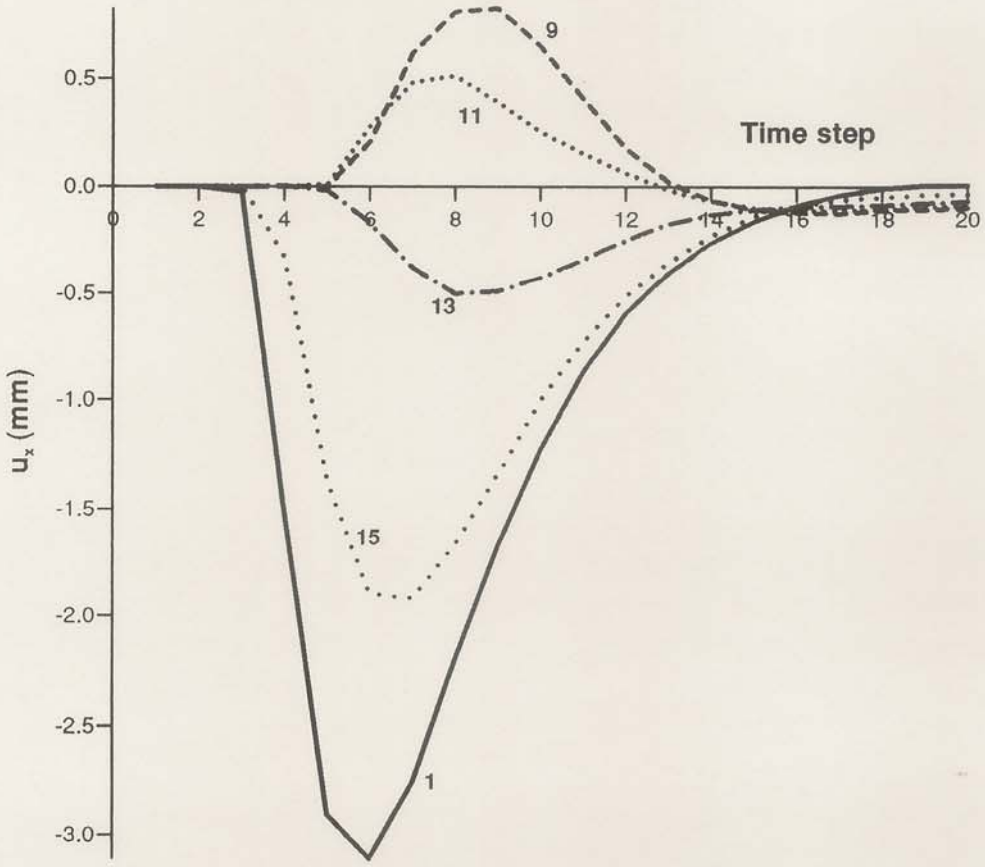


Figure 2.35 x displacements versus time step

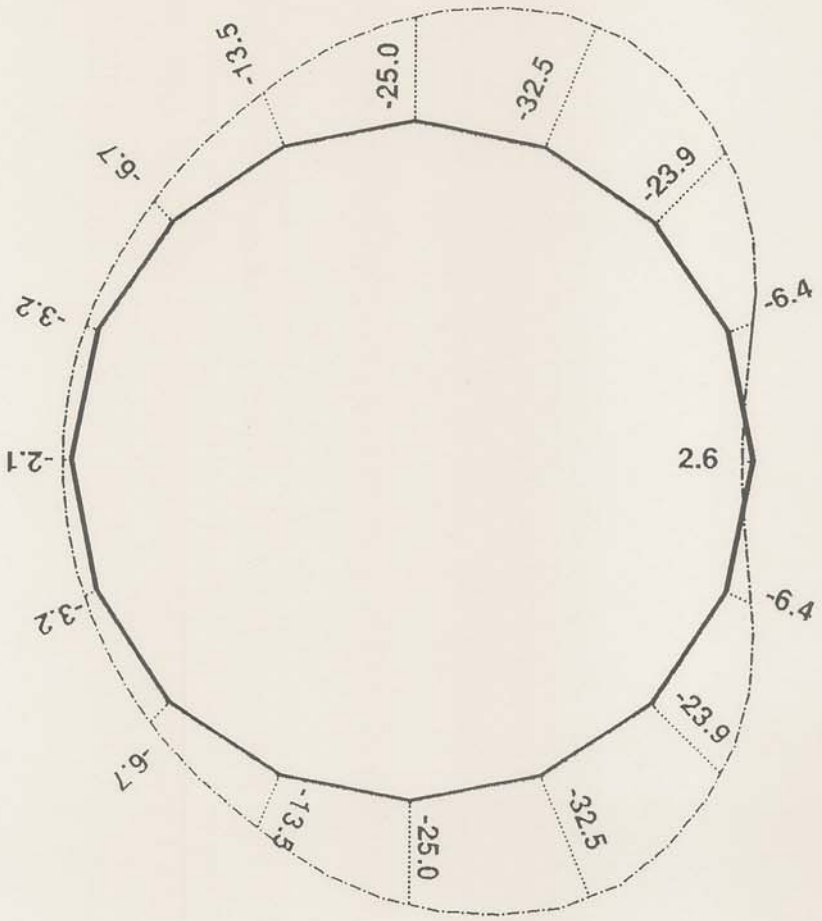


Figure 2.36 Tangential stresses at 6th time step

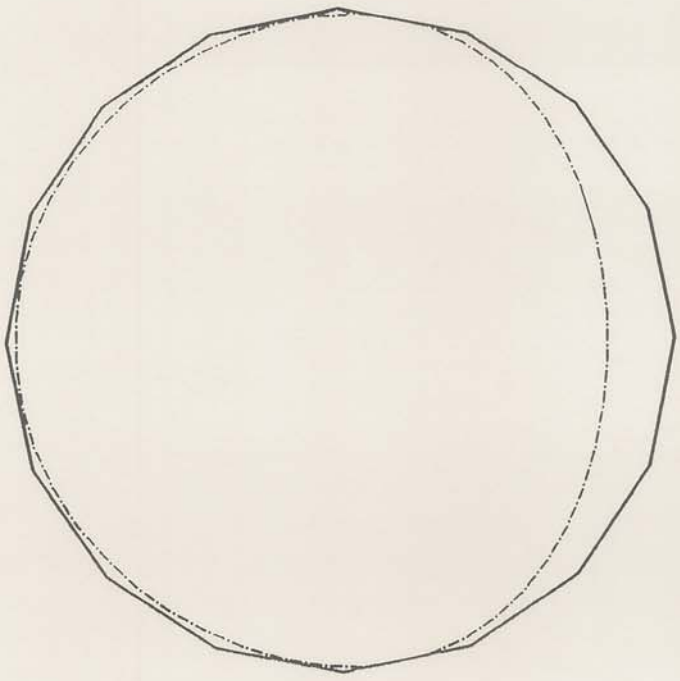


Figure 2.37 Deformed shape at 6th time step

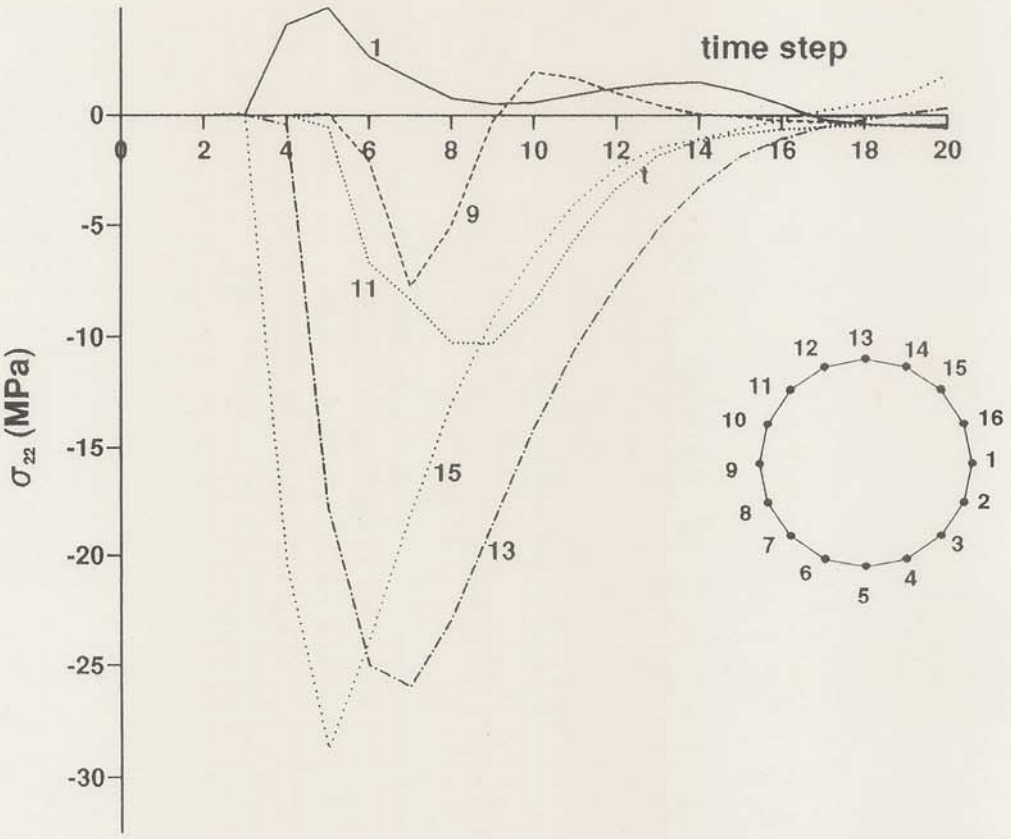


Figure 2.38 Tangential stresses versus time steps

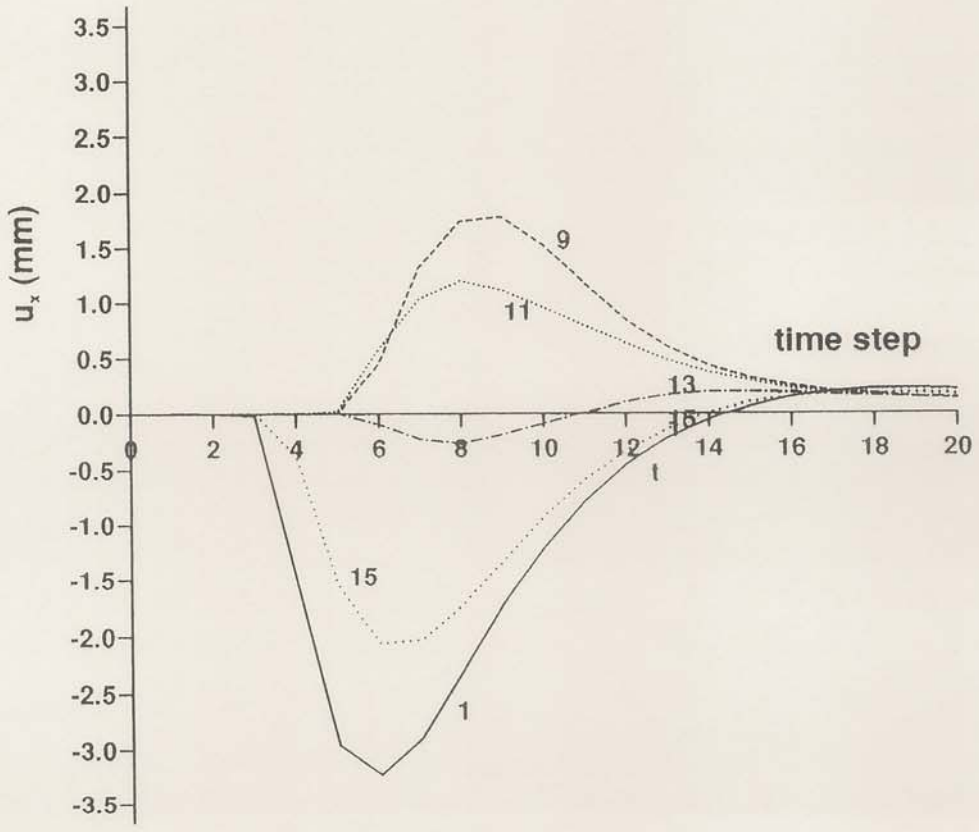


Figure 2.39 x displacements versus time steps

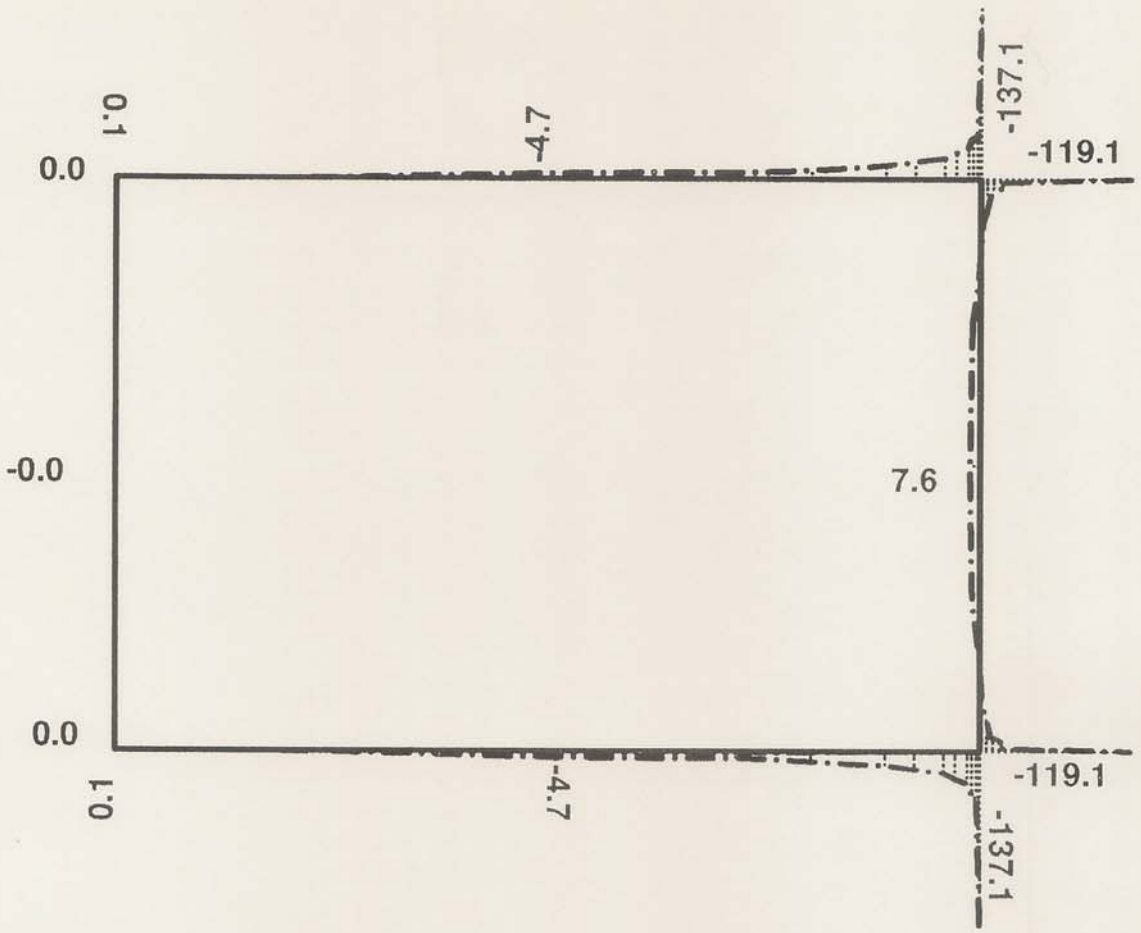


Figure 2.40 Tangential stresses at 6th time step

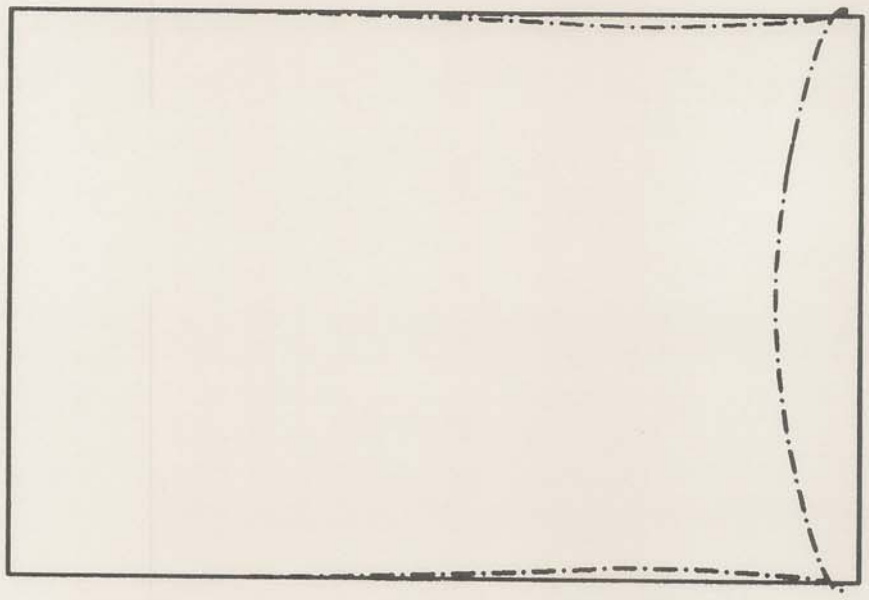


Figure 2.41 Deformed shape at 6th time step

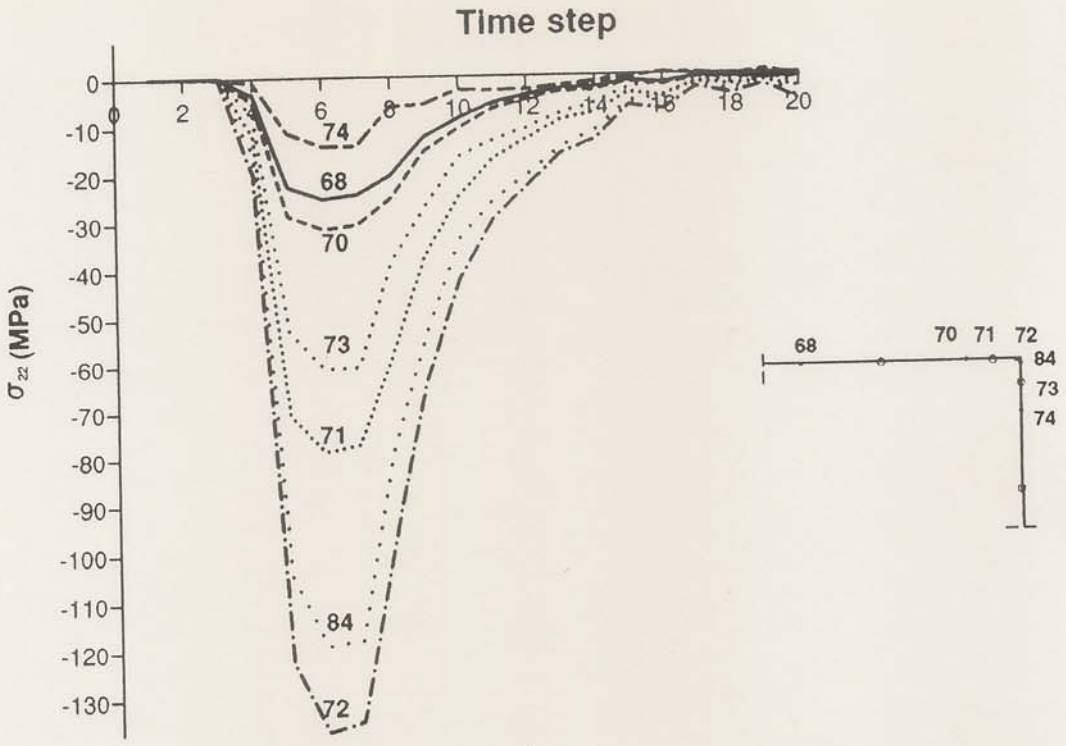


Figure 2.42 Tangential stresses versus time steps

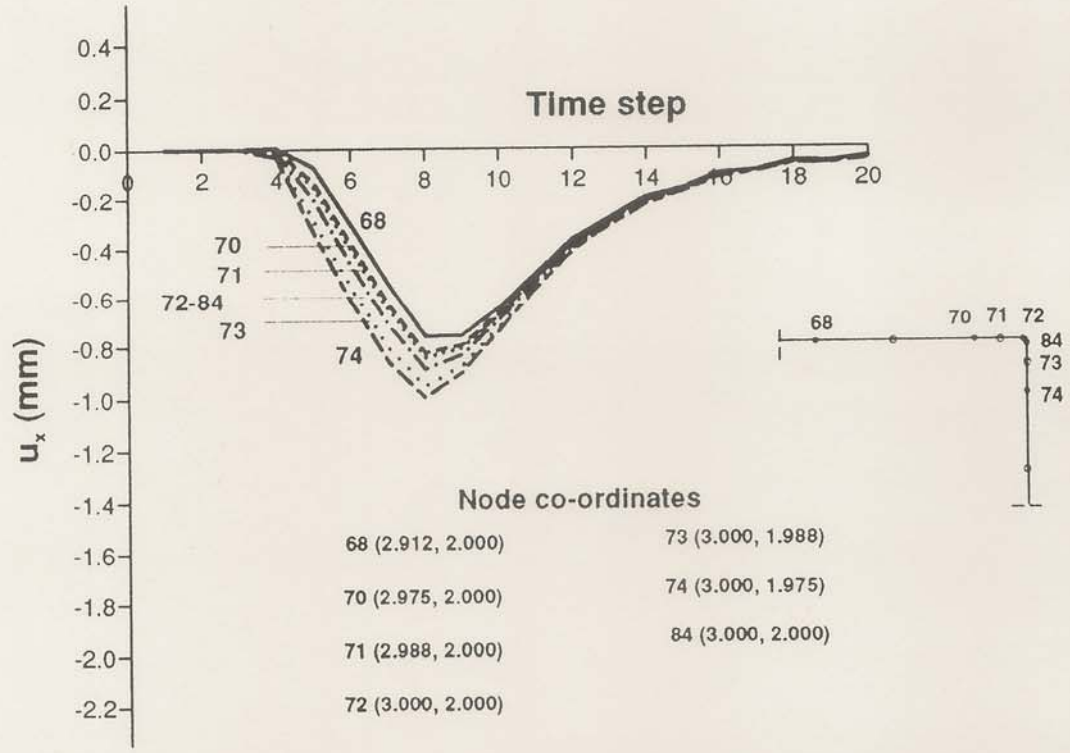


Figure 2.43 x displacements versus time step

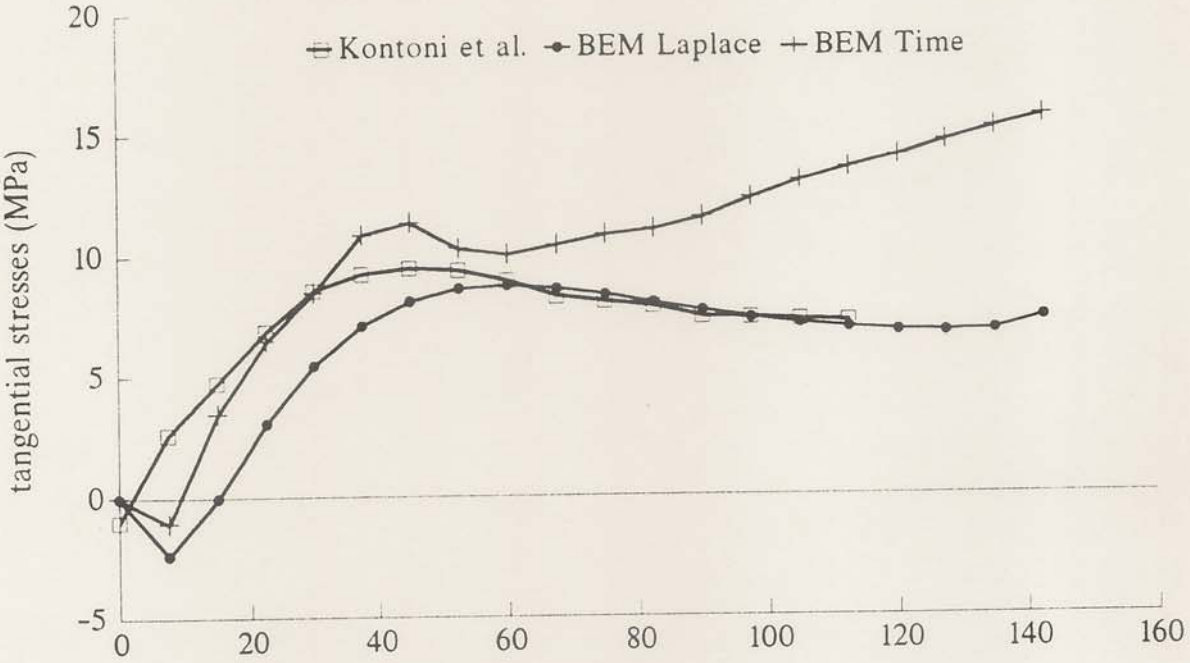


Figure 2.44 Cylindrical cavity under internal pressure

CHAPTER 3

TIME DOMAIN FORMULATION

3.1 INTRODUCTION

In this chapter the implementation of the direct BEM applicable to the time-domain of two dimensional transient elastodynamic problems, is presented.

The methodology used here is similar to the one used by Manolis (1983) and Manolis and Beskos (1988), that employs three-dimensional kernels to solve two-dimensional problems under conditions of plane strain. This is accomplished by integrating the three dimensional kernels along the longitudinal axis of the structure so that the two-dimensional forms of kernels are obtained (Figure 3.1).

A number of applications are presented to demonstrate the accuracy and applicability of the present analysis.

3.2 FORMULATION

The boundary integral equation (2.14) can be written

$$c_{ij}\,u_i = \int_A \left\{G_{ij} * t_i - F_{ij} * u_i\right\} dA = \int_A \left\{ \int_0^T \left[G_{ij}(T-\tau)\, t_i(\tau)\, d\tau \right] - \int_0^T \left[F_{ij}(T-\tau)\, u_i(\tau)\, d\tau \right] \right\} \tag{3.1}$$

and

$$c_{ij}u_i = \int_A \left\{ \int_0^T G_{ij}t_i\, d\tau - \int_0^T F_{ij}u_i\, d\tau \right\} dA = \int_A \left\{ \int_0^T \left[G_{ij}t_i - F_{ij}u_i \right] d\tau \right\} dA \tag{3.2}$$

the total time T is discretized into M equal time steps of time interval

$$\Delta\tau = \frac{T}{M} \tag{3.3}$$

and the current time at the k step is discretized as

$$\tau_\alpha = \sum_{\alpha=1}^{k} \alpha \Delta\tau \tag{3.4}$$

Equation 3.2 can be written as

$$c_{ij}u_i = \int_A \left\{ \int_0^\tau \left[G_{ij}t_i - F_{ij}u_i \right] d\tau \right\} dA = \int_0^\tau \left\{ \int_A \left[G_{ij}t_i - F_{ij}u_i \right] dA \right\} d\tau$$

$$= \sum_{\alpha=1}^{k} \int_{\tau_{\alpha-1}}^{\tau_\alpha} \left\{ \int_A \left[G_{ij}t_i - F_{ij}u_i \right] dA \right\} d\tau \tag{3.5}$$

and

$$c_{ij}u_i - \int_{\tau_{k-1}}^{\tau_k} \left\{ \int_A \left[G_{ij}t_i - F_{ij}u_i \right] dA \right\} d\tau = \sum_{\alpha=1}^{k-1} \int_{\tau_{\alpha-1}}^{\tau_\alpha} \left\{ \int_A \left[G_{ij}t_i - F_{ij}u_i \right] dA \right\} d\tau \tag{3.6}$$

applying the summations to all the elements L and the total number N of the nodes of the mesh, as in chapter 2, results in

$$\sum_{l=1}^{N} c_{ij}u_i - \sum_{l=1}^{N} \sum_{m=1}^{L} \left\{ \int_{\tau_{k-1}}^{\tau_k} \left[\int_A \left(G_{ij}t_i - F_{ij}u_i \right) dA \right] d\tau \right\}$$

$$= \sum_{l=1}^{N} \sum_{m=1}^{L} \left\{ \sum_{\alpha=1}^{k-1} \int_{\tau_{\alpha-1}}^{\tau_\alpha} \left[\int_A \left(G_{ij}t_i - F_{ij}u_i \right) dA \right] d\tau \right\} \tag{3.7}$$

The second term of the equation (3.7) gives

$$\sum_{\alpha=1}^{k-1}\left\{\sum_{l=1}^{N}\sum_{m=1}^{L}\int_{\tau_{\alpha-1}}^{\tau_\alpha}\left[\int_A\left(G_{ij}t_i-F_{ij}u_i\right)dA\right]d\tau\right\}$$
$$=\sum_{\alpha=1}^{k-1}\left\{\sum_{l=1}^{N}\left([G_l]^\alpha\{t_l\}^\alpha-[F_l]^\alpha\{u_l\}^\alpha\right)\right\}$$
$$=\sum_{\alpha=1}^{k-1}R_\alpha=Z_{k-1} \qquad \textbf{(3.8)}$$

that represents the dynamic history of all the previous time steps on the current time step k. The matrices $[G]^\alpha$ and $[F]^\alpha$ contain the summations of the contributions of the kernels G_{ij} and F_{ij} (Appendix B) at the previous time step α. Furthermore, as is shown in Appendices D and E, the matrices $[G]^\alpha$ and $[F]^\alpha$ have the time translation property implying that they retain their values constant between the time steps. For this reason they are calculated in the first time step only, since they remain unchanged between the subsequent time steps $[F]^\alpha$, $[G]^\alpha$ α=2,..,k. So it is

$$[G_l]^\alpha=[G_l]^k,\quad [F_l]^\alpha=[F_l]^k \qquad \textbf{(3.9)}$$

as a result of the expression (3.9) equation (3.8) becomes

$$Z_{k-1}=\sum_{\alpha=1}^{k-1}\left\{\sum_{l=1}^{N}\left([G_l]^k\{t_l\}^\alpha-[F_l]^k\{u_l\}^\alpha\right)\right\} \qquad \textbf{(3.10)}$$

$$[F]^k\{u\}^k-[G]^k\{t\}^k=\sum_{\alpha=1}^{k-1}\left\{\sum_{l=1}^{N}\left([G_l]^k\{t_l\}^\alpha-[F_l]^k\{u_l\}^\alpha\right)\right\} \qquad \textbf{(3.11)}$$

Consequently, replacing the second part of equation (3.7), by equation (3.10) and performing the summations in the first part results in

and equivalently to

$$[F]^k\{u\}^k=[G]^k\{t\}^k+\sum_{\alpha=1}^{k-1}\left\{\sum_{l=1}^{N}\left([G_l]^k\{t_l\}^\alpha-[F_l]^k\{u_l\}^\alpha\right)\right\} \qquad \textbf{(3.12)}$$

which gives for the displacement matrix of the k time step

$$\{u\}^k = ([F])^{-1}\left\{[G]^k\{t\}^k + \sum_{\alpha=1}^{k-1}\left\{\sum_{l=1}^{N}\left([G_l]^k\{t_l\}^\alpha - [F_l]^k\{u_l\}^\alpha\right)\right\}\right\} \qquad (3.13)$$

For the numerical integration of BEM, the integrals of displacement and traction kernels have to be calculated analytically. This analytical integration, depends on the variation of the kernels between the time steps, which can be either constant or linear. Both cases are examined in Appendices D and E.

The numerical integration is performed by Gaussian Quadrature formulae and indirect schemes as in Laplace domain case and as will be discussed in Chapter 4, section 4.3.

The stresses on the boundary and the solution for internal points, were explained in Chapter 2 and are similarly applied here.

3.3 APPLICATIONS

In the following applications the material properties used are: Young's modulus E=20,000 MPa, Poisson's ratio ν=0.25 and density ρ=2.55 Mg/m^3. Furthermore in all applications the total time interval is 0.015 s implemented in 20 time steps.

3.3.1 Plane Wave

3.3.1.1 Circular Cross Section Cavity in an Infinite Medium Swept by a Plane P Wave

This problem has been previously solved,in time domain by direct BEM by Manolis (1983). The same problem has been mentioned in the Laplace domain, in section 2.2.1.1. It is examined here to allow a comparison to take place between the two methodologies.

The radius of the cavity is 5.0 m. The incident wave is a shock compressional plane wave of - 10.0 MPa magnitude that is applied suddenly and maintained constant in time. The angle of incidence of the incoming wave is 0°, measured from the global horizontal x axis. Eight quadratic elements are used to represent the surface of the mesh, corresponding to 16 nodes (Figure 3.2). The position of the

plane wavefront at t=0, is considered as tangent to the circle, at node 1.

The variation of these stresses in time, for the nodes 1,9,11,13,15 which are at angles 0°, 180°, 135°, 90° and 45° respectively from the x axis, is shown in Figure 3.2. The most compressive hoop stress is -30.5 MPa and appears at node 13 at the 12th step, which is in good agreement with the results of Manolis (1983).

The tangential stresses on the periphery of the opening for the time step 12, for all the nodes, is shown in Figure 3.3. The deformed shape of the mesh at time step 12 is shown in Figures 3.4. The displacements in x direction, is shown in Figure 3.5.

It can be seen, similarly to the Laplace domain case, that the stresses and displacements at late times tend to be approximately equal for the symmetric to the y axis nodes (nodes 1 and 9, 15 and 11). The cavity behaves similarly to the elastostatic case loading and the dynamic solution approaches the static one.

The present results conform fairly well with the results in Laplace domain solution.

3.3.1.2 Circular Cross Section Cavity in an Infinite Medium Swept by a Plane SV Wave

In Figure 3.6, the same cavity of radius 5.0m with the same discretization into 8 elements as in the application of section 3.3.1.1 is examined subjected to an incident SV plane wave that carries a shear stress of -10.0 MPa, at 0° angle of incidence. The same problem has been examined in Laplace domain (Chapter 2, section 2.3.1.4).

The variation of the tangential stresses with time is presented in Figure 3.6.

The tangential stresses on the boundary of the cavity, for the 12th time step are depicted in Figure 3.7. The most compressive stress -47.6 MPa appears at the node 10 and -38.6 MPa at node 4, that form an angle of 157.5° and 292.5° with x axis respectively. Furthermore, significant tensile stresses of 47.6 MPa and 38.6 MPa appear at nodes 8 and 16 that form angles of 202.5° and 67.5° with the x axis respectively.

The x and y displacements versus time steps are presented in Figures 3.8 and 3.9 respectively.

At late times, the cavity behaves similarly to the elastostatic case loading.

The results seem to be in good agreement with the Laplace domain solution that

has been examined in section 2.3.1.4.

3.3.1.3 Circular Cross Section Cavity in an Infinite Medium Swept by a Plane P+SV Wave

Figure 3.10 shows the same cavity as the one examined as in section 3.3.1.1 this time loaded by a combination of an incident P of -10.0 MPa compressional and an SV of -10.0 MPa shear plane wave, at 0° angle of incidence. The same problem has been examined in Laplace domain (Chapter 2, section 2.3.1.5).

In Figure 3.10 the variation of the tangential stresses in time are presented.

In Figure 3.11 the tangential stresses on the boundary for the 11th time step are depicted. In this case the most compressive stress -116.9 MPa appears at the 11th step at node 11 and -123.6 MPa at node 4, that form an angle of 135° and 292.5^o respectively with the x axis (Figure 3.11). Also, significant tensile stresses of 56.4 and 55.6 MPa appear at nodes 8 and 16 that form angles of 202.5^o and 22.5^o with x axis respectively. In Figure 3.10 the variation of the hoop stresses in time at nodes 1,9,11,13 and 15 is presented. The x and y displacements versus time steps are depicted in Figures 3.12 and 3.13 respectively.

The results are in relatively good agreement with the Laplace domain solution (see corresponding analysis in section 2.3.1.5).

3.3.2 Explosion at a Point

3.3.2.1 Circular Cross Section Cavity in an Infinite Space Loaded by an Explosion Wave

A circular cross section cavity of radius 2.50 m, with the same material characteristics as in section 3.3.1.1 is examined under the loading of an explosive stress wave generated from an underground explosion at a point. The co-ordinates of the explosion point with respect to the general co-ordinate system with origin at the centre of the cavity are (8.0, 0.0) m. The three types of formulas introduced in Chapter 4, section 4.2 are used here, with or without decay of the wave in time. The equivalent charge weight of a GP 1000 lb bomb (U.S.Army, 1984) is used, which is 188.7 kg of explosive weight. The attenuation coefficient is taken n=2.75, and the coupling factor f=1.0. The total time interval examined here is 0.015 s in 20 time steps.

a. Cubic Root Formula Without Decay in Time

The stress wave generated by the explosive, is calculated from the cubic root formula for every node and is maintained constant without decay in time. The direct compressive stress derived from this formula at node 1 has been calculated analytically and is -33.7 MPa.

The change of the hoop stresses versus the time steps at the nodes 1,15,13,11,14 and 9 is demonstrated in Figure 3.14. The maximum tangential compressional stress is -53.2 MPa and appears at the node 14 at the 6th time step, which forms an angle of 67.5^o with the x global axis, with respect to the maximum tensional stress, a magnitude of 6.9 MPa characterizes node 1 at the 6th time step.

In Figure 3.15, the tangential stresses on the periphery of the opening and the deformed shape of the mesh at the 6th time step are presented.

The x and y displacements versus time steps are pictured at Figure 3.16 and 3.17 respectively. At late times all the deformation curves of the nodes assume a parallel orientation indicating that the cavity undergoes a rigid body translation.

Comparing the present results with those of Laplace domain (Chapter 2, section 2.3.2.1.a), indicates a fairly good agreement between them.

b. Cubic Root Formula With Decay in Time

The direct compression stress computed from this formula at node 1 has been calculated analytically and is 33.7 MPa. Figure 2.32 shows the mesh and the rays of the incident wave.

The change of tangential hoop stresses and displacements versus time steps at nodes 1, 15,13,11 and 9 is demonstrated in Figure 3.18. The maximum tangential compressional stress is -31.3 MPa and appears at node 14 at the 6th time step, which forms an angle of 67.5^o with the x global axis, in contrast the maximum tensional stress is 3.3 MPa and appears at node 1 again at the 6th time step. The stresses rapidly increase to the peak value and drop equally rapidly by tending to zero at late times.

Furthermore, Figure 3.19 shows the results on the periphery of the opening, of the hoop tangential stresses and the deformed shape of the mesh at the 6th time step.

The peak values of x-displacements appear at the 6th time step for node 1, and at the 11th time step for node 9 (Figure 3.20). The y-displacements versus time steps are depicted at Figure 3.20a. At late times the displacements tend to zero values and the cavity returns to its original undeformed shape, following the passage of the shock wave.

The present results, for the early times, are in good agreement, with the respective

results of Laplace domain (Chapter 2, section 2.3.2.1.a).

3.3.3 Explosion Inside a Cylindrical Cavity

The same problem, as in section 2.3.3, of a cylindrical cavity in an infinite medium (R=6.35 m, E=207,000 MPa, ν=0.33, ρ=8.48 Mg/m^3), subjected to a suddenly applied internal explosion of 10.0 MPa is solved here in time domain. The problem has been solved by the direct BEM in time domain by Mansur (1983).

Figure 2.44 shows the variation of the hoop stress in time for the node of the cavity that is at 0° angle with x axis. Also the respective results of Kontoni et al. and of the Laplace domain results as in Chapter 2, section 2.3.3 have been explained. It can be seen that there is a good agreement between the two solutions.

3.3.4 Solution at Internal Points of Half-plane

To demonstrate the precision of the program, in the following examples, the displacements at internal points are calculated in Laplace and time domain and are compared against published applications.

a. Half-plane Under Ramp Type Stress Distribution on the Free Surface

The problem of a half-space initially at rest, that is disturbed by a suddenly applied ramp-like vertical pressure, distributed over a part of the free surface, is examined here Figure 3.21). This problem has been solved with BEM by Mansur (1983) in time domain and by Ahmad (1986) in Laplace domain. The results obtained here are compared against the known solutions and the reliability of the program is examined. The material properties of the half-space are: Young's modulus E=1,379 MPa, Poisson's ratio ν=0.15 and density ρ=2.09 Mg/m^3. The total examining time is 140 ms implemented in 20 time steps. The free surface is discretized into seventeen quadratic elements of 6.10m length each (Figure 3.22).

The time history of the vertical displacements at the internal points D(0.0,-21.3), F(24.4,-18.3), of the free surface are illustrated at Figures 3.23 and 3.24 in Laplace domain, and at points G(45.7,-3.0) and K(0.0,0.0) at Figures 3.25 and 3.26 in Laplace domain and in time domain as well. It is observed that there is very good agreement of the time domain and Laplace domain solutions of the present work in all cases. Also it can be seen that the present work produces results very close to those by Mansur (1983) and Ahmad (1986).

It is worth mentioning here that the jump term, discussed in Chapter 4 section 4.3.2.c, is 0.5 for the nodes on the free surface of the ground.

b. Half-plane Under Suddenly Applied and Maintained Stress Distribution on the Free Surface

The problem of a half-space initially at rest, that loaded by a suddenly applied and maintained vertical pressure, distributed over a part of the free surface, is examined next (Figure 3.27). This problem has been solved with BEM by Cruse (1968) in Laplace domain and by Mansur (1983) in time domain. The results obtained here are compared against the known solutions and the reliability of the program is examined. The material properties of the half-space are Young's modulus E=17250 MPa, Poisson's ratio ν=0.25 and density ρ=3.10 Mg/m^3. The total examining time is 0.45s implemented in 20 time steps. The free surface is discretized in eleven quadratic elements of 152.4m length each (Figure 3.28).

The time history of the vertical displacements at the internal point D(0.0,-76.2) and at the point C(0.0,0.0) of the free surface are illustrated at Figures 3.29 and 3.30 in Laplace domain and time domain as well. A good agreement at early times between the three methods is observed. At the late times, the results start to differ significantly. The reason for these differences could be the different formulas that are used for the numerical inversion of the Laplace transform and the long total time interval that is used.

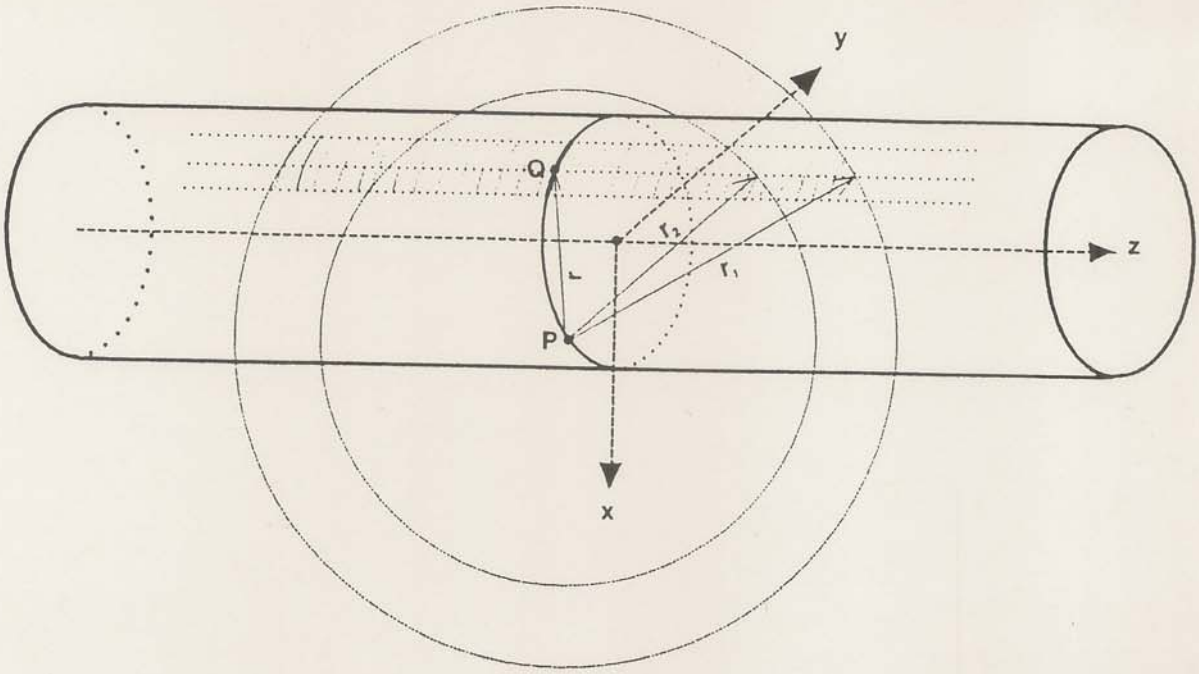


Figure 3.1 Integration of 3d kernels along z axis

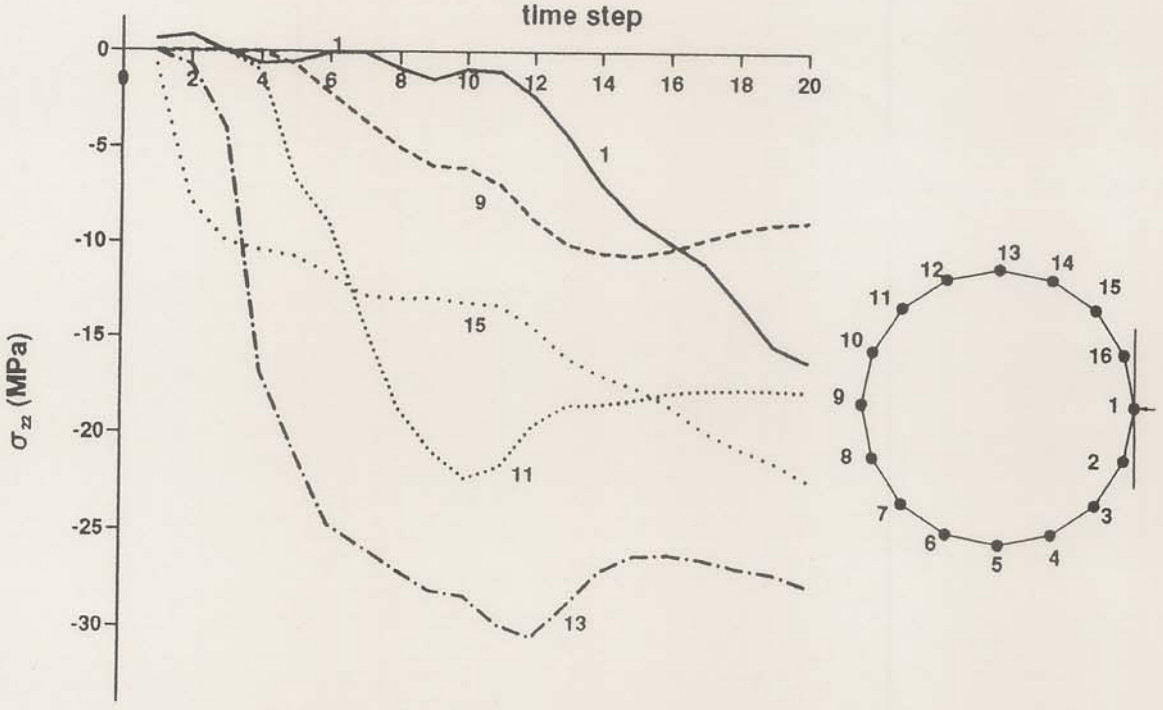


Figure 3.2 Tangential stresses versus time steps

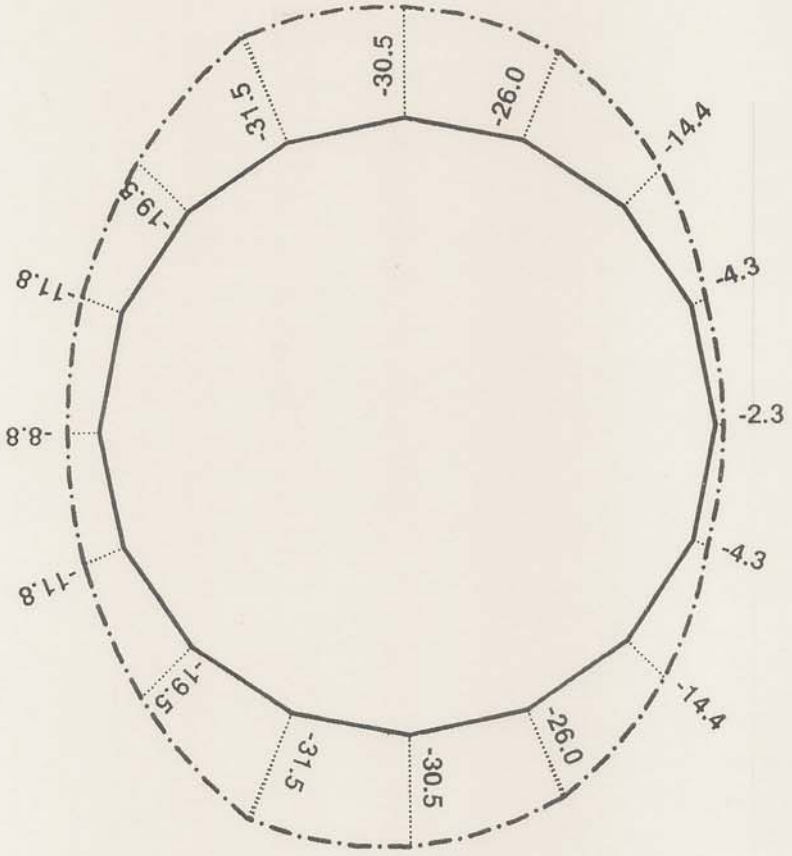


Figure 3.3 Tangential stresses at 12th time step

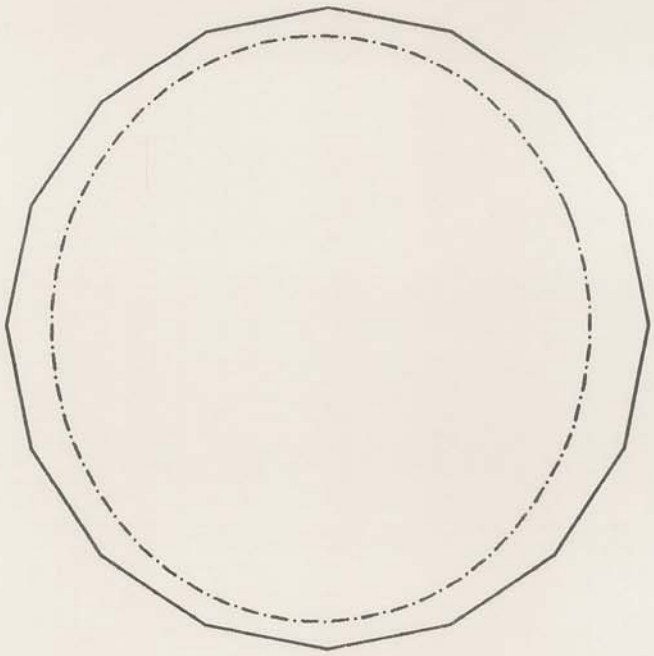


Figure 3.4 Deformed shape at 12th time step

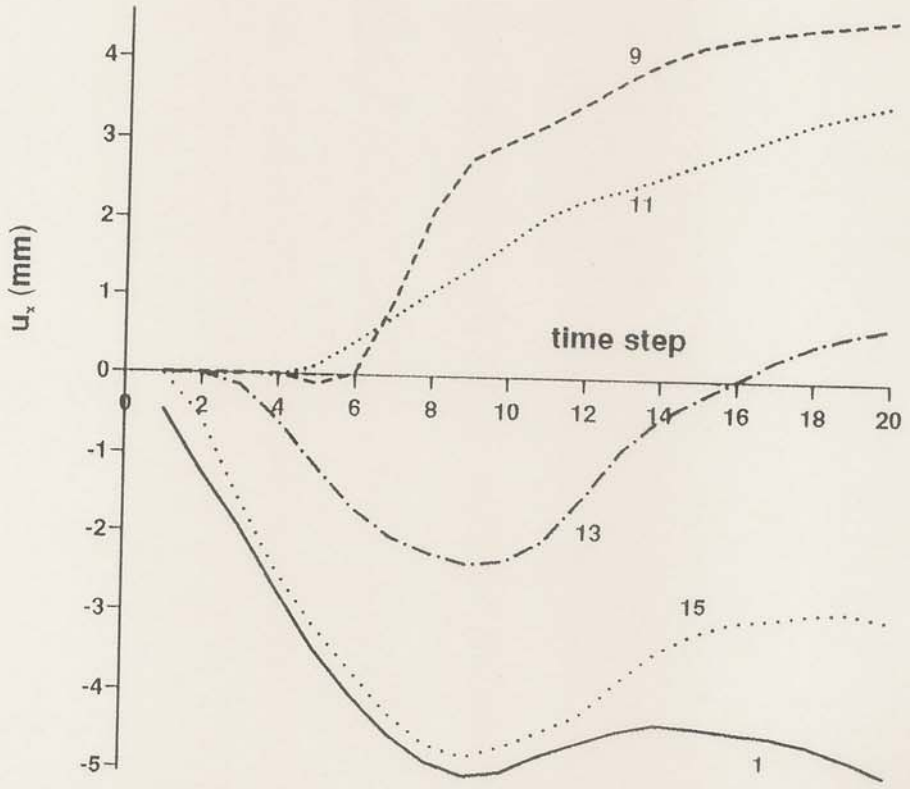


Figure 3.5 x displacements versus time steps

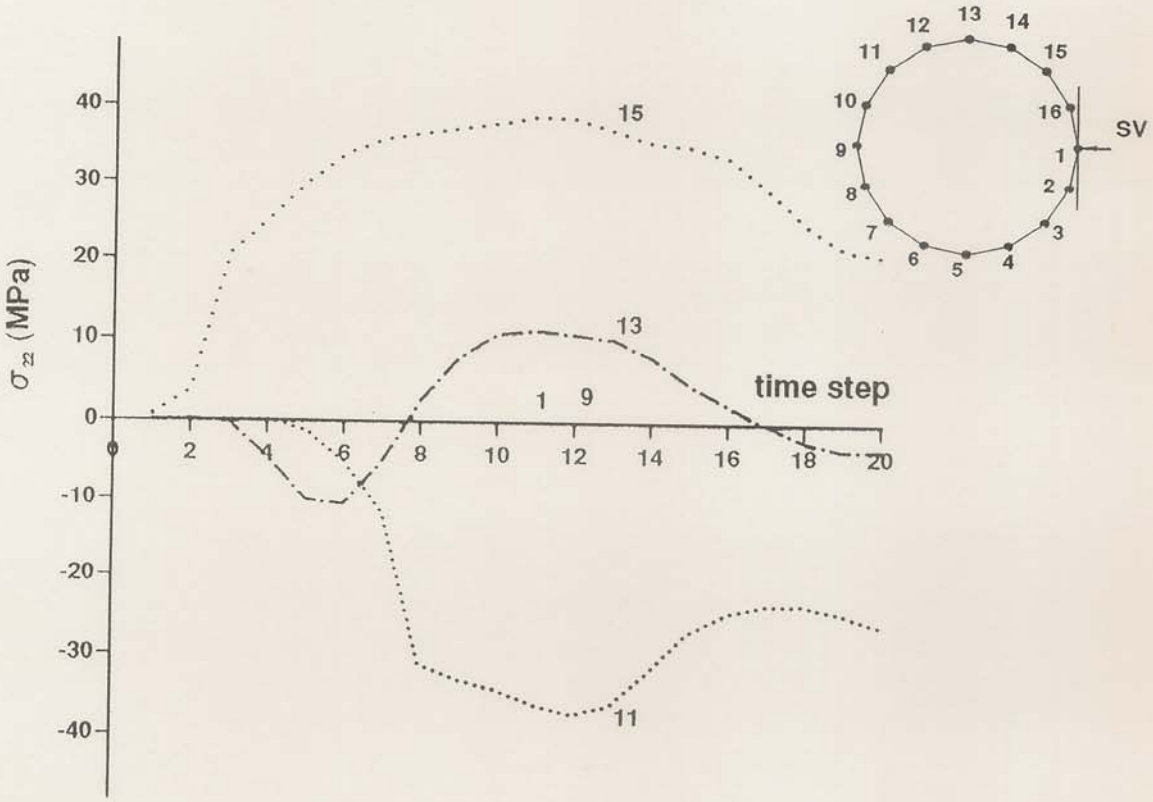


Figure 3.6 Tangential stresses versus time steps

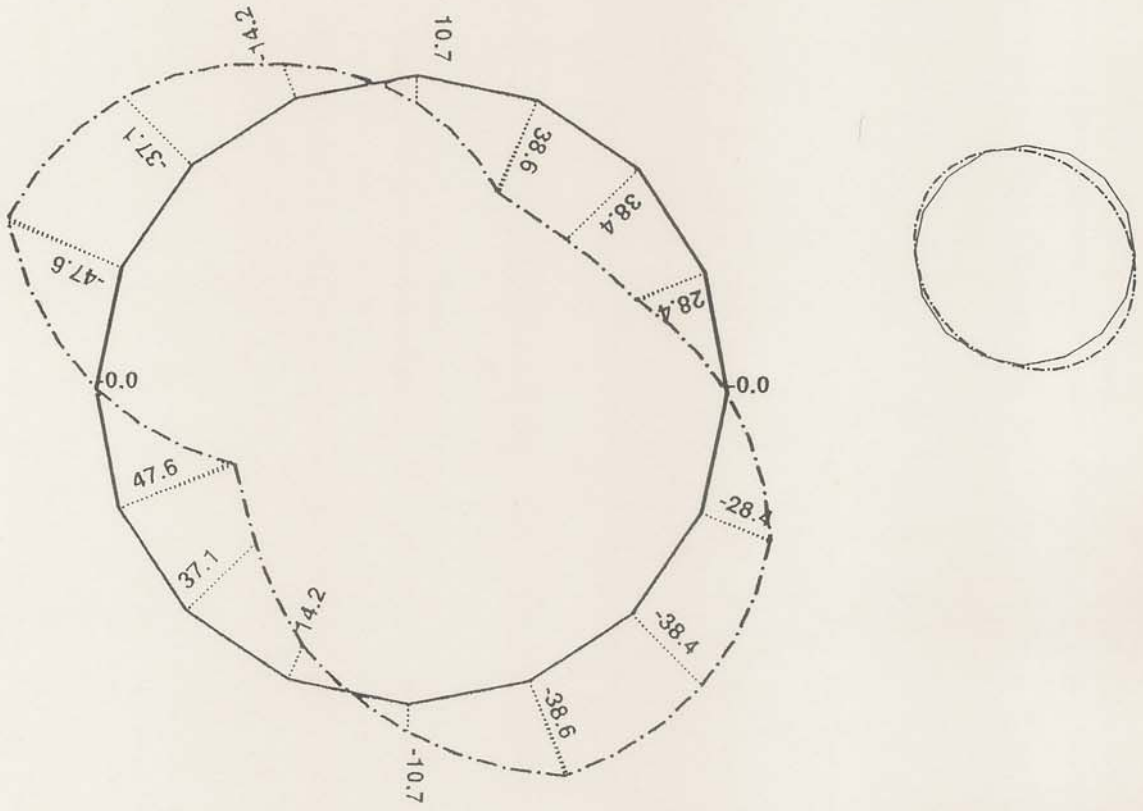


Figure 3.7 Hoop stresses and deformed shape at 12th time step

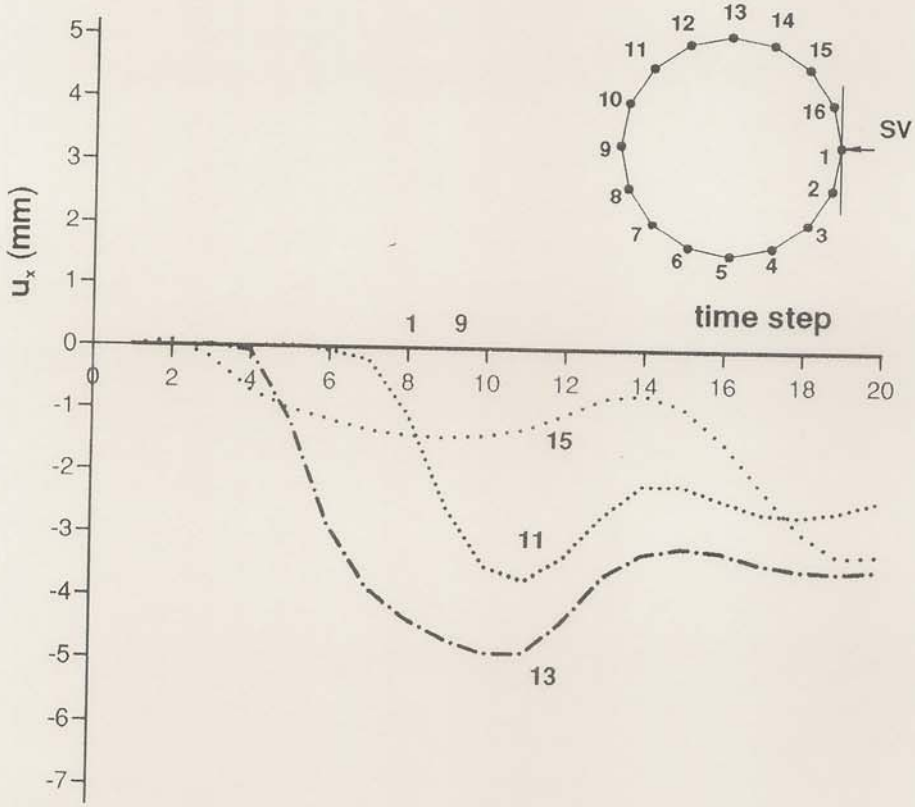


Figure 3.8 x displacements versus time steps

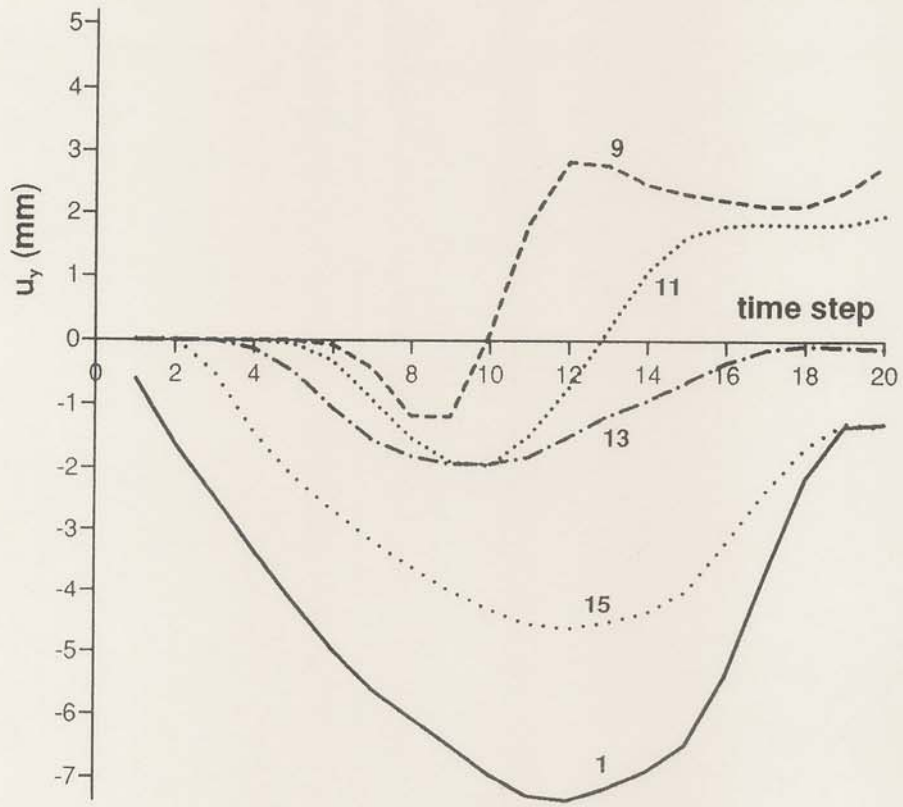


Figure 3.9 y displacements versus time steps

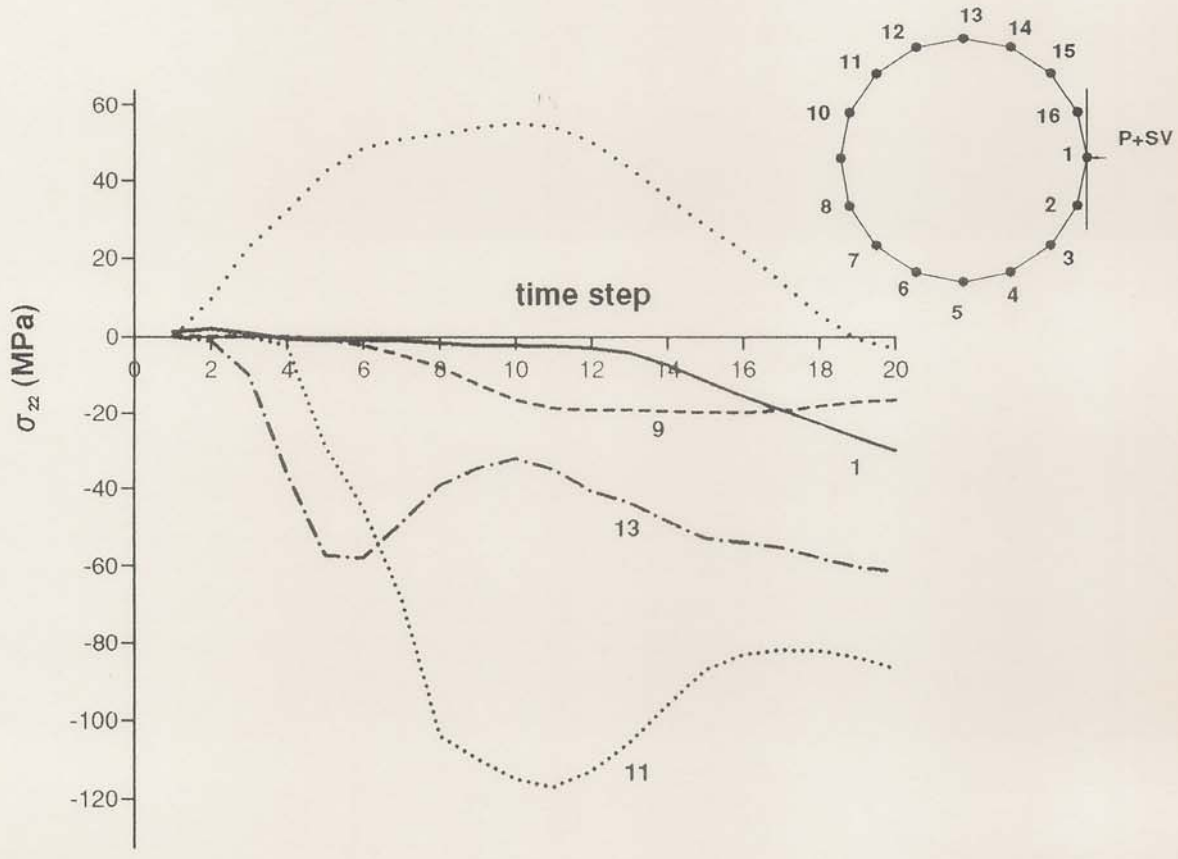


Figure 3.10 Tangential stresses versus time steps

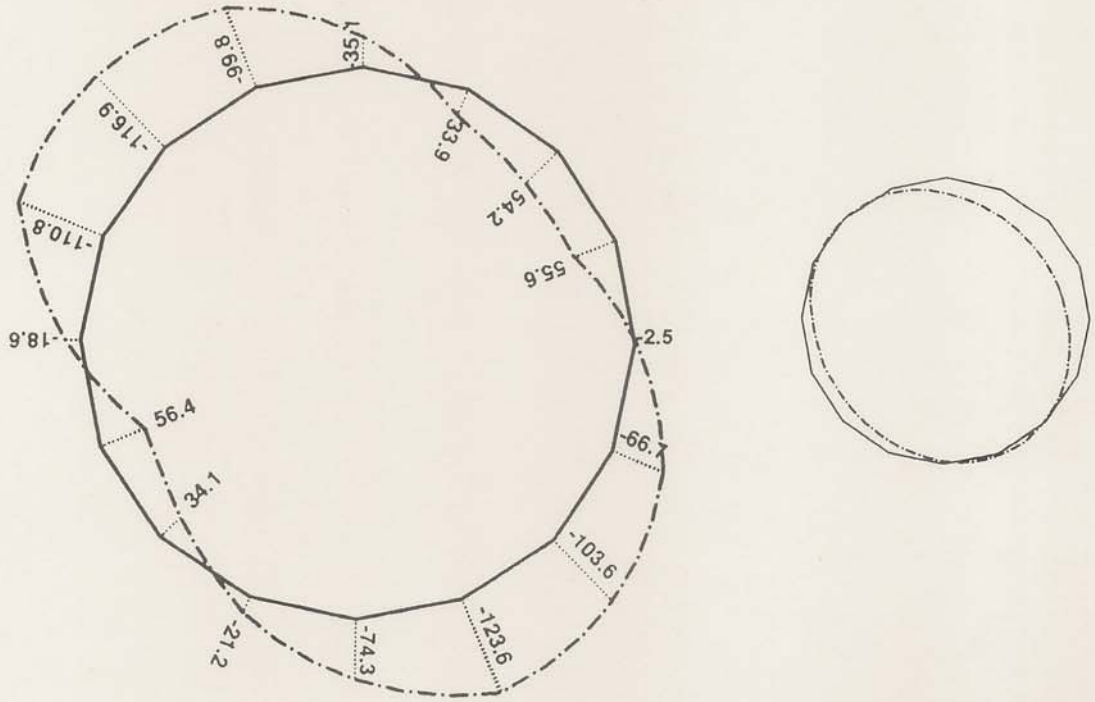


Figure 3.11 Tangential stresses and deformed shape at 11th time step

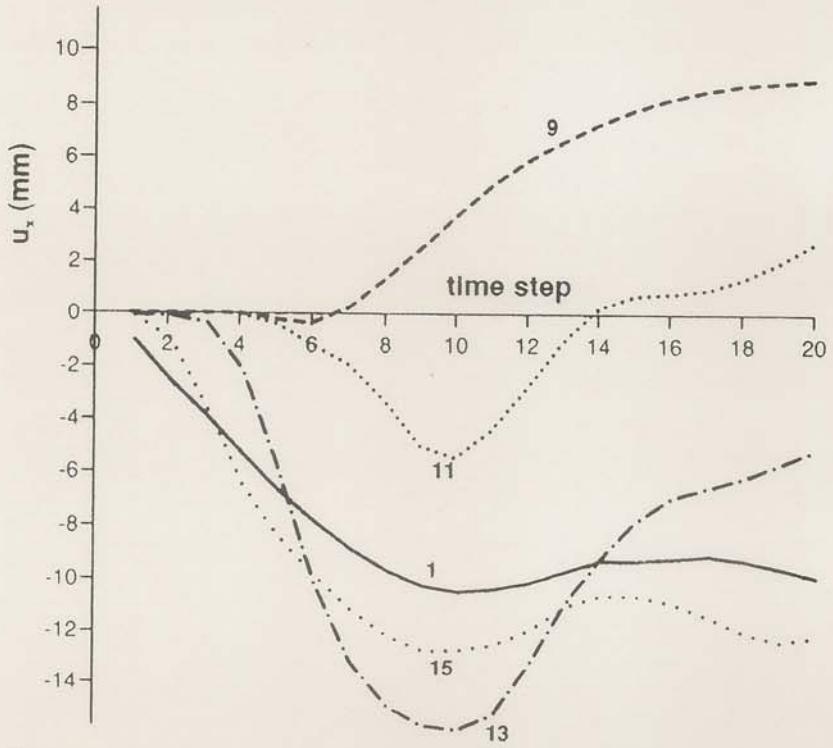


Figure 3.12 x displacements versus time steps

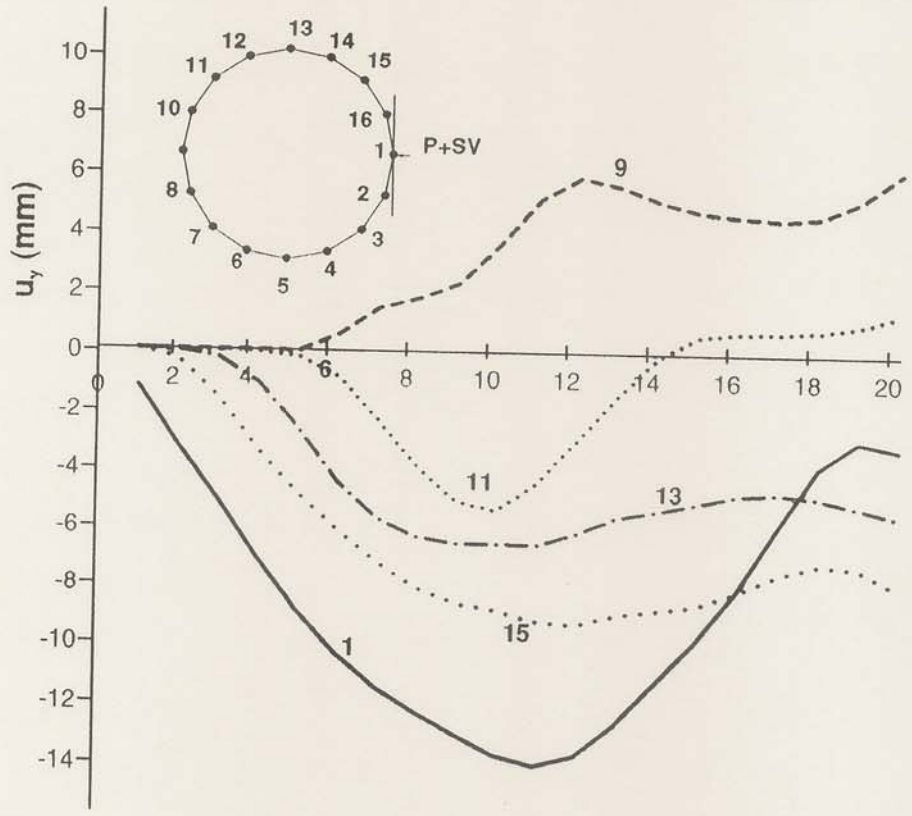


Figure 3.13 **y displacements versus time steps**

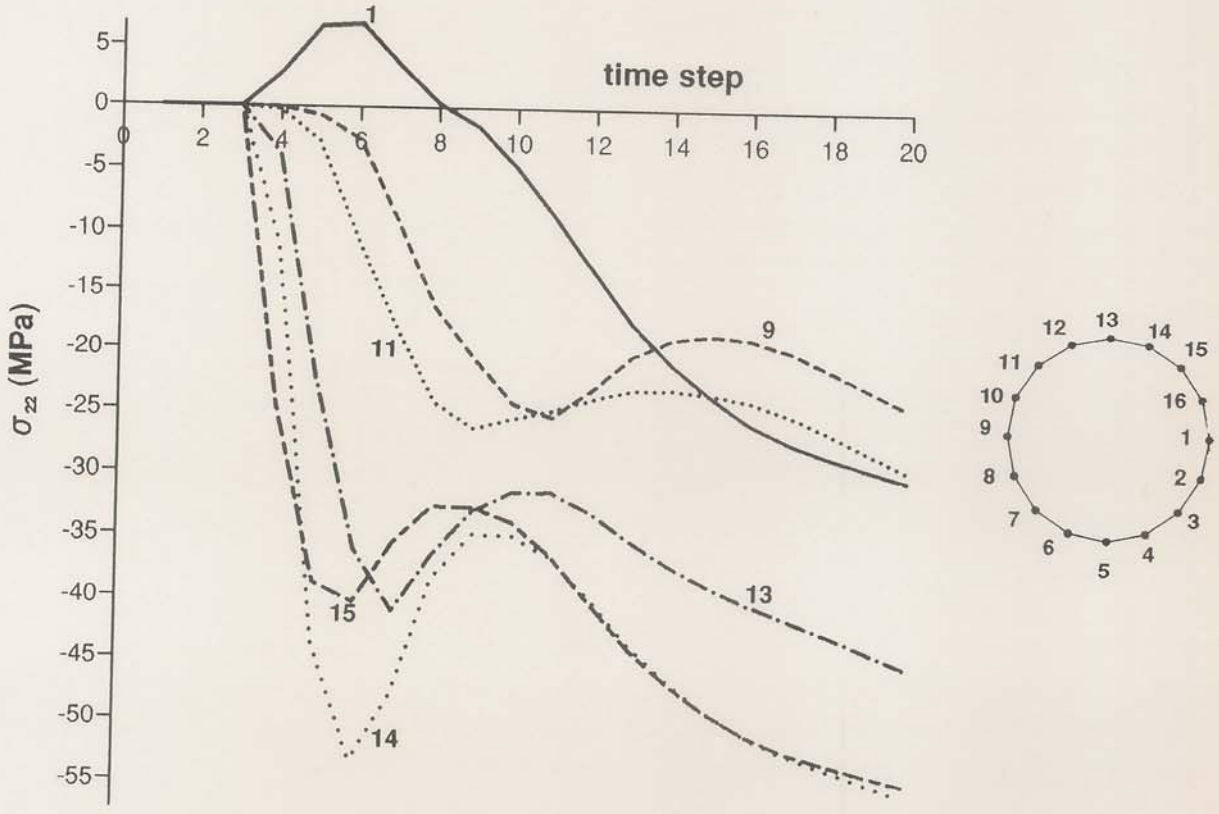


Figure 3.14 **Tangential stresses versus time steps**

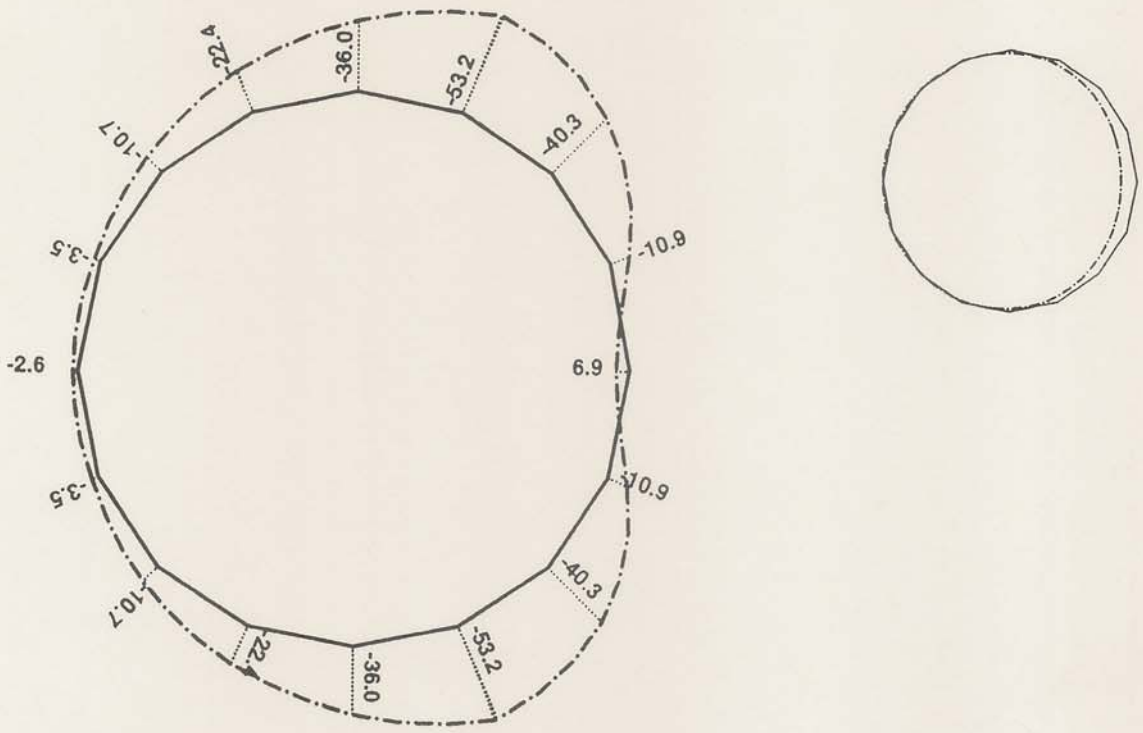


Figure 3.15 Tangential stresses and deformed shape at 6th time step

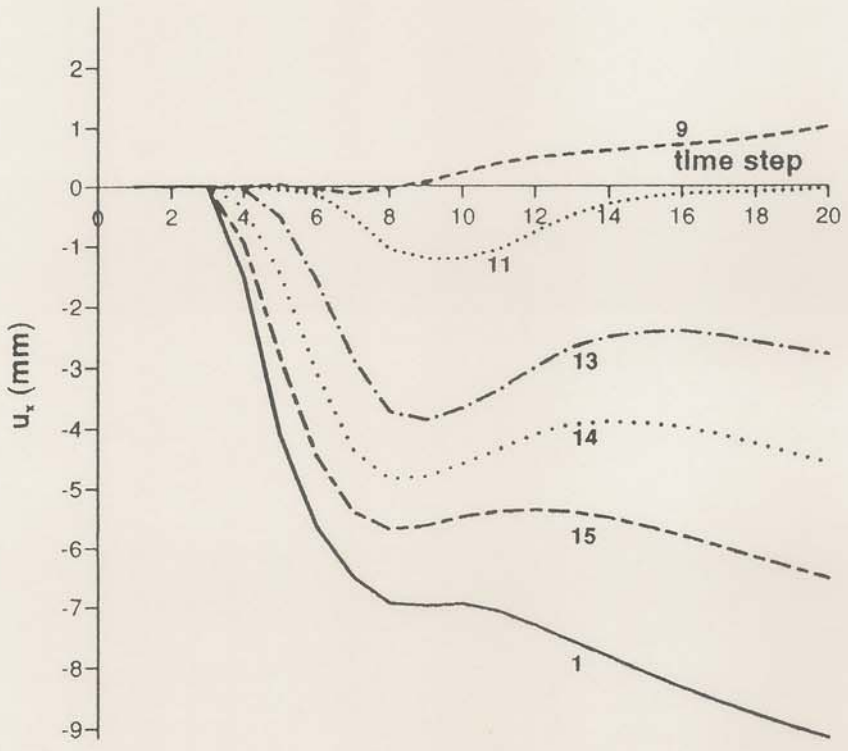


Figure 3.16 x displacements versus time steps

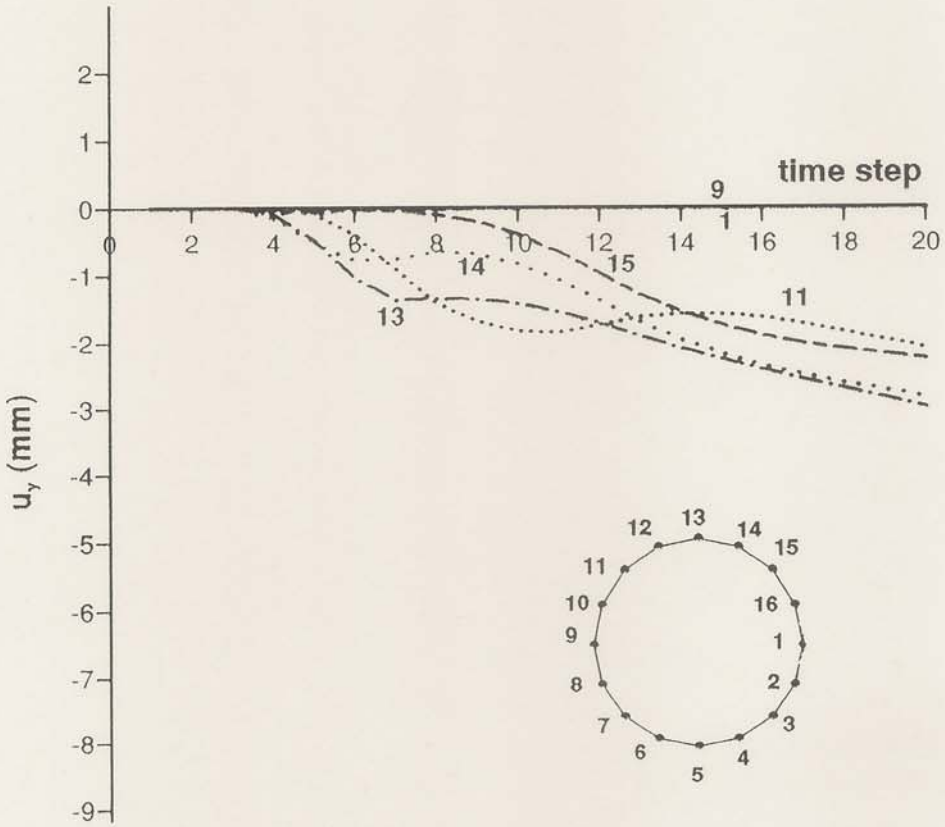


Figure 3.17 y displacements versus time steps

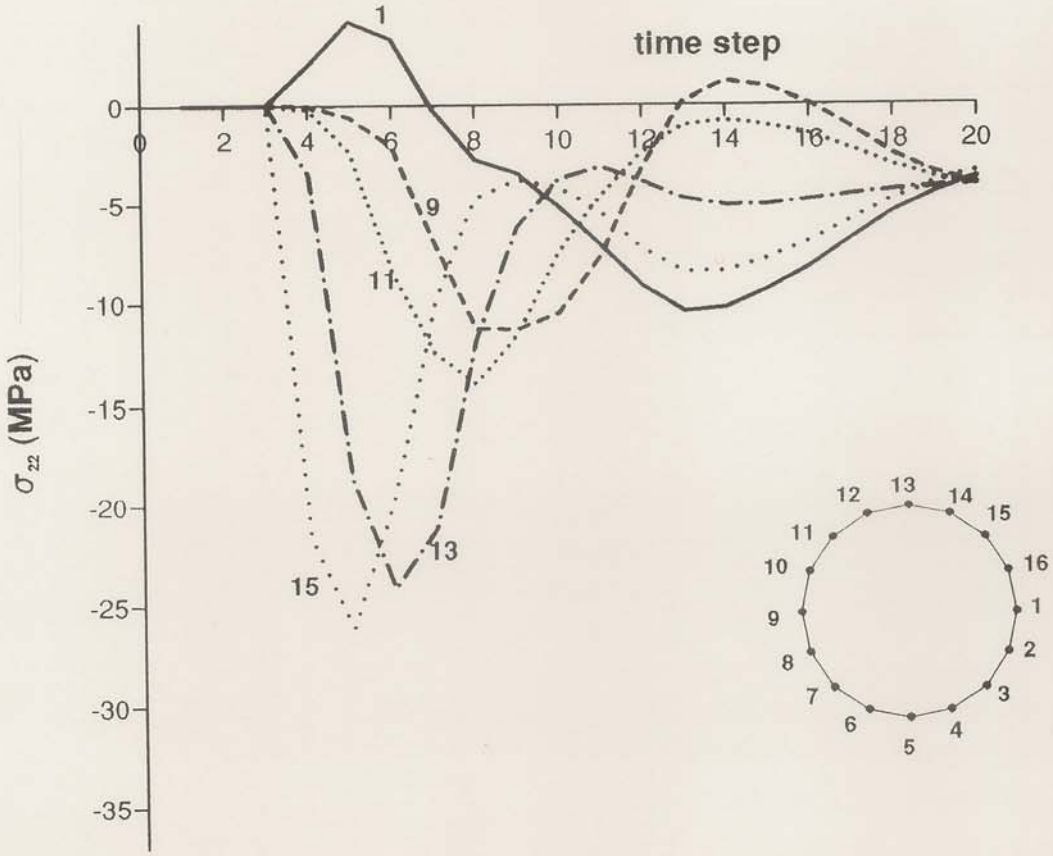


Figure 3.18 Tangential stresses versus time steps

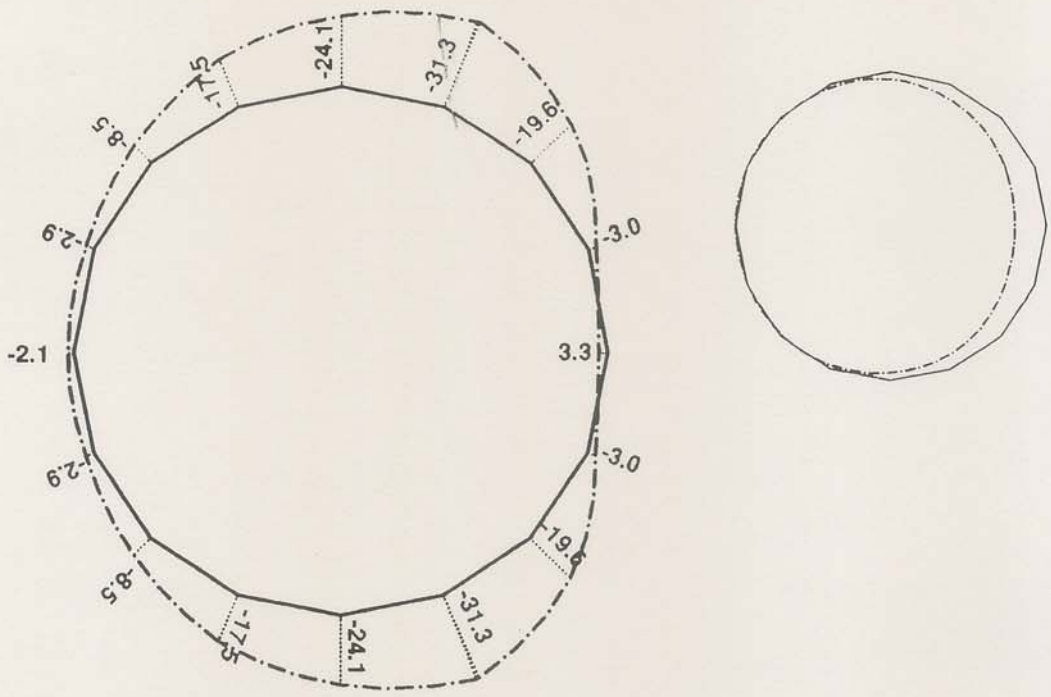


Figure 3.19 Tangential stresses at 6th time step

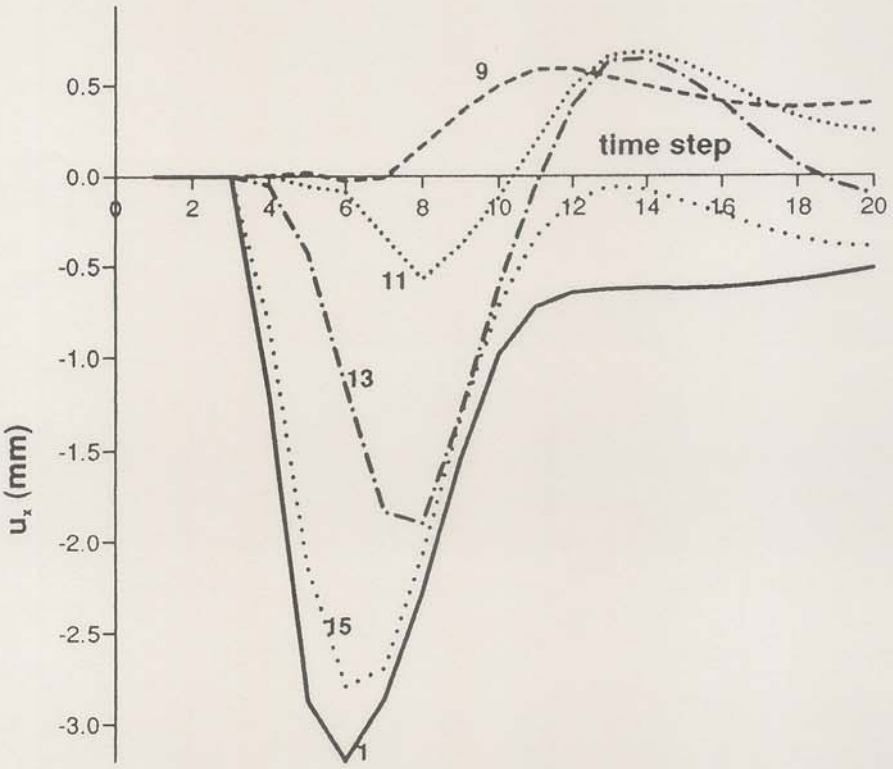


Figure 3.20 x displacements versus time steps

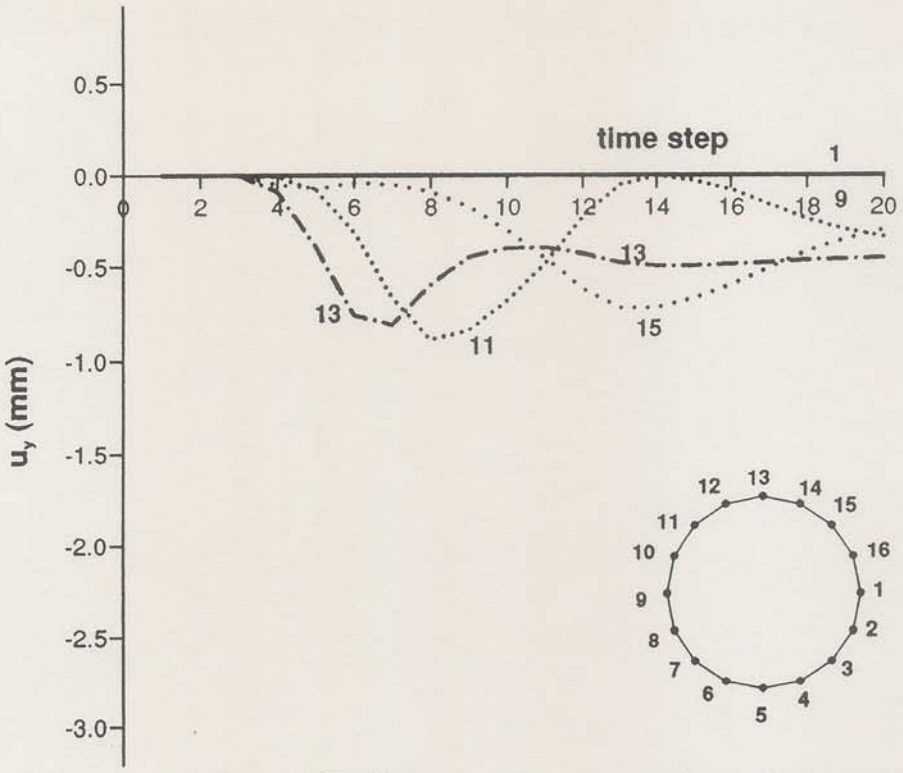


Figure 3.20.a. y displacements versus time steps

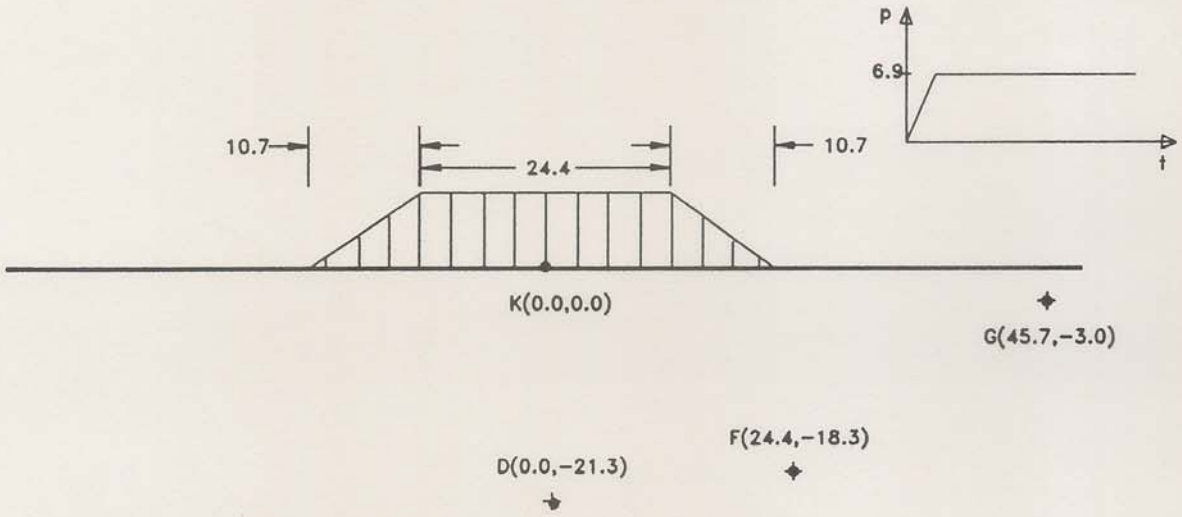


Figure 3.21 Half-space under stress distribution

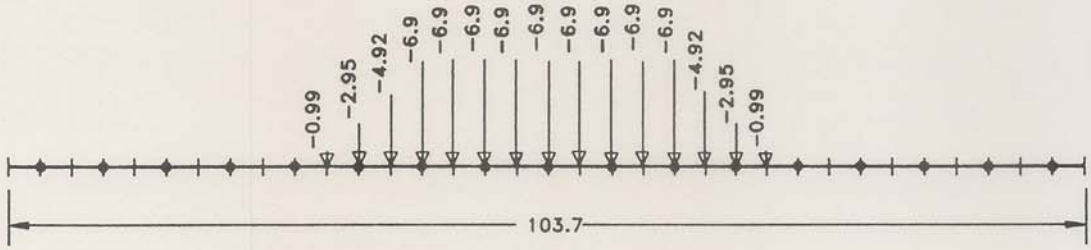


Figure 3.22 Discretization of the free surface and applied tractions

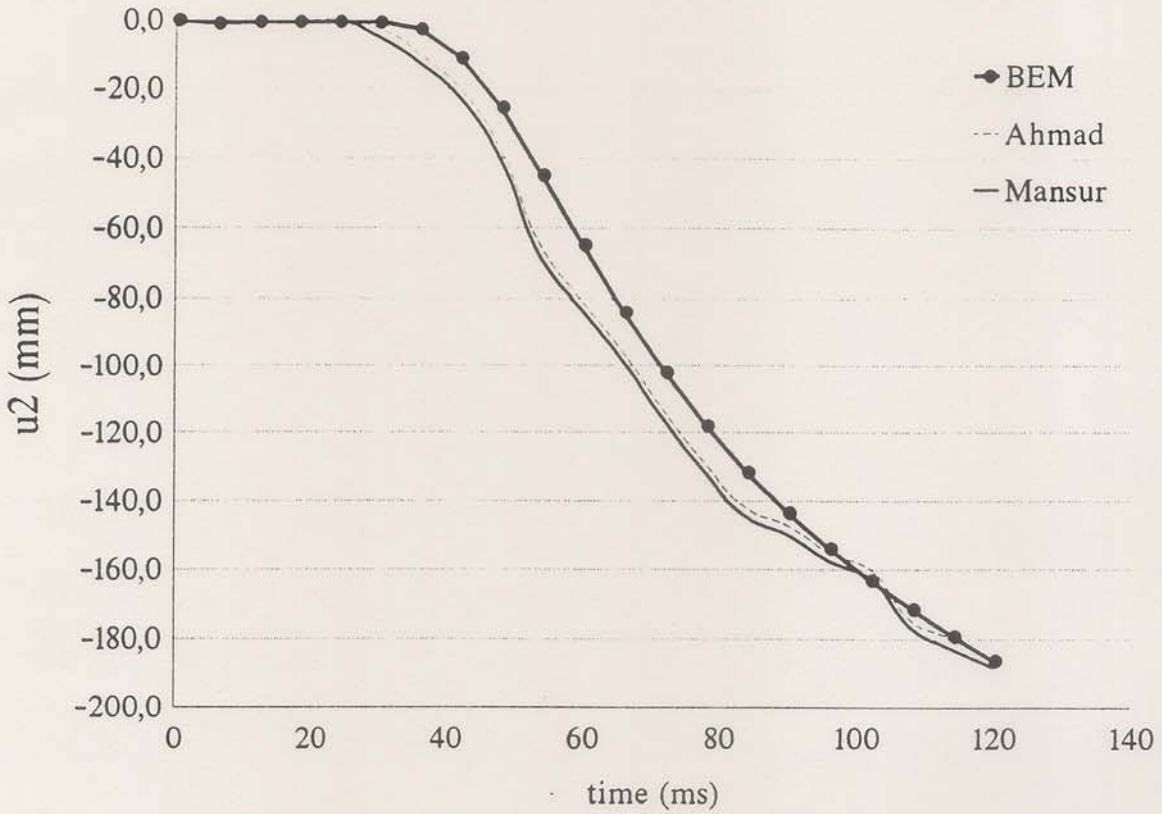


Figure 3.23 Displacements at point D(0.0,-21.3)

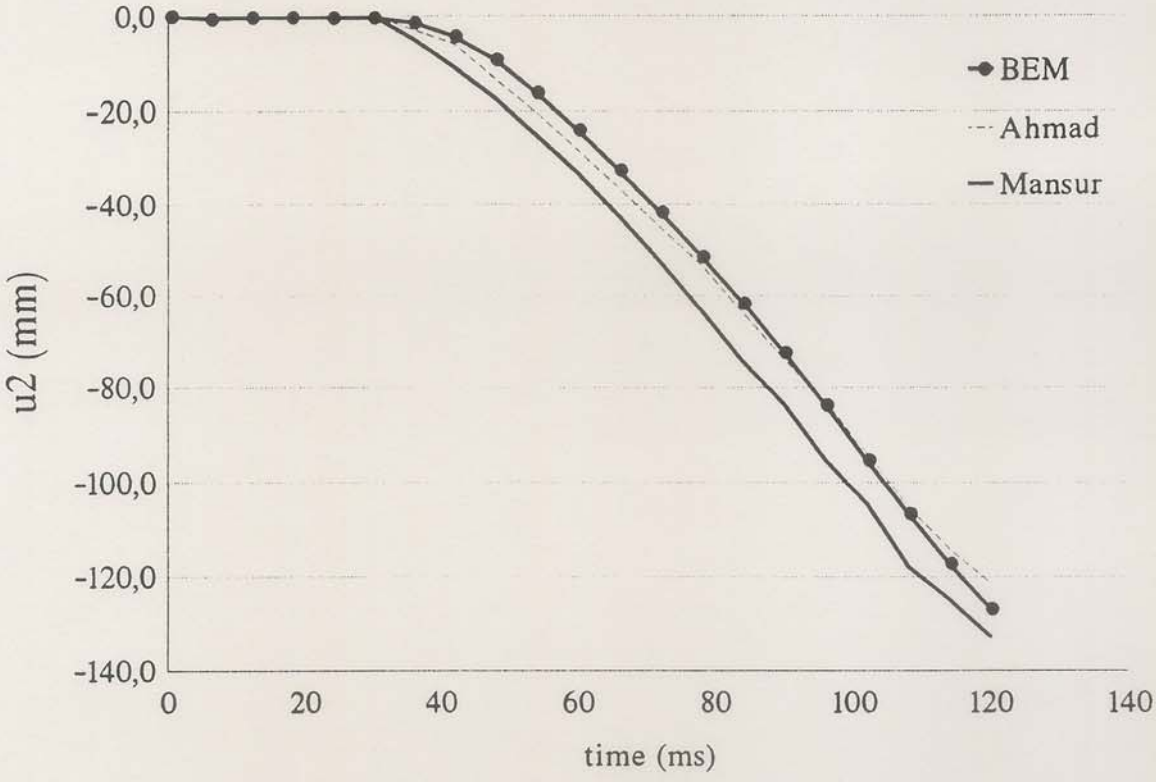


Figure 3.24 Displacement at point F (24.4,-18.3)

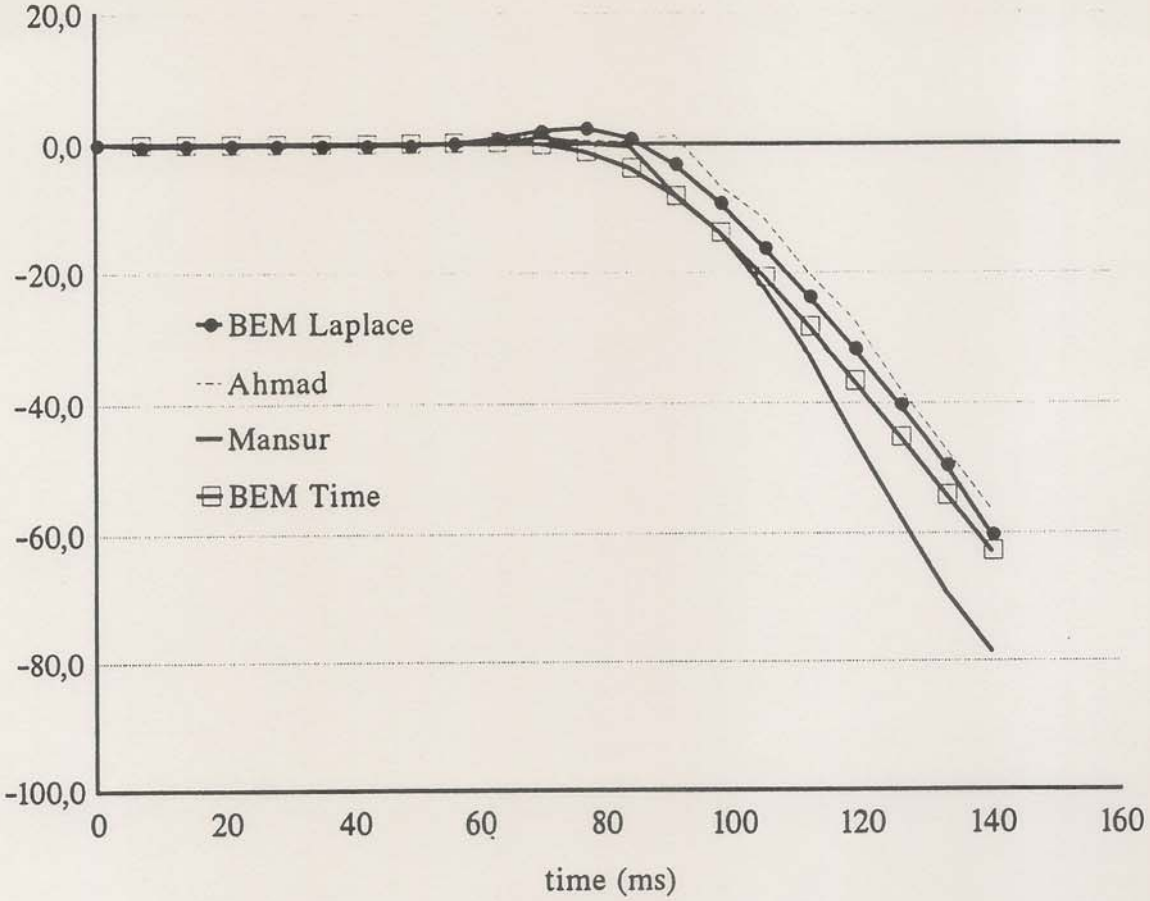


Figure 3.25 Displacement at point G (45.7,-3.0)

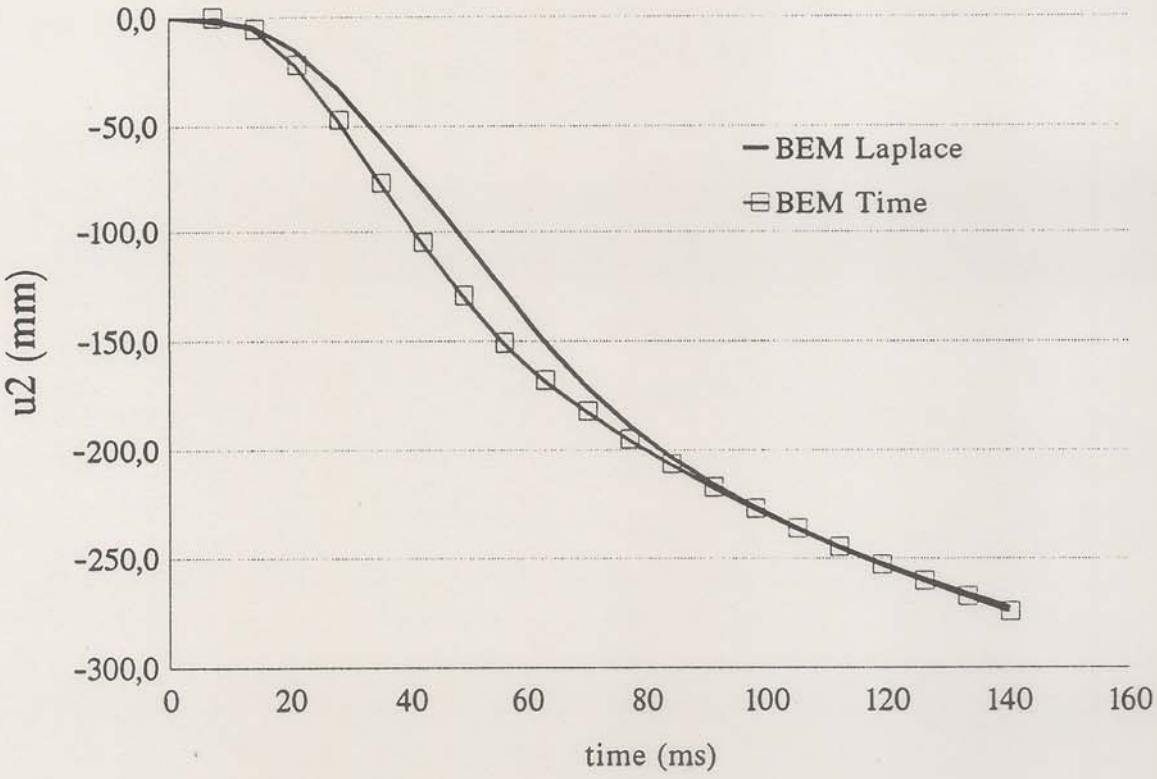


Figure 3.26 Displacement at point K (0.0,0.0)

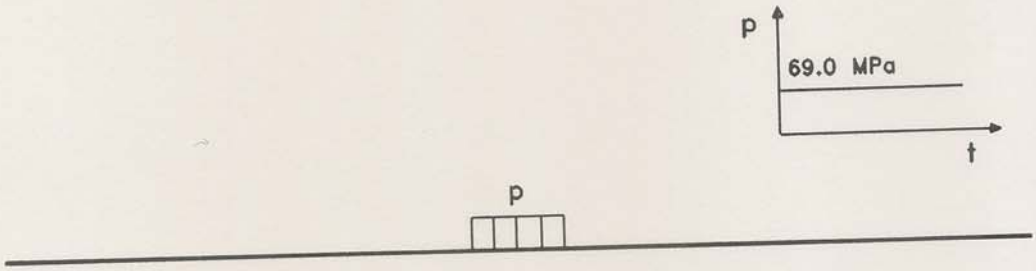


Figure 3.27 Half-space under stress distribution

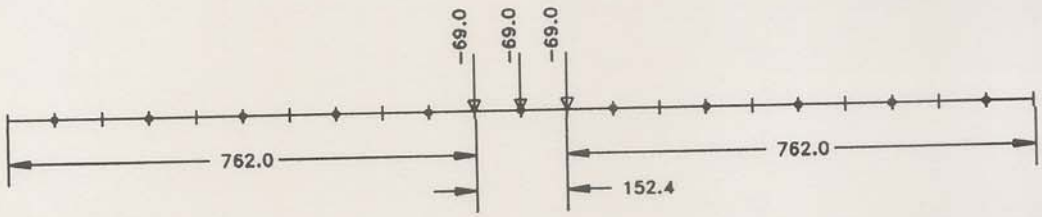


Figure 3.28 Discretization of the free surface and applied tractions

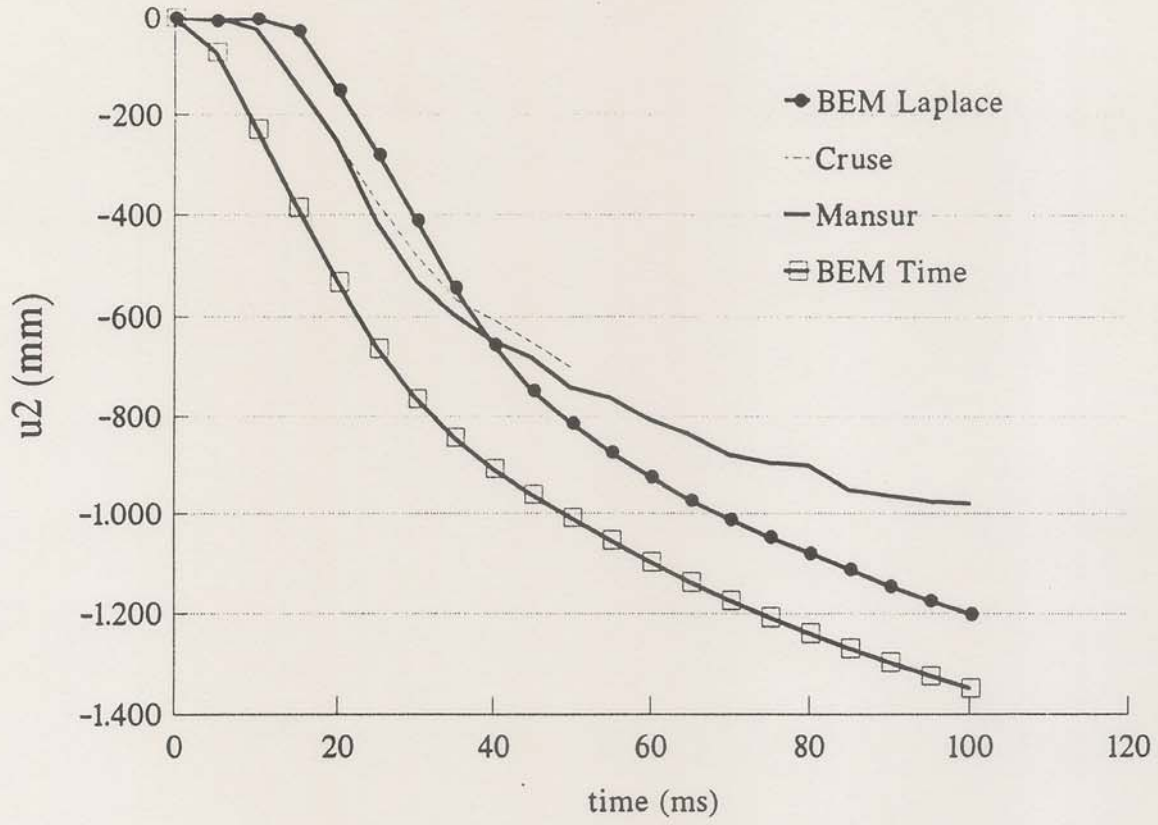


Figure 3.29 Displacement at point D(0.0,-76.2)

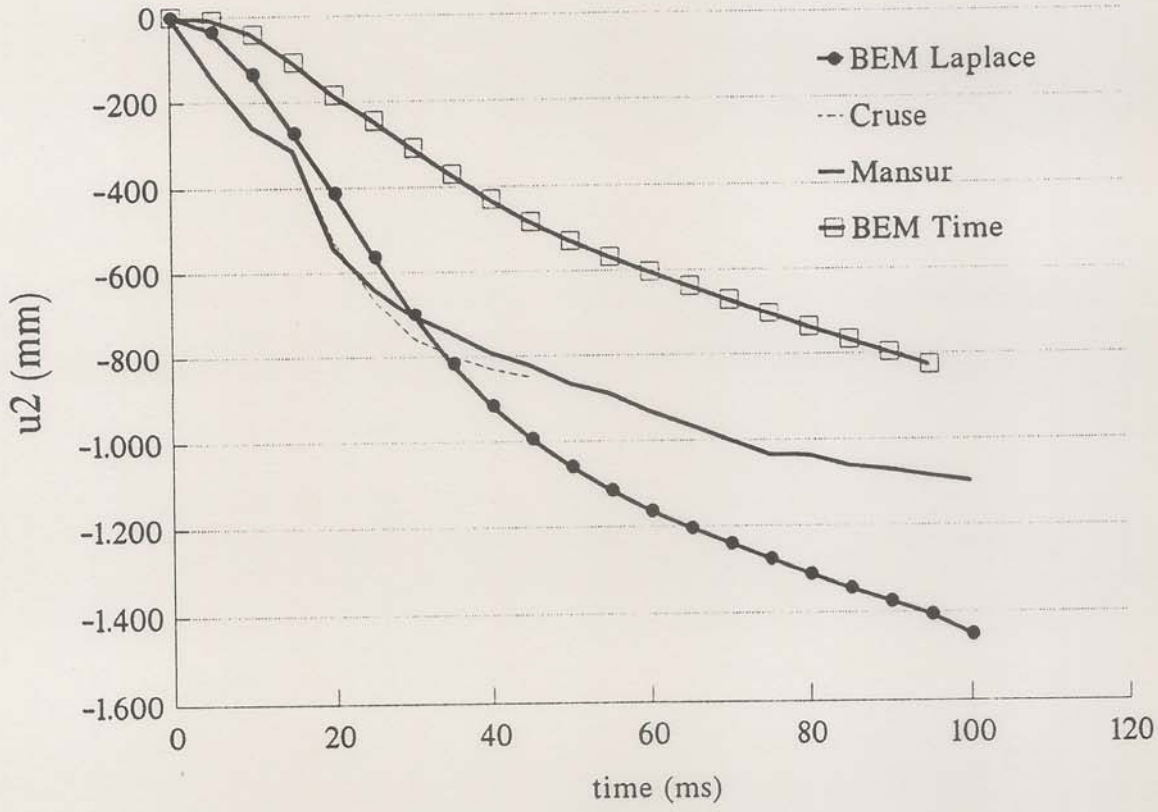


Figure 3.30 Displacement at point C(0.0,0.0)

CHAPTER 4

FUNDAMENTAL TOPICS

4.1 INTRODUCTION

In this chapter, special topics used throughout the thesis are covered. More specifically reference is made here to the following topics:

a. Formulae used for evaluating the pressures due to an explosion at a point.

b. Numerical integration technique for calculating singular and non-singular cases.

c. Isoparametric representation of line elements and technique of numerical calculation of strains.

d. Numerical evaluation of the local to the elements co-ordinate system.

e. Stresses and strain tensors, traction vector and transformations of stresses.

4.2 WAVE GENERATED BY AN EXPLOSION AT A POINT

In assessing the dynamic response of a medium subjected to an explosion, it is important to know the velocity (V) and the pressure (P) at a point. Three types of empirical formulae that give these quantities are examined here.

4.2.1 The Cubic Root Type

This type is mainly used in explosions from conventional weapons (U.S.Army 1984, Weidlinger and Hinman 1987). The velocity of the ground particle and the pressure generated at a node i=1,2,...n of the mesh are given by the expressions:

$$V_{0i}=Bf\left(\frac{R_i}{W^{1/3}}\right)^{-n}, \qquad P_{0i} = K(\rho c)V_{0i} \tag{4.1}$$

where V_{0i}: ground particle velocity at node i.

P_{0i}: the peak pressure at node i.

W : charge weight of the explosive.

R_i : the distance of the explosion point from the examining node i.

ϱc : the acoustic impedance of the ground with ϱ being the mass density and c the wave velocity, taken equal to seismic velocity of the ground.

n : attenuation coefficient of the ground.

f : coupling factor of the explosion energy, taken as 1.0 for fully buried explosions.

B : empirical coefficient that is unit system dependent.

K : constant that is unit system dependent.

Units Special attention should be paid to the system of units used in the above equations.

a. *Imperial units*

The units used are

V_{0i} [in/s] or [ft/s], R_i[ft], W[lbf] (force), ϱc [psi/ft/s]

the constant K takes values

$$K = \begin{cases} B/12 & when \quad V_{0i}\ [in/s] \\ B & when \quad V_{0i}\ [ft/s] \end{cases} \qquad (4.2)$$

using these units and constants in equation (4.1) the pressure is

P_{0i} [psi]

b. *S.I. units*

The units used are

V_{0i}[m/s], R_i[m], W[N], ϱc [MPa s/m]

the constant K takes values

$$K = B\frac{0.3048^{(n+1)}}{4.4482^{(n/3)}} \qquad (4.3)$$

the units of the pressure extracting from equation (4.1) are : P_{0i} [MPa].

The acoustic impedance ϱc is measured in [MPa s/m]. Usually the mass density is given to [Mg/m^3] and can be converted to [MPa s^2/m^2] as

ϱ [Mg/m^3] = ϱ/1000 [MPa s^2/m^2]

4.2.2 The Square Root Type

This type is usually used in blasting operations (Dowding, 1985). The expressions for the pressure remains the same as in equation (4.1) and the velocity is given by:

$$V_{0i} = Bf\left(\frac{R_i}{W^{1/2}}\right)^{-n}, \qquad P_{0i} = K(\rho c)V_{0i} \qquad (4.4)$$

where the meaning of variables and coefficients was explained above and their values according to the unit system that is used are exactly the same as above, with the exception that in SI units the constant K is

$$K = B\frac{0.3048^{(n+1)}}{4.4482^{(n/2)}} \qquad (4.5)$$

4.2.3. Empirical Formula

The following equations for the ground particle velocity is also used for rock blasting operations (Holberg and Persson, 1980; Hoek and Brown, 1980):

$$V_{0i}=Bf\left(\frac{W^{\alpha}}{R_i^{\beta}}\right) \qquad P_{0i} = K(\rho c)V_{0i} \qquad (4.6)$$

where α,β are site dependent empirical coefficients.

Units

The units usually used are

v[mm/s] or [m/s], R[m], W[kgf] (force), ϱc [MPa s/m]

the constant K takes values

$$K=\begin{cases}\dfrac{1000}{B}9.8066^{a} & when \quad V_{0i}\ [mm/s]\\ \dfrac{1}{B}9.8066^{a} & when \quad V_{0i}\ [m/s]\end{cases} \qquad (4.7)$$

using these units and constants in equation (4.6) the pressure units are P_{0i} [MPa]

4.2.4 Decay of the Wave

Irrespective of the velocity formula, the decay of pressure in time, for every node is:

$$P_i = P_{0i}e^{-\frac{t}{t_a}} \qquad (4.8)$$

where t is the elapsed time counted from the time of arriving at node i, and $t_a=R_i/c$.

4.3 NUMERICAL INTEGRATION

For the numerical solution of the boundary integral equation (2.26), which is repeated

here :

one dimensional Gaussian Quadrature scheme is used in this work (Stround and Secrest,

$$\sum_{l=1}^{N} c_{ij}\bar{u}_i = \sum_{l=1}^{N}\sum_{m=1}^{L}\left\{\sum_{\alpha=1}^{3}\bar{T}_{i\alpha}\int_{-1}^{1} G_{ij}M_\alpha |J| d\xi - \sum_{\alpha=1}^{3}\bar{U}_{i\alpha}\int_{-1}^{1} F_{ij}M_\alpha |J| d\xi\right\} \quad (4.9)$$

1966). The computer algorithms used for the numerical integration are those implemented by Manolis and Beskos (1988) that have been subjected to major modifications, to improve their performance.

The Gaussian formula used is the one for non-singular functions which is

$$\int_{-1}^{+1} f(\xi)d\xi = \sum_{i=1}^{n_g} w_i f(\xi_i) \quad (4.10)$$

where n_g is the total number of integration points, ξ are intrinsic co-ordinates (interval [-1,1]). This formula is applied for the numerical integration at all the points, singular and non-singular, with indirect schemes for singular cases that are explained in the forthcoming.

4.3.1 Singular and Non-singular Cases

a. Singular Case

This is the case when the source node (point P), coincides with one of the three nodes of the examining quadratic element (receiver node, point Q). As a result of this, the distance r between source and receiver becomes zero, and the fundamental solutions (Appendices A, B and C), are becoming indeterminate. The singularity for the product of the displacement G_{ij} kernel by the shape function, for the product of traction F_{ij} kernel by the shape function and for the Jacobian J (Equation 4.37) according to Watson (1979) is

$$\overline{G}_{ij}(P,Q(\xi))\,M_\alpha(\xi) \in O\left(\frac{1}{r^2}\ln r\right)$$

$$\overline{F}_{ij}(P,Q(\xi))\,M_\alpha(\xi) \in O\left(\frac{1}{r^2}\right)$$

$$|J(\xi)| = \frac{\partial q}{\partial \xi} \in O(r^2) \qquad \textbf{(4.11)}$$

This reveals that the product of the kernels by the shape functions and by the Jacobian, needed to form the integral equations (4.9), tends to a finite limit (Watson, 1979). Thus the calculation of the kernels at the singular points is attainable. The weakly logarithmic singularity in G_{ij} is integrable by subdividing the element into even number of subelements and as will be shown the strongly singularity in F_{ij} is evaluated by using rigid body solutions.

The quadratic elements of the mesh that are treated as singular are those that contain the source node (as a middle or edge node).

b. Non-singular Case

This is the case when the distance between source and receiver node is not zero. The elements for which the source node is not coincident with one of their three nodes, are treated as non-singular. In this case the numerical integration of both the displacement and traction fundamental solution, presents no difficulty. For better accuracy and greater efficiency, the element is subdivided into subelements according to the distance of the source node from the element and the element length.

4.3.2 Subdivision of Elements and Number of Integration Points

For improved accuracy and singularities problems, each element is divided into subelements as in Watson (1979), Manolis and Beskos (1988).

a. Singular Case

An element is automatically considered as singular by the developed program, when the distance of the source node from any node of the element is lower than a given tolerance that may be controlled by the user (usual case 10^{-3}). In this case, the element is divided into an even number of subelements (Figure 4.1), so that every node of the quadratic element is located at the end of the element instead of within the subelements. For the numerical integration of the displacement kernel G_{ij} the Gaussian formula for non-singular functions can be used (Equation 4.10), since the points of integration are approaching close to the singular node, but they never coincide with it. The subdivision into subelements and the points of Gaussian quadrature may also be controlled by the user (usual case is subdivision into 4 subelements and a six point Gaussian quadrature). The traction kernel F_{ij} is still

indeterminate, and rigid body translation scheme is used, will be shown later in section 4.3.2.c.

b. Non-singular Case

For distance of the source node from any of the nodes of the element greater or equal than the tolerance (usually 10^{-3}), the element is considered automatically by the program as non-singular. In this case, the subdivision of the element into subelements Figure (4.2), is

$$d=\frac{r_{\min}}{l_e}\,, \qquad \begin{aligned} d<1 &\rightsquigarrow n_s=3 \\ 1\le d\le 2 &\rightsquigarrow n_s=2 \\ d>2 &\rightsquigarrow n_s=1 \end{aligned} \tag{4.12}$$

where r_{min} is the minimum distance of the source node from the three nodes of the element, l_e is the length of the element, n_s is the number of the subelements.

The order of integration is found by solving the inequality
where n_g is the number of integration points, c is the desired tolerance controlled by the user

$$\left(\frac{\alpha}{4r_{\min}}\right)^{2n_g}\le c\alpha \;\Rightarrow\; n_g=\frac{1}{2}\frac{\ln(c\alpha)}{\ln\left(\frac{\alpha}{4r_{\min}}\right)} \tag{4.13}$$

(usual case 10^{-4}).

c. Rigid Body Translation

To calculate the fundamental traction solutions F_{ij} at the singular points, a rigid body translation scheme is used (Watson 1979, Manolis et. al. 1986). With this technique the contribution of the terms c_{ij} of the left part of equation (4.9) can be also calculated. Equation (4.9) can be written

$$\sum_{l=1}^{N}\sum_{m=1}^{L}\left\{\sum_{\alpha=1}^{3}\overline{T}_{i\alpha}\int_{-1}^{1}G_{ij}M_{\alpha}|J|d\xi\right\} = \sum_{l=1}^{N}c_{ij}\overline{u}_i+\sum_{l=1}^{N}\sum_{m=1}^{L}\left\{\sum_{\alpha=1}^{3}\overline{U}_{i\alpha}\int_{-1}^{1}F_{ij}M_{\alpha}|J|d\xi\right\} \tag{4.14}$$

Assuming that the singular node is the node β of the singular element q, the kernels G_{ij} and

F_{ij} at this node are identical to static ones, as the distance r of the source to the receiver node tends to zero.

Expressing equation (4.14) for the singular node only and applying unit rigid body motion to the mesh ($u_i=U_{i\alpha}=1$),the tractions become zero, hence:

$$c_{ij}+\sum_{m=1}^{L}\left\{\sum_{\alpha=1}^{3}\int_{-1}^{1}F_{ij}^{s}M_{\alpha}|J|d\xi\right\}=0 \qquad \textbf{(4.15)}$$

where F_{ijs} is the statical kernel for the examining singular case.

$$c_{ij}+\sum_{m=1}^{L}\left\{\sum_{\alpha=1}^{3}\int_{-1}^{1}F_{ij}^{s}M_{\alpha}|J|d\xi\right\}_{(m\neq q)}+\left\{\sum_{\alpha=1}^{3}\int_{-1}^{1}F_{ij}^{s}M_{\alpha}|J|d\xi\right\}_{(m=q)}=0 \qquad \textbf{(4.16)}$$

Separating the singular element q results in

Further, assuming that β is the singular node on the singular element q (β=1,2 or 3), and separating this singular node, yields

$$\begin{aligned}d_{ij}^{s}&=c_{ij}+\left\{\int_{-1}^{1}F_{ij}^{s}M_{\alpha}|J|d\xi\right\}_{(\alpha=\beta)}\\&=-\sum_{m=1}^{L}\left\{\sum_{\alpha=1}^{3}\int_{-1}^{1}F_{ij}^{s}M_{\alpha}|J|d\xi\right\}_{(m\neq q)}-\left\{\sum_{\alpha=1}^{3}\int_{-1}^{1}F_{ij}^{s}M_{\alpha}|J|d\xi\right\}_{(\alpha\neq\beta)}\end{aligned} \qquad \textbf{(4.17)}$$

The behaviour of the transient kernel as r→0 is identical with the static kernel. Moreover, the term c_{ij} is identical for the static and dynamic case depending only on the geometry at the singular node (Banerjee et. al., 1987). Thus the dynamic diagonal block is given by an expression similar to the static solution

$$d_{ij}^{d}=c_{ij}+\left\{\int_{-1}^{1}F_{ij}^{d}M_{\alpha}|J|d\xi\right\}_{(\alpha=\beta)} \qquad \textbf{(4.18)}$$

Eliminating the term c_{ij} between equations (4.15) and (4.18), results in

$$d_{ij}^{d} = d_{ij}^{s} + \left\{ \int_{-1}^{1} \left(F_{ij}^{d} - F_{ij}^{s} \right) M_{\alpha} \, | J | \, d\xi \right\}_{(\alpha=\beta)} \qquad \textbf{(4.19)}$$

The dynamic and static traction kernels are calculated by using normal Gaussian rules, as in the case of the displacement kernel. With this methodology there is no need for the calculation of the terms c_{ij}. The static diagonal block is calculated in the first time step and in the subsequent time steps it remains unchanged.

The above are valid for bounded regions (interior problem). For infinite regions (exterior problems), no rigid body movement is possible. However the above procedure may still be used for calculating the diagonal blocks, by making the infinite domain a finite one. For this reason, the boundary of the mesh with surface S is surrounded by an imaginary circle of radius r_0 tending to infinity and surface Γ_0. This circle contains all the points of the region regularity concept of Kellog). The direction of the surface Γ_0 is anticlockwise, because the outer boundary encloses the boundary of the mesh. The rigid body displacements are applied as before for the boundary $S+\Gamma_0$, obtaining a similar equation to equation (4.17) :

$$d_{ij}^{s} = c_{ij} + \left\{ \int_{-1}^{1} F_{ij}^{s} M_{\alpha} \, | J | \, d\xi \right\}_{(\alpha=\beta)} = - \sum_{m=1}^{L} \left\{ \sum_{\alpha=1}^{3} \int_{-1}^{1} F_{ij}^{s} M_{\alpha} \, | J | \, d\xi \right\}_{(m \neq q)}$$

$$- \left\{ \sum_{\alpha=1}^{3} \int_{-1}^{1} F_{ij}^{s} M_{\alpha} \, | J | \, d\xi \right\}_{(\alpha \neq \beta)} - \lim_{r_0 \to \infty} \int_{\Gamma_0} F_{ij}(P,Q) \, ds \qquad \textbf{(4.20)}$$

the additional term (last term of the second part of the equation) is the azimuthal integral (Watson, 1979). For this integral, the source node P is located on the surface S and the receiver point is on the surface Γ_0 of the infinite sphere. The fundamental solutions of tractions and displacements, from physical consideration, are taken as finite on Γ_0. Further the azimuthal (jump term), is a line integral on a line Γ_0, of a non-zero function always in the anticlockwise direction (Figure 4.3) and its value is

for exterior problems (opening) and infinite domains :

$$\lim_{r_0 \to \infty} \int_{\Gamma_0} F_{ij}(P,Q) \, ds = -\delta_{ij} \qquad \textbf{(4.21)}$$

for exterior problems (opening) and semi-infinite domains :

$$\lim_{r_0 \to \infty} \int_{\Gamma_0} F_{ij}(P,Q)ds = -0.5\delta_{ij} \qquad (4.22)$$

for interior problems (solid) :

$$\lim_{r_0 \to \infty} \int_{\Gamma_0} F_{ij}(P,Q)ds = 0 \qquad (4.23)$$

In the case of multi-subregional (multi-zonal) problems, the integral equations are formed independently for every subregion and the dynamic diagonal blocks are calculated independently for every zone.

For structures in semi-infinite media, the jump term takes different values for the boundaries of the mesh. So it assumes a value of $-0.5\delta_{ij}$ for the ground free surface, or of $-\delta_{ij}$ for the boundaries of external type (clock-wise direction) that are inside the medium or zero for all the boundaries of a zone inside the medium (clock-wise or anti-clock-wise direction).

4.4 ISOPARAMETRIC LINE ELEMENTS

4.4.1 Introduction

The most effective way of dealing with problems employing numerical methods, is the use of isoparametric elements. These elements are extensively used in FEM (Burnett, 1987), and in recent years were also introduced in BEM.

The elements that are used in this work are of isoparametric quadratic type, with three nodes.

4.4.2 Shape Functions

For illustration purposes, the element of Figure (4.4) with the three nodes I,II,III with Cartesian co-ordinates in a global system (x1,y1), (x2,y2) and (x3,y3) respectively is examined. The numbering of the nodes in this example is taken in clockwise direction. Every point on the element is mapping onto a point in the transformed curvilinear plane and vice-versa (Banerjee and Butterfield,1981). This transformation is one to one. The transformation from the curvilinear to the Cartesian co-ordinate system, is accomplished by the following interpolation functions, which are extracted from Lagrange's interpolation formulae, according to Robinson (1973) :

$$M_1(\xi) = \frac{1}{2}\xi(\xi - 1)$$
$$M_2(\xi) = (1 - \xi^2)$$
$$M_3(\xi) = \frac{1}{2}\xi(1 + \xi) \qquad \textbf{(4.24)}$$

where ξ is the local co-ordinate in the transformed curvilinear shape (non-dimensional), that takes values in the interval [-1 ,1]. These functions M_k, accept the unity value at node k and vanish at the rest nodes of the element. For example, for the first (k=1) node of the element is ξ=-1, $M_1(-1)=1$, $M_2(-1)=0$, $M_3(-1)=0$. Similarly for the middle (k=2) node is $\xi=0$, $M_1(0)=0$, $M_2(0)=1$, $M_3(0)=0$ and finally for the last (k=3) node $\xi=1$, $M_1(1)=0$, $M_2(1)=0$, $M_3(1)=1$.

Given the co-ordinate ξ of a node P in the curvilinear plane, counted from the middle node (Figure 4.5), and the Cartesian co-ordinates of the three nodes of the element (X_i,Y_i), (i=1,2,3), the co-ordinates of the same point P in the Cartesian co-ordinate system can be found by using the interpolation functions as

$$x(\xi) = M_\alpha(\xi)X_\alpha = M_1(\xi)X_1 + M_2(\xi)X_2 + M_3(\xi)X_3$$
$$y(\xi) = M_\alpha(\xi)Y_\alpha = M_1(\xi)Y_1 + M_2(\xi)Y_2 + M_3(\xi)Y_3 \qquad \textbf{(4.25)}$$

$$x_k = M_\alpha(\xi)X_{k\alpha} \qquad \textbf{(4.26)}$$

or in tensor notation
where k=1,2 is the index for the two co-ordinates at the node P and α=1,2,3 is the index for the three nodes of the element.

4.4.3 Displacement and Traction Functions

For isoparametric elements the same shape interpolation functions are also used for the displacements and tractions. Thus the interpolation of the displacement (traction) at the point P(x,y), by known values of n and known values of the displacement (traction) at the three nodes of the element U_k (T_k), is given by

$$u_k(\xi) = M_\alpha(\xi)U_\alpha = M_1(\xi)U_1 + M_2(\xi)U_2 + M_3(\xi)U_3$$
$$t_k(\xi) = M_\alpha(\xi)T_\alpha = M_1(\xi)T_1 + M_2(\xi)T_2 + M_3(\xi)T_3 \qquad \textbf{(4.27)}$$

4.4.4 Derivatives of Interpolation Functions

The derivatives of the interpolation functions with respect to ξ, are found by differentiating equation (4.24)

$$M_{1,\xi} = \frac{dM_1}{d\xi} = \xi - \frac{1}{2}$$
$$M_{2,\xi} = \frac{dM_2}{d\xi} = -2\xi$$
$$M_{3,\xi} = \frac{dM_3}{d\xi} = \xi + \frac{1}{2} \qquad \textbf{(4.28)}$$

4.4.5 Strains

The calculation of the strains is a difficult and usually a complicated task. In the following sections an attempt is made to clarify this matter.

Strain in a direction is the change of displacement in this direction and is equal to the derivative of the displacement in that direction. Allowing the point P on the element to coincide with one of the three nodes of an element, leads to the following analysis.

4.4.5.1 Strains in the Tangential and Normal Direction

a. Strains in a Cartesian Co-ordinate System

In Figure (4.6) a quadratic element is represented in the undeformed (a) and deformed (b) position. The displacement of a node, e.g. the middle one, is u with co-ordinates in x and y global axis u_1 and u_2 respectively. Accepting a new infinitesimal deformation to a new position (c), the u_1 and u_2 displacements are deformed to a new position u_1' and u_2'. The change of u_1 in the x and y directions is

$$u_{1,1} = \varepsilon_{11} = \frac{\partial u_1}{\partial x}, \qquad u_{1,2} = \varepsilon_{12} = \frac{\partial u_1}{\partial y} \qquad \textbf{(4.29)}$$

Similarly the change of u_2 in the x and y directions is

$$u_{2,1} = \varepsilon_{21} = \frac{\partial u_2}{\partial x}, \qquad u_{2,2} = \varepsilon_{22} = \frac{\partial u_2}{\partial y} \qquad \textbf{(4.30)}$$

Resolving the strains in the local axis (q), tangential to the surface of the element at node P (Figure 4.7 and 4.8) results in

$$u_{1,q} = u_{1,1}\cos\theta_{q1} + u_{1,2}\cos\theta_{q2} = u_{1,1}q_1 + u_{1,2}q_2$$
$$u_{2,q} = u_{2,1}\cos\theta_{q1} + u_{2,2}\cos\theta_{q2} = u_{2,1}q_1 + u_{2,2}q_2 \qquad \textbf{(4.31)}$$

and in tensor notation:

$$u_{k,q} = u_{k,i}\,q_i\ , \qquad k=1,2\ , \quad i=1,2 \qquad \textbf{(4.32)}$$

similarly the strains in the local axis (n) normal to the surface of the element at node P (Figure 4.9 and 4.10) are:

$$u_{1,n} = u_{1,1}\cos\theta_{n1} + u_{1,2}\cos\theta_{n2} = u_{1,1}n_1 + u_{1,2}n_2$$
$$u_{2,n} = u_{2,1}\cos\theta_{n1} + u_{2,2}\cos\theta_{n2} = u_{2,1}n_1 + u_{2,2}n_2 \qquad \textbf{(4.33)}$$

$$u_{k,n} = u_{k,i}\,n_i\ , \qquad k=1,2, \qquad i=1,2 \qquad \textbf{(4.34)}$$

and in tensor notation:
where in the above, n_i, q_i (i=1,2), are the direction cosines of the n,q local axes with x_1,x_2 global axes.

b. Strains in a Curvilinear Co-ordinate System

Applying the chain rule in differentiation one obtains for the strains in tangential and normal local axes, for every one of the three nodes of the element:

$$u_{k,q}(\xi) = \frac{\partial u_k}{\partial q} = \frac{\partial u_k}{\partial \xi}\frac{\partial \xi}{\partial q} = \frac{\partial u_k}{\partial \xi}\Big/\frac{\partial q}{\partial \xi} = \frac{u_{k,\xi}}{q_{,\xi}} \qquad \textbf{(4.35)}$$

where k=1,2 stands for the x and y co-ordinates of the displacement. Taking into account equation (4.27) it is

$$\frac{\partial u_k}{\partial \xi} = \frac{\partial}{\partial \xi}(M_\alpha(\xi)U_\alpha) = U_\alpha \frac{\partial M_\alpha}{\partial \xi} = U_1 \frac{\partial M_1}{\partial \xi} + U_2 \frac{\partial M_2}{\partial \xi} + U_3 \frac{\partial M_3}{\partial \xi} \quad \textbf{(4.36)}$$

It is

$$\frac{\partial q}{\partial \xi} = |J(\xi)| = \sqrt{\left(\frac{\partial x}{\partial \xi}\right)^2 + \left(\frac{\partial y}{\partial \xi}\right)^2} \quad \textbf{(4.37)}$$

and Equation (4.35) results in

$$u_{k,q} = \frac{u_{k,\xi}}{q_{,\xi}} = \frac{u_{k,\xi}}{|J|} \quad \textbf{(4.38)}$$

where $|\mathrm{J}|$ is the Jacobian (determinant) of the transformation.

c. Comparison Between Cartesian and Curvilinear Tangential Strains

From the above analysis, two types for the strains in the tangential direction have derived, one for Cartesian co-ordinate system (Equation 4.32) and one for curvilinear (Equation 4.38), which can be summarised as

$$u_{k,q} = u_{k,i} q_i = \frac{u_{k,\xi}}{q_{,\xi}} \quad \textbf{(4.39)}$$

To verify the validity of this relation, taking account equation (4.38) the first equality gives

$$u_{k,q} = u_{k,i} q_i = \frac{u_{k,\xi}}{x_{i,\xi}} q_i = \frac{u_{k,\xi}}{x_{i,\xi}} \frac{1}{|J|} x_{i,\xi} = \frac{u_{k,\xi}}{|J|} \quad \textbf{(4.40)}$$

and finally

$$u_{k,i} q_i = \frac{u_{k,\xi}}{|J|} \quad \textbf{(4.41)}$$

which is true, since the transformation of the derivatives of a vector field Ψ from the

Cartesian to curvilinear system is given by (Banerjee and Butterfield, 1981) as

$$\Psi_{,i} = \frac{\Psi_{,\xi}}{|J|} \tag{4.42}$$

4.4.5.2. Strains in the global x,y directions

Strains over x : Applying the chain rule again for the strains over x direction is

$$u_{k,1}(\xi) = \frac{\partial u_k}{\partial x} = \frac{\partial u_k}{\partial \xi}\frac{\partial \xi}{\partial x} = \frac{\partial u_k}{\partial \xi}\Big/\frac{\partial x}{\partial \xi} = \frac{u_{k,\xi}}{x_{,\xi}} \tag{4.43}$$

$u_{k,\xi}$ is given by equation (4.36) and is repeated here

$$\frac{\partial u_k}{\partial \xi} = \frac{\partial}{\partial \xi}(M_\alpha(\xi)U_\alpha) = U_\alpha\frac{\partial M_\alpha}{\partial \xi} = U_1\frac{\partial M_1}{\partial \xi} + U_2\frac{\partial M_2}{\partial \xi} + U_3\frac{\partial M_3}{\partial \xi} \tag{4.44}$$

and

$$\frac{\partial x_k}{\partial \xi} = \frac{\partial}{\partial \xi}(M_\alpha(\xi)X_\alpha) = X_i\frac{\partial M_i}{\partial \xi} = X_1\frac{\partial M_1}{\partial \xi} + X_2\frac{\partial M_2}{\partial \xi} + X_3\frac{\partial M_3}{\partial \xi} \tag{4.45}$$

Strains over y : Similarly for the strains over y direction is

$$u_{k,2} = \frac{\partial u_k}{\partial y} = \frac{\partial u_k}{\partial \xi}\frac{\partial \xi}{\partial y} = \frac{\partial u_k}{\partial \xi}\Big/\frac{\partial y}{\partial \xi} = \frac{u_{k,\xi}}{y_{,\xi}} \tag{4.46}$$

$$\frac{\partial y_k}{\partial \xi} = \frac{\partial}{\partial \xi}(M_\alpha(\xi)Y_\alpha) = Y_\alpha\frac{\partial M_\alpha}{\partial \xi} = Y_1\frac{\partial M_1}{\partial \xi} + Y_2\frac{\partial M_2}{\partial \xi} + Y_3\frac{\partial M_3}{\partial \xi} \tag{4.47}$$

and

or in the abbreviated notation with k=1,2 for x,y direction:

$$u_{k,i}(\xi) = \frac{\partial u_k}{\partial x_i} = \frac{u_{k,\xi}}{x_{i,\xi}} \tag{4.48}$$

or

$$u_{k,\xi} = U_\alpha M_{\alpha,\xi} \tag{4.49}$$

and

$$x_{i,\xi} = X_\alpha M_{\alpha,\xi} \tag{4.50}$$

where indices are k=1,2 for the co-ordinates in x and y global axes, i=1,2 for the derivatives in x and y direction and α=1,2,3 for the three nodes of the examining quadratic element.

4.5 LOCAL CO-ORDINATE SYSTEM AT A NODE

The boundary of the domain is discretized into quadratic elements. Every element has three nodes, at the start, middle and at the end of the element. The direction of numbering the nodes of the element for the external problem (opening), is such that travelling on the boundary the material lies to the left hand (Figure 4.5) while for the internal problem (inclusion), the material lies to the right hand.

The numerical integration is performed by collocating at every node, in a local co-ordinate system with origin the examining node. In this system the fundamental solutions for every element and every node of the element are calculated first, and then are transformed to the general co-ordinate system. These calculations are performed at every time step. So it is crucial to establish in the most effective way this co-ordinate system and the transformation procedure.

In the local system the x axis, is the unit vector (n) normal to the surface of the element, in a right hand direction for the above traveller (n is directed away from the material), as designated at Figure (4.5). The y local axis is the unit vector (q) tangent to the surface at right to n angle. These normal and tangent unit vectors, form an anticlockwise local co-ordinate system. The direction cosines are given by

$$q_1 = \cos\theta_{q1} = \frac{\partial x}{\partial s} = \frac{\partial x}{\partial \xi}\frac{\partial \xi}{\partial s} = \frac{1}{|J(\xi)|}\frac{\partial x}{\partial \xi} = \frac{1}{|J(\xi)|}\frac{\partial}{\partial \xi}(M_\alpha(\xi)X_\alpha)$$
$$= \left(X_1\frac{\partial M_1}{\partial \xi} + X_2\frac{\partial M_2}{\partial \xi} + X_3\frac{\partial M_3}{\partial \xi} \right)\frac{1}{|J(\xi)|} \qquad \textbf{(4.51)}$$

and

$$q_2 = \cos\theta_{q2} = \frac{\partial y}{\partial s} = \frac{\partial y}{\partial \xi}\frac{\partial \xi}{\partial s} = \frac{1}{|J(\xi)|}\frac{\partial y}{\partial \xi} = \frac{1}{|J(\xi)|}\frac{\partial}{\partial \xi}(M_\alpha(\xi)X_\alpha)$$
$$= \left(Y_1\frac{\partial M_1}{\partial \xi} + Y_2\frac{\partial M_2}{\partial \xi} + Y_3\frac{\partial M_3}{\partial \xi} \right)\frac{1}{|J(\xi)|} \qquad \textbf{(4.52)}$$

where θ_{q1}, θ_{q2} are the angles of x and y global axis with the q tangent unit vector (y local axis) respectively.
From simple trigonometry, the direction cosines of x and y global axis with the n normal unit vector (x local axis) are

$$n_1 = \cos\theta_{n1} = \cos\theta_{q2} = q_2 \qquad \textbf{(4.53)}$$

and

$$n_2 = \cos\theta_{n2} = -\cos\theta_{q1} = -q_1 \qquad \textbf{(4.54)}$$

4.6 STRESS-STRAIN TENSORS, TRACTION VECTOR, TRANSFORMATIONS

4.6.1 Strains

The problems examined in this work are considered under plane strain conditions, implying (Crouch and Starfield,1983) that the component of the displacement u_z is zero and the components u_x and u_y are independent of the z co-ordinate. This leads for the strains to

$$e_{xz} = e_{zx} = \frac{1}{2}\left(\frac{\partial u_x}{\partial z} + \frac{\partial u_z}{\partial x} \right) = 0 \ , \qquad e_{yz} = e_{zy} = \frac{1}{2}\left(\frac{\partial u_y}{\partial z} + \frac{\partial u_z}{\partial y} \right) = 0 \quad \textbf{(4.55)}$$

4.6.2 Stresses

The stress tensor consists of stresses acting on planes whose normals are parallel to the axes of the co-ordinate system. The sign convention for the stresses used throughout this work is defined in Figure (4.11). The stress σ_{ij} is positive if it acts in the positive (or negative) j direction on a plane whose outward normal points in the positive (or negative) i direction. With this sign convention, which is used mainly in continuum mechanics, the compression stresses are negative and the tensile stresses are positive. This is opposite from the sign convention normally used in Geomechanics.

4.6.3 Tractions

The traction vector (stress vector), is the vector that represents the stresses acting on a plane of arbitrary orientation. This vector t_i refers to two components t_1 and t_2 that corespond to the x and y axes respectively. These two components t_1 and t_2 can be expressed in terms of the stress tensor elements and the angle of the inward to the material unit vector normal to the plane with the x global axis as shown in the Figure (4.12). This unit normal vector is opposite to the unit outward normal vector (n) to the element, as defined in Appendix B.

$$\begin{aligned} t_x &= \sigma_{xx}\cos\alpha + \sigma_{yx}\sin\alpha = \sigma_{xx} n_1' + \sigma_{yx} n_2' \\ t_y &= \sigma_{xy}\cos\alpha + \sigma_{yy}\sin\alpha = \sigma_{xy} n_1' + \sigma_{yy} n_2' \end{aligned} \qquad \textbf{(4.56)}$$

$$t_i = \sigma_{ij} n_j' \qquad \textbf{(4.57)}$$

and in tensor notation
where

$$n_1' = \cos\alpha = -n_1 = -\cos\theta_{n1} \ , \qquad n_2' = \sin\alpha = -n_2 = -\sin\theta_{n1} \tag{4.58}$$

n_1, n_2 are the direction cosines of the unit normal to the surface of the element vector, in the outward direction (outward the material), and n_1', n_2' are the direction cosines of the opposite to n unit vector (towards the material).

4.6.4 Transformations

The transformation of the stresses from the x,y global co-ordinate system to the x',y' local system is given by

$$\sigma_{ij}' = \lambda_{ik}\lambda_{jm}\sigma_{km} \tag{4.59}$$

where the symbol λ refer to the direction cosines. The expression (4.59) upon exponsion results in the following equations

$$\begin{aligned}
\sigma_{x'x'} &= \sigma_{xx}\cos^2\omega + \sigma_{yy}\sin^2\omega + 2\sigma_{xy}\sin\omega\cos\omega \\
\sigma_{y'y'} &= \sigma_{xx}\sin^2\omega + \sigma_{yy}\cos^2\omega - 2\sigma_{xy}\sin\omega\cos\omega \\
\sigma_{x'y'} &= \sigma_{y'x'} = -(\sigma_{xx} - \sigma_{yy})\sin\omega\cos\omega + \sigma_{xy}(\cos^2\omega - \sin^2\omega)
\end{aligned} \tag{4.60}$$

The transformation from the local co-ordinate system back to the global co-ordinate system is

$$\begin{aligned}
\sigma_{xx} &= \sigma_{x'x'}\cos^2\omega + \sigma_{y'y'}\sin^2\omega - 2\sigma_{x'y'}\sin\omega\cos\omega \\
\sigma_{yy} &= \sigma_{x'x'}\sin^2\omega + \sigma_{y'y'}\cos^2\omega + 2\sigma_{x'y'}\sin\omega\cos\omega \\
\sigma_{xy} &= \sigma_{yx} = (\sigma_{x'x'} - \sigma_{y'y'})\sin\omega\cos\omega + \sigma_{x'y'}(\cos^2\omega - \sin^2\omega)
\end{aligned} \tag{4.61}$$

where ω is the angle of x global with the x' local axis, measured from the x axis in anticlockwise direction.

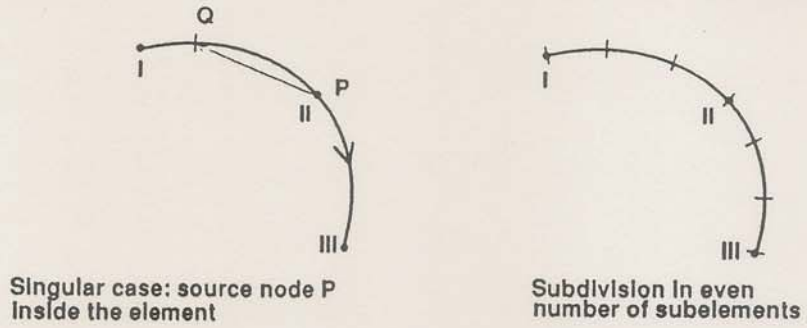


Figure 4.1 Subdivision of an element in subelements for the singular case

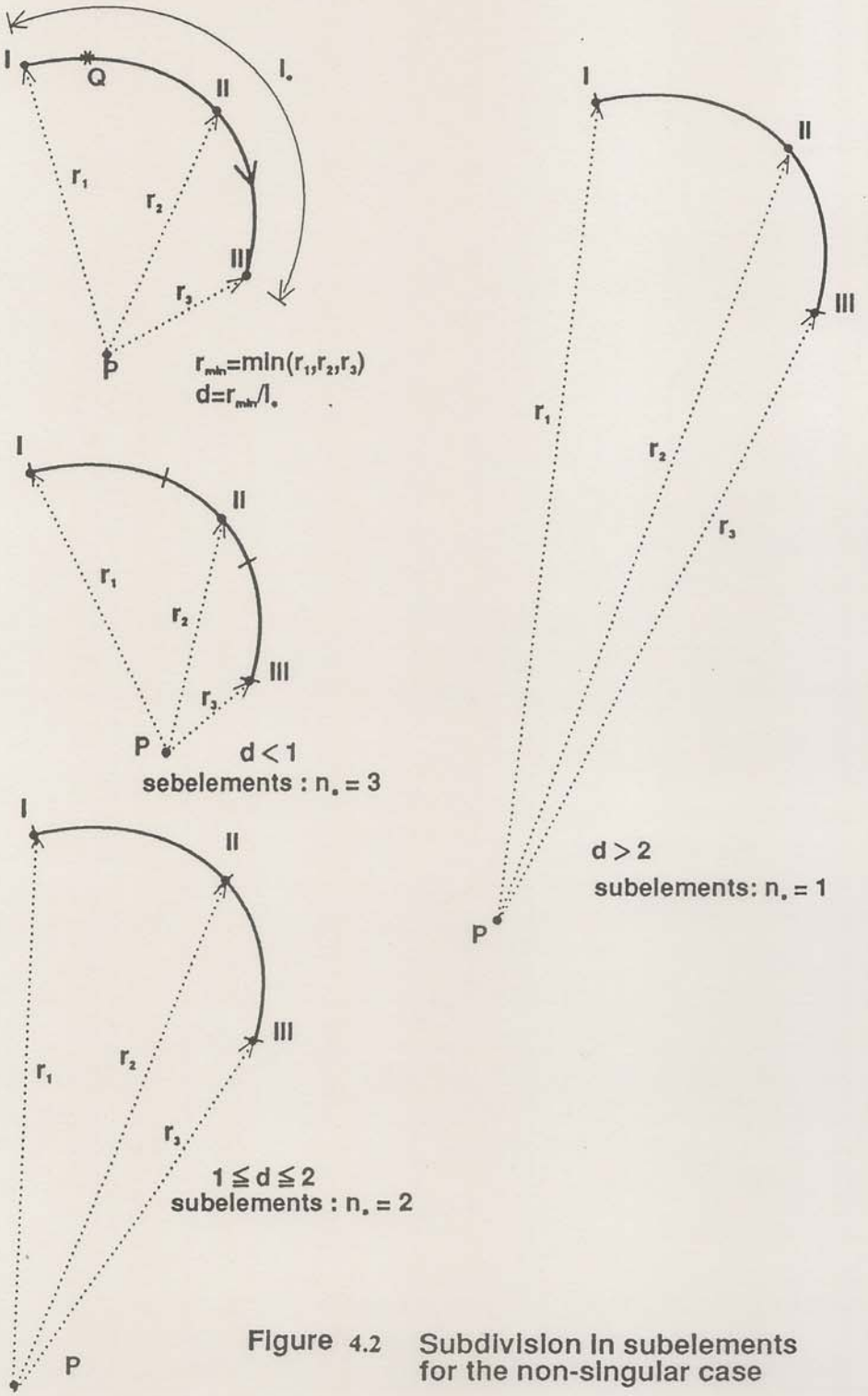
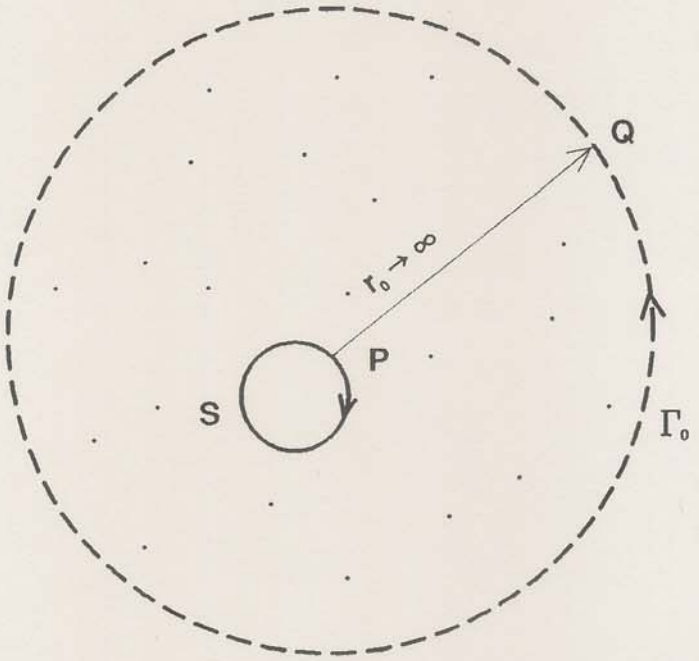
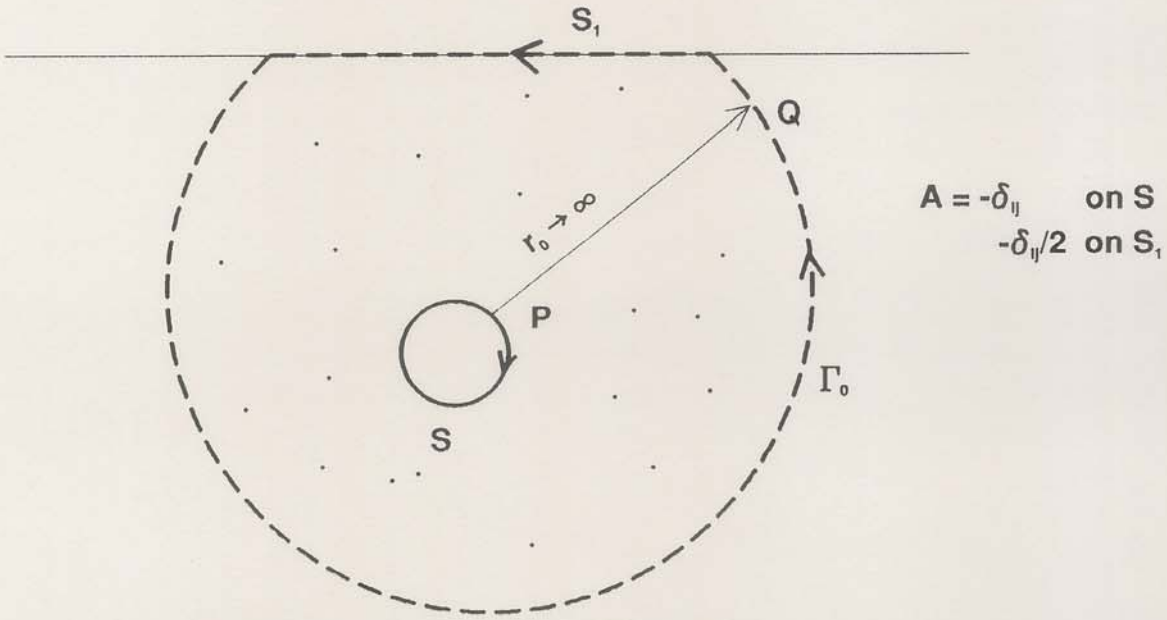


Figure 4.2 Subdivision in subelements for the non-singular case



Exterior problem, infinite region



Exterior problem, semi-infinite region



Interior problem (bounded region)

Figure 4.3 Azimuthal integral (jump term) for infinite regions

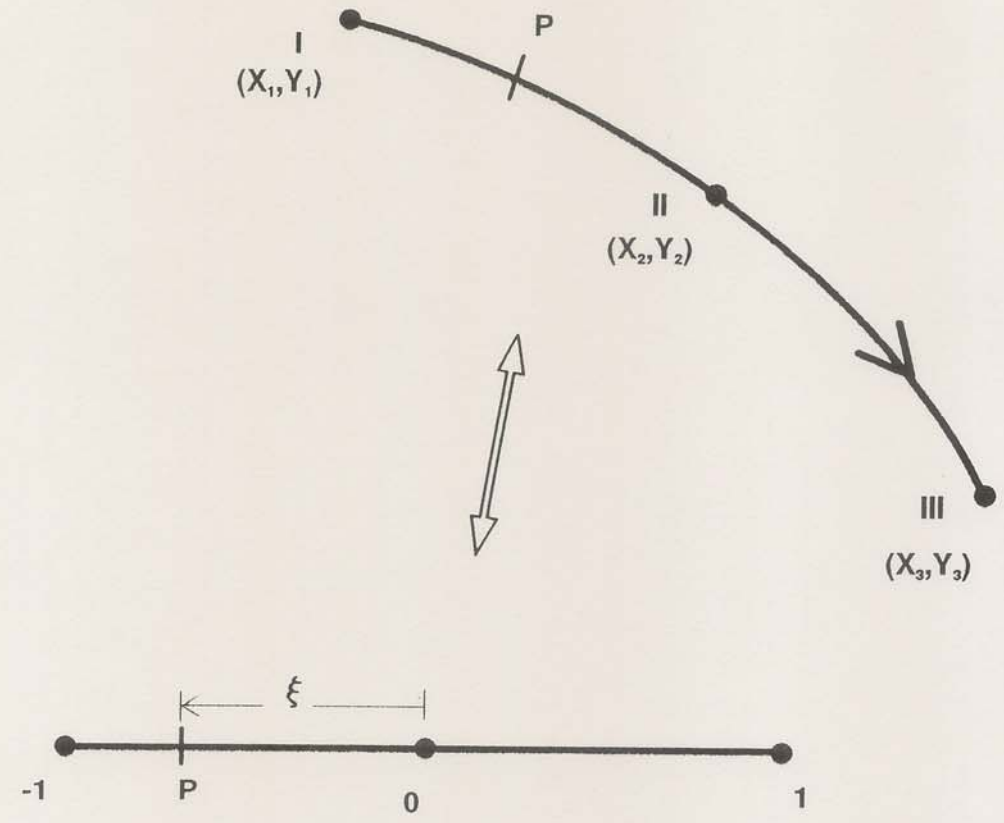
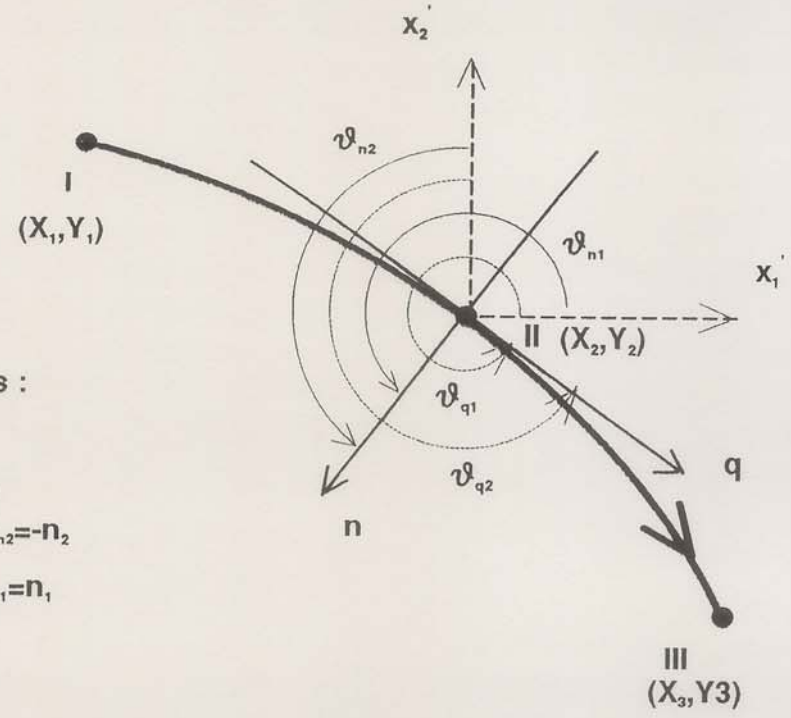


Figure 4.4 Transformation of an element from Cartesian to curvilinear plane



direction cosines :

$n_1=\cos\vartheta_{n1}$

$n_2=\cos\vartheta_{n2}=\sin\vartheta_{n1}$

$q_1=\cos\vartheta_{q1}=-\cos\vartheta_{n2}=-n_2$

$q_2=\cos\vartheta_{q2}=\cos\vartheta_{n1}=n_1$

Figure 4.5 Normal and tangential unit vectors of the local co-ordinate system at a node of an element

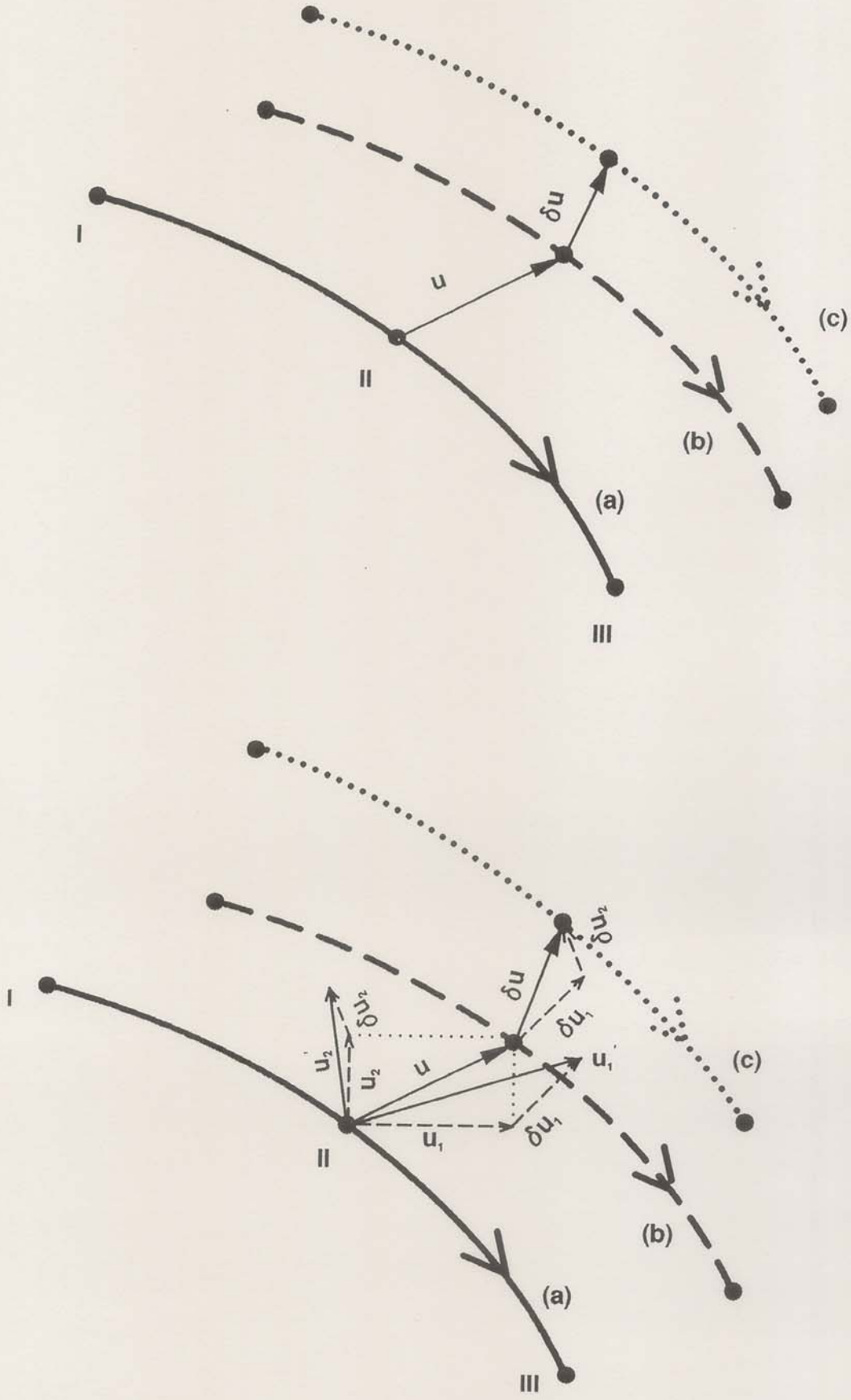
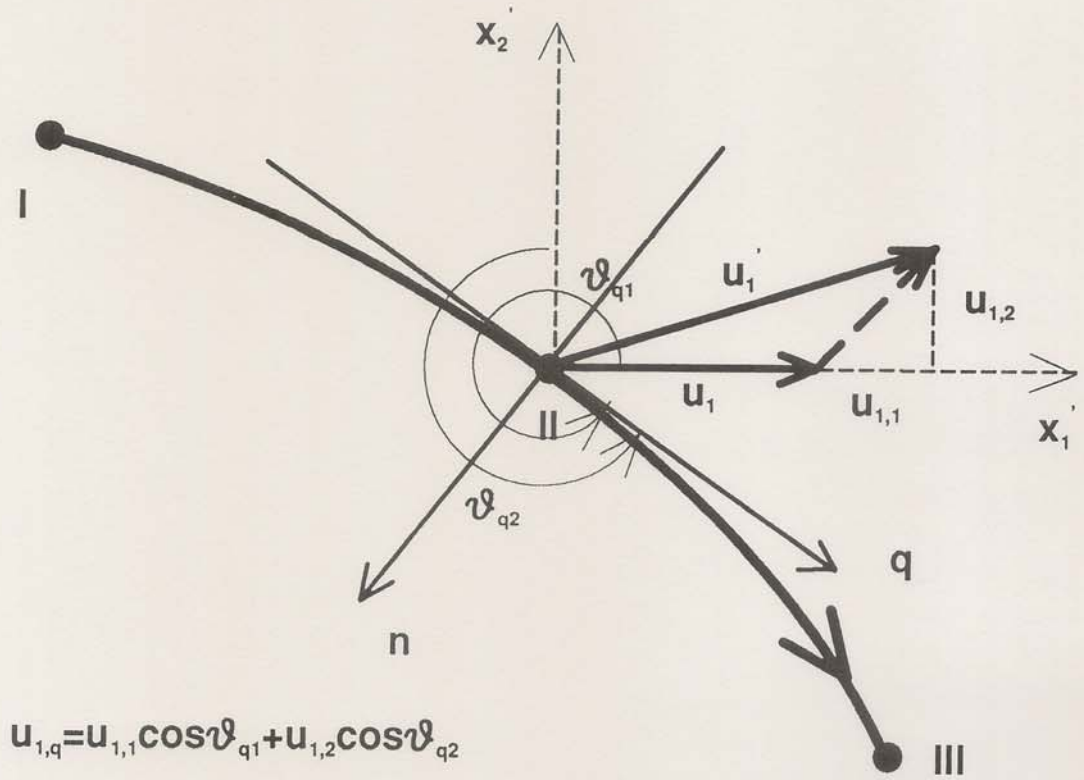
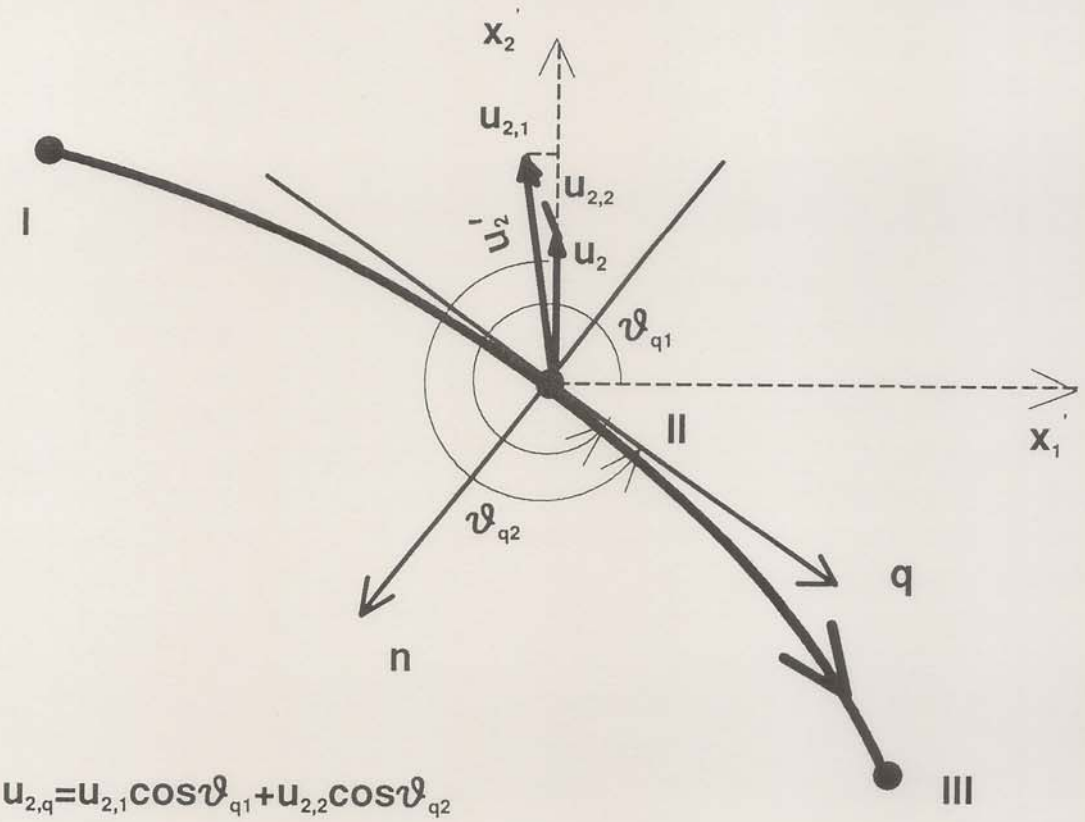


Figure 4.6 Infinitesimal displacements at a node of an element



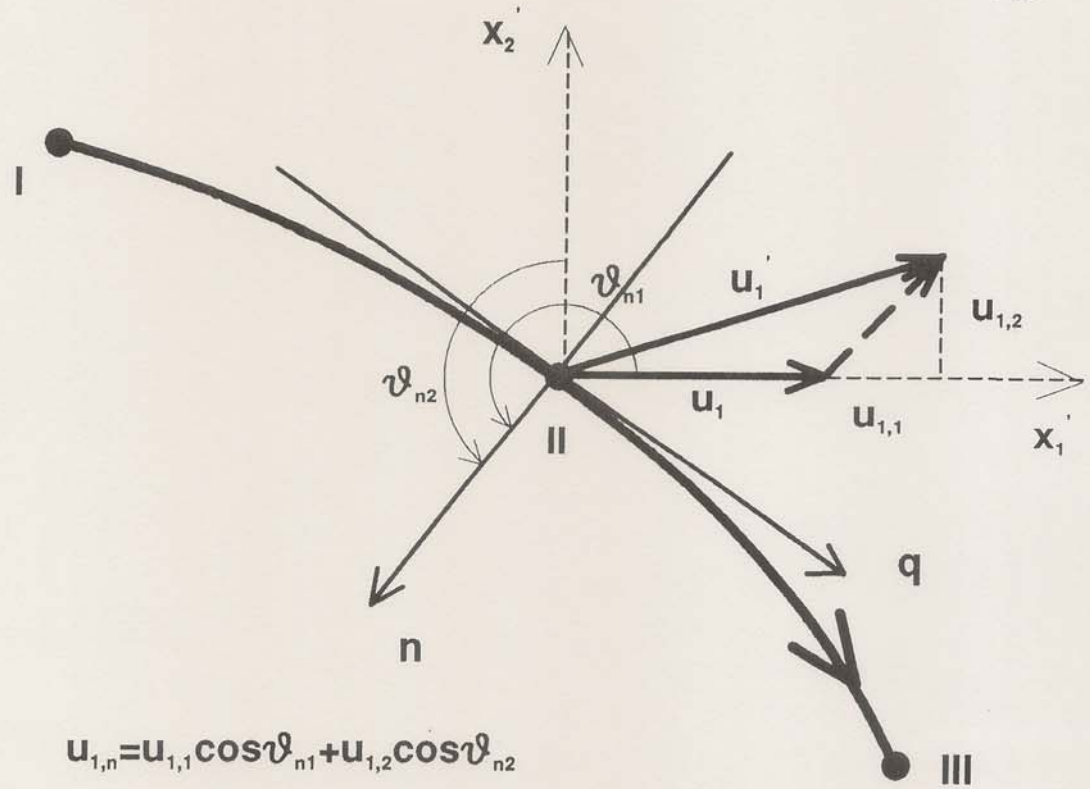
$$u_{1,q}=u_{1,1}\cos\vartheta_{q1}+u_{1,2}\cos\vartheta_{q2}$$

Figure 4.7 Strain of u_1 in the tangential direction



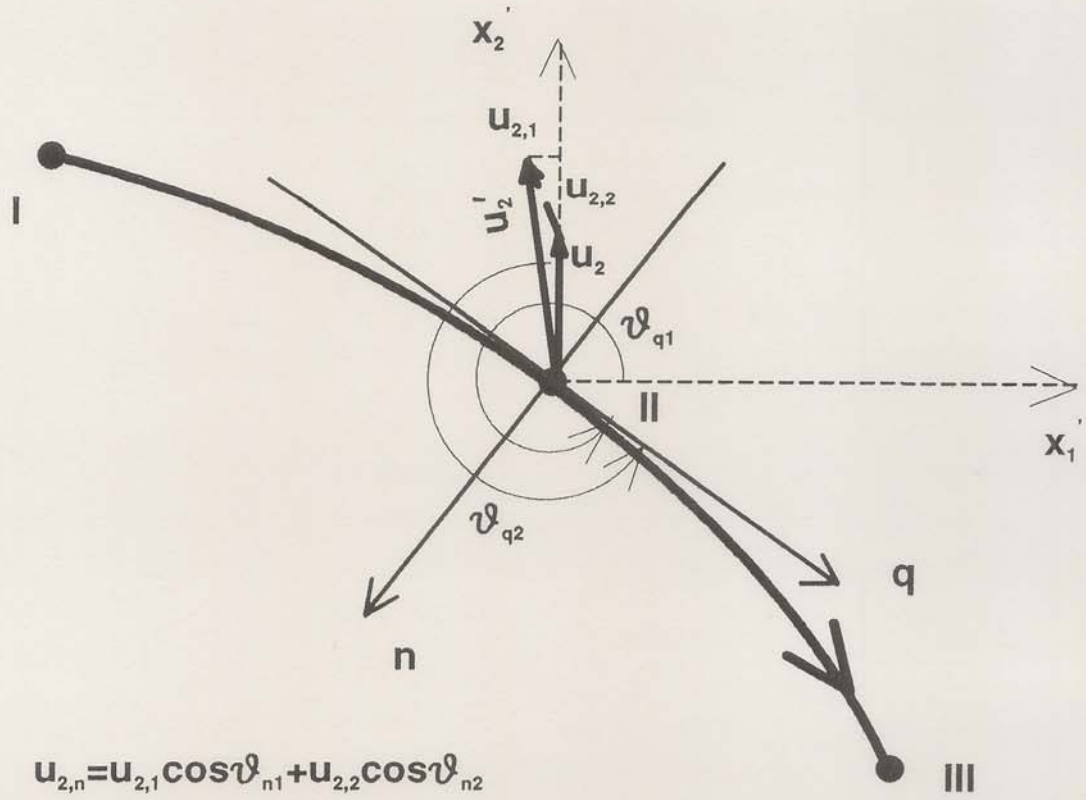
$$u_{2,q}=u_{2,1}\cos\vartheta_{q1}+u_{2,2}\cos\vartheta_{q2}$$

Figure 4.8 Strain of u_2 in the tangential direction



$$u_{1,n}=u_{1,1}\cos\vartheta_{n1}+u_{1,2}\cos\vartheta_{n2}$$

Figure 4.9 Strain of u_1 in the normal direction



$$u_{2,n}=u_{2,1}\cos\vartheta_{n1}+u_{2,2}\cos\vartheta_{n2}$$

Figure 4.10 Strain of u_2 in the normal direction

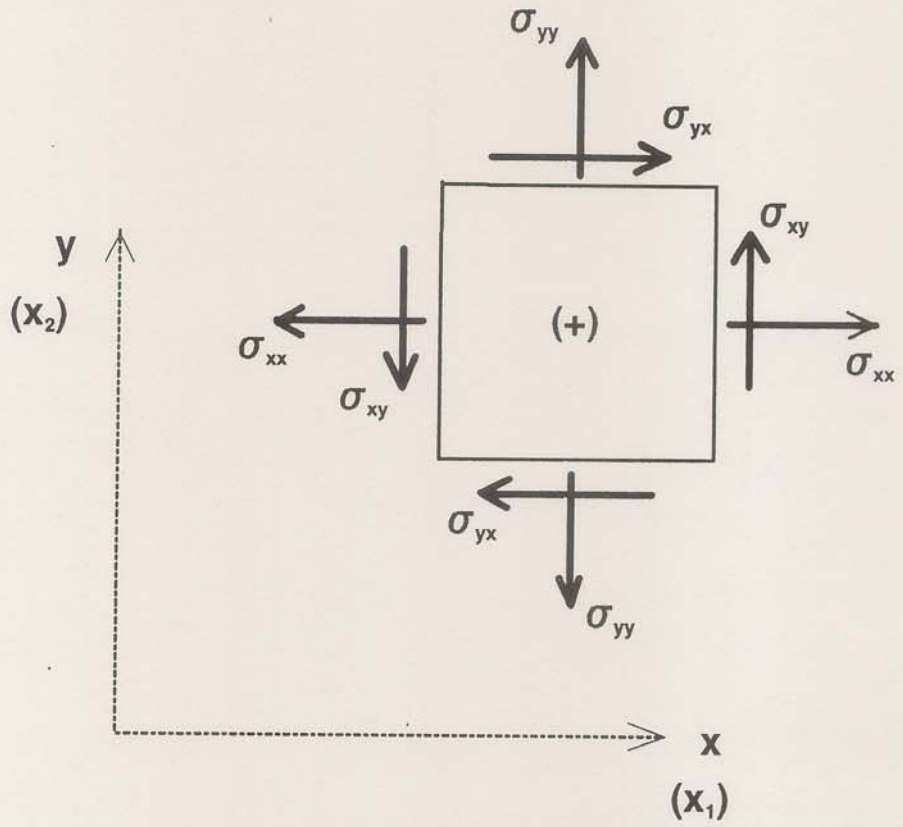


Figure 4.11 Sign convention for the stress tensor

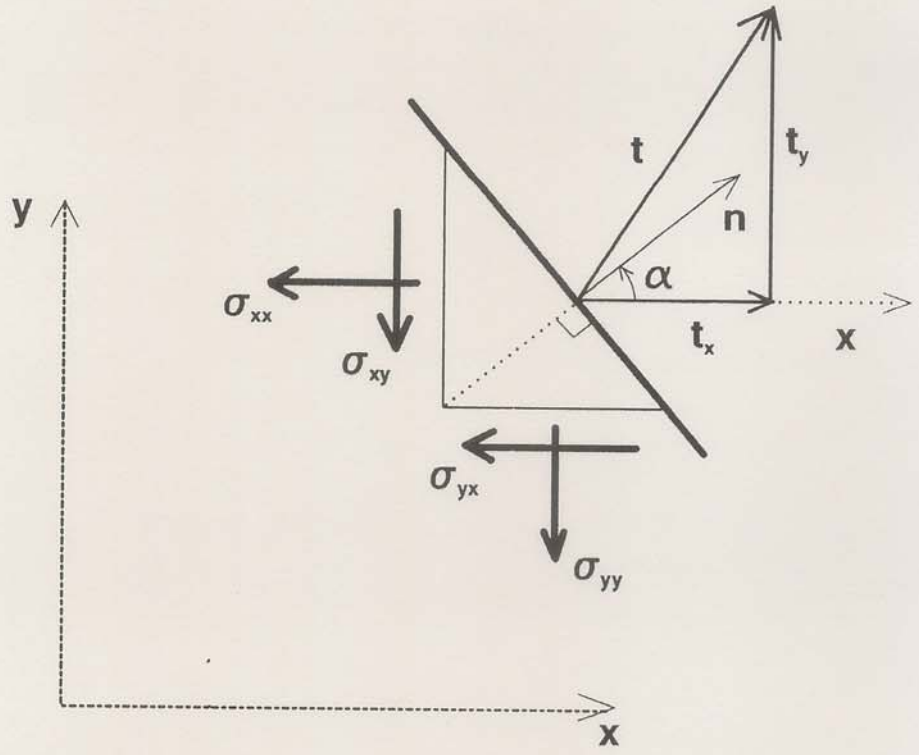


Figure 4.12 Tractions on a plane with unit normal vector at angle α with x axis

CHAPTER 5

MULTI-ZONAL BEM

5.1 INTRODUCTION

An area inside an infinite or semi-infinite, domain consisting of material with characteristics different than the surrounding material, forms an inclusion or a zone.

Wave propagation problems around elastic inclusions have been addressed by Niwa et al. (1984) and Niwa and Hirose (1986). In the first work, the incident wave was a displacement wave field while in the second work the reflected wave in a semi-infinite medium was assumed to be known beforehand.

Furthermore, underground openings with lining subjected to dynamic loading, were examined by Manolis (1980).

In this work the problem of multi-zonal BEM is examined and the formulation and solution of the resulting equations is implemented in a general form suitable for solving a range of similar problems. The solution is achieved by applying the incident tractions or displacements on the surface of the free boundaries. The scattered field is evaluated by solving the system of equations resulting from the application of BEM on the zones and the compatibility and equilibrium conditions on the interface boundary.

5.2. FORMULATION

The methodology for solving multi-domain systems, consists of applying the BEM independently for every zone and introducing the compatibility and equilibrium conditions on the common interface boundaries. The resulting equations can be solved for the unknown quantities (tractions or displacements). This formulation is applicable for Laplace and time domain as well.

5.2.1 Inclusion in an Infinite Medium

The case of an inclusion in an infinite domain is examined first (Figure 5.1). The common interface boundary encloses two domains, the one inside the inclusion with the outer boundary (a) traversing in the counter-clock-wise direction, and the remaining part of the medium that is surrounded by the same interface surface (1) which is traversing now in the clock-wise direction. In addition, the compatibility

and equilibrium conditions on the common interface are applied, and the resulting system may be solved for the unknown quantities. More specifically, the boundary integral equation (2.27) is rewritten for time step (n) :

$$[F]\{u\} = [G]\{t\} \tag{5.1}$$

which gives for the inclusion (domain (a)) :

$$\left[F^{(1)}\right]\left\{u^{(1)}\right\} = \left[G^{(1)}\right]\left\{t^{(1)}\right\} \tag{5.2}$$

and for the remaining part of the outer to the inclusion region (domain (1)) :

$$\left[F^{(a)}\right]\left\{u^{(a)}\right\} = \left[G^{(a)}\right]\left\{t^{(a)}\right\} \tag{5.3}$$

The compatibility and equilibrium conditions on the interface boundary are

$$\begin{aligned} \left\{u^{(a)}\right\} &= \left\{u^{(1)}\right\} \\ \left\{t^{(a)}\right\} &= \left\{-t^{(1)}\right\} \end{aligned} \tag{5.4}$$

Equations (5.2) and (5.3) in view of (5.4) yield

$$\begin{aligned} \left[F^{(1)}\right]\left\{u^{(1)}\right\} &= \left[G^{(1)}\right]\left\{t^{(1)}\right\} \\ \left[F^{(a)}\right]\left\{u^{(1)}\right\} &= \left[G^{(a)}\right]\left\{-t^{(1)}\right\} \end{aligned} \tag{5.5}$$

For known applied conditions on the boundary, the above system of equations may be solved for the unknowns.

It is worth mentioned here, that the jump term (chapter 6, section 6.3) for calculating $F^{(a)}$ is taken zero while the jump term for $F^{(1)}$ is taken as δ_{ij}. Also the dynamic diagonal block is calculated independently for every zone.

5.2.2 Inclusion and Opening in an Infinite Medium

The case of simultaneous existence of an inclusion and opening is examined (Figure 5.2). For simplicity only one boundary of inclusion or opening is taken, but the following are also applicable for multiple boundaries. The common interface surface is traversing once in the counter-clock-wise direction (a) for the inclusion

and once in clock-wise direction for the remaining portion of the domain.

The boundary integral equation (2.27) can be written for the inclusion

$$\left[F^{(a)}\right]\left\{u^{(a)}\right\}=\left[G^{(a)}\right]\left\{t^{(a)}\right\} \qquad \textbf{(5.6)}$$

and for the remaining part

$$\left[F^{(\beta)}\right]\left\{u^{(\beta)}\right\}=\left[G^{(\beta)}\right]\left\{t^{(\beta)}\right\} \qquad \textbf{(5.7)}$$

where β represents the boundaries (1) and (2) (Figure 5.2). The tractions $t^{(\beta)}$ are the opposite to the incident tractions (unit normal inward to the material). By appropriate partitioning the terms of the last equation gives

$$\begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{bmatrix}\begin{Bmatrix} u^{(1)} \\ u^{(2)} \end{Bmatrix}=\begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}\begin{Bmatrix} t^{(1)} \\ t^{(2)} \end{Bmatrix} \qquad \textbf{(5.8)}$$

where F_{ij} and G_{ij} are the parts of the matrices that multiply the tractions $u^{(j)}$ and $t^{(j)}$ respectively.

Combining equations (5.6), (5.8) and (5.4) in a common matrix notation becomes

$$\begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \\ F^{(a)} & 0 \end{bmatrix}\begin{Bmatrix} u^{(1)} \\ u^{(2)} \end{Bmatrix}=\begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \\ -G^{(a)} & 0 \end{bmatrix}\begin{Bmatrix} t^{(1)} \\ t^{(2)} \end{Bmatrix} \qquad \textbf{(5.9)}$$

The tractions $t^{(2)}$ are known to be equal to the opposite to the incident tractions on the boundary of the opening (in order to maintain a free surface by superpositioning the incident and scattered field). The displacements on both boundaries and the tractions on the interface boundary $t^{(1)}$ of the inclusion can be found by re-arranging equation (5.9):

$$\begin{bmatrix} F_{11} & F_{12} & -G_{11} \\ F_{21} & F_{22} & -G_{21} \\ F^{(a)} & 0 & G^{(a)} \end{bmatrix}\begin{Bmatrix} u^{(1)} \\ u^{(2)} \\ t^{(1)} \end{Bmatrix}=\begin{bmatrix} G_{12} \\ G_{22} \\ 0 \end{bmatrix}\left\{ t^{(2)}\right\} \qquad \textbf{(5.10)}$$

which can be solved for the unknowns $u^{(1)}$, $u^{(2)}$ and $t^{(1)}$.

The jump term (chapter 6, para. 6.3.2) for calculating $F^{(a)}$ is taken zero while the jump term for F_{ij} is taken as δ_{ij} .

5.2.3 Underground Openings With Lining

The case of underground openings with lining is examined here. For simplicity only one lining is examined, but the following are also applicable for multiple linings. The lining is regarded as a zone and the structure is divided into two regions, the lining and the surrounding to the lining medium. The lining is enclosed by the inner boundary (b) traversed in a clock-wise direction and the outer boundary (a) in a counter-clock-wise direction. The medium is determined by the boundary (c) transverse in a counter-clock-wise direction (Figure 5.3). The boundary integral equation (5.1) can be written for the two regions, independently:

a. For the lining

$$\left[F^{(l)}\right]\left\{u^{(l)}\right\}=\left[G^{(l)}\right]\left\{t^{(l)}\right\} \qquad \textbf{(5.11)}$$

by appropriate partitioning the terms of the equation (5.11) gives

$$\begin{bmatrix} F_{aa} & F_{ab} \\ F_{ba} & F_{bb} \end{bmatrix}\begin{Bmatrix} u^{(a)} \\ u^{(b)} \end{Bmatrix}=\begin{bmatrix} G_{aa} & G_{ab} \\ G_{ba} & G_{bb} \end{bmatrix}\begin{Bmatrix} t^{(a)} \\ t^{(b)} \end{Bmatrix} \qquad \textbf{(5.12)}$$

where F_{ij} and G_{ij} are the parts of the matrices that multiply the tractions $u^{(j)}$ or $t^{(j)}$ respectively (i=a,b and j=a,b).

b. For the remaining part of the surrounding medium

$$\left[F^{(c)}\right]\left\{u^{(c)}\right\}=\left[G^{(c)}\right]\left\{t^{(c)}\right\} \qquad \textbf{(5.13)}$$

Combining equations (5.12), (5.13) and (5.4) results in

$$\begin{bmatrix} F_{aa} & F_{ab} \\ F_{ba} & F_{bb} \\ F^{(c)} & 0 \end{bmatrix}\begin{Bmatrix} u^{(c)} \\ u^{(b)} \end{Bmatrix}=\begin{bmatrix} G_{aa} & G_{ab} \\ G_{ba} & G_{bb} \\ -G^{(c)} & 0 \end{bmatrix}\begin{Bmatrix} t^{(c)} \\ t^{(b)} \end{Bmatrix} \qquad \textbf{(5.14)}$$

the tractions $t^{(b)}$ are known beforehand to be equal to the opposite to the incident tractions on the boundary of the opening (in order to obtain a free surface by superpositioning the incident and scattered field). The displacements on both boundaries and the tractions on the interface boundary $t^{(c)}$ of the inclusion can be found by re-arranging equation (5.14):

$$\begin{bmatrix} F_{aa} & F_{ab} & G_{aa} \\ F_{ba} & F_{bb} & G_{ba} \\ F^{(c)} & 0 & -G^{(c)} \end{bmatrix} \begin{Bmatrix} u^{(c)} \\ u^{(b)} \\ t^{(c)} \end{Bmatrix} = \begin{bmatrix} G_{ab} \\ G_{bb} \\ 0 \end{bmatrix} \left\{ t^{(b)} \right\} \qquad \textbf{(5.15)}$$

which can be solved for the unknowns $u^{(c)}$, $u^{(b)}$ and $t^{(c)}$

The jump term (Chapter 6, para. 6.3.2) for calculating the tractions matrix of the zone $F^{(l)}$ is taken zero while the jump term for $F^{(c)}$ is taken as δ_{ij} .

5.2.4 Inclusion in a Semi-infinite Medium

The case of an inclusion in a semi-infinite medium may be resolved in a manner similar to the one employed for the case of an inclusion and opening in an infinite medium (see previous section 5.2.2). The boundary (2) in this case is the surface of the ground (Figure 5.2). The tractions $t^{(1)}$ and $t^{(2)}$ are also the opposite to the incident tractions in order to maintain traction free surfaces after superposing the incident and scattered field.

5.2.5 Internal Points

The displacements and stresses at internal points may be found by using the above methodology of inclusions. The aim behind this is to create an inclusion inside the material, with a closed boundary passing through the internal points where stresses and displacements need to be calculated. The material of the inclusion is the same as the surrounding material. With this technique, one may avoid the calculation of the complex internal kernels that are normally required.

5.3 APPLICATIONS

5.3.1 Infinite Domain Solutions

5.3.1.1 Circular Cross Section Cavity with Liner of the Same Material in an Infinite Medium Swept by a Plane P Wave

At first the liner is assumed to be of the same material as the infinite domain. The analysis is performed in order to examine the accuracy of the solution of the proposed method, and to compare the results with known solutions of unlined openings.

The material properties adopted in this problem are the same as in the application

of section 2.3.1.1 (E=20,000 MPa, v=0.25, ρ=2.55 Mg/m^3). The total time interval is 0.015 s implemented in 20 time steps.

The incident wave is also a shock compressional plane wave of -10.0 MPa magnitude that is applied suddenly and maintained constant in time. The angle of incidence of the incoming wave is 0° from the global horizontal x axis. The position of the plane wavefront at t=0, is considered as tangent to the inner circle, at node 33.

The radius of the cavity is 2.50 m (Figure 5.3), and the ring has thickness of 0.20 m extending from the free surface of the cavity (b) to an imaginary boundary (a) that forms the interface surface of the ring and the surrounding material (Figure 5.4). Sixteen quadratic elements per boundary are used to represent the surface of the mesh, corresponding to a total number of 64 nodes (Figure 5.5). For the formation of the kernels during the numerical integration the inner free surface (b) of the ring is traversed clock-wise (cw), while the interface surface of the ring (a) is traversed counter-clock-wise (ccw). The surrounding medium extends from the interface surface (c) to infinity and the interface boundary is traversed in a cw fashion.

The variation of the tangential stresses versus time steps for the nodes 33, 41 and 49 on the inner surface and 1, 9 and 17 on the outer surface, which are at angles 0°, 270° and 180° from the x axis respectively, is shown in Figure (5.6). Furthermore, the displacements in x and y directions, are shown in Figures (5.7) and (5.8) respectively. Comparison of hoop stresses between Figures (5.6) and (2.1) and of displacements between Figures (5.7) and (2.2) show that the results for the inner surface are in very good agreement with those obtained for an unlined cavity for the respective nodes.

5.3.1.2 Circular Cross Section Cavity in an Infinite Stiff Medium with a Stiffer Liner Swept by a Plane P Wave

The same problem has been solved by Garnet and Crouzet-Pascal (1966) by superpositioning of appropriate steady state solutions to the wave equations. More recently, Manolis and Beskos (1983) solved the same problem by combining the BEM and the Laplace transform. This problem is solved here in order to investigate the accuracy of the developed program.

The material properties used in this problem are a relatively stiff medium (granite) and a stiffer liner (steel) with the following characteristics :

For the surrounding medium: Young's modulus E=62,055 MPa, Poisson's ratio v=0.25 and density ρ=2.67 Mg/m^3.

For the liner: Young's modulus E=206,750 MPa, Poisson's ratio v=0.25 and

density ρ=7.50 Mg/m^3

The total time interval is 0.015 s implemented in 20 time steps.

The incident wave is a shock compressional plane P wave of -10.0 MPa magnitude that is applied suddenly and maintained constant in time. The angle of incidence of the incoming wave is 0° from the global horizontal x axis. The position of the plane wavefront at t=0, is considered as tangent to the inner circle, at the node 33.

The radius of the cavity is 5.18 m (Figure 5.9) and the ring has a thickness of 0.20 m, extending from the free surface of the cavity (b) to the boundary (a) that forms the interface surface of the ring and the surrounding material (Figure 5.4).

Sixteen quadratic elements per boundary are used here to represent the surface of the opening, corresponding to a total number of 64 nodes (Figure 5.5). For the formation of the kernels during the numerical integration and inside the ring, the inner free surface is traversed cw, while the interface surface is traversed ccw. The boundary (c) of the surrounding medium is traversed in a cw fashion.

The tangential stresses versus time steps for nodes 33 and 41 of the inner surface and nodes 1 and 9 of the outer surface of the cavity, which are at angles 0° and 270° from the x axis respectively, is shown in Figure (5.10). It is observed that the stiff liner causes significant stress comparing with the unlined cavity. In the same Figure, the respective results for node 41 of Manolis and Beskos (1983) are presented and they appear to be in a good agreement with the results of the present work.

5.3.1.3 Circular Cross Section Cavity with a Stiff Liner in an Infinite Soft Medium Swept by a Plane P Wave

The same problem as in application of section 5.3.1.2 but with a soft medium is solved here.

The material properties used in this problem are a soft medium (slow granite) and a stiff liner (steel) with the following characteristics :

For the surrounding medium: E=7,467 MPa, ν=0.25 and ρ=2.67 Mg/m^3.

For the liner: E=206,750 MPa, ν=0.25 and ρ=7.50 Mg/m^3

The total time interval is 0.015 s implemented in 20 time steps.

The incident wave is a shock compressional plane P wave of -10.0 MPa magnitude that is applied suddenly and maintained constant in time.

The inner radius of the cavity is R_i=5.18 m and the outer R_o=5.38 m.

Sixteen quadratic elements per boundary are used also here to represent the surface of the opening (Figure 5.5).

The variation of the tangential stresses versus time steps for nodes 33 and 41 of the inner surface and nodes 1 and 9 of the outer surface of the cavity, which are at angles 0° and 270° from the x axis respectively, is shown in Figure (5.11). It is observed that the stiff liner in conjunction with the soft medium, causes significant stress concentration comparing with the previous Application of a soft medium (section 5.3.1.2). The respective results for node 41 are in a good agreement with the results of Garnet and Crouzet-Pascal (1966).

5.3.1.4 Circular Cross Section Cavity with a Stiff Liner in an Infinite Soft Medium Loaded by an Explosion at a Point P Wave

A circular lined cavity under an explosion wave is examined here.

The material properties used in this problem are a soft medium (slow granite) and a stiff liner (steel) with the following characteristics :

For the surrounding medium: E=7,464 MPa, v=0.25 and ρ=2.67 Mg/m^3.

For the liner: E=206,750 MPa, v=0.25 and ρ=7.50 Mg/m^3

The internal radius of the cavity is r_i=5.18 m and the external r_o=5.38 m.

The total time interval is 0.015 s implemented in 20 time steps.

The incident wave is a shock compressional P wave generated from an explosion at a point 8.0 m from the centre of the opening, on the x axis. The equivalent charge weight of a GP 1000 lb bomb (U.S.Army,1984) corresponding 188.7 kg of explosive weight is used. The attenuation coefficient is taken n=2.75, and the coupling factor f=1.0. The cubic root formula with decay in time is used here (Chapter 4, para. 4.2).

The discretization consists of 16 elements per boundary (Figure 5.5). The traversing of the boundaries is shown in Figure (5.4).

The tangential stresses versus time steps for the nodes 37 and 41 of the inner surface and 5 and 9 of the outer surface of the lining, which form an angle of 315° and 270° from the x axis respectively, are shown in Figure (5.13). The stresses, (as was also mentioned in the case of the unlined opening, chapter 2 section 2.3.2.1.b) rapidly increase to the peak value and consequently drop to zero at the 20th time step. It is observed that the presence of the stiff liner causes significant stress concentration of magnitude -108.3 MPa at node 37 which appears at the 7th time step and -76.9 MPa at node 41 at the 10th time step. The duration of the phenomenon is relatively long, since the cavity returns to its original unstressed state at the 20th time step, following the passage of the shock wave.

The same as above cavity with a softer liner is examined next. The material characteristics of the liner are: E=20,000 MPa, v=0.20 and ρ=2.30 Mg/m^3 corresponding to concrete. In Figure (5.14) the tangential stresses versus time

steps are presented. The stresses are lower than the previous case, with peak values of -30.0 MPa at node 37 which appears at 8th time step and -23.1 at node 41 at 11th time step. The cavity returns to the unstressed state at late times and the duration of the phenomenon is even longer than previously.

5.3.1.5 Circular Cross Section Cavity with a Soft Liner in an Infinite Stiff Medium Loaded by an Explosion at a Point P Wave

The material properties used in this problem are a stiff medium and a soft liner with the following characteristics :
For the surrounding medium: E=60,000 MPa, v=0.25 and ρ=2.70 Mg/m^3.
For the liner: E=20,000 MPa, v=0.20 and ρ=2.30 Mg/m^3

The internal radius of the cavity is r_i=5.18 m and the external r_o=5.38 m.

The total time interval is 0.015 s implemented in 20 time steps.

The incident wave is a shock compressional P wave generated from an explosion at a point 8.0 m from the centre of the opening, on the x axis. The equivalent charge weight of a GP 1000 lb bomb (U.S.Army,1984) which is 188.7 kg of explosive weight is used. The attenuation coefficient is taken n=2.75, and the coupling factor f=1.0. The cubic root formula with decay in time is used here (Chapter 6, para. 6.2).

The discretization consists of 16 elements per boundary (Figure 5.5).

The tangential stresses versus time steps for the nodes 37 and 41 of the inner surface and 5 and 9 of the outer surface of the lining, are shown in Figure (5.15). The stresses, (as in the case of the unlined opening was mentioned, chapter 2 section 2.3.2.1.b) rapidly increase to the peak value and consequently drop to zero at the 10th time step. The peak values of the stresses are -29.8 MPa at node 37 which appears at 3nd time step and -18.8 at node 41 at 4th time step. The cavity returns to the unstressed state at early times and the duration of the phenomenon is shorter than previously.

The same as above cavity but with slightly different material characteristics (softer liner, harder medium) is examined next. These characteristics are:
For the surrounding medium: E=70,000 MPa, v=0.25 and ρ=2.70 Mg/m^3.
For the liner: E=10,000 MPa, v=0.20 and ρ=2.30 Mg/m^3 . The tangential stresses versus time steps for the nodes 37 and 41 of the inner surface and 5 and 9 of the outer surface of the lining, are shown in Figure (5.16). The stresses rapidly increase to the peak value also and consequently drop to zero at the 10th time step. The peak values of the stresses are -29.3 MPa at node 37 which appears at

3nd time step and -15.6 MPa at node 41 at 4th time step. The cavity returns to the unstressed state at early times and the duration of the phenomenon as short as previously.

Comparing the above two applications, it is observed that the change of stresses is not influenced significantly from the small change to the material characteristics.

From the applications presented in this Chapter, the following useful conclusions result for a lined cavity:

(1). The stiffer than the medium liner causes high stress concentration at the liner. The stiffer liner causes larger stresses and longer duration of the phenomenon.

(2). The softer than the medium liner, causes reduction of the stresses at the liner.

5.3.1.6 Two Circular Cross Section Lined Cavities in an Infinite Stiff Medium Swept by a Plane P Wave

The problem of two lined cavities in an infinite medium under an incident P plane wave are examined here. The centres of the two cavities are lying on the x axis at a distance of 8.0 m between them. The outer and inner radius of both cavities are 2.70 and 2.50 m (Figure 5.21). Two cases are presented: the case of same concrete liners in both cavities and the case of a stiff liner (steel) in the one cavity and a concrete liner in the other one. In both cases the characteristics of the surrounding material are E=60,000 MPa, v=0.25 and ρ=2.70 Mg/m^3 corresponding to stiff granite. The loading is an incident shock compressional plane P wave of -10.0 MPa magnitude that is applied suddenly and maintained constant in time. The angle of incidence of the incoming wave is 270° from the global horizontal x axis (vertical incidence). The position of the plane wavefront at t=0, is considered as tangent to the outer circle, at the nodes 5 and 21 (Figure 5.22). The total time interval for both cases is 0.015 s implemented in 20 time steps.

a. Concrete Liners

Both liners have the following characteristics : E=20,000 MPa, v=0.25 and ρ=2.30 Mg/m^3 corresponding to concrete.

Eight quadratic elements per boundary are used here to represent the surface of the openings, corresponding to a total number of 64 nodes (Figure 5.22). For the formation of the kernels during the numerical integration and inside the rings, the inner free surfaces of the liners are traversed cw, while the interface surfaces

are traversed ccw. The boundary (c) of the surrounding medium is traversed in a cw fashion (Figure 5.3).

The tangential stresses versus time steps for nodes 33 and 49 of the inner surface of the liners is shown in Figure (5.23). It is observed that the stress concentration is higher for the nodes of the two cavities closer to each other (nodes 41 and 49 of Figure 5.22) and lower for the nodes further to each other (nodes 33 and 57). In Figure (5.24) the variation of y-displacements versus time steps for nodes 33, 37, 41 and 45 is presented.

b. Steel and Concrete Liner

The material characteristics of the liner of (b) cavity (Figure (5.21)) are E=206,000 MPa, v=0.25 and ρ=7.50 Mg/m^3 corresponding to steel, while the characteristics for (a) cavity are E=20,000 MPa, v=0.25 and ρ=2.30 Mg/m^3

Eight quadratic elements per boundary are also used here to represent the surface of the opening, corresponding to a total number of 64 nodes (Figure 5.21). The inner free surfaces of the liners are traversed cw, while the interface surfaces are traversed ccw. The boundary (c) of the surrounding medium is traversed in a cw fashion (Figure 5.3).

The tangential stresses versus time steps for nodes 33 and 41 of (a) cavity and for nodes 49 and 57 of (b) cavity is shown in Figure (5.25). It is observed that the stress concentration is higher for the stiff liner with maximum magnitude -70.0 MPa at node 49 at 6th time step. The stresses in the soft liner are significantly lower with value -17.0 at node 41 at 6th time step. It is also evident here that the nodes of the two cavities closer to each other (nodes 41 and 49) are stressed more than the nodes farther to each other (nodes 33 and 57).

In Figure (5.26) the variation of y-displacements versus time steps for nodes 33 and 37 of (a) cavity, and 53 and 61 of (b) cavity is presented. It is observed that the displacements at the soft liner (a) are greater than the respective displacements of the stiff liner (b).

5.3.2 Semi-Infinite Domain Solutions

5.3.2.1 Circular Cross Section Cavity with a Stiff Liner in a Semi-infinite Medium Loaded by an Explosion at a Point P Wave

The case of a lined opening in a semi-infinite medium is examined here.

The material characteristics used in this problem are :

For the surrounding medium: E=20,000 MPa, v=0.25 and ρ=2.55 Mg/m^3.

For the liner: E=40,000 MPa, v=0.25 and ρ=2.70 Mg/m^3

The internal radius of the cavity is r_i=2.50 m and the external r_o=2.70 m.

The total time interval is 0.015 s implemented in 20 time steps.

The incident wave is a shock compressional P wave generated from an explosion at a point 8.0 m from the centre of the opening verically on the y axis. The equivalent charge weight of a GP 1000 lb bomb (U.S.Army,1984) corresponding to 188.7 kg of explosive weight is used. The attenuation coefficient is taken as n=2.75, and the coupling factor is assumed equal to f=1.0. The cubic root formula with decay in time is used here (chapter 4 section 4.2).

The discretization consists of 8 elements per boundary (Figure 5.17).

The variation of the tangential stresses versus time steps for the nodes 34, 36, 38 and 46 of the inner surface of the lining, which form an angle of 0°, 315°, 270° and 90° is shown in Figure (5.18). The stresses, (as in the case of the unlined openings was mentioned, chapter 2 section 2.3.2.1.b) rapidly increase to the peak value and consequently drop to zero at the 10th time step. The peak values of the stresses are -28.0 MPa at node 36 which appears at 5th time step and -20.0 at node 34 at 6th time step. The change of x and y displacements versus time steps is depicted in Figures (5.19) and (5.20).

5.3.2.2 Two Circular Cross Section Cavities with Liner of the Same Material in a Semi-Infinite Medium Swept by a Plane P Wave

At first, the liners are assumed to be of the same material as the semi-infinite domain. The material properties adopted in this problem are the same as in the application of section 5.3.1.7 (E=60,000 MPa, v=0.25, ρ=2.55 Mg/m^3). The total time interval is 0.015 s implemented in 20 time steps.

The incident wave is also a shock compressional plane wave of -10.0 MPa magnitude that is applied suddenly and maintained constant in time. The angle of incidence of the incoming wave is 270° from the global horizontal x axis (vertical incidence). The position of the plane wavefront at t=0, is considered as tangent to the inner circles, at nodes 54 and 70 (Figure 5.27).

The inner radius of the cavities is 2.50 m, and the outer 2.70 m (Figure 5.27). The ground surface is represented by an open line of total length 32.0 m separated into eight quadratic elements. Eight quadratic elements for the rest of the boundaries are also used to represent the surface of the mesh, corresponding to a total number of 81 nodes (Figure 5.27).

The variation of the tangential stresses versus time steps for the nodes 50 and 58 of cavity (a) and for the node 9 of the ground surface (Figure 5.27), is shown in Figure (5.28). Furthermore, the variation of the displacements in x and y directions, are shown in Figures (5.29) and (5.30) respectively.

5.3.2.3 Two Circular Cross Section Cavities in a Semi-Infinite Stiff Medium with a Stiffer Liner Swept by a Plane P Wave

The same problem as in the previous application 5.3.1.1 is examined here with the same material properties for the medium (granite) but with stiffer liners (steel) with the following characteristics :
For the surrounding medium: Young's modulus E=60,000 MPa, Poisson's ratio v=0.25 and density ρ=2.70 Mg/m^3.
For the liner: Young's modulus E=206,000 MPa, Poisson's ratio v=0.25 and density ρ=7.30 Mg/m^3

The total time interval is 0.015 s implemented in 20 time steps and the loading is also a P wave of -10.0 MPa striking vertically as previously.
The geometry of the problem is also depicted in Figure (5.27).

The tangential stresses versus time steps for nodes 50 and 58 of the inner surface of cavity (a), is shown in Figure (5.31). Furthermore, the variation of the displacements in x and y directions, are shown in Figures (5.32) and (5.33) respectively.

Comparing Figures (5.28), (5.29), (5.30) and (5.31), (5.32), (5.33), is observed that the stiff liner causes significant stress (-30 MPa) comparing with the unlined cavity (-17.0 MPa), while the displacements remain about the same for the two cases.

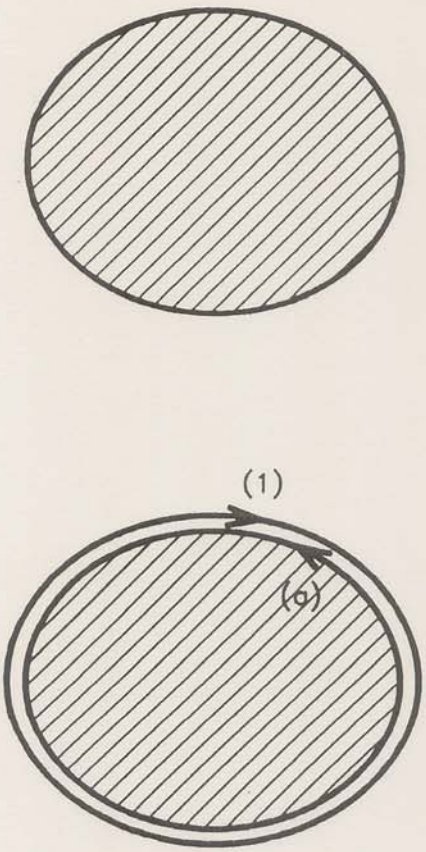


Figure 5.1 Inclusion in infinite medium

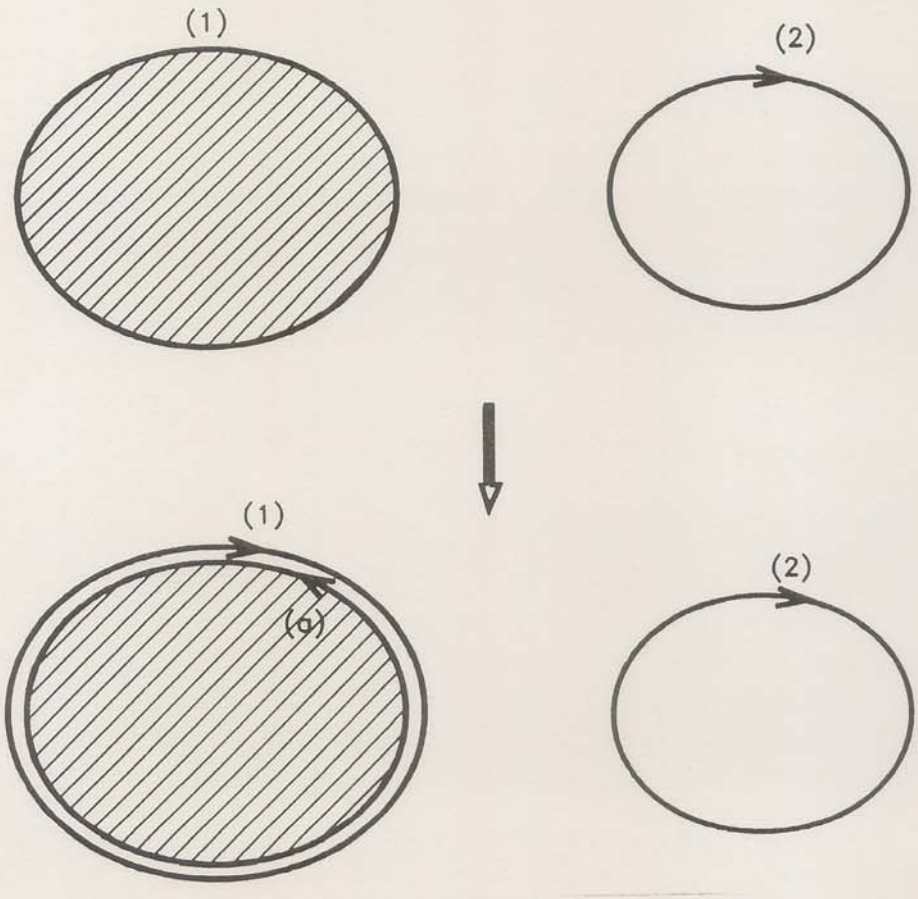


Figure 5.2 Inclusion and cavity in infinite medium

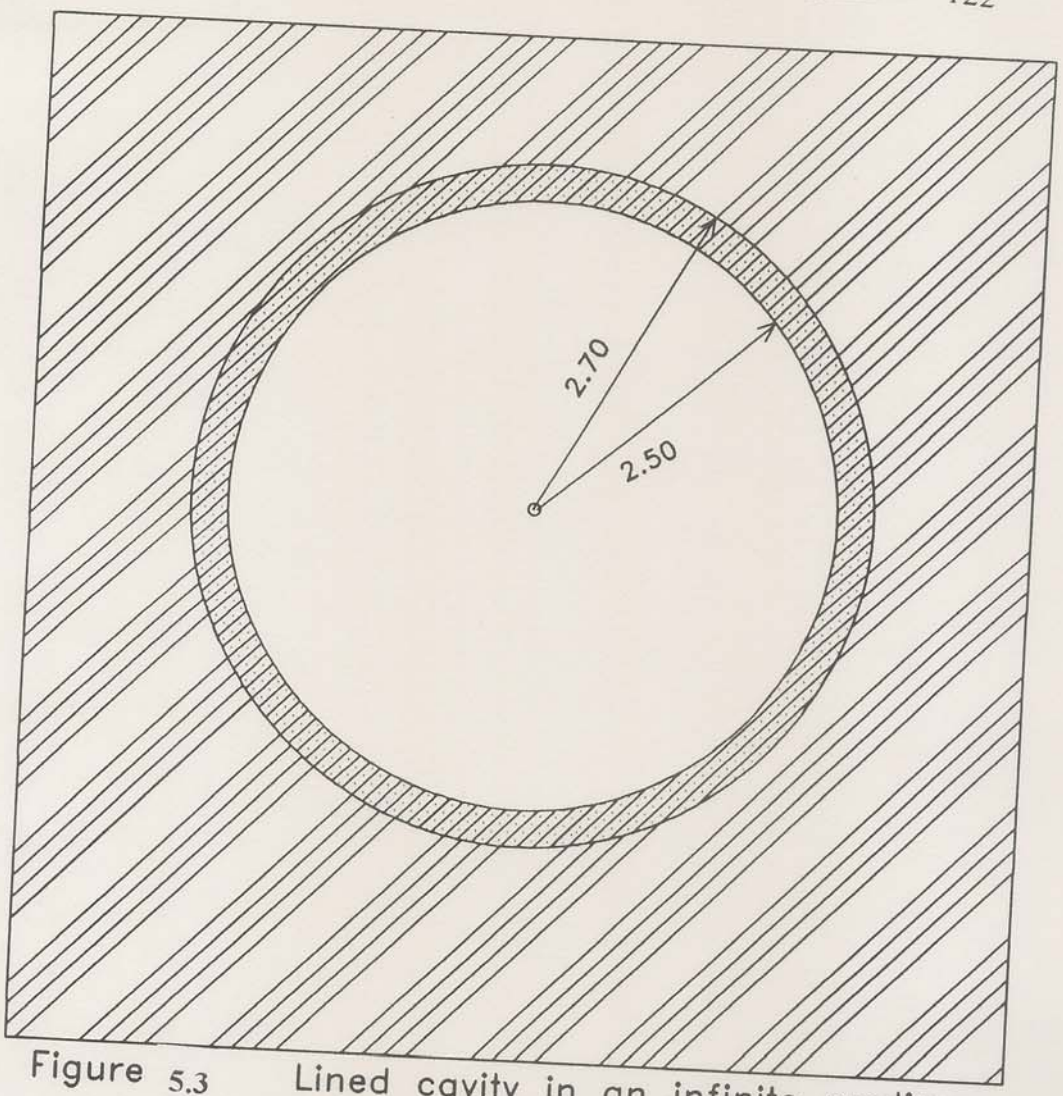


Figure 5.3 Lined cavity in an infinite medium

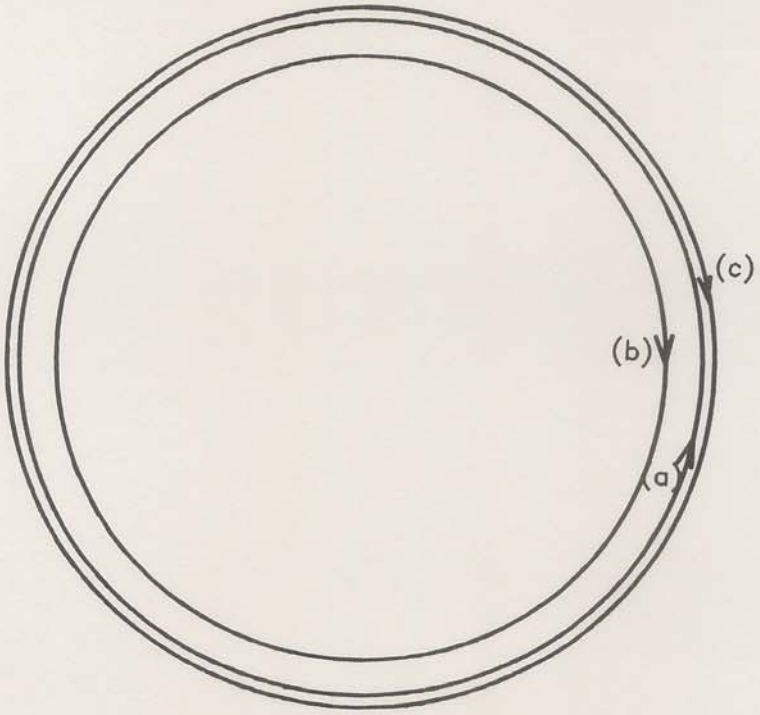


Figure 5.4 Underground opening with lining

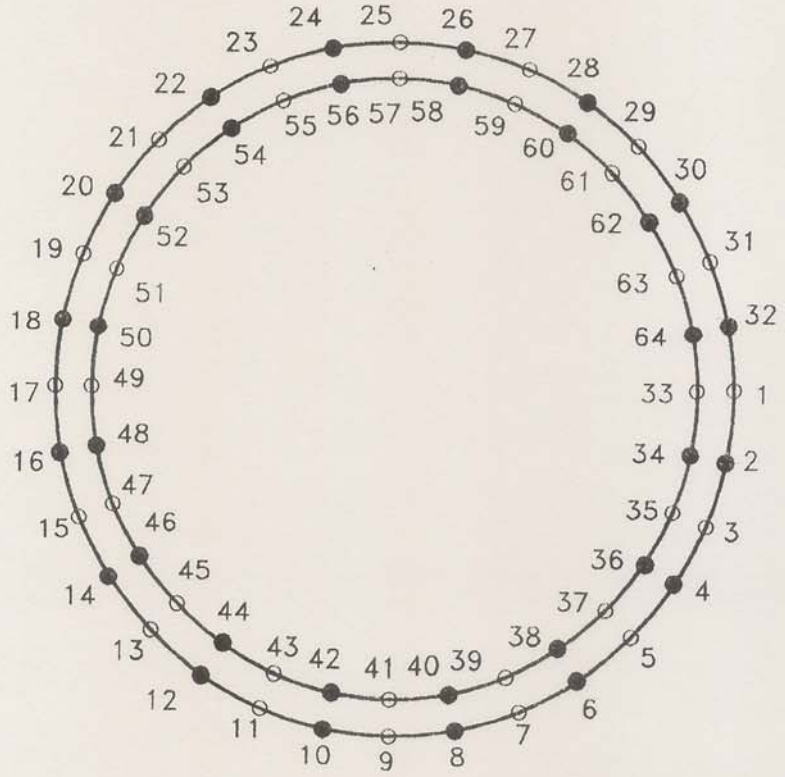


Figure 5.5 Discretization of the lining into 16 quadratic elements per boundary

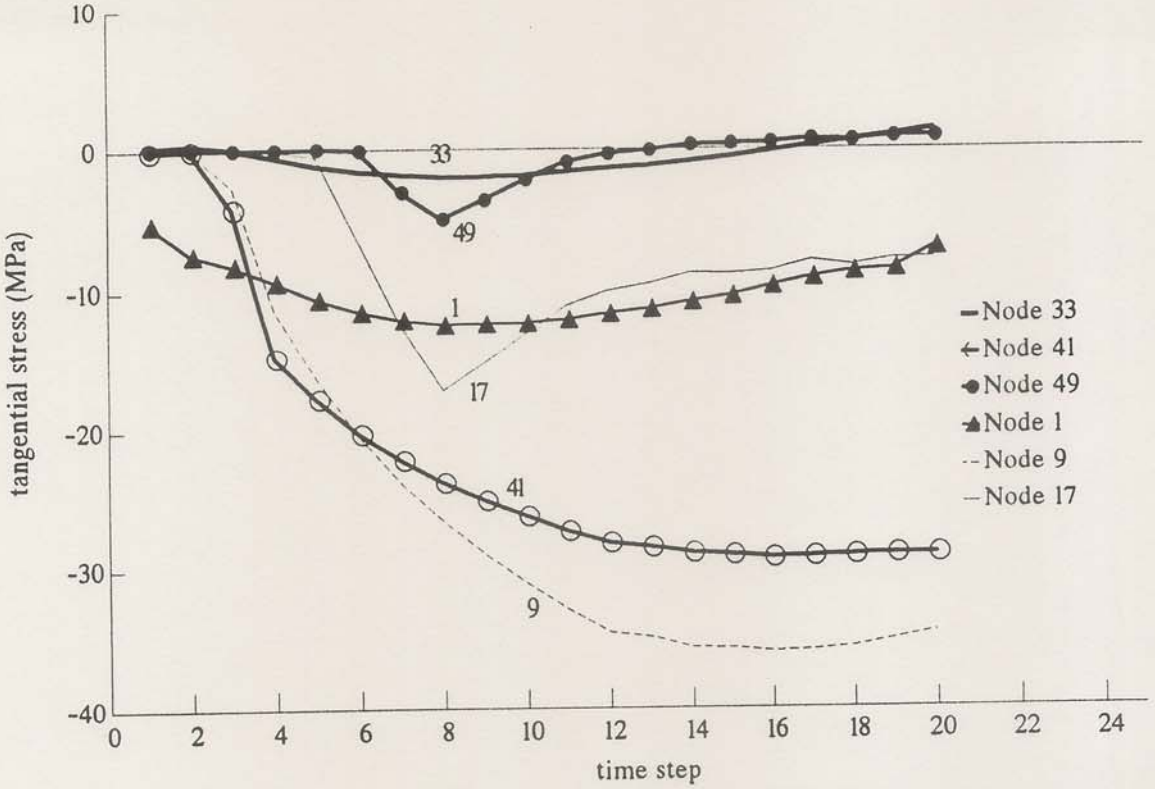


Figure 5.6 Time history of tangential stresses

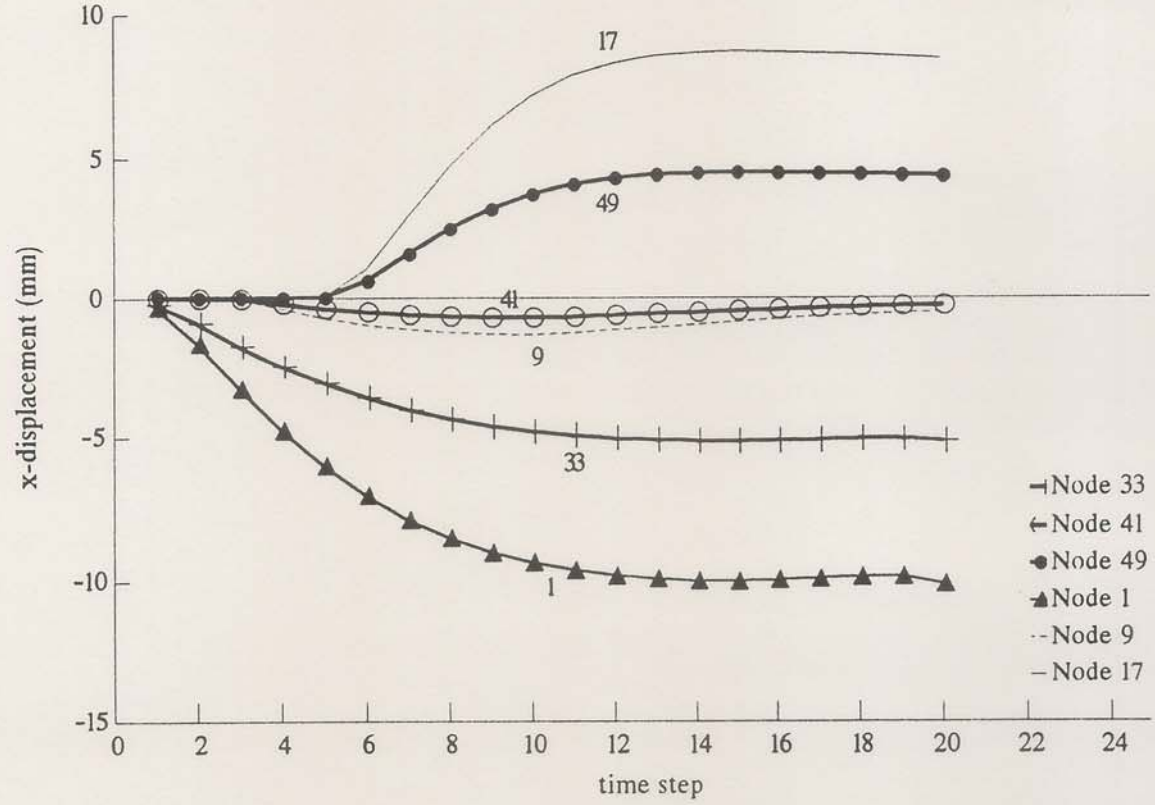


Figure 5.7 Time history of x-displacements

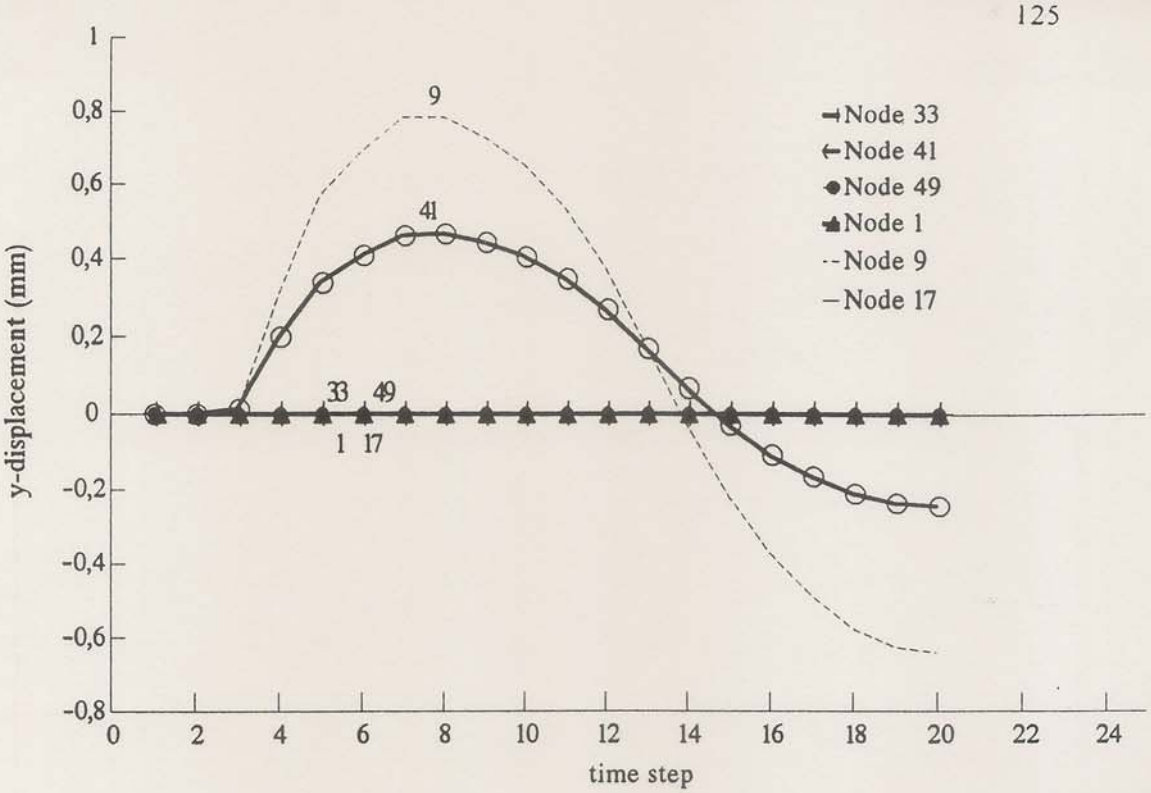


Figure 5.8 Time history of y-displacements

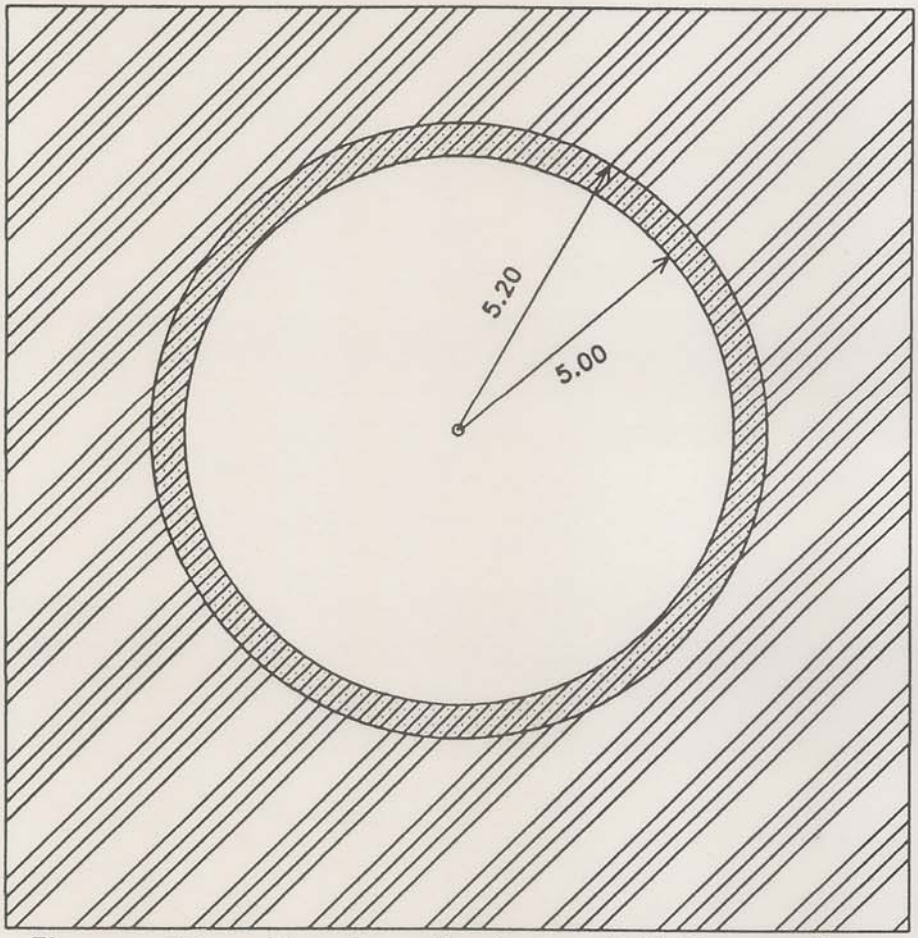


Figure 5.9 Lined cavity in an infinite medium

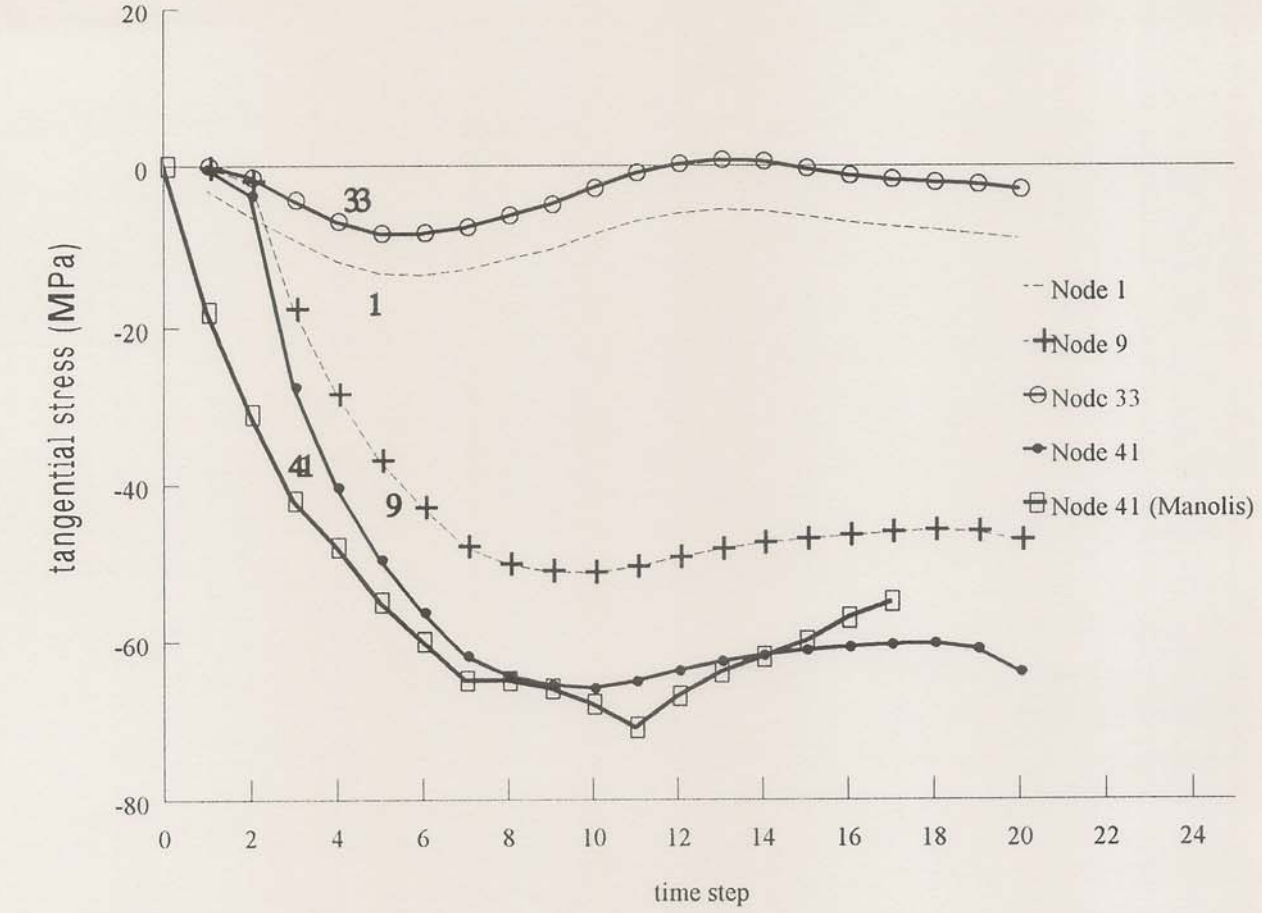


Figure 5.10 Tangential stresses, stiff medium-stiffer liner

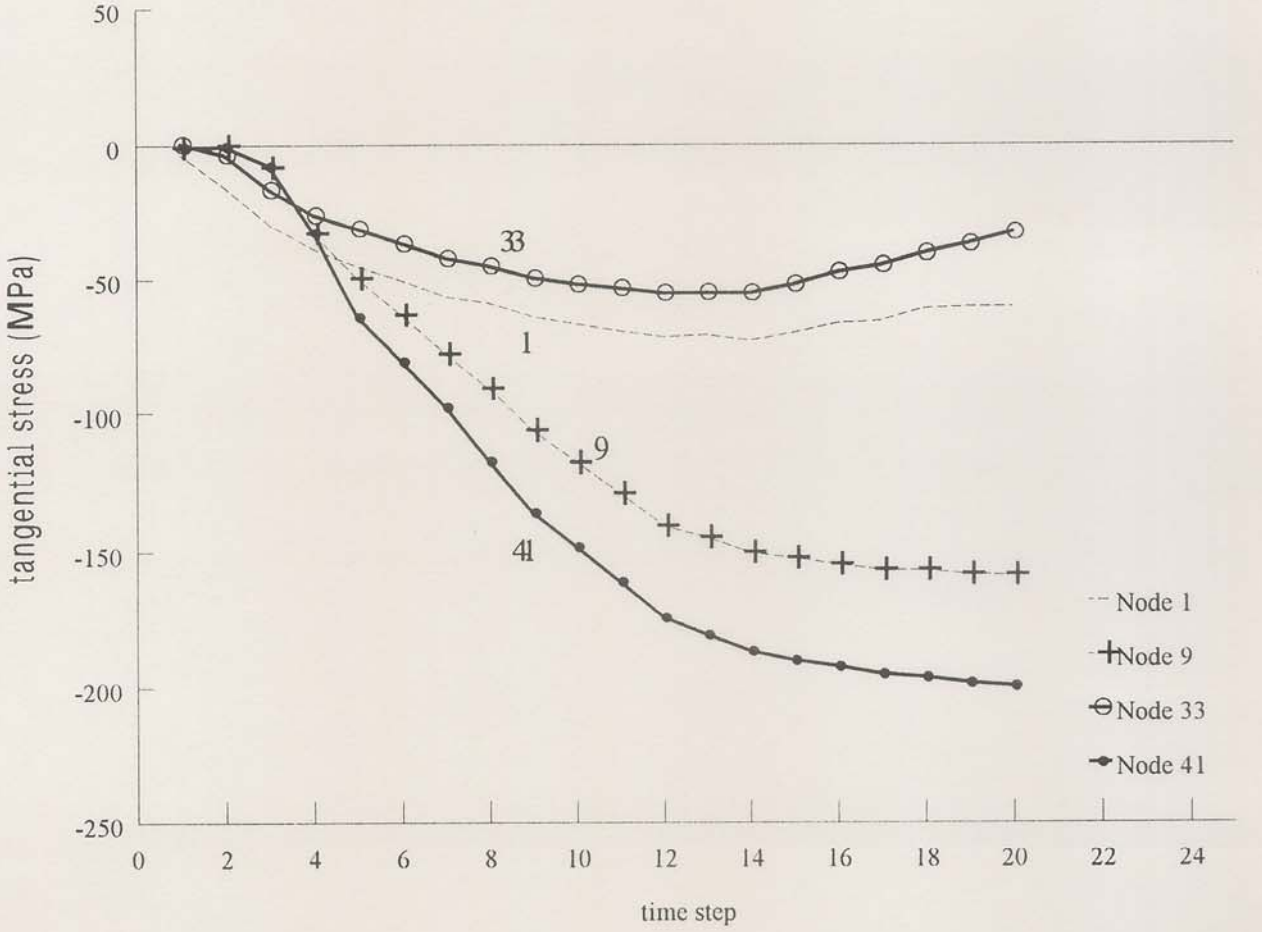


Figure 5.11 Tangential stresses, stiff liner-soft medium

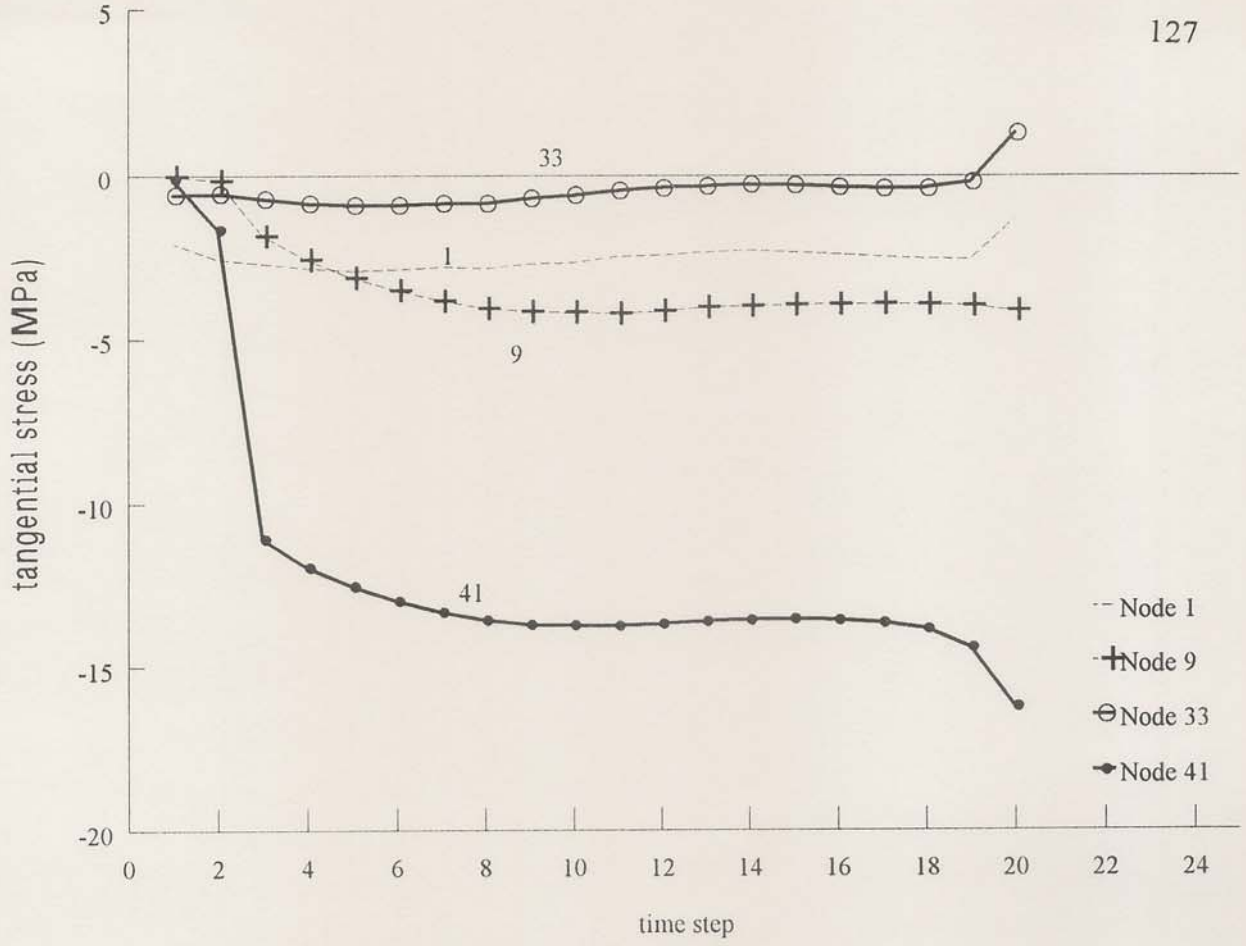


Figure 5.12 Tangential stresses, soft liner-stiff medium

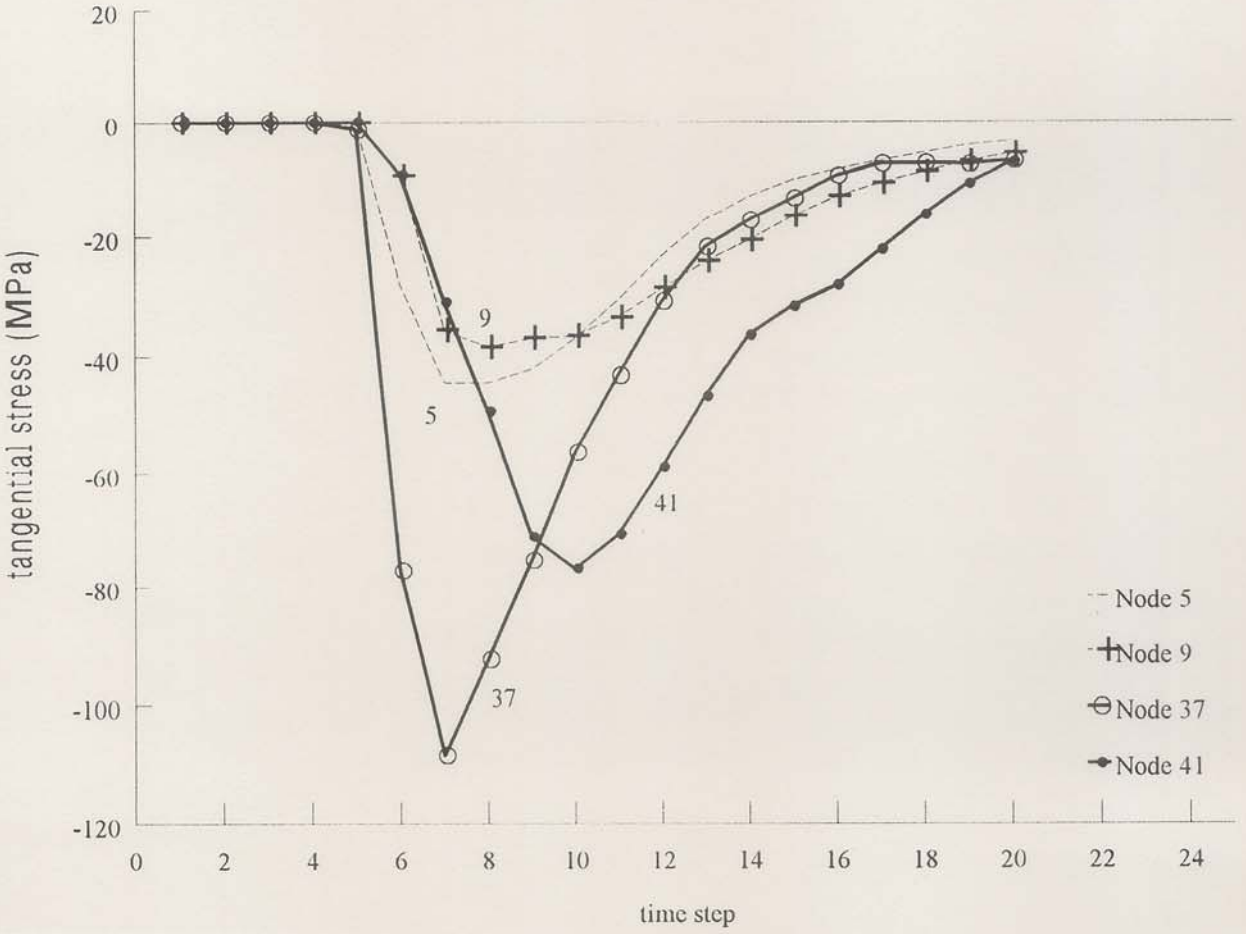


Figure 5.13 Tangential stresses, stiff liner-soft medium

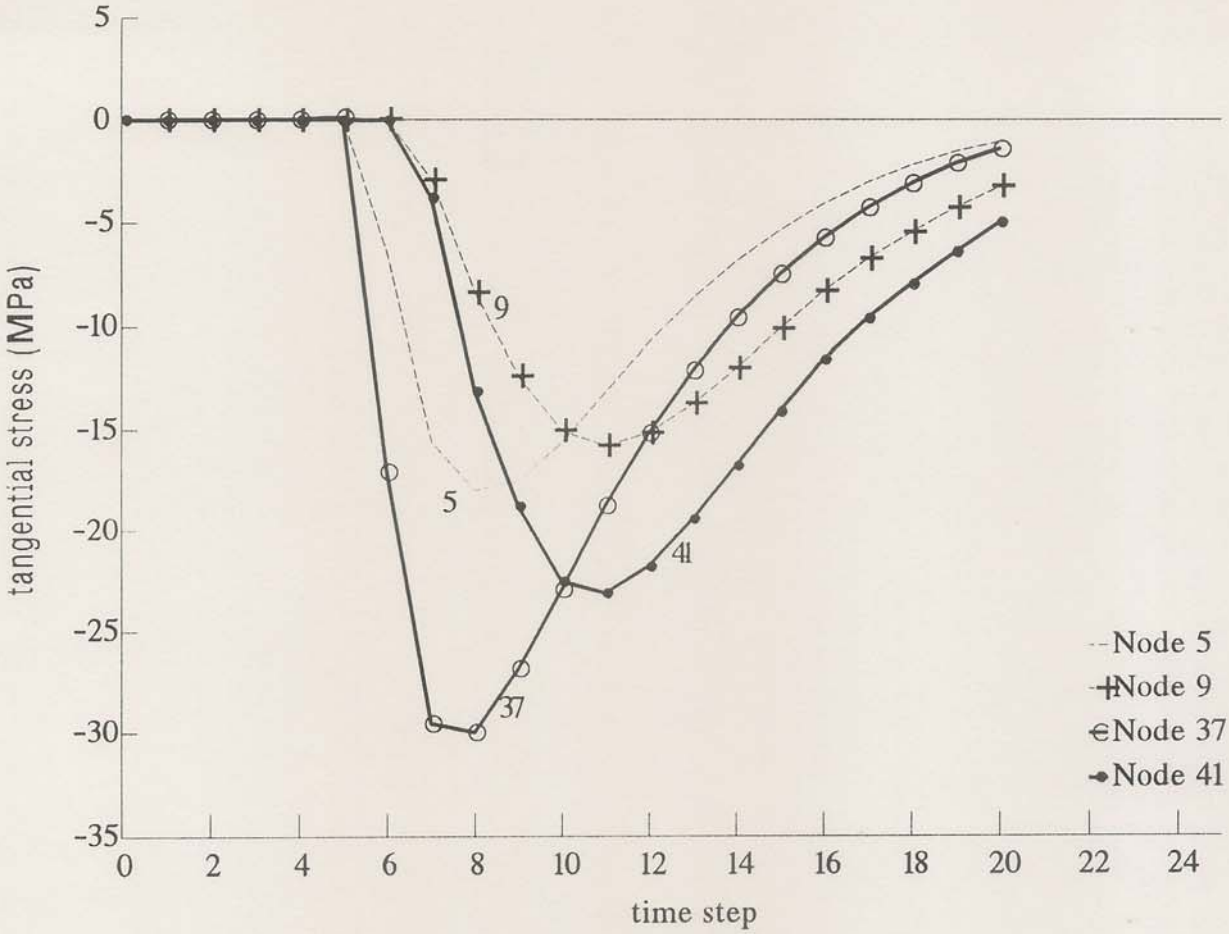


Figure 5.14 Tangential stresses, soft medium

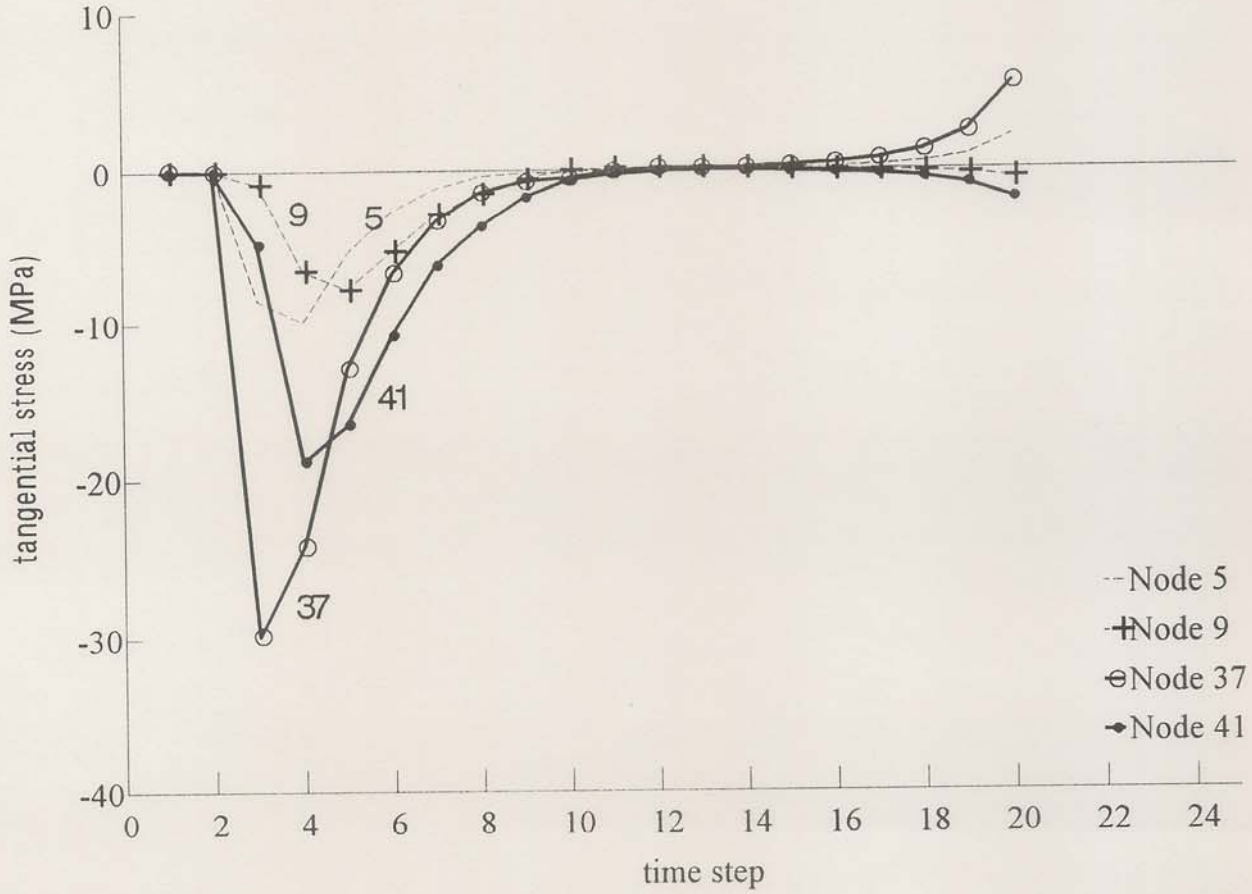


Figure 5.15 Tangential stresses, stiff medium

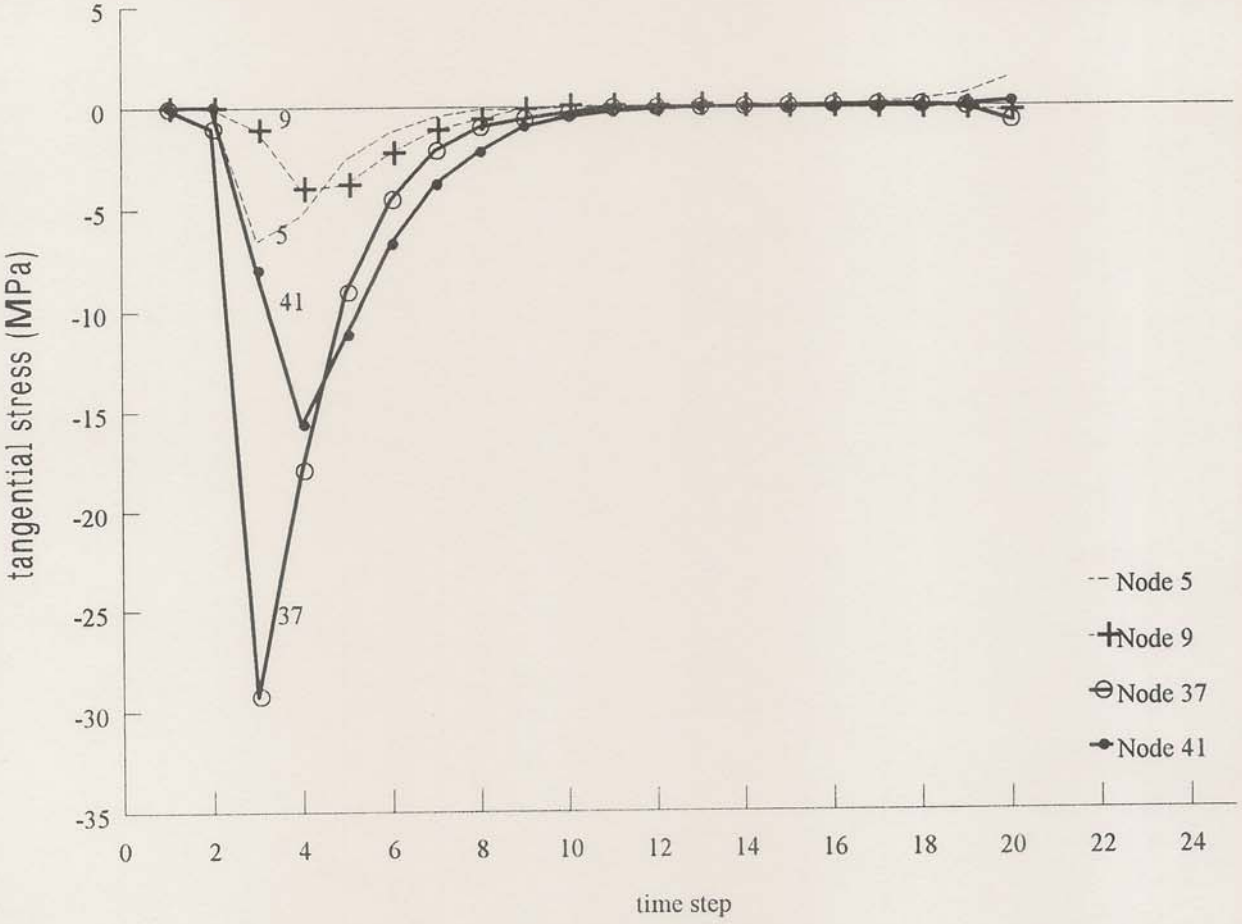


Figure 5.16 Tangential stresses, soft liner-stiff medium

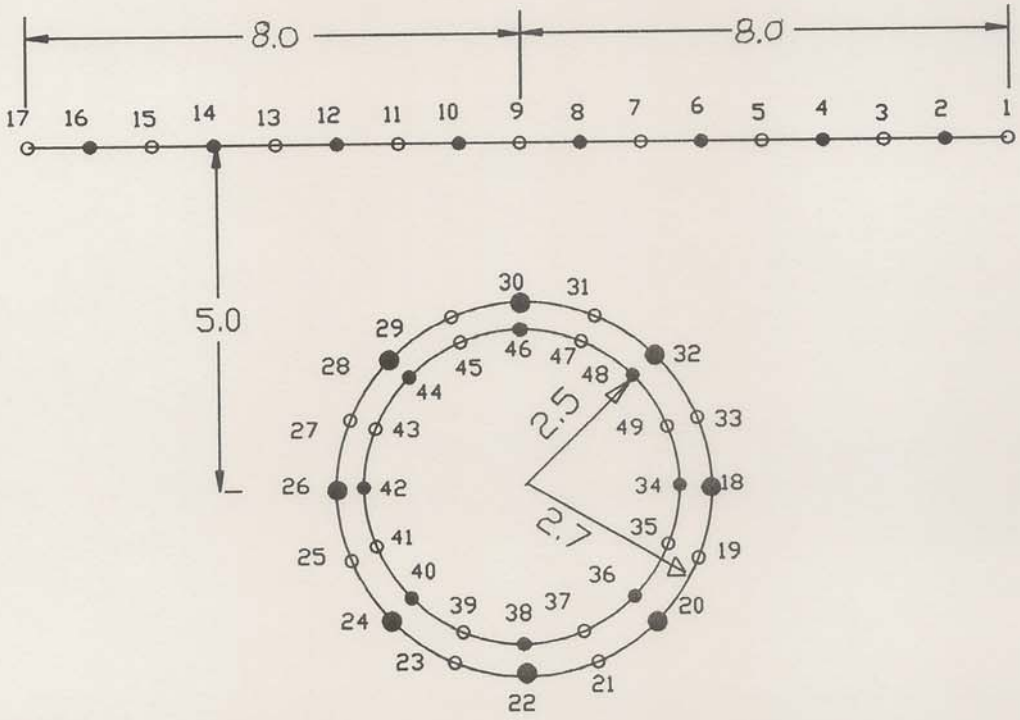


Figure 5.17 Discretization of the mesh

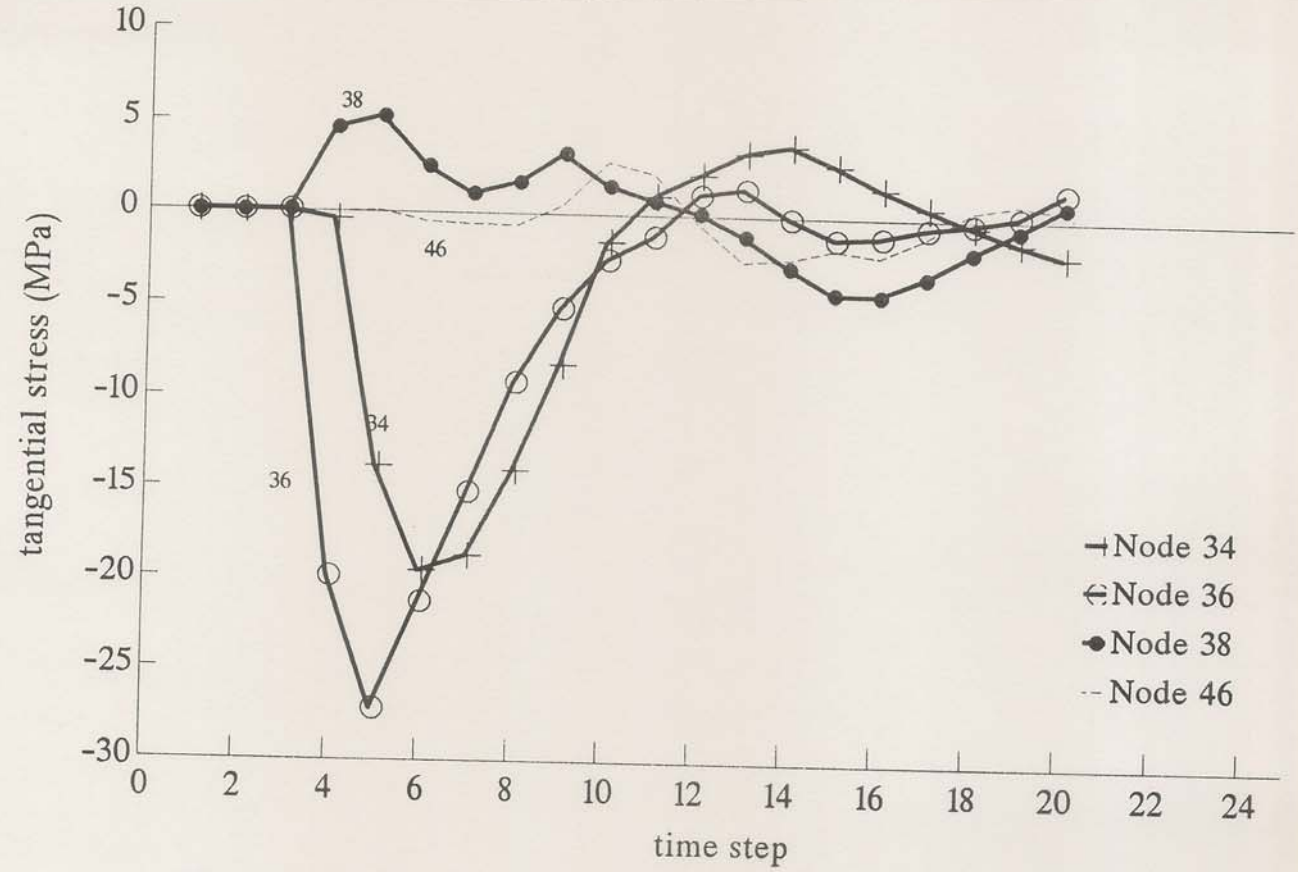


Figure 5.18 Tangential stresses, by discretizing the ground surface

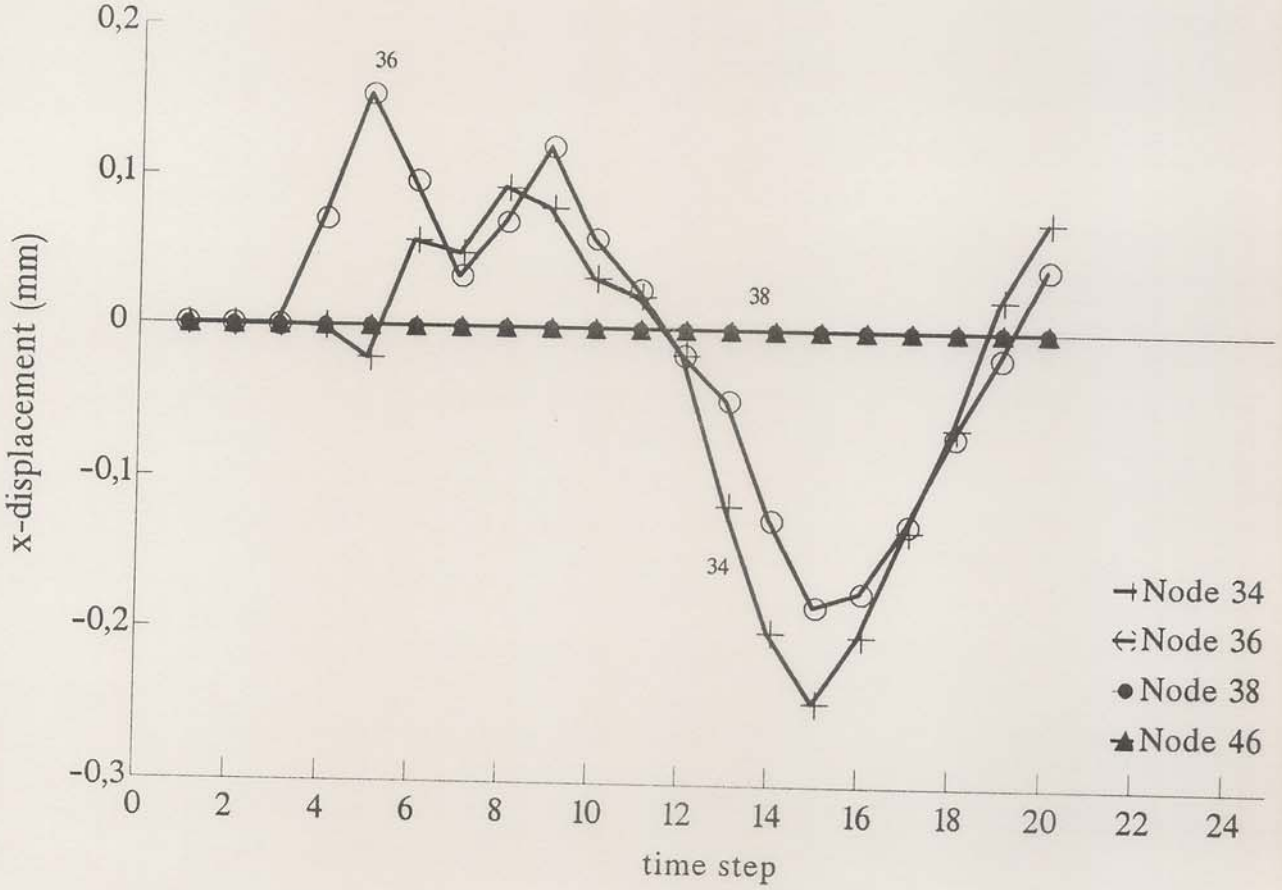


Figure 5.19 x-displacements, by discretizing the ground surface

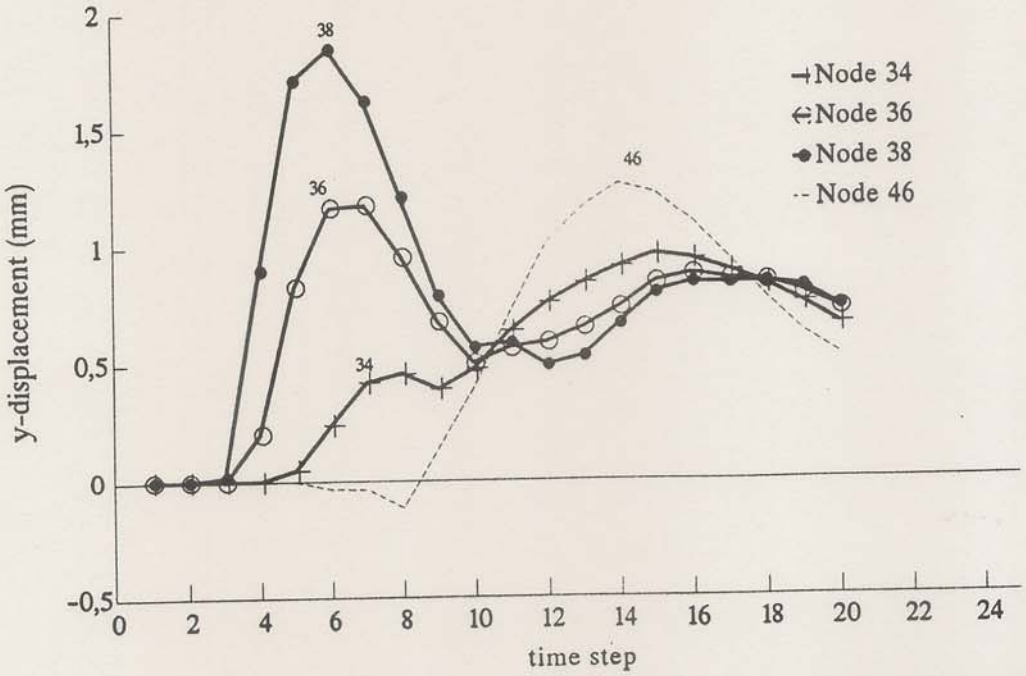


Figure 5.20 y-displacements, by discretizing the ground surface

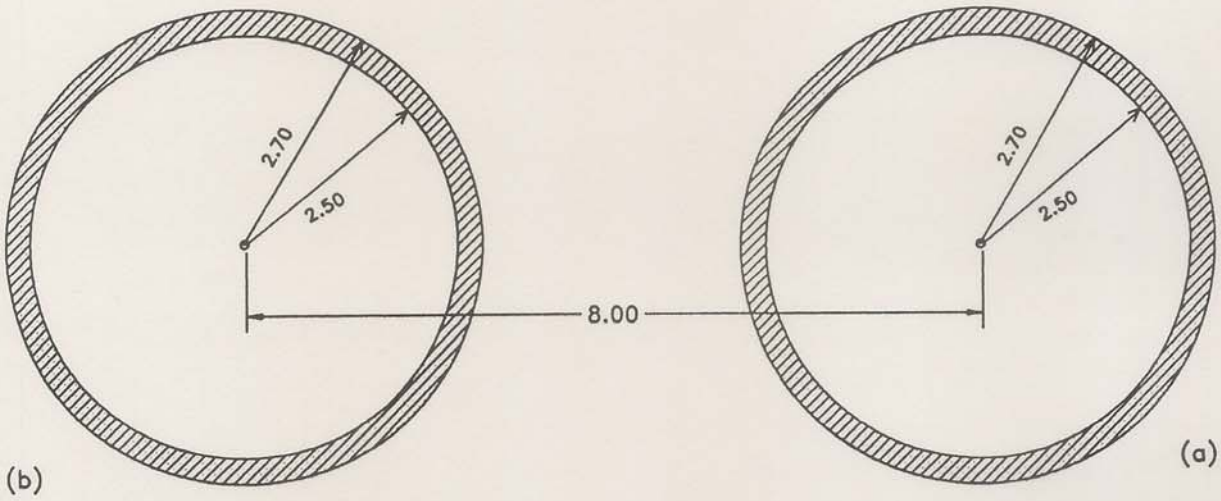


Figure 5.21 Two lined cavities in an infinite medium

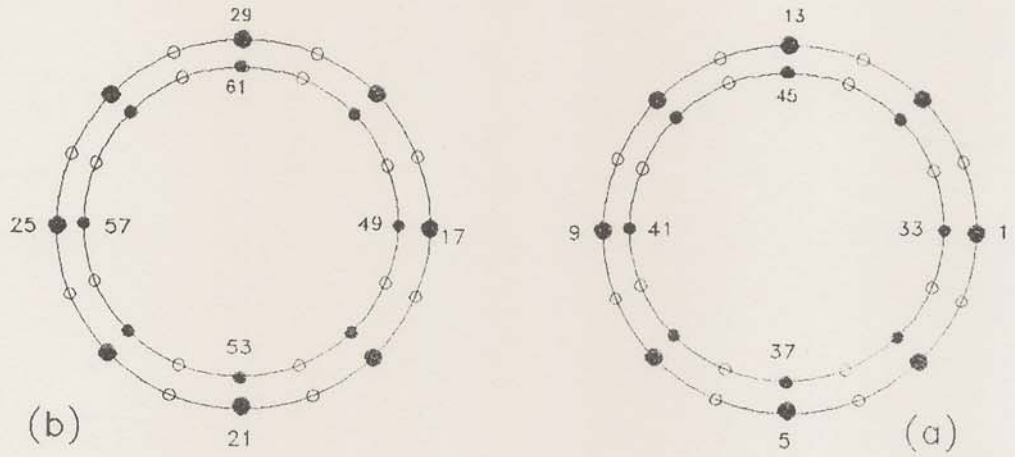


Figure **5.22** Discretization of the mesh

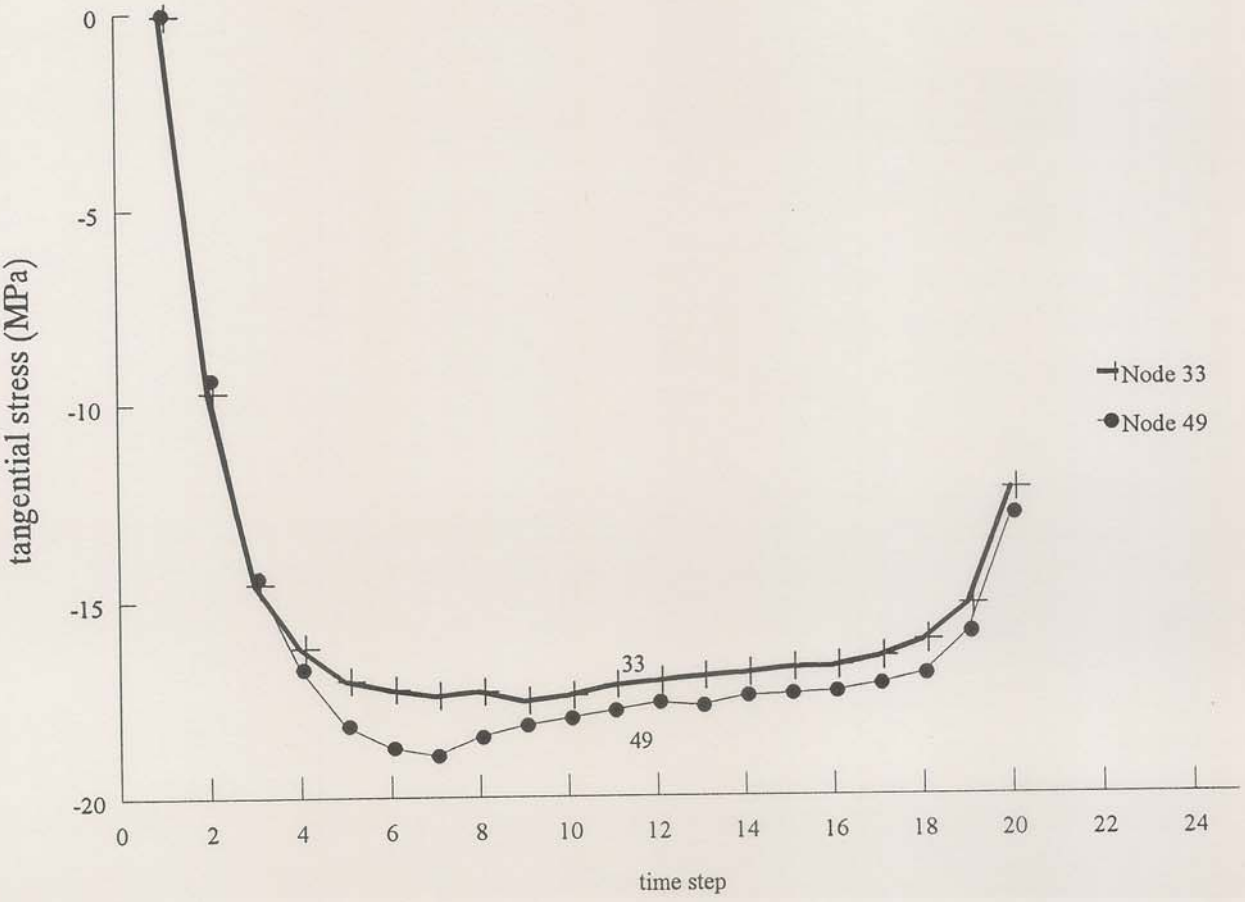


Fig **5.23** : Time history of tangential stresses

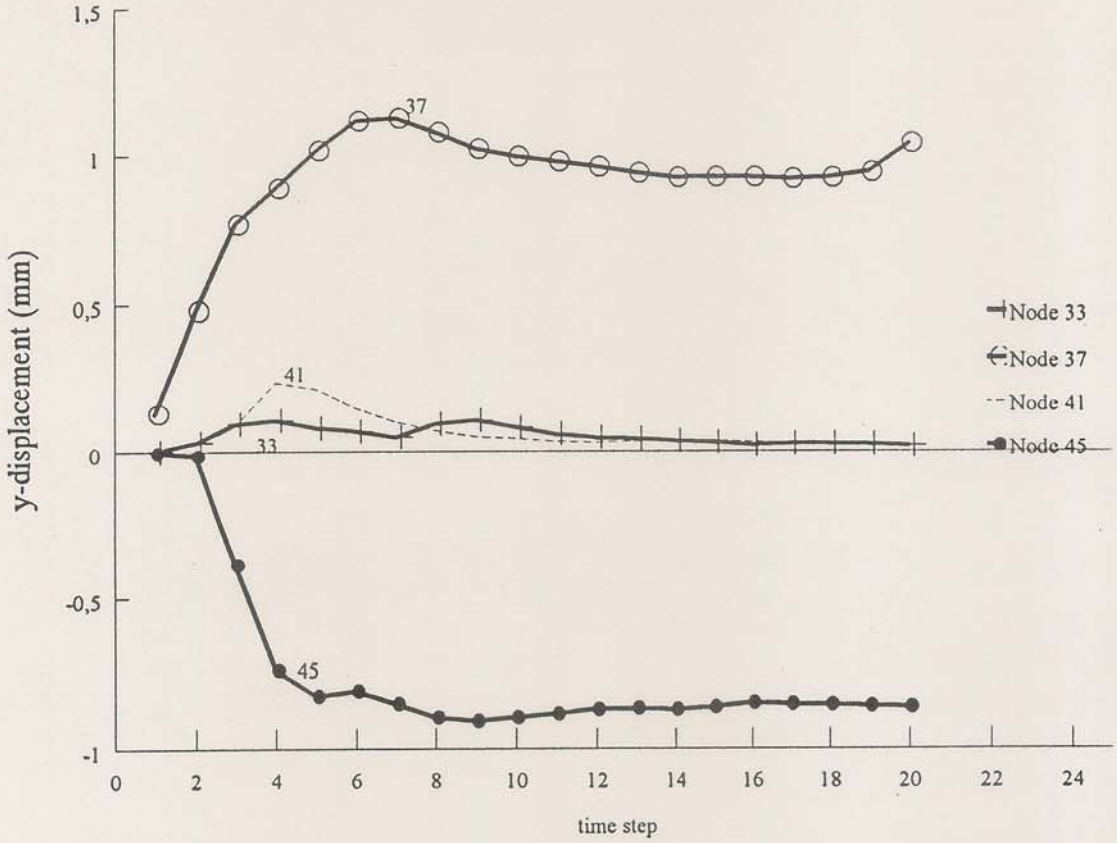


Fig 5.24 Time history of y-displacements

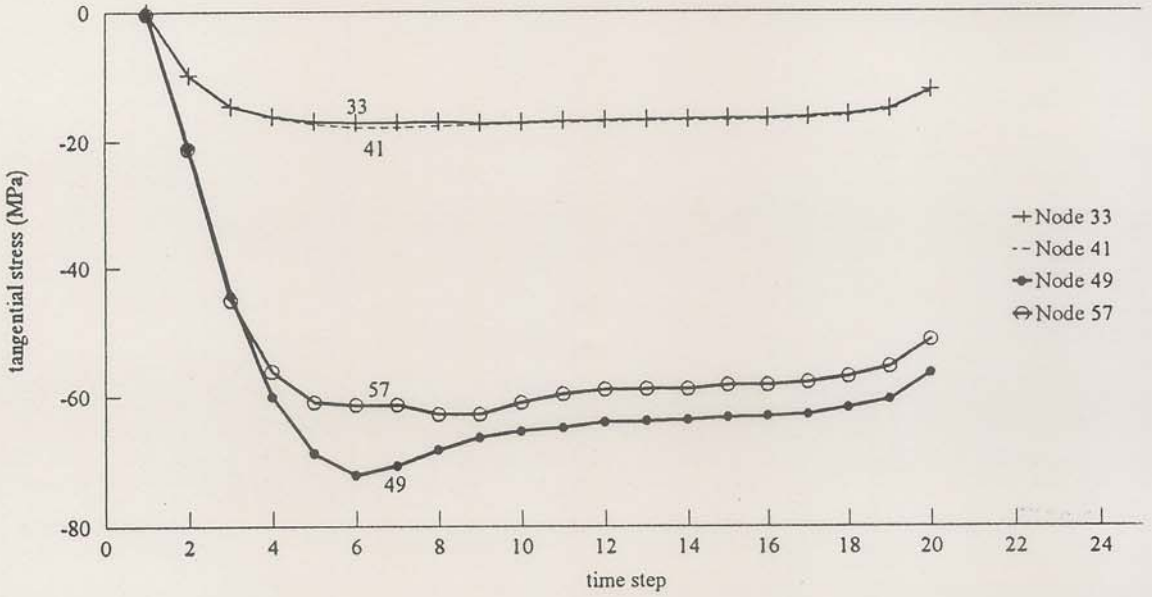


Fig 5.25 Time history of tangential stresses

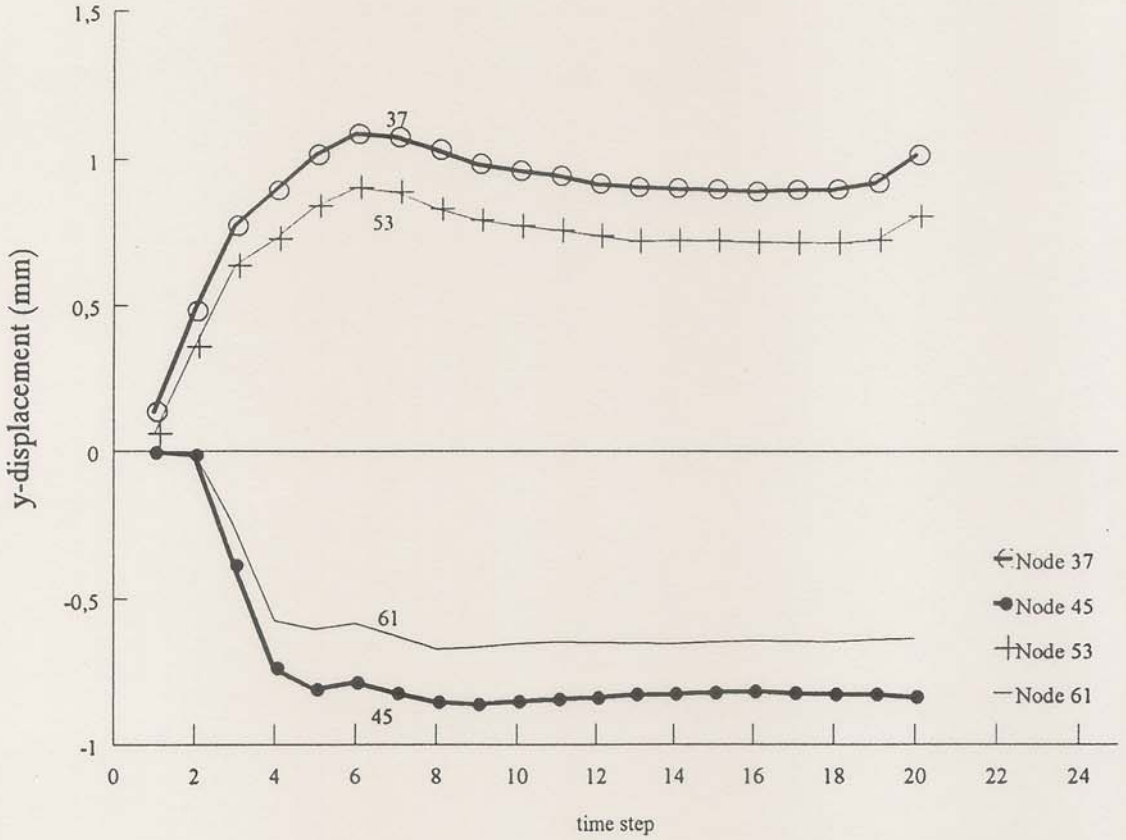


Fig 5.26 Time history of y-displacements

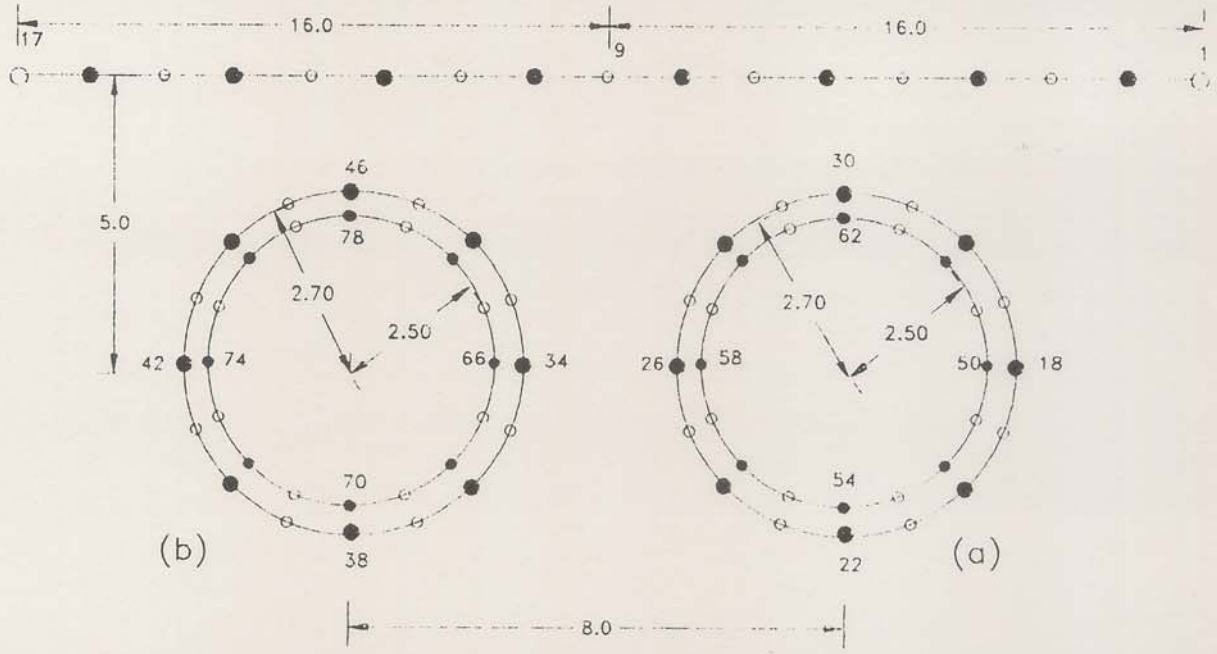


Figure 5.27 Discretization of the mesh

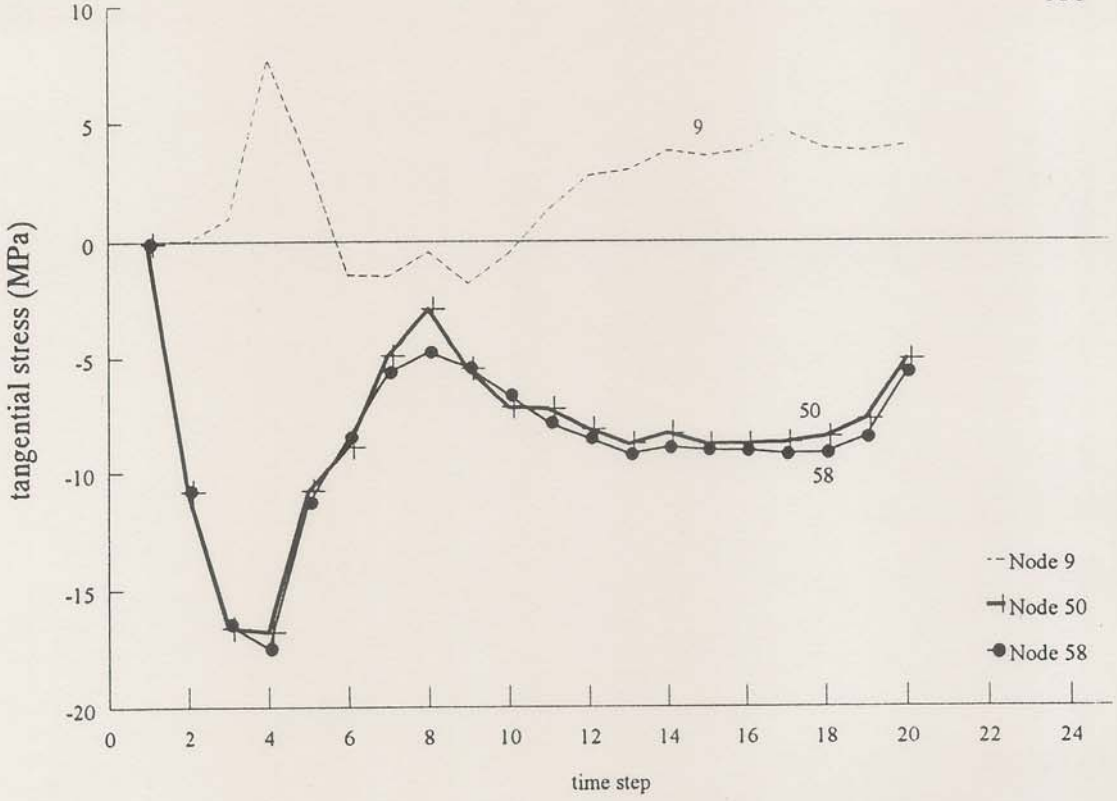


Fig **5.28** Time history of tangential stresses

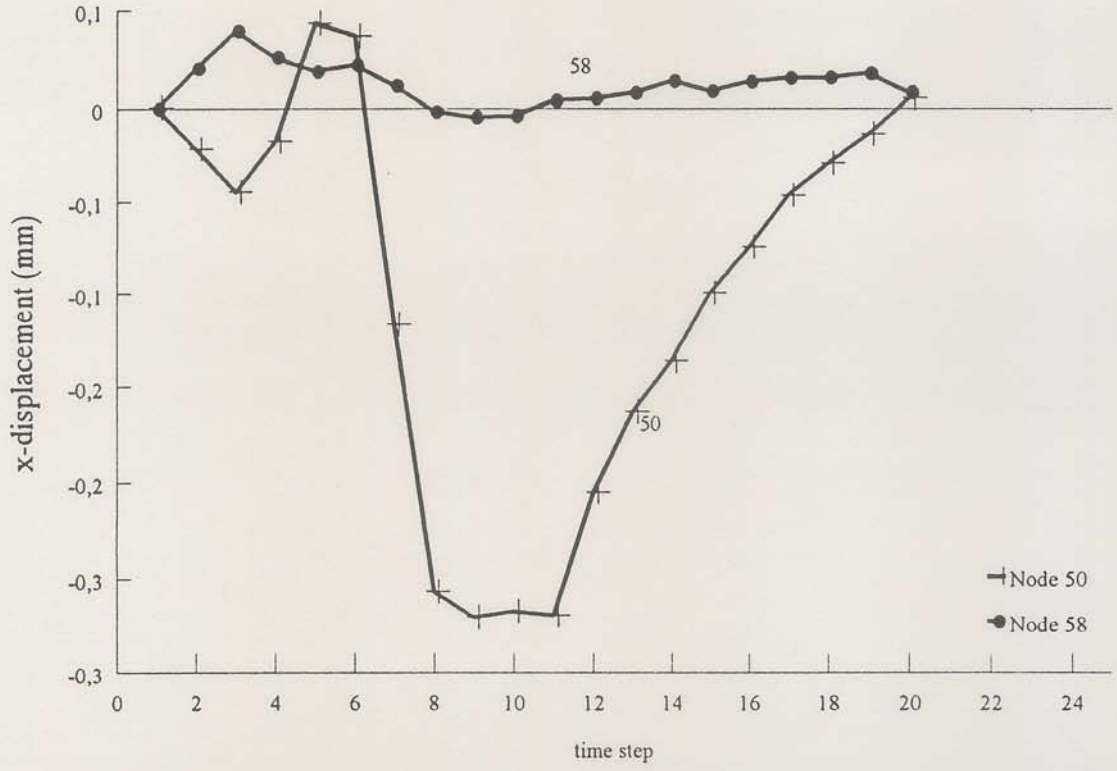


Fig **5.29** Time history of x-displacements

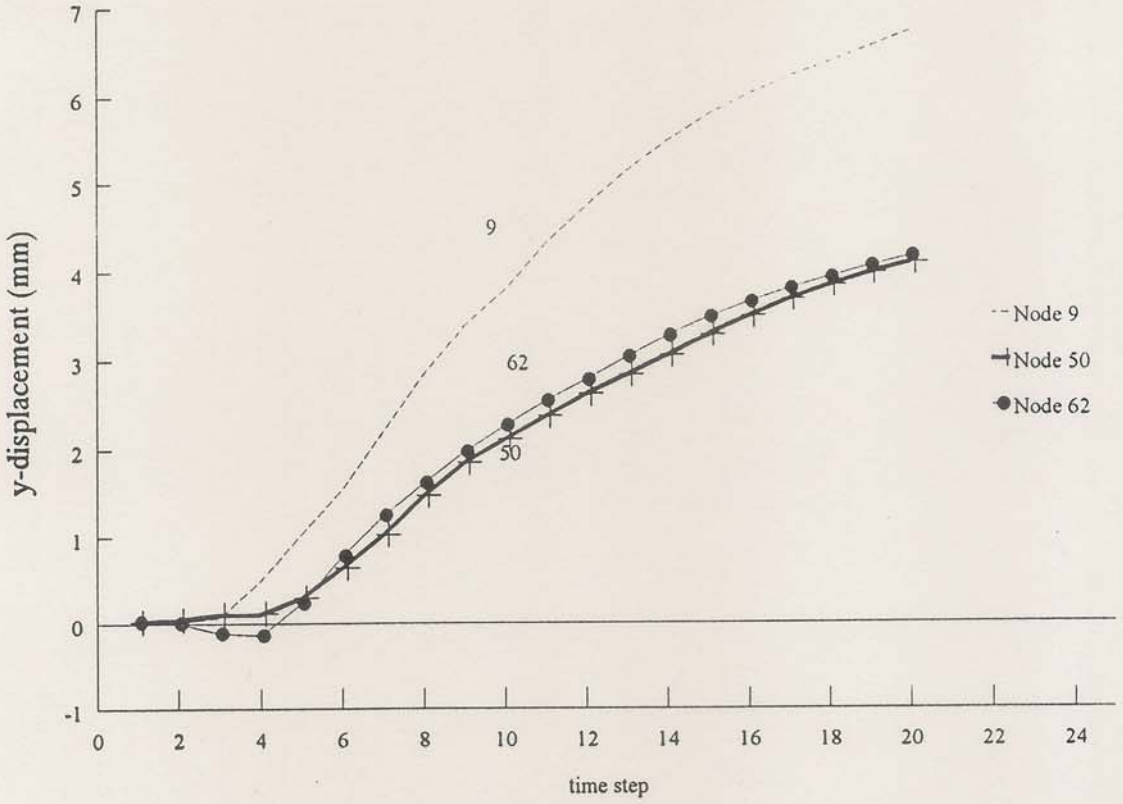


Fig 5.30 Time history of y-displacements

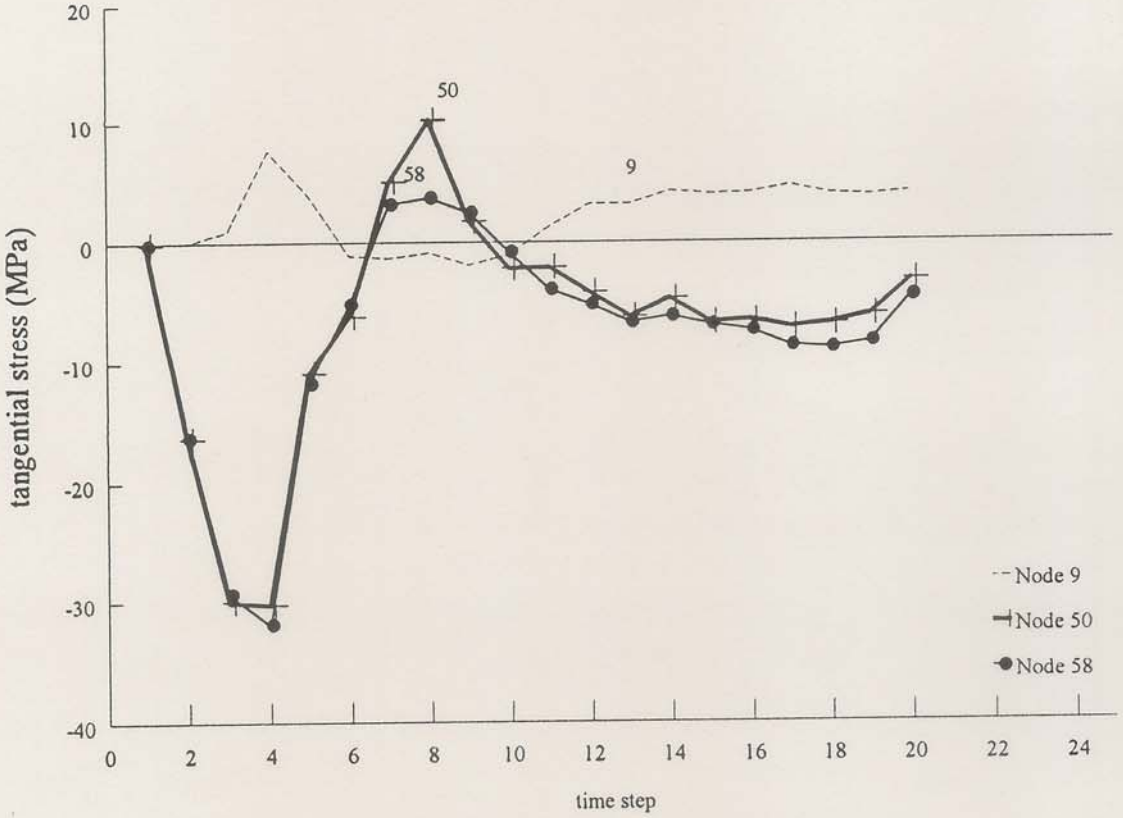


Fig 5.31 Tangential stresses on the outer surface

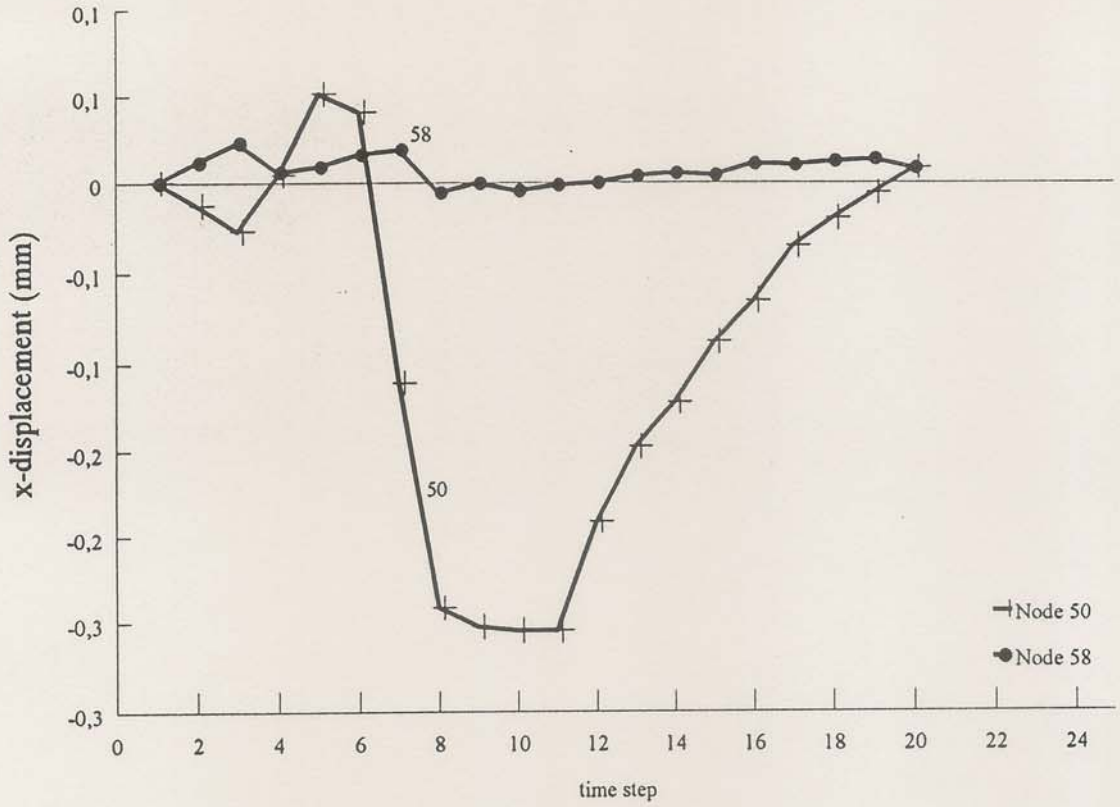


Fig 5.32 Time history of x-displacements

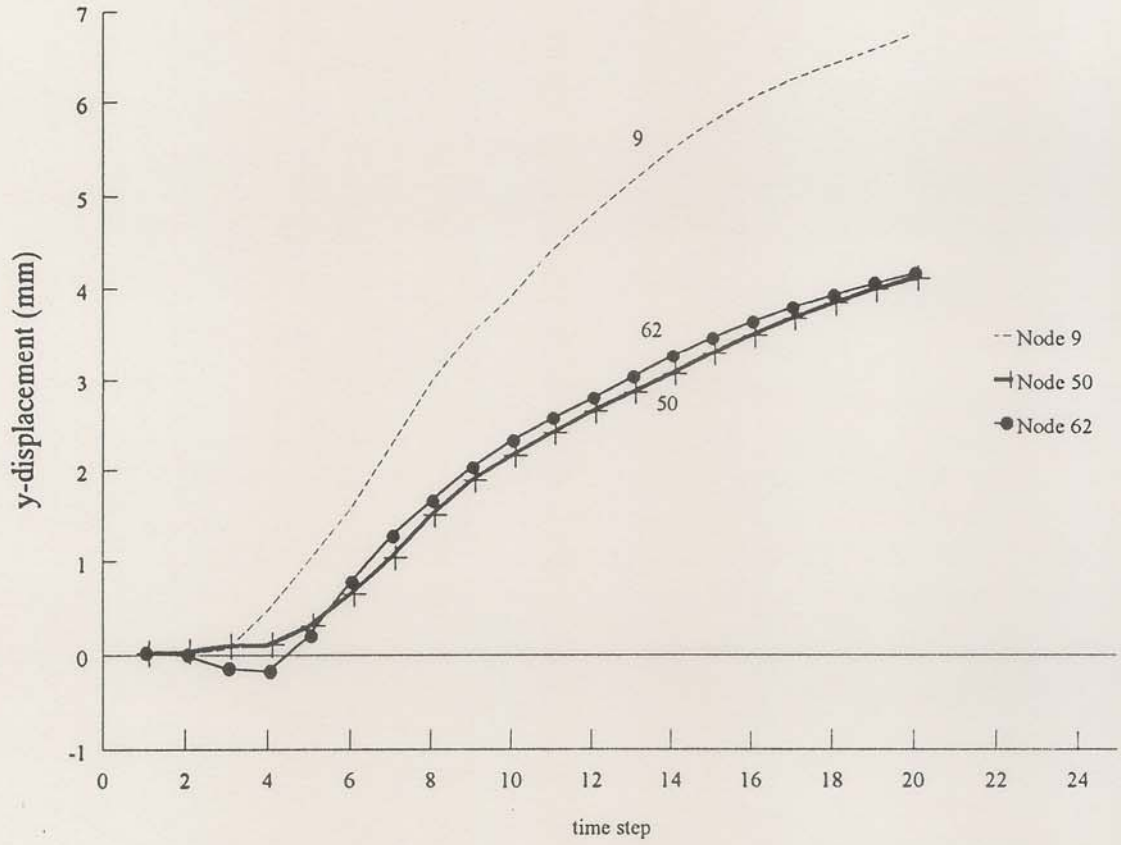


Figure 5.33 Time history of y-displacements

CHAPTER 6

WAVE PROPAGATION

6.1 INTRODUCTION

Two methods dealing with semi-infinite media are examined in this work.

The first method consists in discretizing the free surface of the ground in the usual way of discretizing any other boundary of the mesh. The solution is performed for all the boundaries including the boundary of the free surface. Attention should be paid only to tranversing the ground surface boundary in such a way that the material should be laid to the left hand of the traveler (cownter-clock-wise direction). The main advantages of this method are that the ground surface can be of arbitrary geometry and the solution is more accurate since the exact elastodynamic kernels are used. This method is treated as in the previous Chapters 2, 3 and 5 was stated and would not be utilised here.

The second method consist in using the formulae of reflection of harmonic waves on the ground surface. The reflected wave is calculated and is superimposed with the incident one. The ground surface is not taken into account any more. This method presuposes that the ground surface is a straight line.

6.2 REFLECTION OF WAVES

In this section, the conditions of reflection of waves on the free surface of the ground are investigated by employing the laws of reflection of harmonic waves.

The conditions of reflection of waves have been examined in the literature extensively (Achebach,1973; Eringen and Suhubi,1975). The analyses in those papers have been restricted in some special cases, e.g. horizontal free surface of the ground, special angle of incidence.

In this work the problem is examined in a general sense, for arbitrary orientation of the free surface of the ground and for any angle of incidence of P or SV wave. Also the reflection conditions of the waves on the free surface are examined and the reflection coefficients of stresses and amplitudes are derived. The analysis has been performed in a general form suitable for programming and automatic checking of the above conditions is performed from the developed program.

The following analyses are valid for Laplace domain and time domain as well.

6.2.1 Reflection Conditions

If a free ground surface exists, the developed program automatically investigates if the reflection conditions are fulfilled. For this purpose and for the case of the plane and explosive point waves, the reflection conditions are presented in Appendix E.

6.2.2 Reflection of P Waves

When an incident P wave, emanating from underground, strikes on the free surface, it reflects as P and SV wave (Figure 6.1). A local co-ordinate system is used with x axis parallel to the unit vector normal to the free surface line and direction to the right hand, travelling from the first defined node of the free line in the mesh (node 1), to the second (node 2). The y axis is tangent to the free surface line at right angles to the x axis.

The angles of the normal unit vector with the x axis of the general co-ordinate system

$$\theta_n \ , \quad \theta_P^i \ , \quad \theta_P^r \ , \quad \theta_{SV}^r \tag{6.1}$$

and the angle of incident P, reflected P and reflected SV waves are respectively.

The ground surface is considered free of tractions and thus the stresses in the normal direction to the surface line and the shear stresses in the local system must be zero, which gives

$$\sum \sigma_{11} = 0 \ , \qquad \sum \sigma_{12} = 0 \tag{6.2}$$

The equations of harmonic waves are used in order to extract the reflection stress coefficients. The mathematics behind this, are presented in Appendix F.

The ratio of the amplitudes of the reflected P to the incident P wave is :

$$\frac{A_1}{A_0} = \frac{k^2\cos^2(2\theta_S^r) + \sin(2\theta_P^i)\sin(2\theta_S^r)}{k^2\cos^2(2\theta_S^r) + \sin(2\theta_P^r)\sin(2\theta_S^r)} \frac{\varphi_0}{\varphi_1} \tag{6.3}$$

and the ratio of the amplitudes of the reflected SV to the incident P wave is :

$$\frac{A_2}{A_0} = \frac{-2k\sin(2\theta_S^r)\cos(2\theta_P^r)}{k^2\cos^2(2\theta_S^r) + \sin(2\theta_P^r)\sin(2\theta_S^r)} \frac{\varphi_0}{\varphi_2} \tag{6.4}$$

the stress reflection coefficients, for the normal to the free surface stress, are :

$$\frac{\sigma_{11}^{(1)}}{\sigma_{11}^{(0)}} = -\frac{A_1}{A_0}\frac{\varphi_1}{\varphi_0}$$

$$\frac{\sigma_{11}^{(2)}}{\sigma_{11}^{(0)}} = \frac{k\mu \sin(2\theta_S^r)}{\lambda+2\mu \cos^2(\theta_P^i)}\frac{A_2}{A_0}\frac{\varphi_2}{\varphi_0} \qquad \textbf{(6.5)}$$

for the tangent to the free surface stress :

$$\frac{\sigma_{22}^{(1)}}{\sigma_{22}^{(0)}} = -\frac{A_1}{A_0}\frac{\varphi_1}{\varphi_0}$$

$$\frac{\sigma_{22}^{(2)}}{\sigma_{22}^{(0)}} = \frac{-k\mu \sin(2\theta_S^r)}{\lambda+2\mu \cos^2(\theta_P^i)}\frac{A_2}{A_0}\frac{\varphi_2}{\varphi_0} \qquad \textbf{(6.6)}$$

and for the shear stress in the local system :

$$\frac{\sigma_{12}^{(1)}}{\sigma_{12}^{(0)}} = \frac{-\sin(2\theta_P^r)}{\sin(2\theta_P^i)}\frac{A_1}{A_0}\frac{\varphi_1}{\varphi_0}$$

$$\frac{\sigma_{12}^{(2)}}{\sigma_{12}^{(0)}} = \frac{-k \cos(2\theta_S^r)}{\sin(2\theta_P^i)}\frac{A_2}{A_0}\frac{\varphi_2}{\varphi_0} \qquad \textbf{(6.7)}$$

6.2.3 Reflection of SV Wave

When an incident SV wave, emanating from underground, strikes on the free ground surface, it reflects (Figure 6.2). The reflected wave according to the angle of incoming wave with the normal to the ground surface line (Figure 6.3) is:

(a). Angle lower to the critical angle, a P and SV wave is reflecting.

(b). Angle equal to the critical angle, a SV wave is only reflecting.

(c). Angle greater to the critical angle, a SV wave is reflecting.

The same local co-ordinate system that was introduced in section 6.2.2 is also used here.

The angles of the normal unit vector with the x axis of the general co-ordinate system and the angle of incident P, reflected
P and reflected SV waves are

$$\theta_n \ , \quad \theta_S^i \ , \quad \theta_P^r \ , \quad \theta_{SV}^r \tag{6.8}$$

respectively.

The equations of harmonic waves are used again, in order to extract the reflection stress coefficients. The mathematics behind this, are also presented in Appendix F.

The ratio of the amplitudes of the reflected and incident wave is :

$$\frac{A_1}{A_0} = \frac{k \sin(4\theta_S^r)}{k^2\cos^2(2\theta_S^r) + \sin(2\theta_P^r)\sin(2\theta_S^r)} \frac{\varphi_0}{\varphi_1} \tag{6.9}$$

and the ratio of the amplitudes of the reflected SV to the incident P wave:

$$\frac{A_2}{A_0} = \frac{k^2\cos^2(2\theta_S^r) + \sin(2\theta_S^i)\sin(2\theta_P^r)}{k^2\cos^2(2\theta_S^r) + \sin(2\theta_P^r)\sin(2\theta_S^r)} \frac{\varphi_0}{\varphi_2} \tag{6.10}$$

the stress reflection coefficients, for the normal to the free surface stress, are :

$$\begin{aligned} \frac{\sigma_{11}^{(1)}}{\sigma_{11}^{(0)}} &= \frac{\lambda+2\mu \cos^2(\theta_P^r)}{k\mu \sin(2\theta_S^i)} \frac{A_1}{A_0} \frac{\varphi_1}{\varphi_0} \\ \frac{\sigma_{11}^{(2)}}{\sigma_{11}^{(0)}} &= \frac{A_2}{A_0} \frac{\varphi_2}{\varphi_0} \end{aligned} \tag{6.11}$$

for the tangent to the free surface stress

$$\begin{aligned} \frac{\sigma_{22}^{(1)}}{\sigma_{22}^{(0)}} &= -\frac{\lambda+2\mu \sin^2(\theta_P^r)}{k\mu \sin(2\theta_S^i)} \frac{A_1}{A_0} \frac{\varphi_1}{\varphi_0} \\ \frac{\sigma_{22}^{(2)}}{\sigma_{22}^{(0)}} &= \frac{A_2}{A_0} \frac{\varphi_2}{\varphi_0} \end{aligned} \tag{6.12}$$

and for the shear stress in the local system

$$\begin{aligned} \frac{\sigma_{12}^{(1)}}{\sigma_{12}^{(0)}} &= \frac{-\sin(2\theta_P^r)}{k \cos(2\theta_S^i)} \frac{A_1}{A_0} \frac{\varphi_1}{\varphi_0} \\ \frac{\sigma_{12}^{(2)}}{\sigma_{12}^{(0)}} &= -\frac{A_2}{A_0} \frac{\varphi_2}{\varphi_0} \end{aligned} \tag{6.13}$$

6.3 WAVE PATTERNS

It is accepted here that the ground consists of a uniform layer and the wave is propagating in straight line rays. Straight lines are also the rays of the reflected wave. Refractions caused by changes in the P and SV wave speeds due to non-uniform material or to depth are not considered here.

If a free ground surface exists, two families of ray waves are encountered at every node of the mesh, the incident rays and the reflected rays. These two waves follow different patterns and arrive at different times at the examining nodes. Furthermore, the stresses that they carry are superimposed.

The patterns of the incident wave rays are direct. The distance of every node from the wavefront is calculated analytically. For the plane wave case, the wavefront is placed automatically from the program at the most appropriate node, or at a selected distance specified by the user. In the case of a wave generated from an explosion point, the wavefront at t=0 is at the explosion point and so the distances of the nodes from this point are calculated.

The patterns of the reflected on the ground surface rays are indirect. For every node two ray patterns are encountered. In the case of P incident wave, one for the P-P and one for the P-SV wave. In the case of SV incident wave, one for the SV-SV and one for SV-P wave.

All the distances are calculated numerically, from the given geometrical characteristics of the mesh and the reflection formulas.

The sign convention in this work for positive angles is counterclockwise.

6.4 INCIDENT PLANE WAVE

For the case of a plane wave, at every node two families of ray waves are encountered, the incident rays and the reflected rays. These two waves follow different patterns and arrive at different times at the examining nodes of the mesh. The stresses that they carry are superimposed (subroutine WINCID, see Appendix H, section H.1.3).

6.4.1 Wavefront

The incident wavefront is accepted as a straight line, which is propagating in the medium, transferring the shock stresses. The equation of the wavefront is calculated analytically.

For illustration purposes, a mesh with one boundary only, is considered with a local co-ordinate system (x"O_iy") with centre $O_i(x_{0i},y_{0i})$ the centre of the boundary, measured from the origin O (0.0,0.0) of a general co-ordinate system (Figure 6.4). An incident plane wave impinges on the ground line at an angle φ with the x" axis (angle of the tail of the wave). The equation of the wavefront line in the local system is given from analytical geometry as
where d is the distance of the origin O_i from the wavefront line. This distance either is

$$x''\cos\varphi + y''\sin\varphi - d = 0 \qquad \textbf{(6.14)}$$

given from the user or is calculated automatically from the developed program.

For the numerical calculation of the distance d, all the nodes P_j (j=1,2,...,N) of the mesh are examined, and the distance O_iQ_k of the node P_k with polar angle θ_k closer to φ is taken as the first approximation. The equation of the wavefront is written

$$x''\cos\varphi + y''\sin\varphi - d_k = 0 \qquad \textbf{(6.15)}$$

In order to assure that all the nodes are in front of the wavefront, a second inspection is performed by examining the distance of all the nodes from the new wavefront, which is given by (Figure 6.6)

$$d_j = x''\cos\varphi + y''\sin\varphi - d_k \qquad \textbf{(6.16)}$$

The distance d_j is negative when the origin O_i and the node j are at the same side of the wavefront line and positive if they are at opposite sides. So if any node j is found with positive distance d_j, it iplies that it is behind the wavefront and O_iP_j is accepted as the new approximation of the distance d.

For calculation of d in the case of multiple boundaries in the same mesh, the wavefront line is determined as previously for every boundary separately in a local co-ordinate system. The final wavefront is the local wavefront line with minimum distance from the centre G of all the boundaries. For clarity, the compound mesh of Figure (6.5) consisting of one rectangle, one ellipse and one circle is depicted, with corresponding local centres $O_1(x_{01},y_{01})$, $O_2(x_{02},y_{02})$ and $O_3(x_{03},y_{03})$, and local wavefront positions (ε_1), (ε_2) and (ε_3) respectively. The distance of the local centres O_1, O_2 and O_3 from the separate wavefronts (ε_1), (ε_2) and (ε_3) are d_1, d_2 and d_3 respectively. Moreover, the distances dg_1, dg_2 and dg_3 of the centre $G(x_g,y_g)$ of all the boundaries from the

separate wavefronts is estimated. The line (ε_2) is excluded as lying inside the mesh, since the distance dg_2 is positive and the origins G and O_2 are at the opposite side of the line (ε_2).

The representative wavefront line is (ε_3) since $|dg_3| > |dg_1|$ (or $dg_3 < dg_1 < dg_2$), with equation

$$x^{(G)}\cos\varphi + y^{(G)}\sin\varphi - d_G = 0 \qquad \textbf{(6.17)}$$

where $x^{(G)}$, $y^{(G)}$ are the co-ordinates of the wavefront line in the system x'Gy' and d_G is the distance of the centre of the mesh G from the wavefront line. The above have been included in subroutine WAVEFIR, see Appendix H, section H.1.2.

6.4.2 P Wave

a. Free field stresses

The stresses in the medium without the existence of the opening are the free field stresses and are calculated as in the following.

The incident wave is a shock P plane wave that carries a compressive stress in the direction of motion of magnitude σ_{11}. In the perpendicular to the motion direction at the wavefront, the lateral movement of the material is restricted, because of the nature of the instantaneous shock wave that is moving with high speed. This results in a uniaxial strain condition. Applying the stress-strain relations for isotropic linear elastic material with ε_{11} the only non vanishing component, yields

$$\sigma_{ij} = \lambda\varepsilon_{kk}\delta_{ij} + 2\mu\varepsilon_{ij} = \lambda\varepsilon_{11}\delta_{ij} + 2\mu\varepsilon_{ij} \qquad \textbf{(6.18)}$$

and

$$\sigma_{11} = \lambda\varepsilon_{11} + 2\mu\varepsilon_{11} \Rightarrow \varepsilon_{11} = \frac{1}{\lambda+2\mu}\sigma_{11}$$
$$\sigma_{22} = \lambda\varepsilon_{11} \Rightarrow \sigma_{22} = \frac{\lambda}{\lambda+2\mu}\sigma_{11} = \frac{\nu}{1-\nu}\sigma_{11}$$
$$\sigma_{12} = 0 \qquad \textbf{(6.19)}$$

For the formulation of the BEM equations, the tractions at every node of the mesh

are needed.

For every node P_i a local co-ordinate system is established, with unit vectors the normal outward to the material n and tangential q to the surface of the element vectors, as in Figure (4.5). The node P_i is considered as the middle node of a quadratic element, with start and end nodes of this element are the previous P_{i-1} and next P_{i+1} node of the mesh. The direction cosines are given by equations (4.51), (4.52), (4.53) and (4.54) in Chapter 4, section 4.5.

For the case of Laplace domain, the stresses are transformed (subroutine WAVPLAN, see Appendix H, section H.1.3) by multiplying them by the factor

$$\frac{e^{-t_a s}}{s} \tag{6.20}$$

for the constant loading function and by

$$\left(\frac{1-e^{-t_c s}}{t_c s}\right)\left(\frac{e^{-t_a s}}{s}\right) \tag{6.21}$$

for the ramp-like loading function according to equations (2.2) and (2.3). t_a is the arriving time of the wavefront at node P_i that is found from

$$t_a = \frac{d_i}{c_1} \tag{6.22}$$

where d_i is the distance of the node from the wavefront at time t=0, c_1 is the P wave velocity. The distance d_i is found from analytical geometry from the equation of the wavefront line

$$x'_\varepsilon \cos\varphi + y'_\varepsilon \sin\varphi - d_g = 0 \tag{6.23}$$

by substituting in this equation the co-ordinates $(x'_\epsilon, y'_\epsilon)$ of the wavefront line (ε), with

$$d_i = x'_i \cos\varphi + y'_i \sin\varphi - d_g \tag{6.24}$$

the co-ordinates (x_i', y_i') results

where (x_i', y_i') are the co-ordinates of the node P_i in a co-ordinate system with centre the centre $G(x_g, y_g)$ of all the boundaries that form the mesh (Figure 6.5).

b. Tractions on the boundary

The incident tractions on the surface of the boundary are calculated from the free field stresses, on the tangential at node P_i plane with an outward unit normal vector n (Figure 4.5). They are given according to equation (4.56) of Chapter 4, section 4.6.3 by

$$t_x = \sigma_{xx}\cos\theta_{n1} + \sigma_{yx}\sin\theta_{n1} = \sigma_{xx}n_1 + \sigma_{yx}n_2$$
$$t_y = \sigma_{xy}\cos\theta_{n1} + \sigma_{yy}\sin\theta_{n1} = \sigma_{xy}n_1 + \sigma_{yy}n_2 \qquad \textbf{(6.25)}$$

However, in order to produce a traction free surface on the boundary the opposite tractions are taken (which are acting on the same tangential plane but with unit normal vector inward to the material and opposite to n) (Figure 6.6.a). These tractions are given by the expressions

$$t_x' = \sigma_{xx}\cos\alpha + \sigma_{yx}\sin\alpha = \sigma_{xx}n_1' + \sigma_{yx}n_2'$$
$$t_y' = \sigma_{xy}\cos\alpha + \sigma_{yy}\sin\alpha = \sigma_{xy}n_1' + \sigma_{yy}n_2' \qquad \textbf{(6.26)}$$

where

$$n_1' = -n_1\ , \qquad n_2' = -n_2$$

Applying these tractions on the boundary, the problem is solved by the BEM, and the stresses on the boundary are found. Finally, the total stress field is obtained by superimposing these stresses with the free field stresses (Baron and Mathews, 1961)

6.4.3 SV Wave

The above for the P wave, are also valid for the SV wave, with the exception that for the estimation of the arriving times, the SV wave velocity is used :

$$t_a = \frac{d_i}{c_2} \qquad \textbf{(6.28)}$$

6.5 REFLECTED PLANE WAVE

When a propagating P wave in an infinite medium encounters the free ground surface, a reflection takes place. Thus the wave changes direction and the stresses that it carries also change. The changes in the stresses were examined previously in section 6.2 while

here the wave patterns are examined.

6.5.1 Ground Surface

In the absence of external loading, the ground surface is a traction free surface and remains such after the reflection of the wave. In this work, the surface is considered as a straight line of an arbitrary inclination. The line (subroutine SURFLINE, see Appendix H, section H.1.3) is defined from the co-ordinates (x_1,y_1) and (x_2,y_2) of the start (node 1) and end (nodes 2), (Figure 6.16), that also determine the positive direction of the line. The equation of this line is

$$(y_2 - y_1)x + (x_1 - x_2)y - (x_1 y_2 - y_1 x_2) = 0 \qquad \textbf{(6.29)}$$

that may be written as

$$-\frac{y_1 - y_2}{d}x + \frac{x_1 - x_2}{d}y - \frac{x_1 y_2 - y_1 x_2}{d} = 0 \Rightarrow$$
$$-x\sin\theta_{n2} + y\cos\theta_{n2} - c = 0 \qquad \textbf{(6.30)}$$

and finally

$$x\cos\theta_{n1} + y\cos\theta_{n2} - c = 0 \qquad \textbf{(6.31)}$$

where θ_{n1} and θ_{n2} are the angles of the unit outward normal vector with the x and y global axes respectively and c and d are

$$c = \frac{x_1 y_2 - y_1 x_2}{d} \quad , \qquad d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \qquad \textbf{(6.32)}$$

The direction cosines of the unit outward normal vector are found by comparing the line of the surface with a quadratic element with the three nodes as the start, middle and end node of the line of the surface and using shape functions as in section 4.4. In all the following calculations, a local co-ordinate system is used with x and y axes the unit normal and tangential vector to the line of the surface of the ground.

The developed program automatically searches all the boundaries, and when a free surface boundary is found, it investigates if the reflection conditions are fulfilled for the

reflection of the plane wave.

6.5.2 Local Co-ordinate System

For every node of the mesh a local co-ordinate system x'Ey' is established, with origin at the node and axes parallel to the system that was fixed on the ground surface (axes normal and parallel to the free surface).

6.5.3 Angles of Reflection

The wave rays are reflected only on certain angles of incidence on the free ground surface. In the following, φ is the angle of the axis of the incident plane wave (tail of the wave) with the x global axes and θ_n is the angle of the outward normal to the free surface unit vector n with the x global axis.

Further the angles of incident P, reflected P and SV wave counted in a local co-ordinate system with x axis the unit normal outward vector n, are θ_P^i, θ_P^r and θ_{SV}^r. The magnitude of the angles, without any orientation, of the incident P, reflected P and SV wave are θ_0, θ_1 and θ_2 respectively. According to Achenbach (1973), is $\theta_0=\theta_1$.

At every node two reflected wave rays are encountered, the reflected P and the reflected SV wave. Their lines form angles φ_1 and φ_2 respectively with the x' local axis.

The reflection conditions can be expressed in terms of the angles φ and θ_n. Further, the angles θ_P^i, θ_P^r and θ_{SV}^r can be expressed in terms of θ_0 and θ_2 and the angles φ_1 and φ_2 in terms of θ_0 and θ_2 also. These relations are given in a simple and easy for programming purposes form, for P and SV wave, in the following.

6.5.4 Reflection of P Wave

(subroutine REFP, see Appendix H, section H.1.3).

6.5.4.1 Reflection Conditions-Angles of Waves

The incident wave is reflected on the surface of the ground, if the following conditions are fulfilled (subroutine REFLCOND, see Appendix H, section H.1.3):

a. Case $0\le\theta_n<\pi/2$, reflection occurs only when $\pi/2<\varphi-\theta_n<3\pi/2$ (Figure 6.7.a) For this case the angles of incident P and reflected SV wave are illustrated in Figure (6.8). These angles are given by

(i). For $\pi/2<\varphi-\theta_n<\pi$ is (Figure 6.8.i)

$$\theta_P^i=\pi-\theta_0\ ,\qquad \theta_P^r=\pi+\theta_0\ ,\qquad \theta_{SV}^r=\pi+\theta_2$$
$$\varphi_1=\theta_0\ ,\qquad \varphi_2=\theta_2 \tag{6.33}$$

(ii). For $\pi<\varphi-\theta_n<3\pi/2$ (Figure 6.8.ii)

$$\theta_P^i=\pi+\theta_0\ ,\qquad \theta_P^r=\pi-\theta_0\ ,\qquad \theta_{SV}^r=\pi-\theta_2$$
$$\varphi_1=2\pi-\theta_0\ ,\qquad \varphi_2=2\pi-\theta_2 \tag{6.34}$$

b. Case $\pi/2\le\theta_n<\pi$, reflection occurs in the cases that $\pi/2<\varphi-\theta_n<2\pi-\theta_n$ or when $-\pi/2<\varphi-\theta_n<0$ (Figure 6.7.b). It is also

(i). For $\pi/2<\varphi-\theta_n<\pi$ (Figure (6.9.i)

$$\theta_P^i=\pi-\theta_0\ ,\qquad \theta_P^r=\pi+\theta_0\ ,\qquad \theta_{SV}^r=\pi+\theta_2$$
$$\varphi_1=\theta_0\ ,\qquad \varphi_2=\theta_2 \tag{6.35}$$

(ii). For $\pi<\varphi-\theta_n<2\pi-\theta_n$ or $-\pi/2<\varphi-\theta_n<0$ (Figure (6.9.ii)

$$\theta_P^i=\pi+\theta_0\ ,\qquad \theta_P^r=\pi-\theta_0\ ,\qquad \theta_{SV}^r=\pi-\theta_2$$
$$\varphi_1=2\pi-\theta_0\ ,\qquad \varphi_2=2\pi-\theta_2 \tag{6.36}$$

c. Case $\pi\le\theta_n<3\pi/2$, reflection occurs in the cases that $\pi/2<\varphi-\theta_n<2\pi-\theta_n$ or $-\pi/2<\varphi-\theta_n<0$ (Figure 6.7.c). In this case is

(i). For $\pi/2<\varphi-\theta_n<2\pi-\theta_n$ or $-\theta_n<\varphi-\theta_n<-\pi$ is (Figure 6.10.i)

$$\theta_P^i=\pi-\theta_0\ ,\qquad \theta_P^r=\pi+\theta_0\ ,\qquad \theta_{SV}^r=\pi+\theta_2$$
$$\varphi_1=\theta_0\ ,\qquad \varphi_2=\theta_2 \tag{6.37}$$

(ii). For $-\pi<\varphi-\theta_n<0$ is (Figure 6.10.ii)

$$\theta_P^i=\pi+\theta_0\ ,\qquad \theta_P^r=\pi-\theta_0\ ,\qquad \theta_{SV}^r=\pi-\theta_2$$
$$\varphi_1=2\pi-\theta_0\ ,\qquad \varphi_2=2\pi-\theta_2 \tag{6.38}$$

d. Case $3\pi/2\le\theta_n<2\pi$, reflection occurs only when $-3\pi/2<\varphi-\theta_n<-\pi/2$ (Figure 6.7.d). For this case it is

(i). For $-3\pi/2<\varphi-\theta_n<-\pi$ is (Figure 6.11.i)

$$\theta_P^i=\pi-\theta_0\ ,\qquad \theta_P^r=\pi+\theta_0\ ,\qquad \theta_{SV}^r=\pi+\theta_2$$
$$\varphi_1=\theta_0\ ,\qquad \varphi_2=\theta_2 \tag{6.39}$$

(ii). For $-\pi<\varphi-\theta_n<-\pi/2$ (Figure 6.11.ii)

$$\theta_P^i=\pi+\theta_0\ ,\qquad \theta_P^r=\pi-\theta_0\ ,\qquad \theta_{SV}^r=\pi-\theta_2$$
$$\varphi_1=2\pi-\theta_0\ ,\qquad \varphi_2=2\pi-\theta_2 \tag{6.40}$$

In case that no reflection conditions are fulfilled, the incident only wave is examined.

6.5.4.2 Ray Patterns-Arriving Times

The ray patterns are calculated numerically (subroutine PATP, see Appendix H, section H.1.3). At every node E, two wave rays are encountered, the P incident and P reflected (P-P) ray which follows the pattern ABE and the P-SV ray that follows the pattern BDE as is illustrated in Figures (6.7), (6.8), (6.9), (6.11). The points A and B are on the wavefront line at time t=0 and the points B and D are lying on the ground surface.

The distance of a node E from the line of the ground is found from analytical geometry, by replacing the x and y co-ordinates in equation (6.31) by the co-ordinates of the node x_E, y_E and begomes

$$h_E=\left|x_E\cos\theta_{n1}+y_E\cos\theta_{n2}-c\right| \tag{6.41}$$

the absolute value of the distance is taken, because its value is negative. This is due to the fact that the node E and the origin of the co-ordinate system are lying from the same side of the surface line.

Furthermore, it is

$$(BE)=\frac{h_E}{\cos\varphi_1}\ ,\qquad (DE)=\frac{h_E}{\cos\varphi_2} \tag{6.42}$$

The co-ordinates of the points B and D in the local system x'y' are

$$x_B'=h_E\ ,\qquad x_D'=h_E$$
$$y_B'=(BE_1)=(BE)\sin\varphi_1\ ,\qquad y_D'=(DE_1)=(DE)\sin\varphi_2 \tag{6.43}$$

These co-ordinates are transformed to a co-ordinate system xEy, with origin at the node E and axes x,y parallel to the global axes

$$x_B'' = x_B' \cos\theta_n - y_B' \sin\theta_n \;, \qquad x_D'' = x_D' \cos\theta_n - y_D' \sin\theta_n$$
$$y_B'' = x_B' \sin\theta_n + y_B' \cos\theta_n \;, \qquad y_D'' = x_D' \sin\theta_n + y_D' \cos\theta_n \tag{6.44}$$

and in the global co-ordinate system

$$x_B = x_B'' + x_E \;, \qquad x_D = x_D'' + x_E$$
$$y_B = y_B'' + y_E \;, \qquad y_D = y_D'' + y_E \tag{6.45}$$

These co-ordinates are saved in a file for using them later in the drawings of the preprocessing phase (subroutine RAYPLANE, see Appendix H, section H.1.4).

Finally the co-ordinates of B and D in a co-ordinate system with centre the centre of the whole mesh G (Figure 6.5) are

$$x_B^{(G)} = x_B - x_g \;, \qquad x_D^{(G)} = x_D - x_g$$
$$y_B^{(G)} = y_B - y_g \;, \qquad y_D^{(G)} = y_D - y_g \tag{6.46}$$

The distances (AB) and (CD) are found analytically as the distance of points B and D from the wavefront line (subroutine WAVEFIR, see Appendix H, section H.1.2), by using equation (6.42) :

$$(AB) = | x_B^{(G)} \cos\varphi + y_B^{(G)} \sin\varphi - d_G |$$
$$(CD) = | x_D^{(G)} \cos\varphi + y_D^{(G)} \sin\varphi - d_G | \tag{6.47}$$

$$t_E^P = \frac{(AB) + (BE)}{c_1} \;, \qquad t_E^{SV} = \frac{(CD)}{c_1} + \frac{(DE)}{c_2} \tag{6.48}$$

The arriving time of the reflected P and SV wave at the node E, is
where c_1 and c_2 pressure and shear wave velocities.

6.5.4.3 Reflected Tractions

The reflected stresses are found from the incident in the local co-ordinate system with x and y axes the normal outward and tangential to the surface of the ground, by multiplying them by the stress reflection coefficients presented in section 6.2.2 and 6.2.3 for the reflection of P and SV waves respectively. It is mentioned that uniaxial strain conditions exist in the normal to the direction of the wave rays, for the incident and the reflected wave as well.

The transformed stresses in Laplace domain are found by multiplying the stresses by

the factors of equations (2.2) and (2.3) for the case of constant and ramp-like loading.

The stresses are converted from the local co-ordinate system to the global system, by applying the equations (4.60) of Chapter 4, section 4.6.4.

From the resulting stresses, the reflected tractions on the boundary are derived by applying the equation (4.56) of Chapter 4, section 4.6.3. The incident and reflected tractions are superimposed to give the final stresses for the BEM formulation.

6.5.5 Reflection of SV Wave
(subroutine REFSV, see Appendix H, section H.1.3).

The same reflection conditions, as in the case of P wave (section 6.5.4.1) are valid. The calculations of the ray patterns are similar as in the case of P wave (subroutine PATSV, see Appendix H, section H.1.3), as illustrated in Figure (6.12) and the arriving times of the waves and the Laplace transformation of loading are extracted in a similar manner also.

6.6 INCIDENT EXPLOSION POINT WAVE
(subroutine WINCID, see Appendix H, section H.1.3).

It is accepted here that the explosion takes place at an underground point. The wave rays are propagating in radial directions from the explosion point and they carry a pressure P in the direction of propagation, described by the expressions given in Chapter 4, section 4.2.

6.6.1 P Wave

a. Free Field Stresses (subroutine WAVEXPL, see Appendix H, section H.1.3).
The incident wave is a compressional shock P plane wave that carries a pressure in the direction of motion at a node E, of magnitude P_{0E} (subroutine ZEXPLOSQ, see Appendix H, section H.1.3).

The transformed stresses in Laplace domain for a constant or decaying in time wave, are found by using the equations of Chapter 4, section 4.2 (subroutine DECEXPL, see Appendix H, section H.1.3). The arriving time is found from

$$t_a = \frac{d_i}{c_1} \qquad \textbf{(6.49)}$$

where d_i is the distance of the examining node from the explosion point. By applying equation (4.6.4) of Chapter 4, one obtains the stresses from the local co-ordinate system

to the global system.

The stress in the ray direction at the position of a node is σ_{11}=-$P_{0E.}$ In the perpendicular to the ray line direction, a uniaxial strain condition exists at the node, which yields (subroutine EXPLMOR, see Appendix H, section H.1.3):

$$\sigma_{22} = \frac{v}{1-v}\sigma_{11} \ , \qquad \sigma_{12} = 0 \qquad \textbf{(6.50)}$$

These stresses are transformed to the global co-ordinate system by using the equation (4.6.4).

b. Tractions on the Boundary

The tractions on the boundary are calculated in the same way as in the plane incident wave case (section 6.4.2.b).

6.6.2 SV Wave

The above for the P wave, are valid also for the SV wave, with the exception that for the estimation of the arriving times, the shear wave velocity c_2 is used.

6.7 REFLECTED EXPLOSION POINT WAVE

6.7.1 Angles of Reflection

In the following, φ_B and φ_D are the angles of the incident at point B and D line rays (tail of the rays) with the x global axis and θ_n is the angle of the outward normal to the free surface unit vector n with the x global axis.

Furthermore, for point B the angles of incident P, reflected P and SV wave counted in a local co-ordinate system with x axis the unit normal outward vector n, are θ_{BP}^{i}, θ_{BP}^{r} and θ_{BSV}^{r}. Also for point D are θ_{DP}^{i}, θ_{DP}^{r} and $\theta_{DSV}^{r.}$ The magnitude of the angles, without any orientation, for point B of the incident P, reflected P and SV wave are θ_1, θ_2 and θ_5. For point D the P incident and SV reflected wave are θ_3 and θ_4 respectively. According to Achenbach (1972) is $\theta_1=\theta_2=\theta_0$.

At every node E two reflected wave rays are encountered, the reflected P and the reflected SV wave. Their lines form angles φ1 and φ2 respectively with the x' local axis.

The angles θ_P^{i}, θ_P^{r} and θ_{SV}^{r} may be expressed in terms of θ_0 and θ_2 and similarly the angles φ_1 and φ_2 in terms of θ_0 and θ_2. These relations are given in the following in a simple and easy for programming purposes form, for P and SV wave.

The angle of the line from the explosion point A to the node E counted from the local x' axis is ψ.

6.7.2 Reflection of P Wave

The nomenclature for the reflection of explosive point P wave is depicted in Figure (6.13), (subroutine REFEXP, see Appendix H, section H.1.3).

6.7.2.1. Reflection Conditions-Angles of Waves

In case of an undergroun explosion, the incident wave is reflected on the surface of the ground (subroutine REFCONEX, see Appendix H, section H.1.3). The distance of the explosion point A from the free surface is
if h_a is positive or zero, the explosion point is over or on the free surface line

$$h_A = x_A \cos\theta_{n1} + y_A \cos\theta_{n2} - c \tag{6.51}$$

respectively. This is so, because the point A and the geometrical centre of the mesh (centre of the global system), are lying from opposite sides or on the free surface line.

At every node E, two wave rays are encountered, the P incident and P reflected (P-P) ray follows the pattern ABE and the P-SV ray follows the pattern ADE as in Figure (6.13) is illustrated.

The distance of a node E from the line of the ground is
and the distance (AE) from the explosion point to the node E is

$$h_E = \left| x_E \cos\theta_{n1} + y_E \cos\theta_{n2} - c \right| \tag{6.52}$$

$$(AE) = \sqrt{(x_E - x_A)^2 + (y_E - y_A)^2} \tag{6.53}$$

a. Incident and Reflected Angles at Points B and D

(i). For $0 \le \psi \le \pi$ is (Figure 6.14)

$$\begin{aligned} &(\theta_P^i)_B = \pi - \theta_0 \,, \quad (\theta_P^r)_B = \pi + \theta_0 \,, \quad (\theta_{SV}^r)_B = \pi + \theta_5 \\ &(\theta_P^i)_D = \pi - \theta_3 \,, \quad (\theta_P^r)_D = \pi + \theta_3 \,, \quad (\theta_{SV}^r)_D = \pi + \theta_4 \\ &\varphi_1 = \theta_2 = \theta_0 \,, \quad \varphi_2 = \theta_4 \end{aligned} \tag{6.54}$$

(ii). For $\pi < \psi < 2\pi$ is (Figure 6.14)

$$
\begin{aligned}
&(\theta_P^i)_B = \pi + \theta_0 \ , \qquad (\theta_P^r)_B = \pi - \theta_0 \ , \qquad (\theta_{SV}^r)_B = \pi - \theta_5 \\
&(\theta_P^i)_D = \pi + \theta_3 \ , \qquad (\theta_P^r)_D = \pi - \theta_3 \ , \qquad (\theta_{SV}^r)_D = \pi - \theta_4 \\
&\varphi_1 = 2\pi - \theta_2 \ , \qquad \varphi_2 = 2\pi - \theta_4
\end{aligned}
\tag{6.55}
$$

b. Angles $\theta_1=\theta_2=\theta_0$ and θ_5 (subroutine REFANGEX, see Appendix H, section H.1.3).

The co-ordinates of the explosion point A in the local system xEy are

$$
x_A' = x_A - x_E \ , \qquad y_A' = y_A - y_E \tag{6.56}
$$

and transforming the co-ordinates in the local system x'Ey' is

$$
\begin{aligned}
x_E'' &= x_E' \cos\theta_n + y_E' \sin\theta_n \\
y_E'' &= -x_E' \sin\theta_n + y_E' \cos\theta_n
\end{aligned}
\tag{6.57}
$$

it is also

$$
\psi = \arctan\left(\frac{y_A''}{x_A''} \right) \tag{6.58}
$$

From simple geometry it is

$$
(AE_2) = (AE)\sin\psi \ , \qquad (AB_2) = (AE_2)\frac{h_a}{h + h_a} \tag{6.59}
$$

and

$$
\theta_0 = \arctan\frac{|(AB_2)|}{h_a} \tag{6.60}
$$

Moreover, it is

$$
k = \frac{\sin\theta_0}{\sin\theta_5} \quad \Rightarrow \quad \theta_5 = \left| \arcsin\left(\frac{\sin\theta_0}{k} \right) \right| \tag{6.61}
$$

c. Angles θ_3 and θ_4 (subroutine TH34EX, see Appendix H, section H.1.3).

These angles are calculated by a numerical technique similar to Whitehouse (1983) whereby is that for a given value of θ_3 the angle θ_4 can be found and consequently the

distance AE'_2. An iterative loop starts, in which at every step the angle θ_3 is increased by a certain amount until the length AE'_2 becomes approximately equal to AE_2 that is derived from equation (6.60).

More specifically it is

$$AD_2 = h_a \tan\theta_3 \ , \qquad E_2D_2 = ED_1 = h_e \tan\theta_4$$
$$AE_2' = AD_2 + E_2D_2 \qquad \textbf{(6.62)}$$

and

$$k = \frac{\sin\theta_3}{\sin\theta_4} \quad \Rightarrow \quad \theta_4 = \left| \arcsin\left(\frac{\sin\theta_3}{k} \right) \right| \qquad \textbf{(6.63)}$$

The value of AE'_2 is compared with AE_2 of equation (6.60) and is adjusted accordingly until it is approximately equal to AE_2.

6.7.2.2 Ray Patterns-Arriving Times (subroutine PATEXP, see Appendix H, section H.1.3).

Similarly to the plane wave case (section 6.5.4), one may express

$$(BE) = \frac{h_E}{\cos\varphi_1} \ , \qquad (DE) = \frac{h_E}{\cos\varphi_2} \qquad \textbf{(6.64)}$$

The co-ordinates of the points B and D in the local system x'y' are

$$x_B' = h_E \ , \qquad x_D' = h_E$$
$$y_B' = (EB_1) = (EB)\sin\varphi_1 \ , \qquad y_D' = (ED_1) = (ED)\sin\varphi_2 \qquad \textbf{(6.65)}$$

these co-ordinates are transformed to a co-ordinate system with xEy, with origin at the

$$x_B'' = x_B' \cos\theta_n - y_B' \sin\theta_n \ , \qquad x_D'' = x_D' \cos\theta_n - y_D' \sin\theta_n$$
$$y_B'' = x_B' \sin\theta_n + y_B' \cos\theta_n \ , \qquad y_D'' = x_D' \sin\theta_n + y_D' \cos\theta_n \qquad \textbf{(6.66)}$$

node E and axes x,y parallel to the global axes

$$x_B = x_B'' + x_E \ , \qquad x_D = x_D'' + x_E$$
$$y_B = y_B'' + y_E \ , \qquad y_D = y_D'' + y_E \qquad \textbf{(6.67)}$$

and in the global co-ordinate system

These co-ordinates are also saved in a file for using them later for drawings in the preprocessing phase (subroutine RAYPLANE, see Appendix H, section H.1.4).

Finally the co-ordinates of B and D in a co-ordinate system with centre the centre of whole the mesh G (Figure 6.5) are

$$x_B^{(G)} = x_B - x_g \ , \qquad x_D^{(G)} = x_D - x_g$$
$$y_B^{(G)} = y_B - y_g \ , \qquad y_D^{(G)} = y_D - y_g \qquad \textbf{(6.68)}$$

The distance (AB) and (AD) are found analytically by using the following expressions

$$(AB) = \sqrt{(x_A - x_B)^2 + (y_A - y_B)^2} \ , \qquad (AD) = \sqrt{(x_A - x_D)^2 + (y_A - y_D)^2} \qquad \textbf{(6.69)}$$

:

The arriving time of the reflected P and SV wave at the node E, is

$$t_E^P = \frac{(AB) + (BE)}{c_1} \ , \qquad t_E^{SV} = \frac{(AD)}{c_1} + \frac{(DE)}{c_2} \qquad \textbf{(6.70)}$$

where c_1 and c_2 are pressure and shear wave velocities respectively.

6.7.2.3 Reflected Tractions

The reflected stresses are found similarly in the local co-ordinate system, as in the case of the plane wave, from the incident at points B and D, by multiplying them by the stress reflection coefficients (Reflection of P - SV waves). The transformed stresses in Laplace domain are found by multiplying the stresses by the factors of equations (6.21) and (6.22) for the case of constant and ramp-like loading. By applying equation (4.61) of Chapter 4, one obtains the stresses from the local co-ordinate system to the global system.

From the resulting stresses, the tractions on the boundary are derived, according to section 6.4.2.b. The incident and reflected tractions are superimposed to give the final stresses for the BEM formulation.

6.7.3 Reflection of SV Wave

(subroutine REFEXSV, see Appendix H, section H.1.3).

For an absolutely spherical geometry of the charge, only compressive P waves are produced. Any non-spherical geometry of the charge gives rise to shear SV waves. In this work, the generated SV stresses are taken as a percentage of P stress waves.

The same reflection conditions, as in the case of P wave are valid (subroutine REFCONEX, see Appendix H, section H.1.3). The calculations of the ray's patterns are similar as in the case of P wave (section 6.6.1) , and as is illustrated in Figure (6.15). The arriving times of the waves and the Laplace transformation of loading are also extracted in a similar manner.

6.8 APPLICATIONS

6.8.1 Half-space Solutions

In the following applications, openings in semi-infinite domains are examined.

6.8.1.1 Circular Cross Section Cavity in a Half Space Loaded by a P Plane Wave

This problem has been examined by Niwa et al. (1986). In this work a circular cavity of radius 2.5m is considered, with centre at depth 5.0 m from the free surface of the ground. The material characteristics are Young's modulus E=20000 MPa, Poisson's ratio ν=0.25 and density ρ=2.55 Mg/m^3. The total examining time interval is 0.015 s implemented in 20 time steps. The loading is a P incident stress wave of magnitude -10.0 MPa striking at a vertical direction and normal to the free surface. The surface of the

cavity has been discretized into 8 quadratic elements and part of the free surface of the ground of length 16.0 m, also into 8 elements.

In Figure (6.17) the discretization of the surfaces into elements and the nodes numbering are shown.

The problem is solved here by both methods of dealing with semi-infinite, either by using the laws of reflection of harmonic waves on the free surface or by discretizing the ground surface.

Figure (6.18) illustrates the change of the tangential stresses at point C versus time steps by using the laws of reflection of harmonic waves in Laplace domain and in time domain as well. The respective results of Niwa et al. (1980) are presented also in the same picture. A good agreement is observed between this work and the work of Niwa et al.

In Figure (6.19) the tangential stresses at the same point C are presented by using the method of dicsretizing the free surface of the ground. Comparing Figures (6.18) and (6.19) it is observed that the two methods give identical results until the 7th time step. This is the moment that the reflected wave arrives at the node. After this moment the stress begins to diminish due to the fact of superposition of the incident and reflected stresses. In the same Figure the results by using the multi-zonal BEM presented in Chapter 5 are presented.

6.8.1.2 Circular Cross Section Cavity in a Half Space Loaded by an Explosion Point P Wave

The same as previously circular cross section cavity of radius 2.50 m, with the same material characteristics as above, is examined under the loading of an explosive stress wave generated by an underground explosion at a point. The co-ordinates of the explosion point to a general co-ordinate system with origin at the centre of the cavity are (0.0, -8.0) m. The cubic root formula of Chapter 4, section 4.2.1, with decay in time, is used here. The equivalent charge weight of a GP 1000 lb bomb (U.S.Army, 1984) which is 188.7 kg of explosive weight is used here. The attenuation coefficient is taken n=2.75, and the coupling factor f=1.0. The total time interval examined is 0.015 sec separated into 20 time steps.

The change of the tangential hoop stresses versus the time steps at nodes 18,20,22 and 30 by using the laws of reflection of harmonic waves, is demonstrated in Figure (6.20). The x for the same nodes are depicted in Figure (6.21).

Furthermore, in Figure (6.22) the respective results are depicted for the case of discretization of the free surface of the ground. The x and y displacements for this case are illustrated in Figures (6.23) and (6.24). The two methods seem to give results in a good agreement between them.

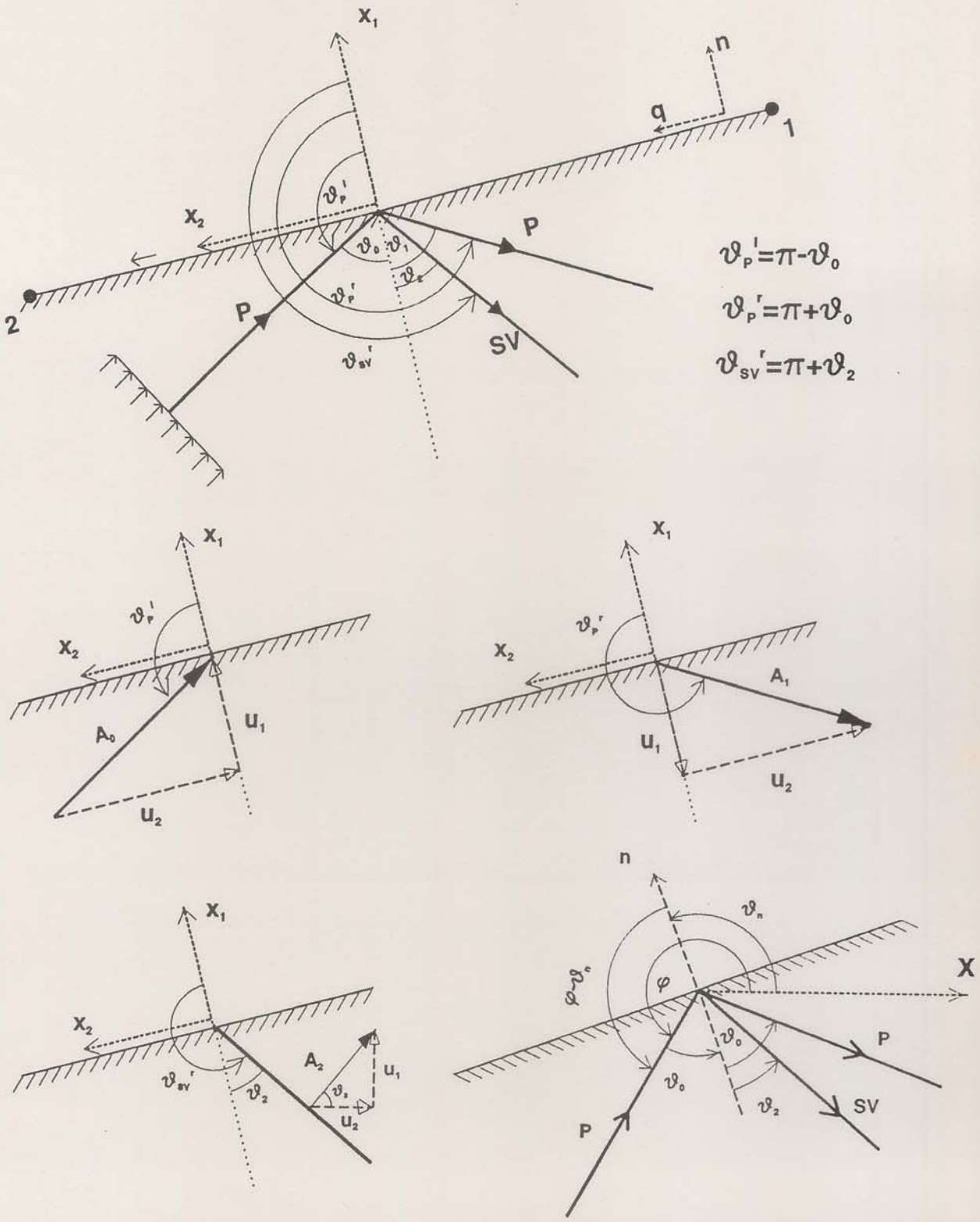


Figure 6.1 Reflection of P wave

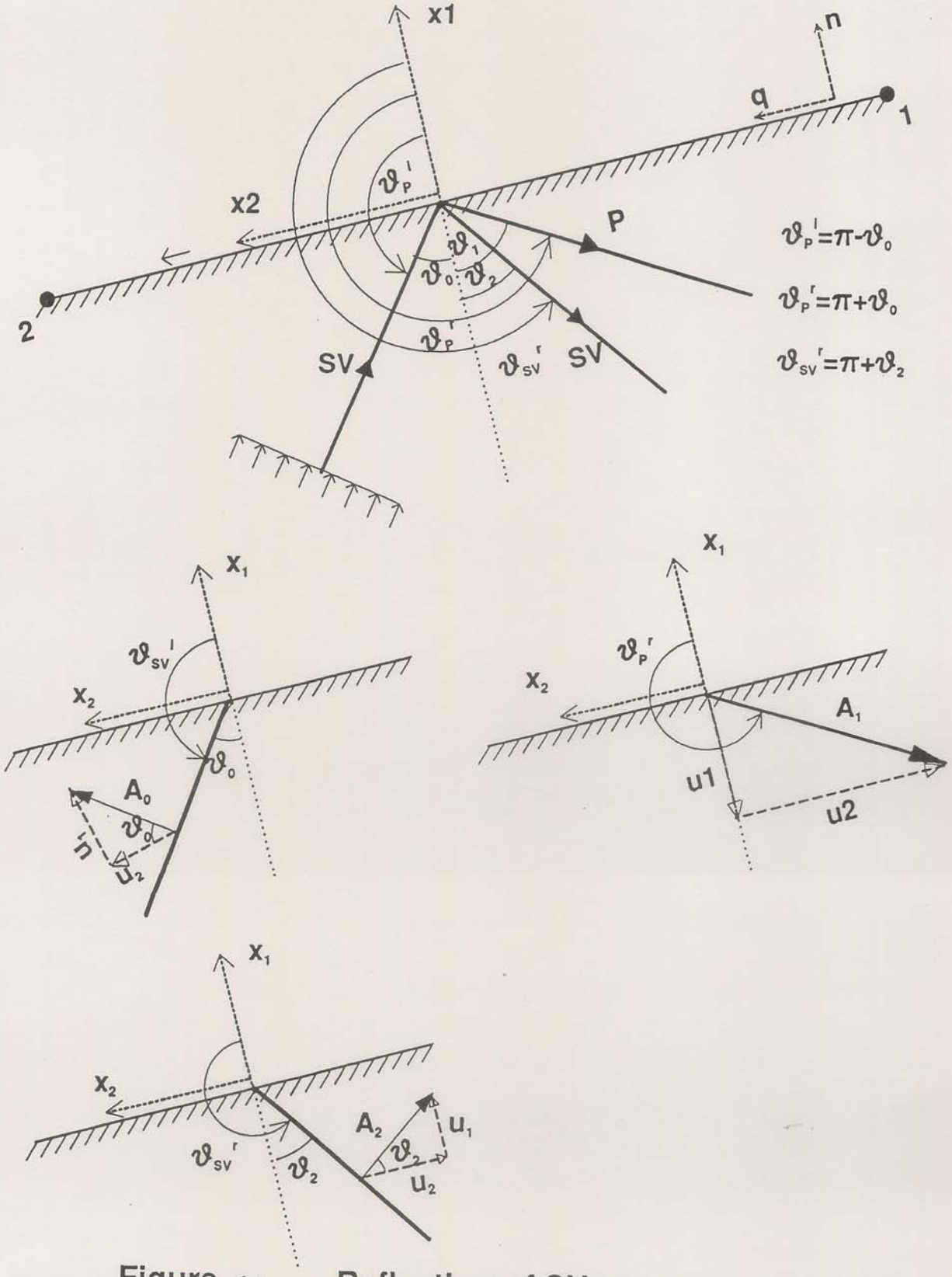


Figure 6.2 Reflection of SV wave

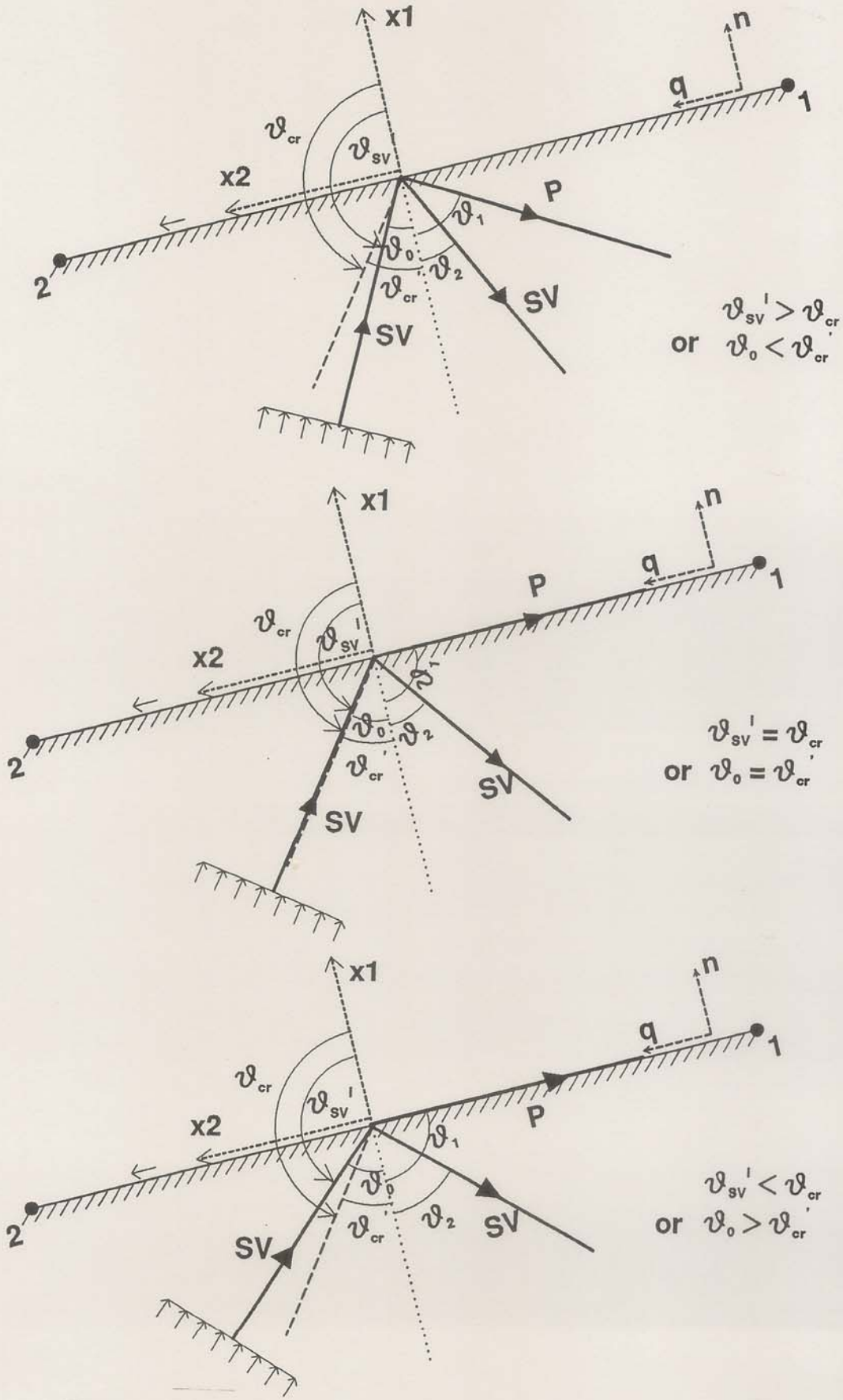
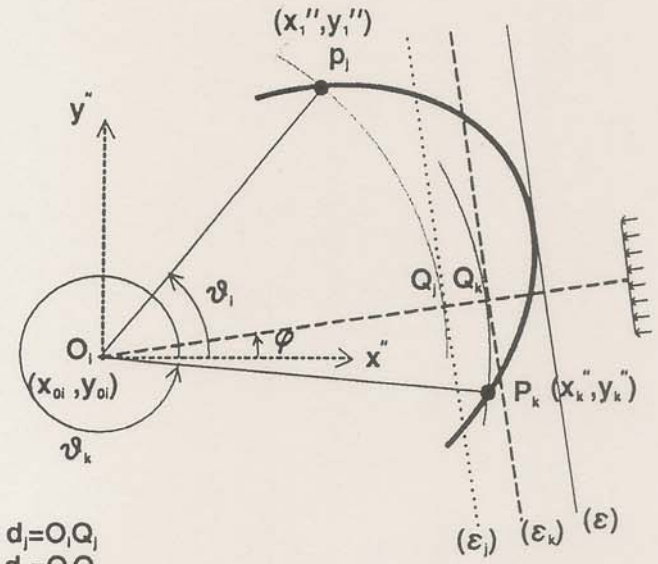


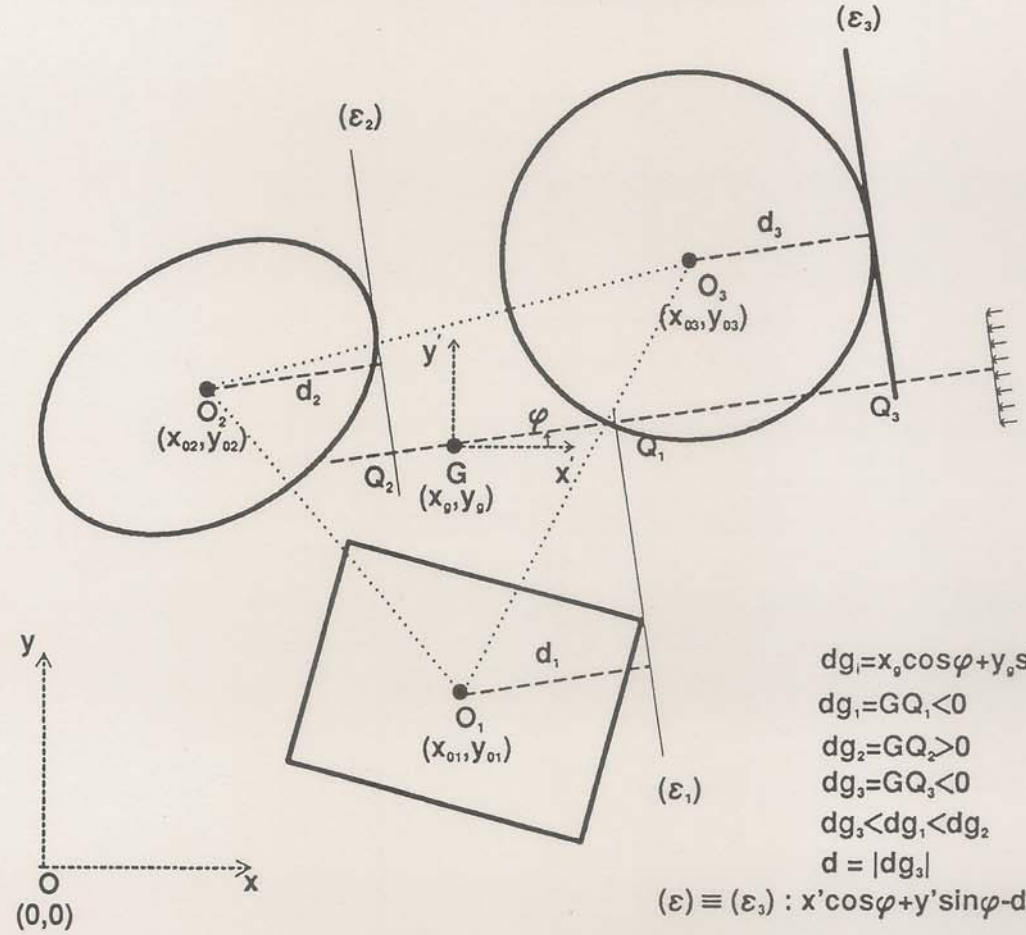
Figure 6.3 Reflection of SV wave



$d_j=O_iQ_j$
$d_k=O_iQ_k$

(ε_k) : $x''\cos\varphi+y''\sin\varphi-d_k=0$

Figure 6.4 Equation of wavefront line, for one boundary



$dg_i=x_g\cos\varphi+y_g\sin\varphi-d_i$

$dg_1=GQ_1<0$

$dg_2=GQ_2>0$

$dg_3=GQ_3<0$

$dg_3<dg_1<dg_2$

$d=|dg_3|$

$(\varepsilon)\equiv(\varepsilon_3)$: $x'\cos\varphi+y'\sin\varphi-d=0$

Figure 6.5 Equation of wavefront line, for many boundaries

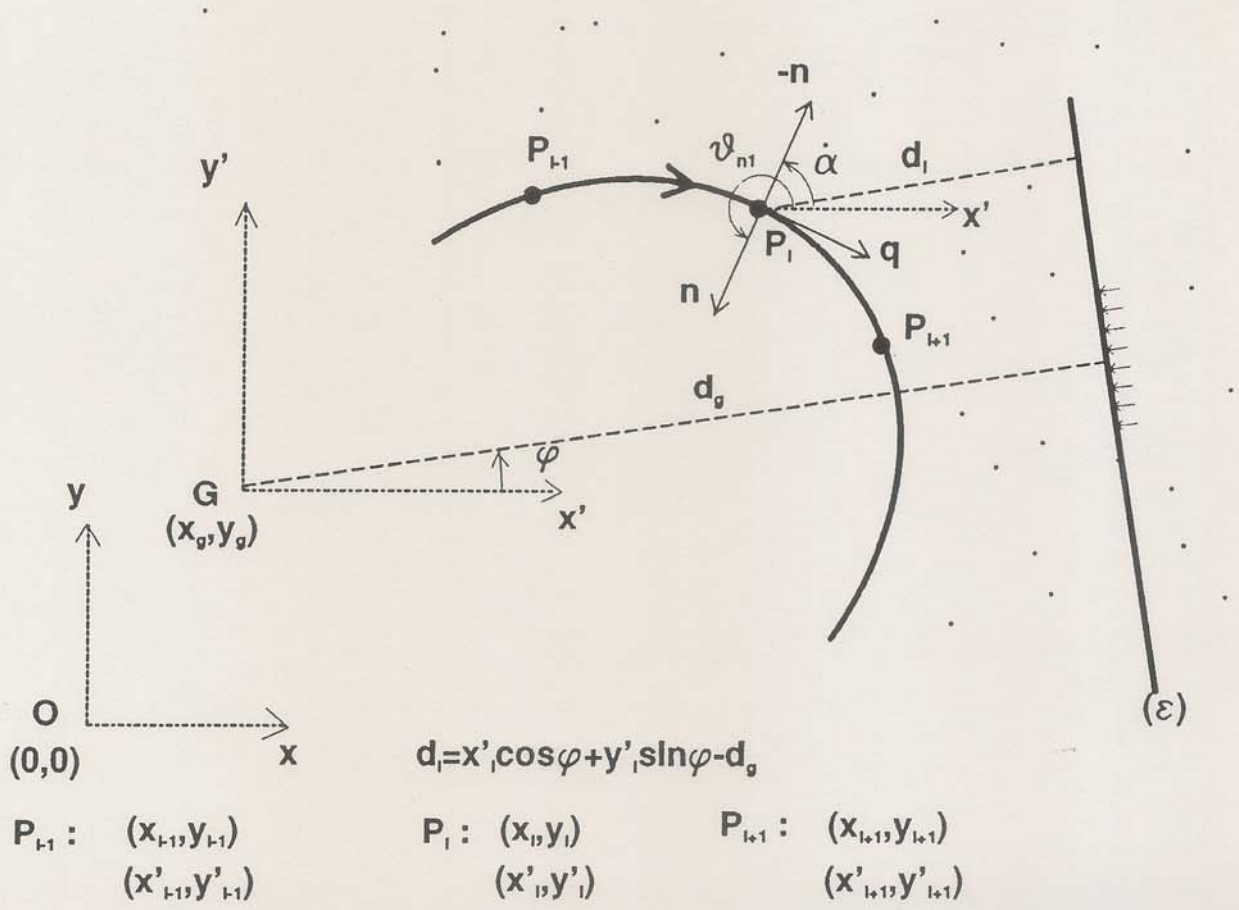
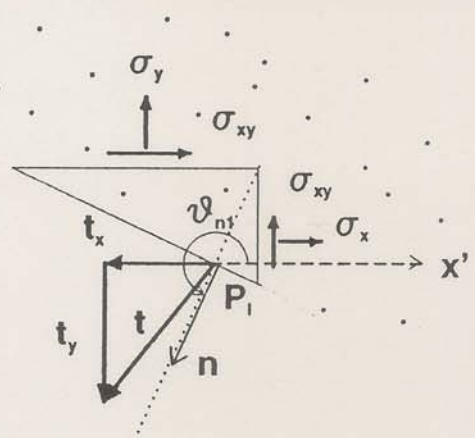
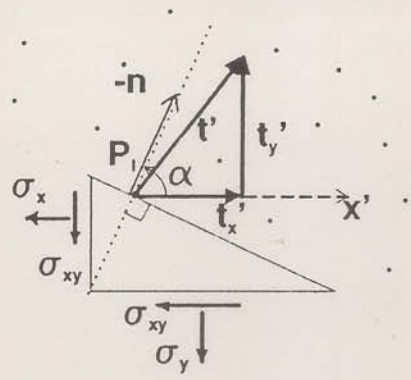


Figure 6.6 Distance of a node from the wavefront line and unit normal and tangent vectors

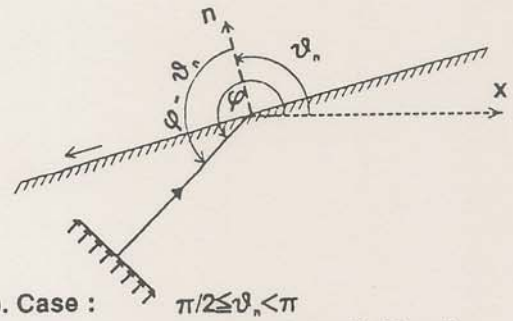


Insident tractions on a tangential plane with outward unit normal vector



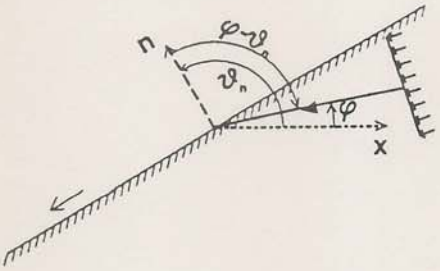
Tractions opposite to the insident tractions

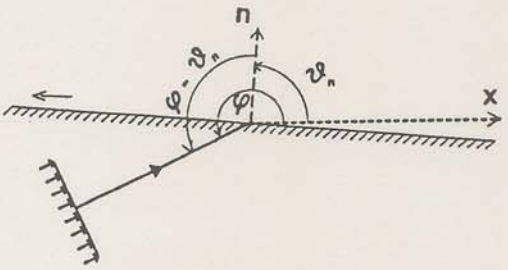
Figure 6.6.a Incident tractions at a node



(b). Case : $\pi/2 \leq \vartheta_n < \pi$

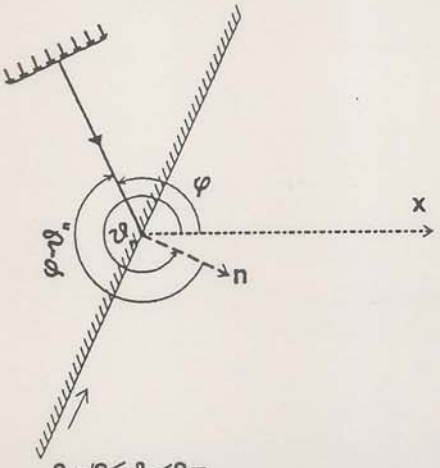
reflection when $\pi/2 < \varphi - \vartheta_n < 2\pi - \vartheta_n$ or when $-\pi/2 < \varphi - \vartheta_n < 0$





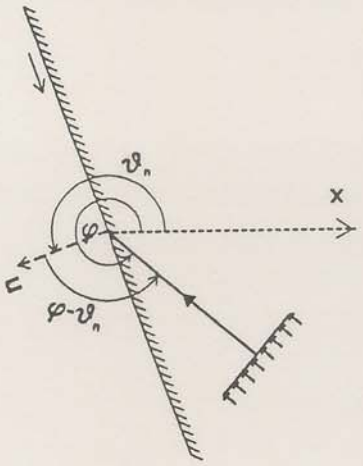
(a). Case : $0 \leq \vartheta_n < \pi/2$

reflection when $\pi/2 < \varphi - \vartheta_n < 3\pi/2$



(d). Case : $3\pi/2 \leq \vartheta_n < 2\pi$

reflection when $-3\pi/2 < \varphi - \vartheta_n < -\pi/2$



(c). Case : $\pi \leq \vartheta_n < 3\pi/2$

reflection when $\pi/2 < \varphi - \vartheta_n < 2\pi - \vartheta_n$ or when $-\pi/2 < \varphi - \vartheta_n < 0$

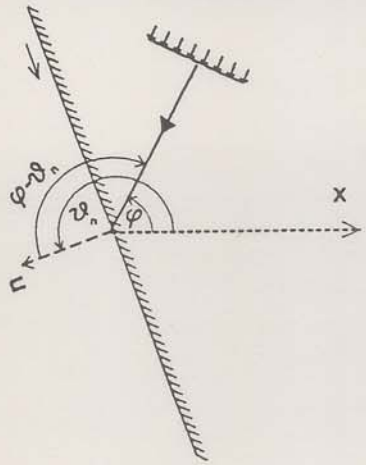
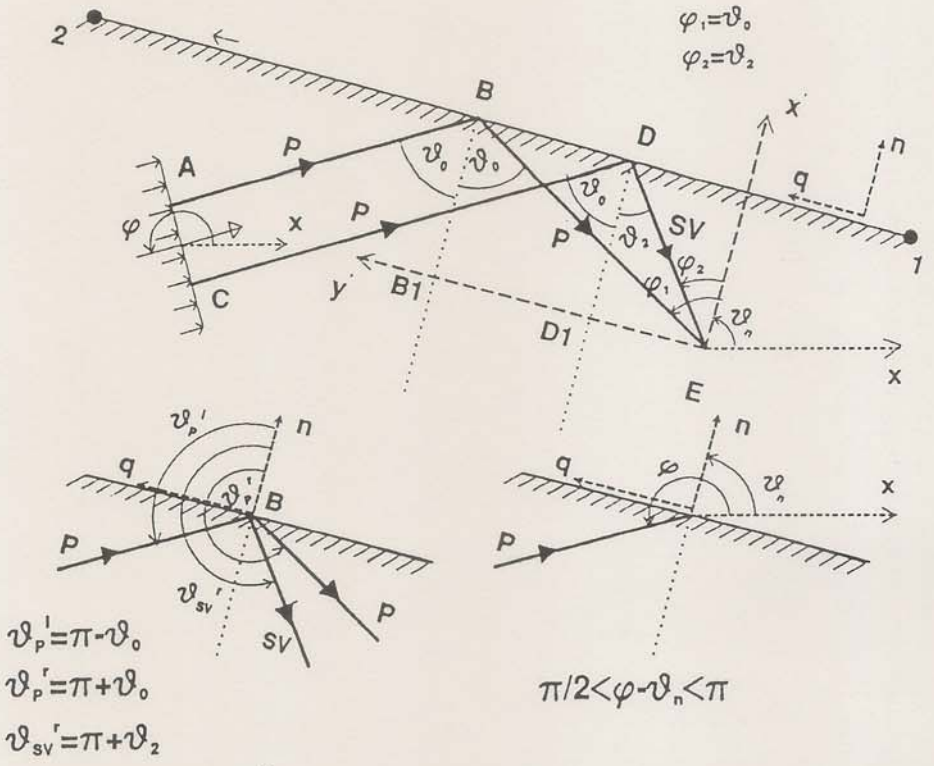


Figure 6.7 Reflection conditions of P wave

Case : (i)



Case : (ii)

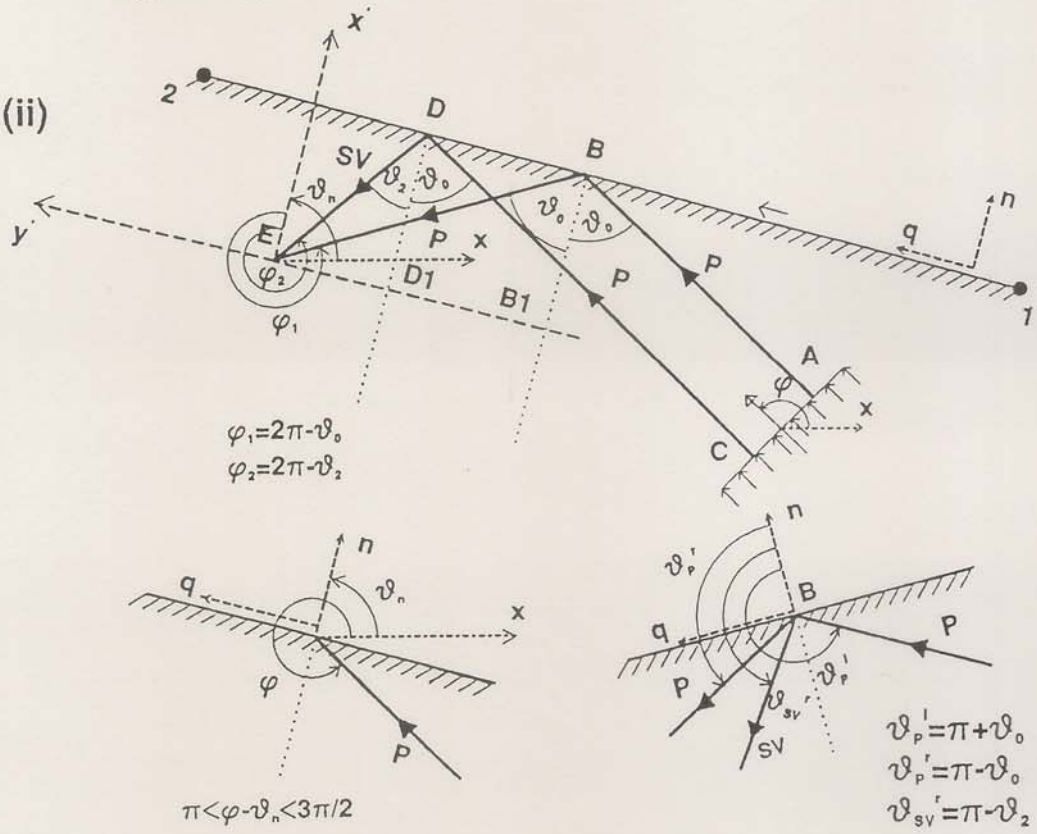
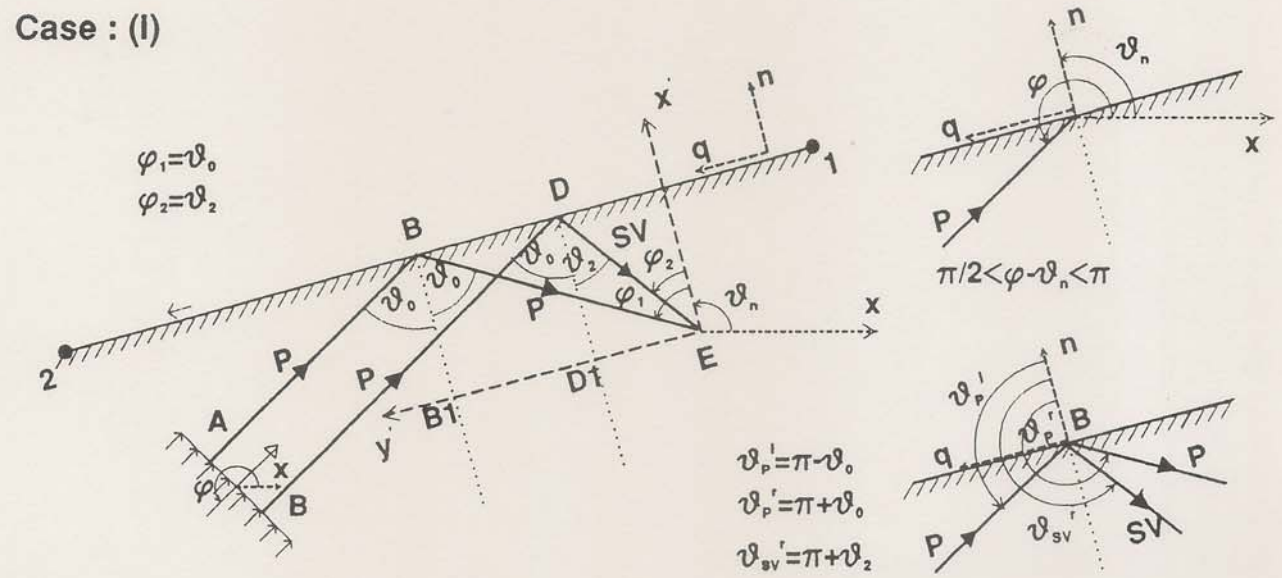


Figure 6.8 Reflection of P plane wave, case $0\leq\vartheta_n<\pi/2$

Case : (I)



Case : (II)

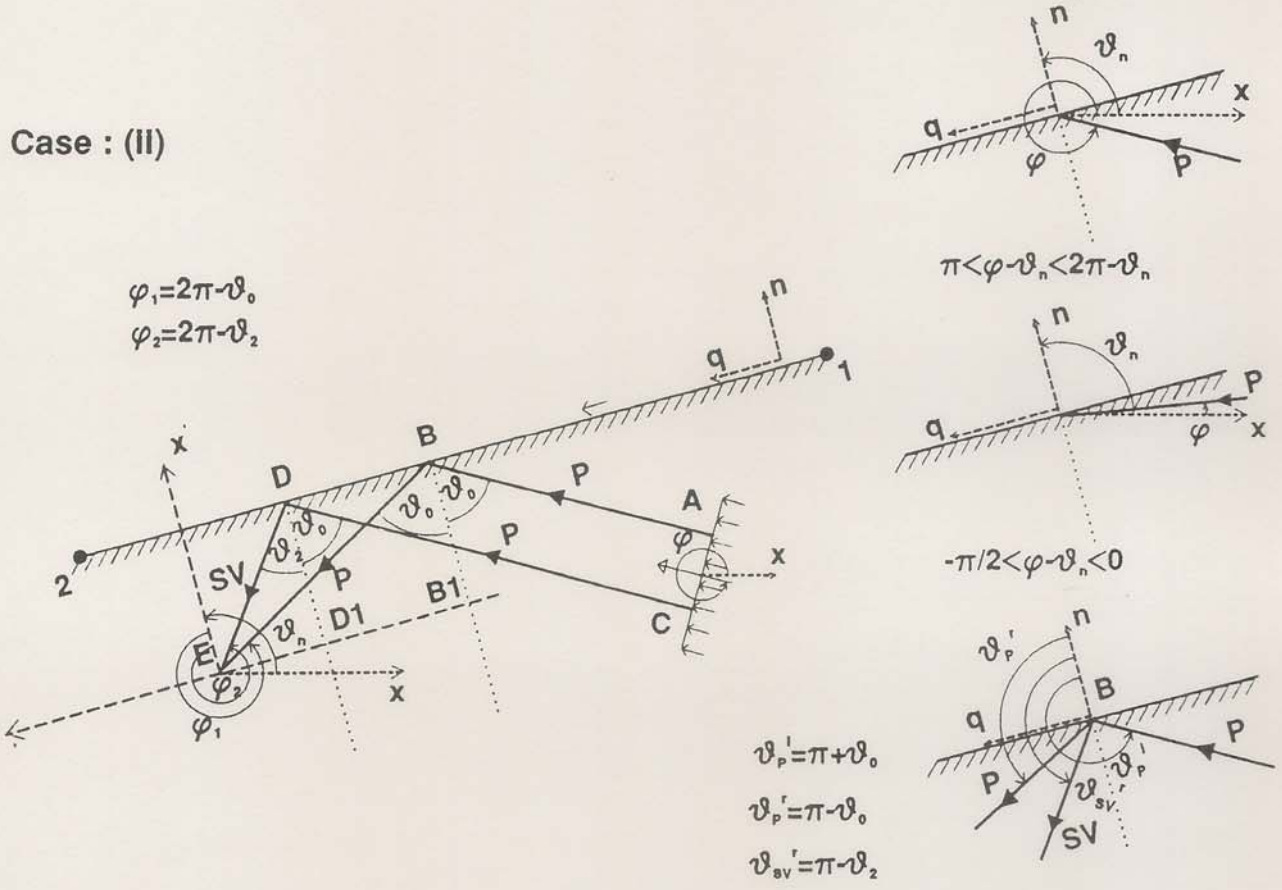


Figure 6.9 Reflection of P plane wave, case $\pi/2\leq\vartheta_n<\pi$

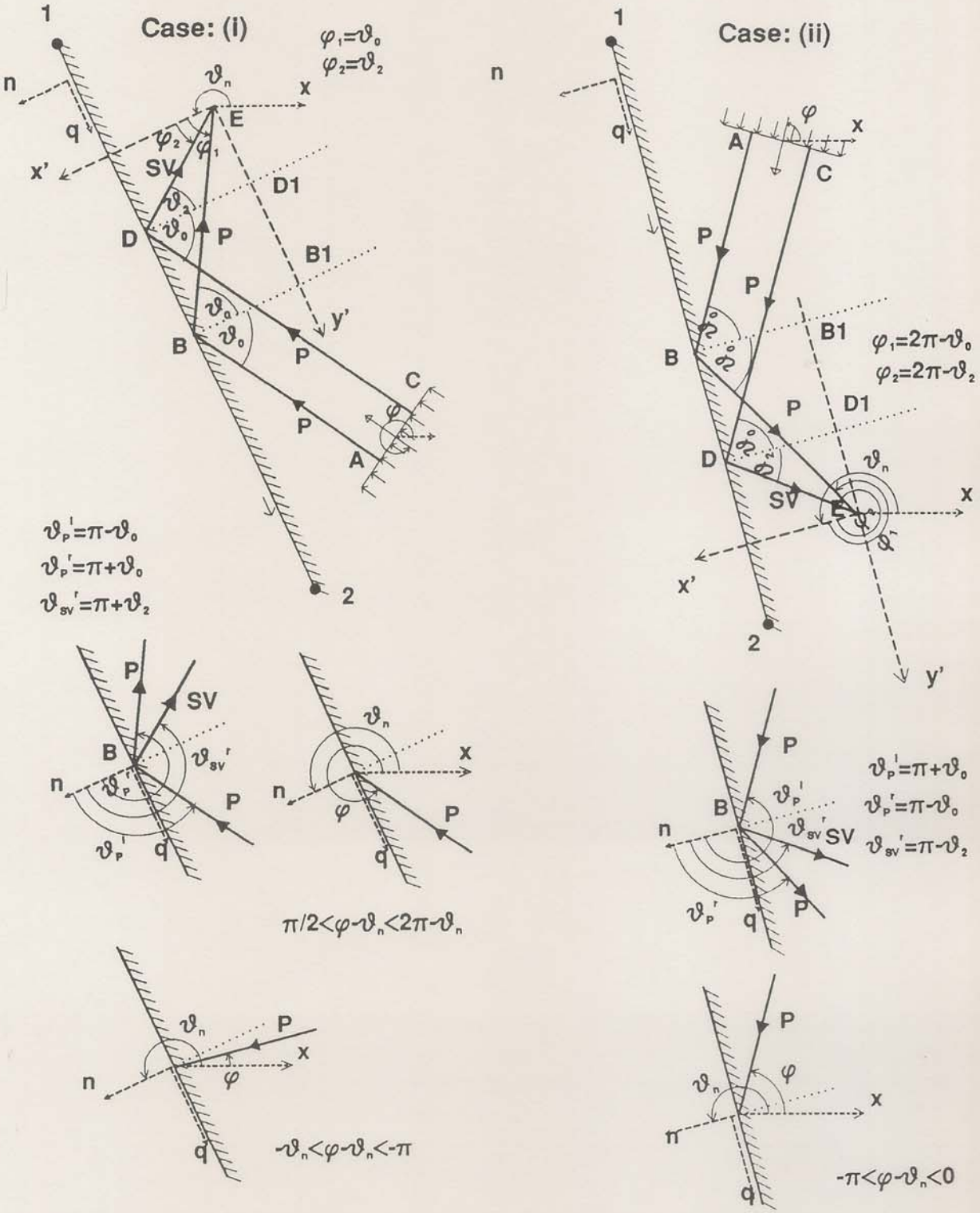


Figure 6.10 Reflection of P plane wave, case $\pi \leq \vartheta_n < 3\pi/2$

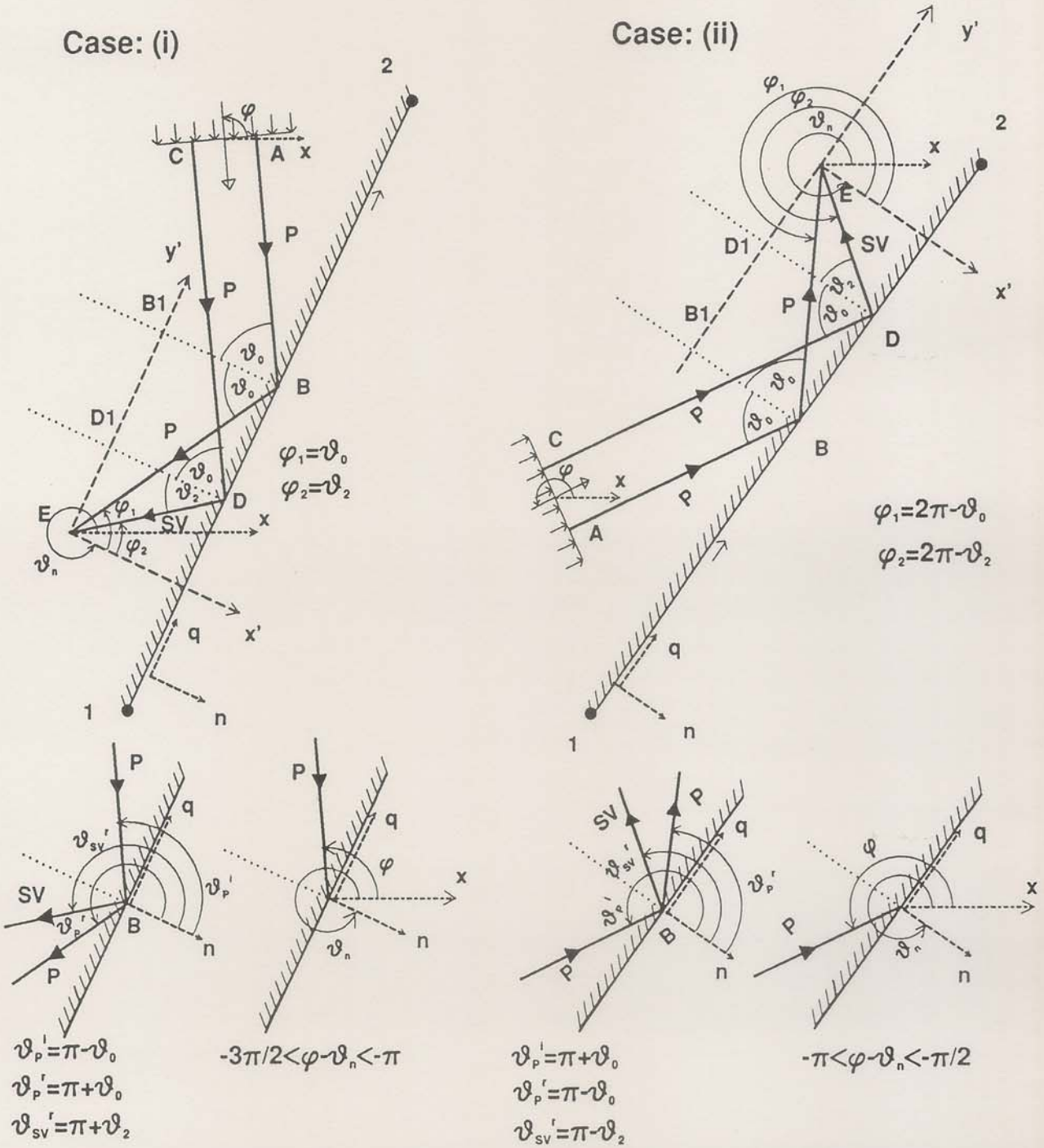


Figure 6.11 Reflection of P plane wave, case $3\pi/2\leq\vartheta_n<2\pi$

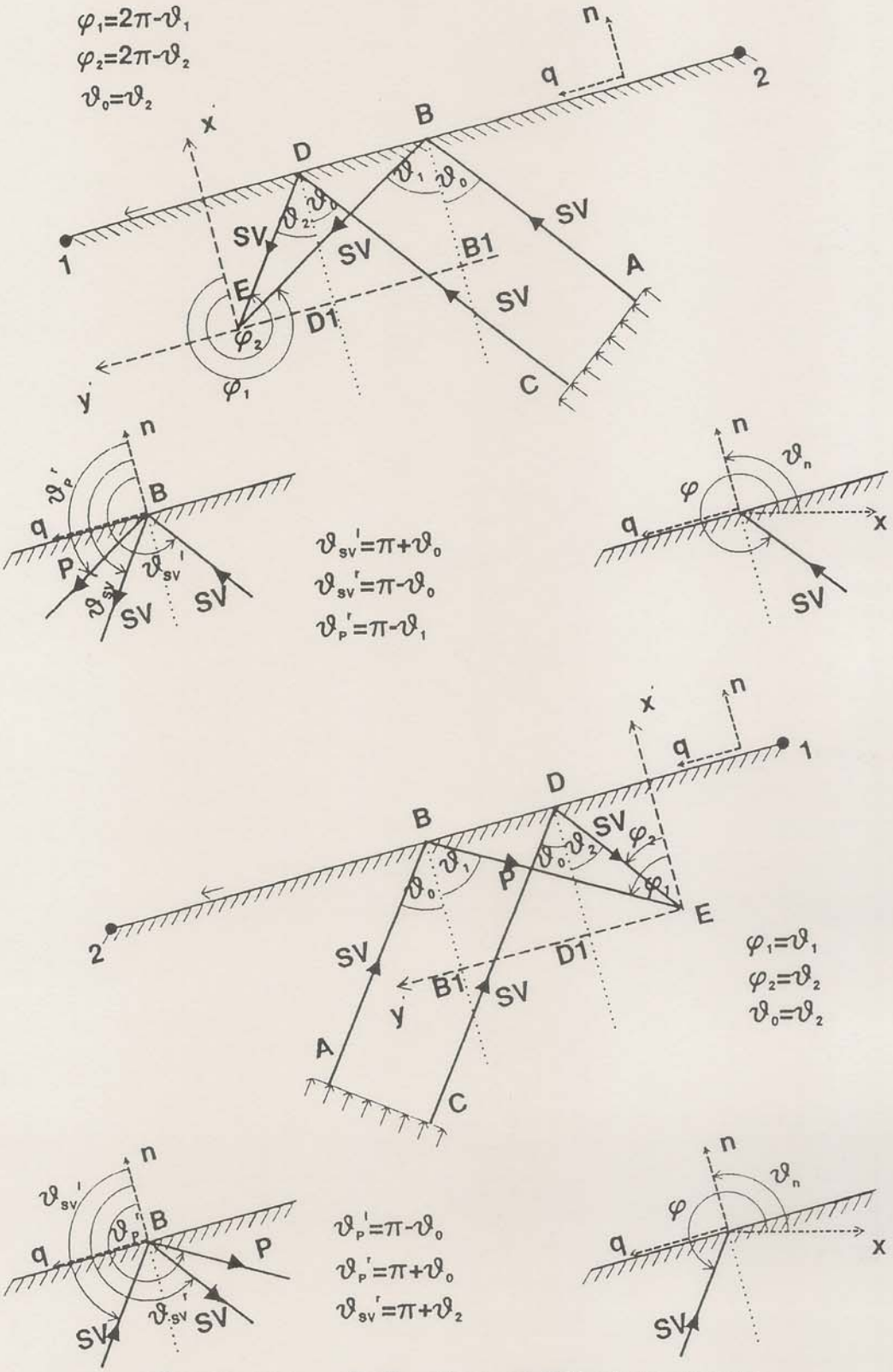


Figure 6.12 Reflection of SV plane wave

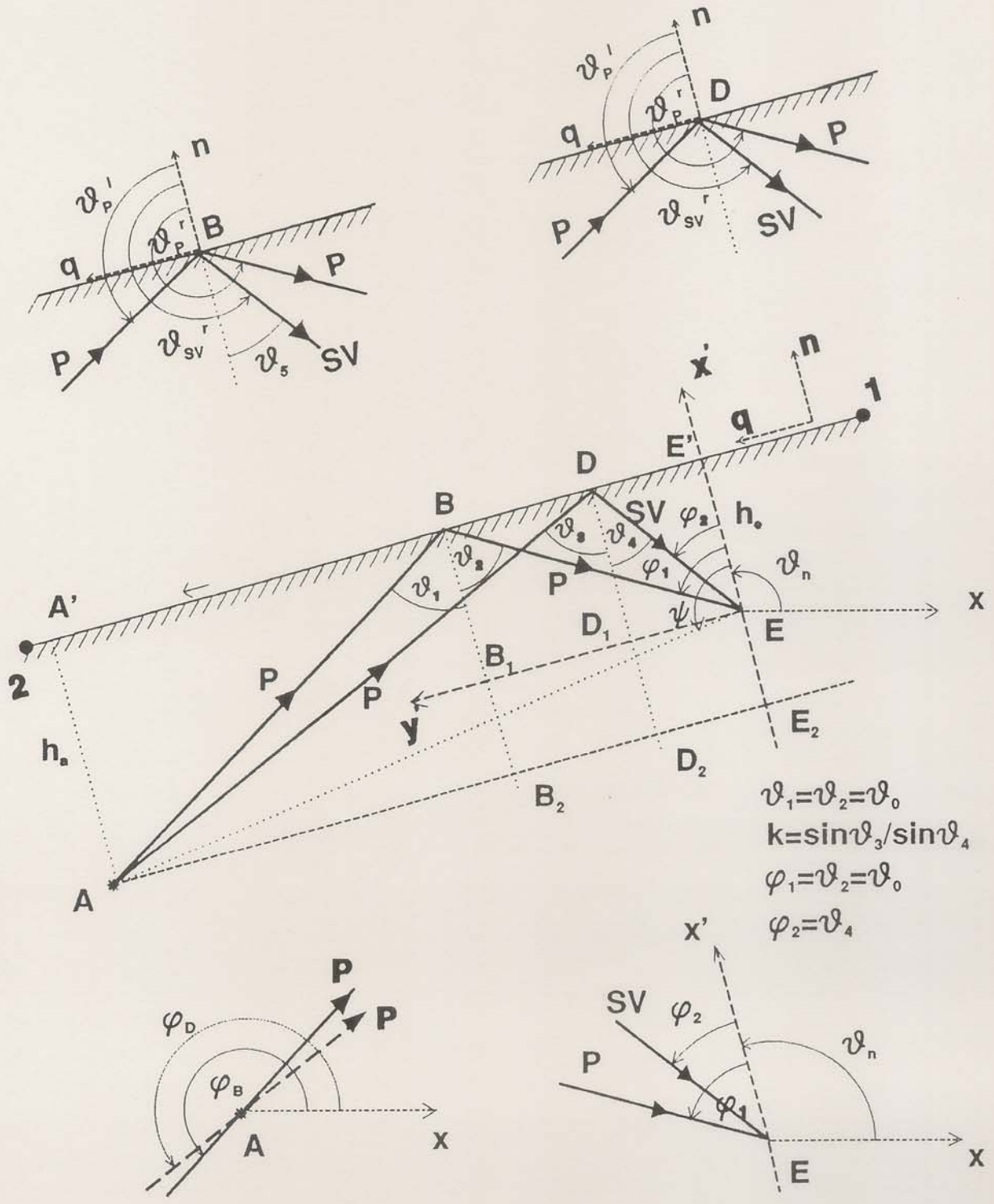


Figure 6.13 Reflection of explosive point P wave

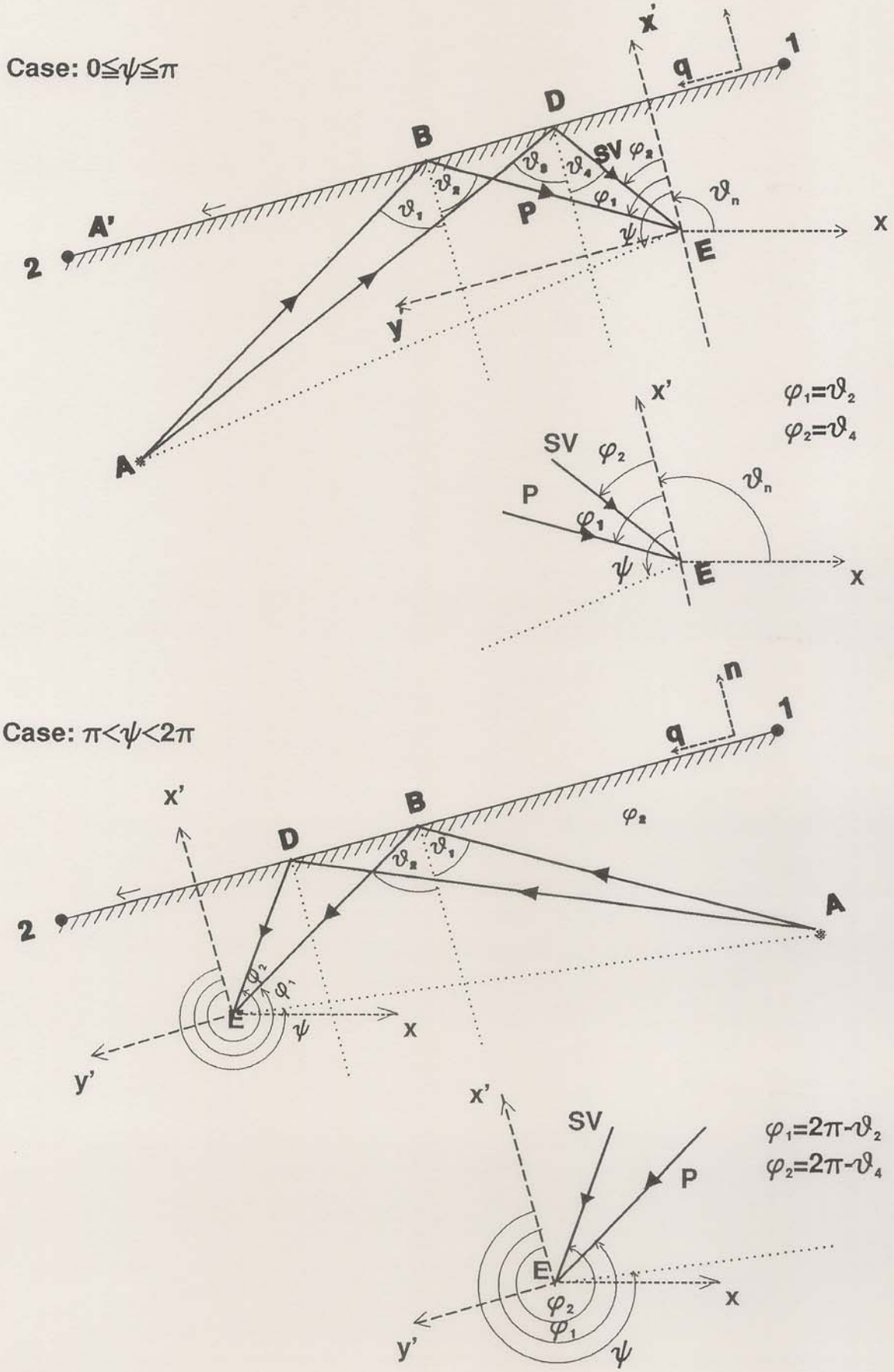


Figure 6.14 Reflection of explosive P or SV wave

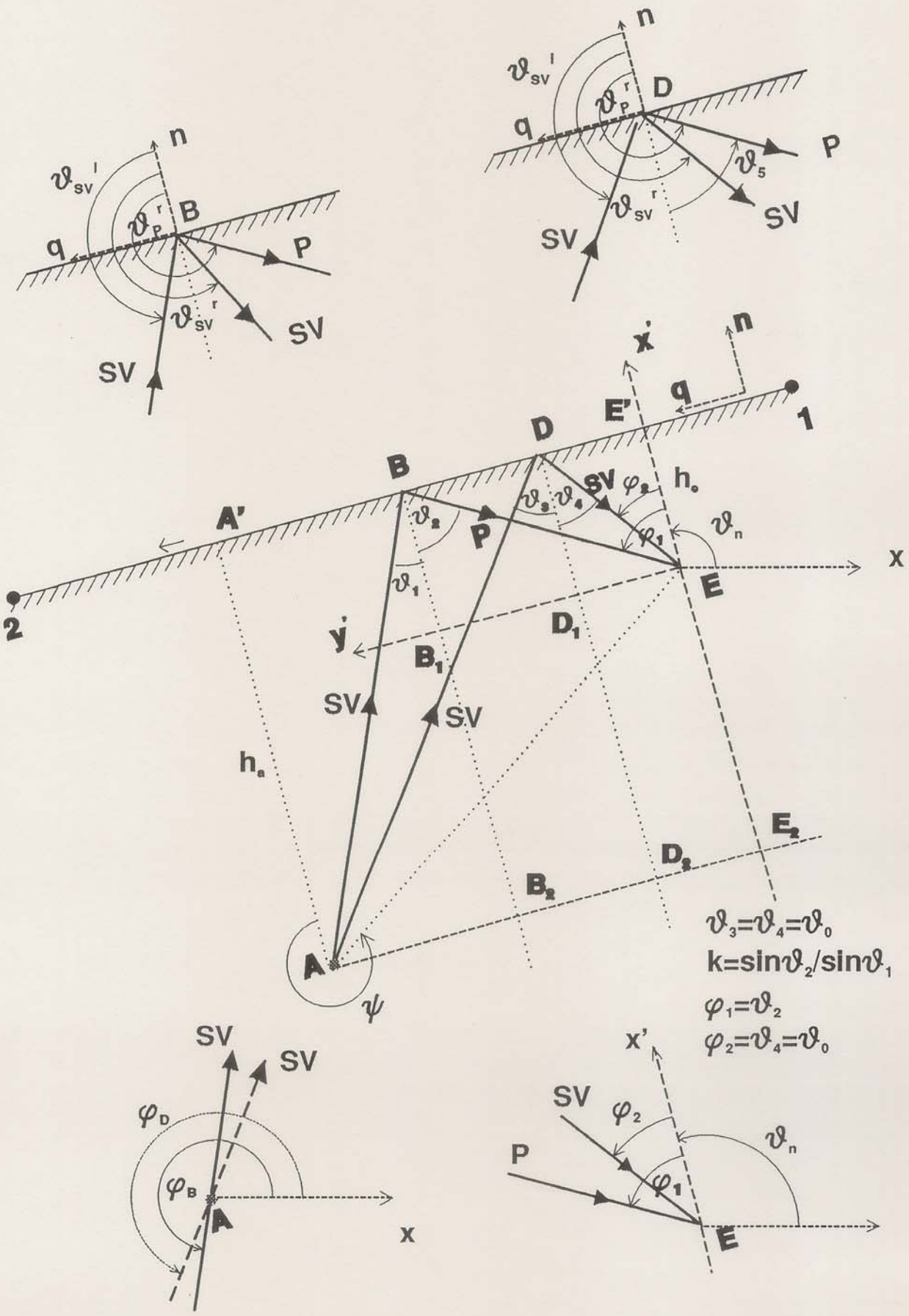


Figure 6.15 Reflection of explosive point SV wave

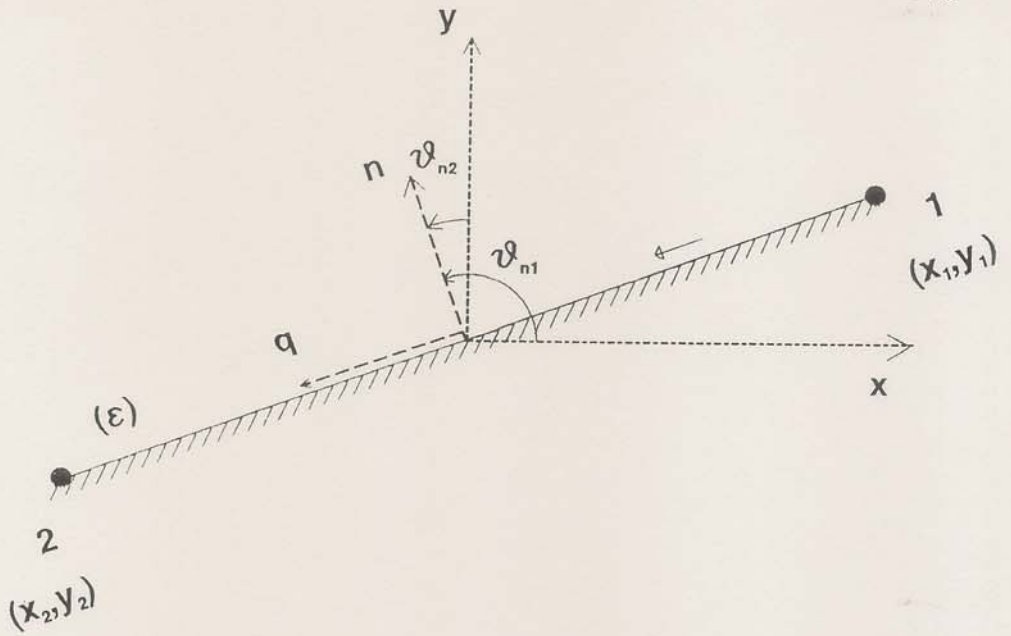
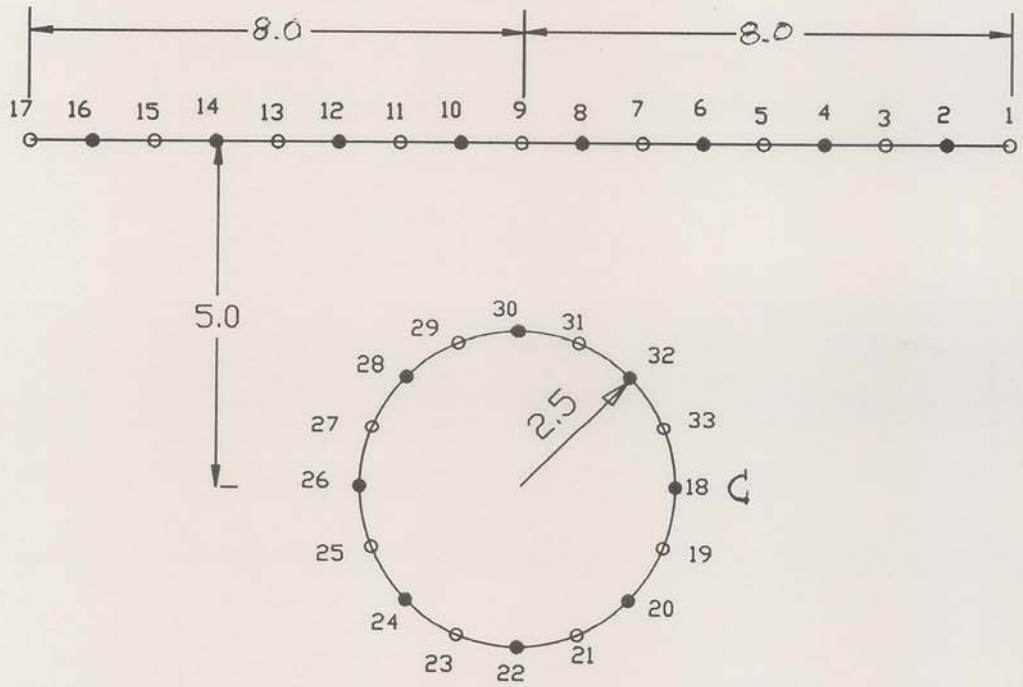
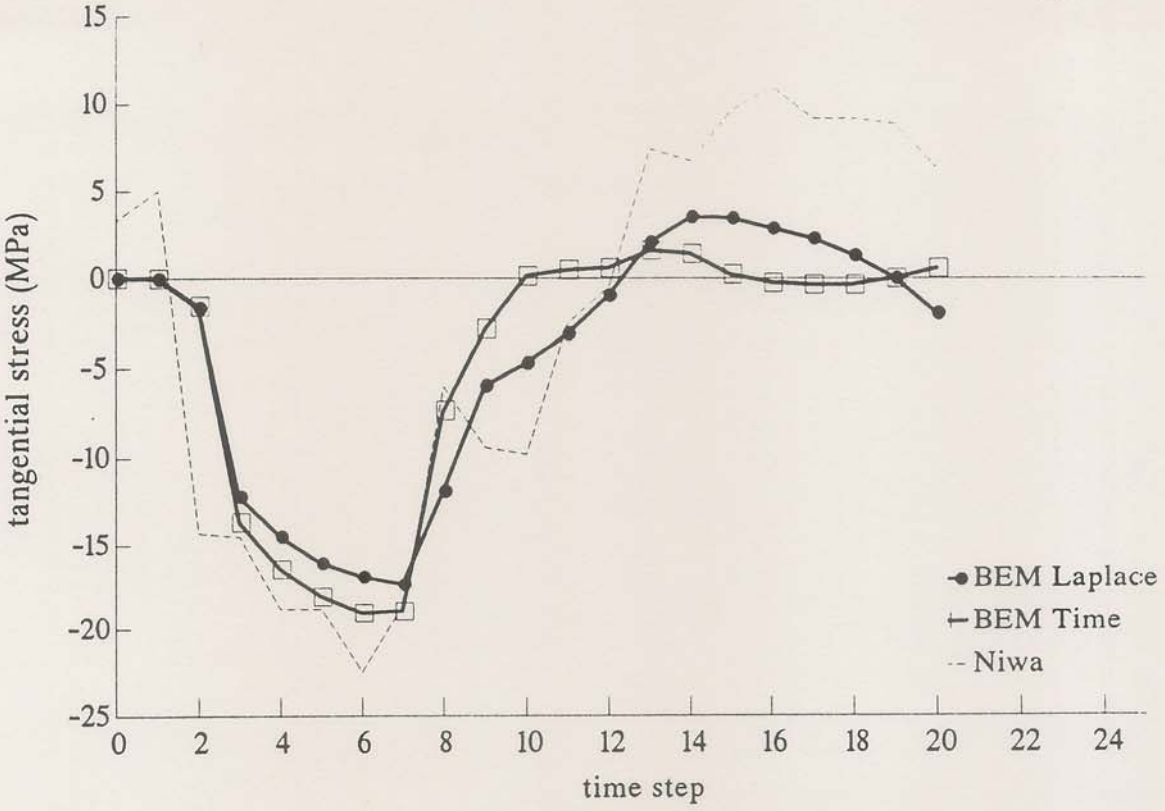


Figure 6.16 Line of ground surface



Figur6.17 Geometry details and discretization of the boundaries



Figur 6.18 Tangential stresses at point C, reflection by laws of harmonic waves

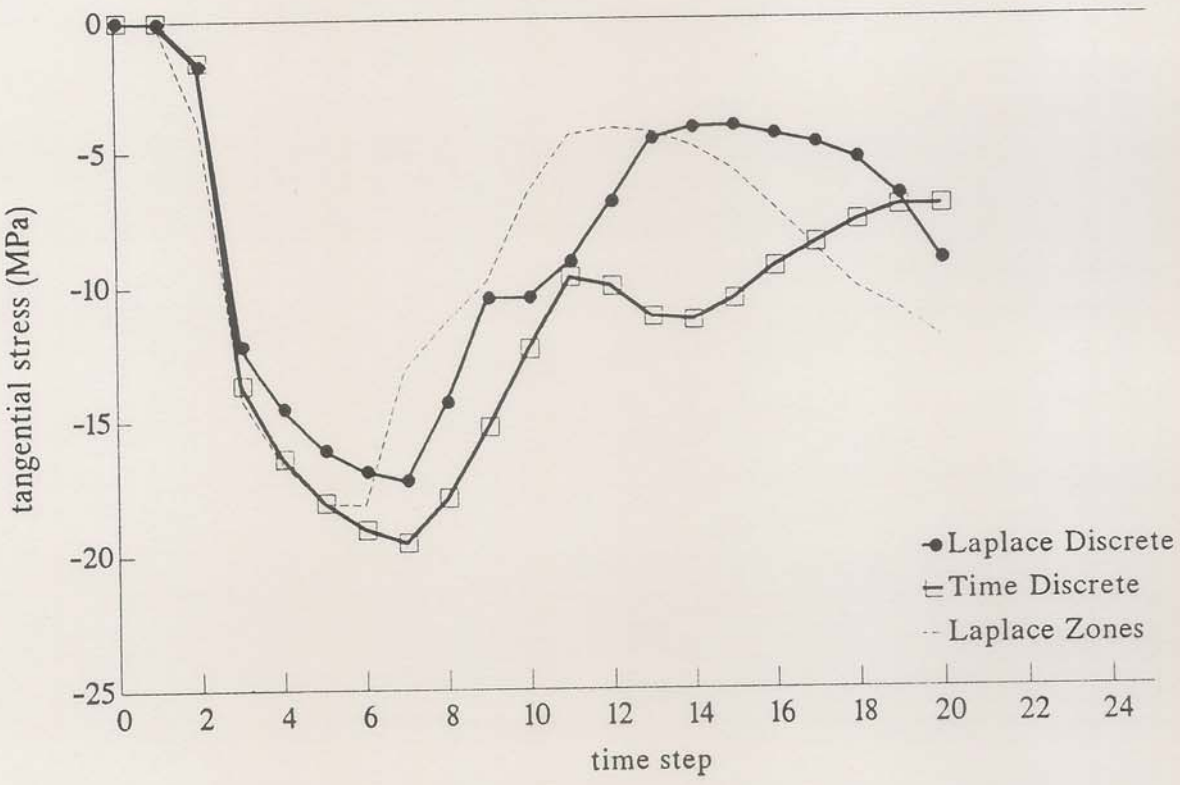


Figure 6.19 Tangential stresses at point C, by discretizing the ground surface

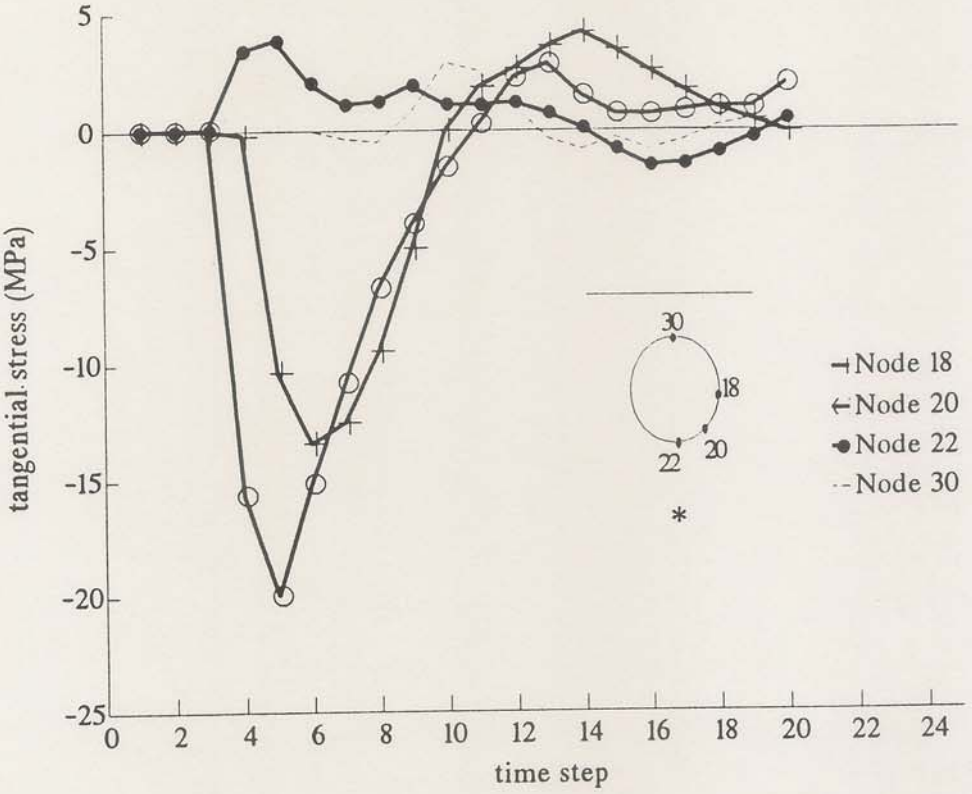


Figure 6.20 Tangential stresses, reflection by laws of harmonic waves

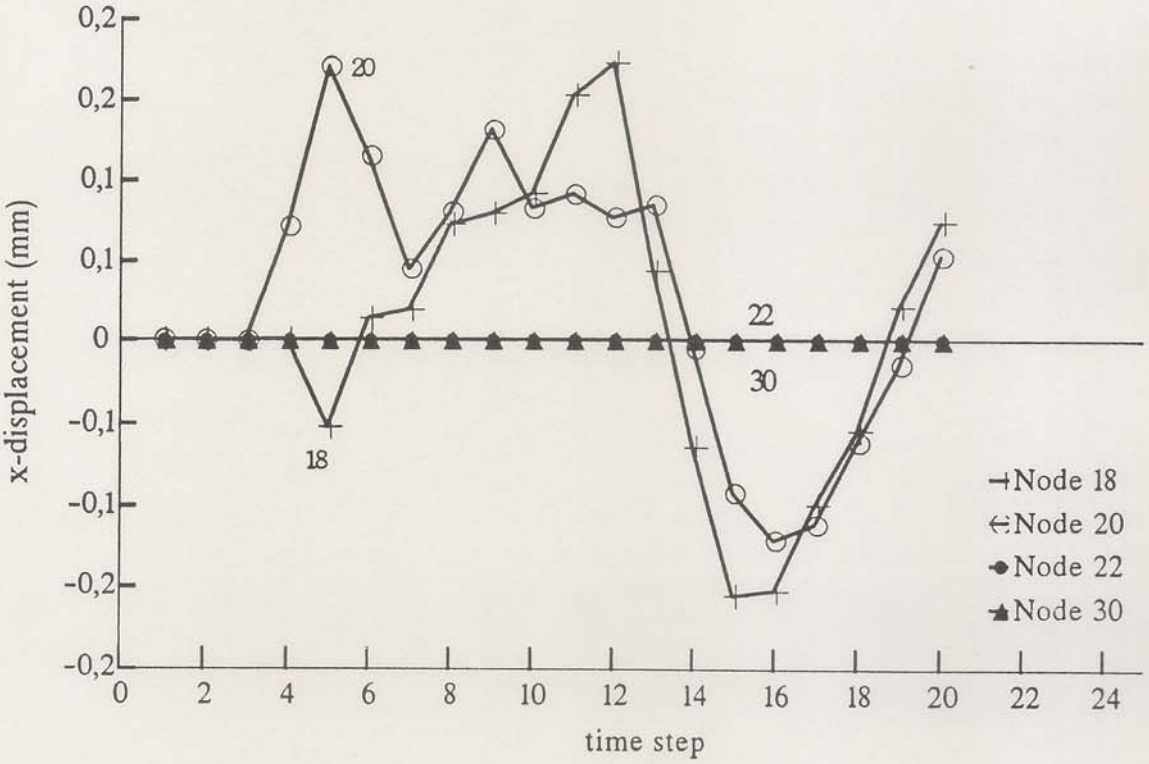


Figure 6.21 x-displacements, reflection by laws of harmonic waves

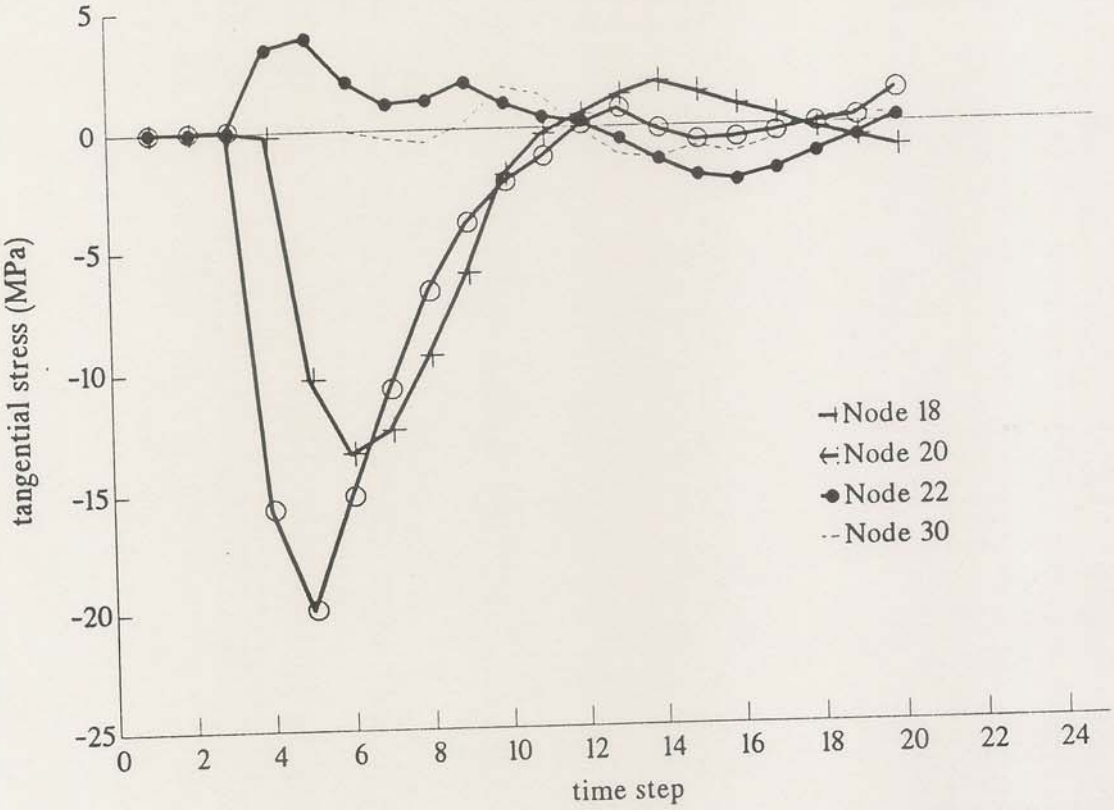


Figure 6.22 Tangential stresses, by discretizing the ground surface

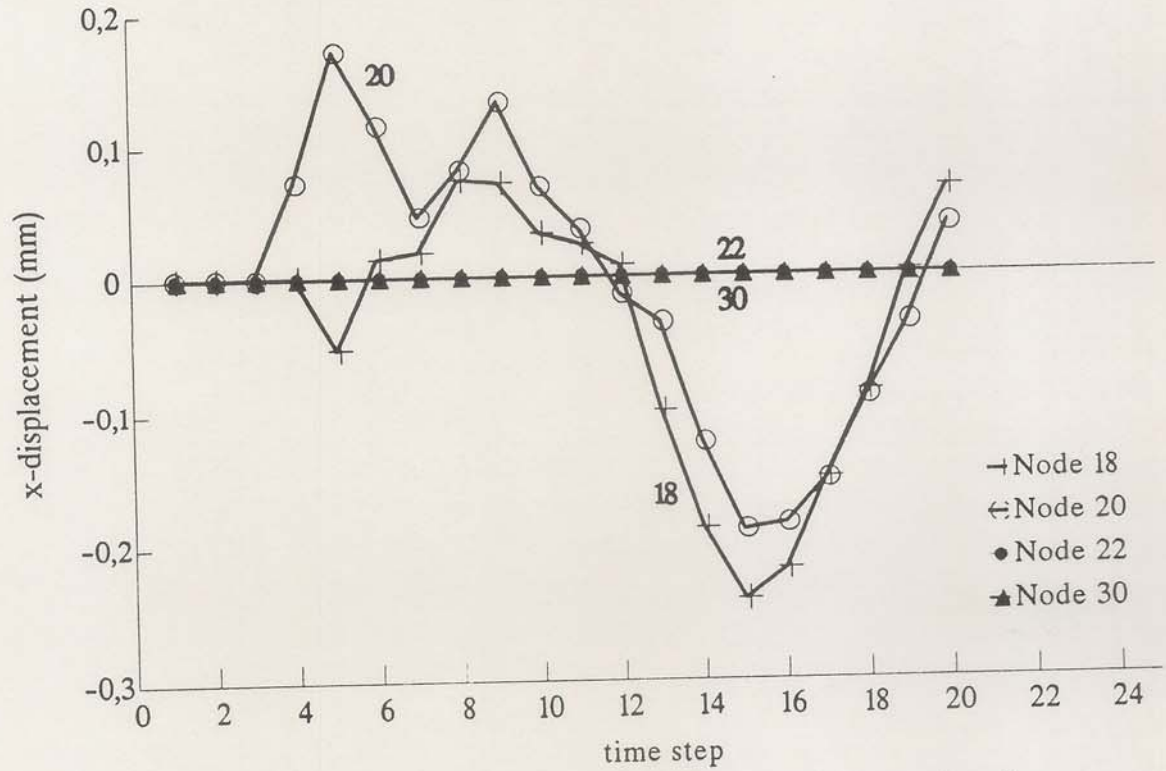


Figure 6.23 x-displacements , by discretizing the ground surface

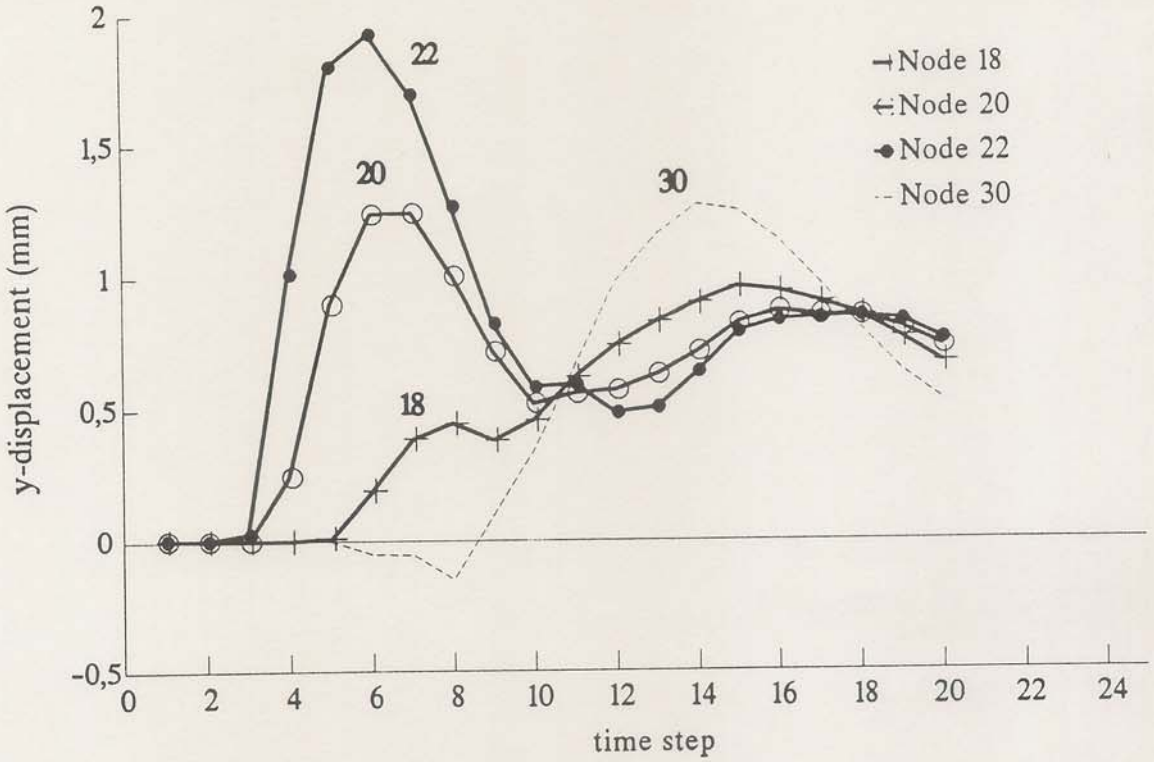


Figure 6.24 y-displacements, by discretizing the ground surface

CHAPTER 7

DEVELOPMENT OF THE PACKAGE

7.1 INTRODUCTION

An integrated package of programs named "DYNAMICS" has been developed in the context of this work, addressing the problem of underground structures subjected to dynamic loading.

As was explained in the previous chapters, the direct BEM method is used and the solution is performed in the Laplace domain and time domain as well.

"DYNAMICS" was developed originally in a UNIX environment, by using the FORTRAN 77 programming language, SUN workstation release. For the creation of the relevant graphs the UNIRAS graphical package was used.

Taking into consideration that the PC environment appears to be more popular, "DYNAMICS" was transferred to an 80486 powerful PC machine. The computer language used in this new release is the Microsoft FORTRAN version 5.1.

As a first attempt, the MSDOS 5.0 operating system was used that imposed a limitation of 640KB in the available memory for running applications. In order to use "DYNAMICS" to solve large problems, there was a requirement for more available memory. In an attempt to overcome this problem, the Microsoft Windows 3.1 operating environment was used and the package of programs was re-compiled for running under MS Windows. As a result, the memory limitations are restricted only on the overall available memory of the available hardware.

The development of "DYNAMICS", was a difficult and time consuming task and the detection of some errors involved in the numerical integration subroutines introduced by Manolis and Beskos (1988), was one of the primary tasks. In connection with this, the help provided by Manolis (1991) in confirming the identification of the errors, was very usefull. In order to facilitate the introduction of the data, a preprocessing phase for the data input and the automatic mesh generation was necessary. Significant effort was spent in developing a free format reader, capable of reading and interpreting the input data file. The mesh generation facilities were introduced next maining to facilitate the parametric analysis studies. The main BEM program in the processing phase was extensively modified again and again until it reached a configuration capable of dealing with a variety of problems.

In the following sectors some details on the capabilities of "DYNAMICS" and a general description of the package are given. Also a description of the most important subroutines used in the different phases of the program are presented. Specific instructions on the way

of running "DYNAMICS" and the preparation of the input data file, are given in Appendix F which can in effect be adopted as a users manual for a person wishing to run "DYNAMICS".

7.2 ANALYSIS OF THE PROGRAM "DYNAMICS"

7.2.1 Capabilities of the Program

The program "DYNAMICS" is capable of solving problems associated with two dimensional idealization of unlined or lined underground openings in infinite and semi-infinite space. The geometry of the structure and the orientation of the free surface of the ground is arbitrary and the materials used are linear elastic or linear viscoelastic. Furthermore, zones of different materials can also be included.

The type of analysis employed, is two dimensional plane strain or plane stress elastodynamic analysis. The properties of the ground are assumed constant inside a specified region and the boundary elements used are quadratic.

The maximum number of elements of the mesh depends only on the available memory (RAM) of the system, for instance 8Mb RAM. The size of the arrays used are dynamically allocated during the running of the program and according to the size of the problem.

The package "DYNAMICS" consists of three main parts: the preprocessing, the processing and the postprocessing phase. These parts are independent programs that can be executed either through a main guide program or independently. The main guide program starts by issuing the command DYNAMICS and allows the user to select the desired phase from a menu that appears on the screen.

The flowchart of the program is illustrated in Figures (7.1), (7.2), (7.3), (7.4), (7.5),(7.6) and (7.7). The program can accept either SI units or Imperial units.

Special attention has been given in developing the program in modular form, to allow for the incorporation of new applications and the modification for a different graphical environment.

7.2.2 Preprocessing

The largest part of the package is the preprocessing phase. It consists of several subroutines that are described in section 7.3.

7.2.2.1 Data Input

For the input of the data, the user has to create a text file in ASCII form. The data are introduced in free format and although particular items must be introduced in the correct

order on a data module, they are not required to be positioned at specific column positions. The term "module" describes the data that are expected to be typed and saved in a disk file. Data items are indicated below by mnemonic names ie. names which suggest the data item required by the program. Only the four first letters of these names are significant with the rest of them being optional. A list of these names with the corresponding descriptions is presented in Appendix F. Integer data items must not contain a decimal point but real data items may optional do so. Also real data items may be entered in a FORTRAN exponent format if desired ie. 0.025 may be entered as 2.5E-2. Individual data items must not contain spaces and are separated from each other by at least one space.
Comments may be included in the input file by including in the first column of the line the asterisk "*". Any line that has the character "*" in column 1 is ignored by the program. The program reads data from only the first 80 columns of each line.

7.2.2.2 Mesh Generation

The generation of the mesh usually is one of the most difficult and error prone tasks both in FEM and BEM programs. For this reason, in this package, a considerable effort has been given for including automatic mesh generation capabilities.

For simple shapes the mesh is generated by giving code numbers as explained in Appendix F is e.g. 1 for an ellipsis, 2 for square, 3 for a circle and 4 for a straight line. For more complicated shapes, the mesh is generated by combining simple elliptical or straight line segments. From the total number of elements the program automatically generates the co-ordinates of the nodes, by using shape interpolation functions.
The whole process is schematically depicted for a square opening in Figure (7.8) From the length of the side and the rotation of the square with respect to the x axis, the co-ordinates of the four nodes and the middle points of the sides are found. The nodes of the quadratic elements on the sides of the square are determined from interpolation and are numbered in anticlockwise direction. Moreover, a new numbering of the nodes is introduced, with the first node being the intersection of the first axis of the square with the side. The remaining nodes are numbered clockwise or anti-clockwise depending whether the boundary is exterior (cavity) or interior (solid).

In Figure (7.9) the relative position of an elliptical segment is examined for the case of the exterior problem, according to the (m,n) local co-ordinate system that the two axes of the ellipses form with the global (x,y) axes. In Figure (7.10) the case of the interior problem is examined. The coefficients g_c, g_e and g_f are used in the program for the calculation of the node co-ordinates.

In Figure (7.11) the transformation of the mesh from the local to a global system for an orthogonal opening is depicted. In Figure (7.12) the initially calculated nodes are renumbered according to the axis of the incident wave.

For the straight line segments the quadratic elements can be of equal or unequal length, as is indicated in Figure (7.13).

The unequal elements can be either progressively decreasing or increasing towards the edges of the line, with a specific ratio determined by the user. There is also capability for progressively increasing or decreasing size of elements towards the middle of the line. This type of elements are very useful for stress concentration studies e.g. corners of a rectangle.

To solve the problem of high stress concentrations at sharp corners, the corner nodes are duplicated, so that the joining at the corner elements becomes independent. Moreover, for linear segments of the mesh, the program is capable of introducing unequal size elements. The size of these elements is either progressively decreased or increased towards the edges of the line, with a specific ratio determined by the user. This technique of gradually decreasing size is illustrated in Figure (7.13) for a line containing five elements and a decreasing rate of 2. For rectangular shapes in specific, the number of elements per side is provided by the user, with a facility either to introduce a different ratio for both sides or to introduce a ratio for one side and allow the program to determine the optimum ratio for the other side. This optimisation is achieved by adjusting the length of the corner elements, both in the x and y direction, so that their corresponding length is equal. The stresses on the boundary are subsequently estimated by employing a numerical differentiation scheme. This technique allows the user to determine the stress distribution over any desirable length of the boundary close to an existing corner. An important detail of this scheme is that the improvement in the stress calculations is not achieved at the expense of an increased computational effort. This is possible because the execution time of the solution phase depends mainly on the number of nodes and elements that are involved in the calculations.

7.2.2.3 Calculation of Loads.

The imposed conditions on the boundary of the mesh are either applied tractions or applied displacements and "DYNAMICS" can handle both cases. In this work attention is focused on the applied tractions, which is the most common case in transient elastodynamics.

The loads can be either in Laplace domain or in time domain.

Furthermore, for the case of semi-infinite media, the solution can be accomplished either by discretization of the free surface of the ground, or by using the laws of reflection of harmonic waves. In the latter case, the loading is created by adding the incident and the actual reflected wave from the free surface of the ground. From the given boundary shapes, the prescribed angle of incidence of the plane wave or the depth of burial of the explosion source, the reflection conditions are investigated automatically and reflections are taken if it is needed. At the end of this phase the user obtains on the screen plots of the mesh and the

applied loading conditions before one may proceed to the next stage.

7.2.3 Processing

The processing is performed by separate programs for Laplace domain and time domain, because in the Laplace transform domain complex variables are used, while in the time domain double precision real variables are required. Furthermore, a separate program also performs the processing of the multi-zonal BEM.

In Laplace transform domain and for the numerical inversion of Laplace transform and the numerical integration in Laplace and time domain, the subroutines by Manolis and Beskos (1988) have been significantly modified to meet the program requirements.

The processing is performed by the BEM and the results for tractions and displacements are saved in an ASCII file for use in the postprocessing phase.

7.2.4 Postprocessing

In the postprocessing phase, the graphics results are presented on the screen or printer/plotter. The stresses and displacements can be drawn in Cartesian co-ordinate system, and/or on the periphery of the opening for every time step. Furthermore, drawing the distribution curves for more than one node, is a facility which is available within the program.

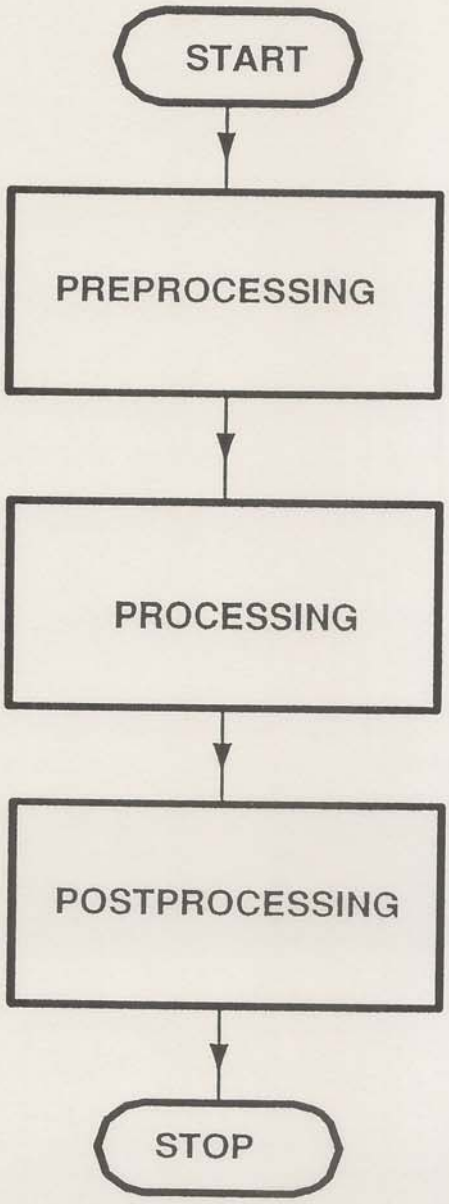


Figure 7.1 Three phases of the Laplace transform and time domain program

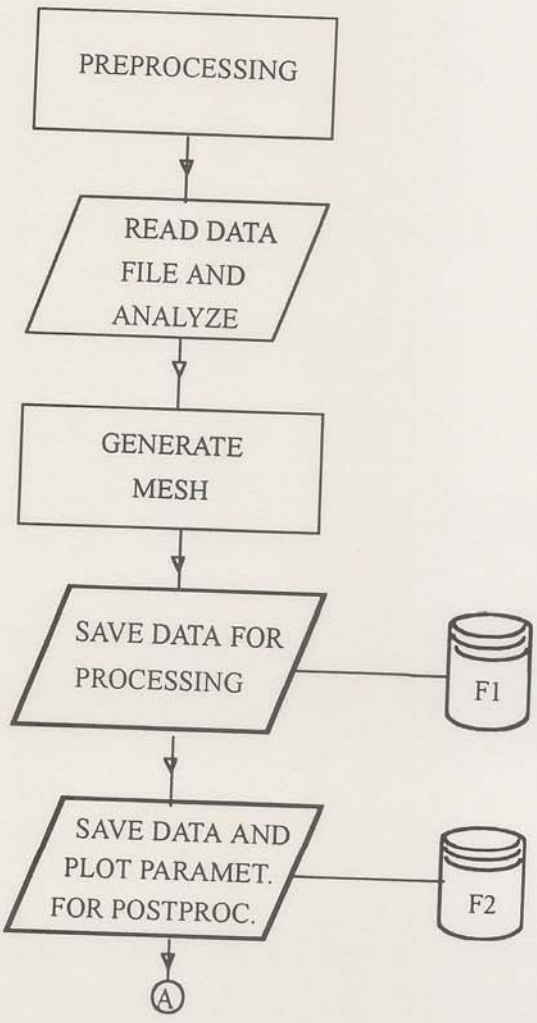


Figure 7.2 Preprocessing phase

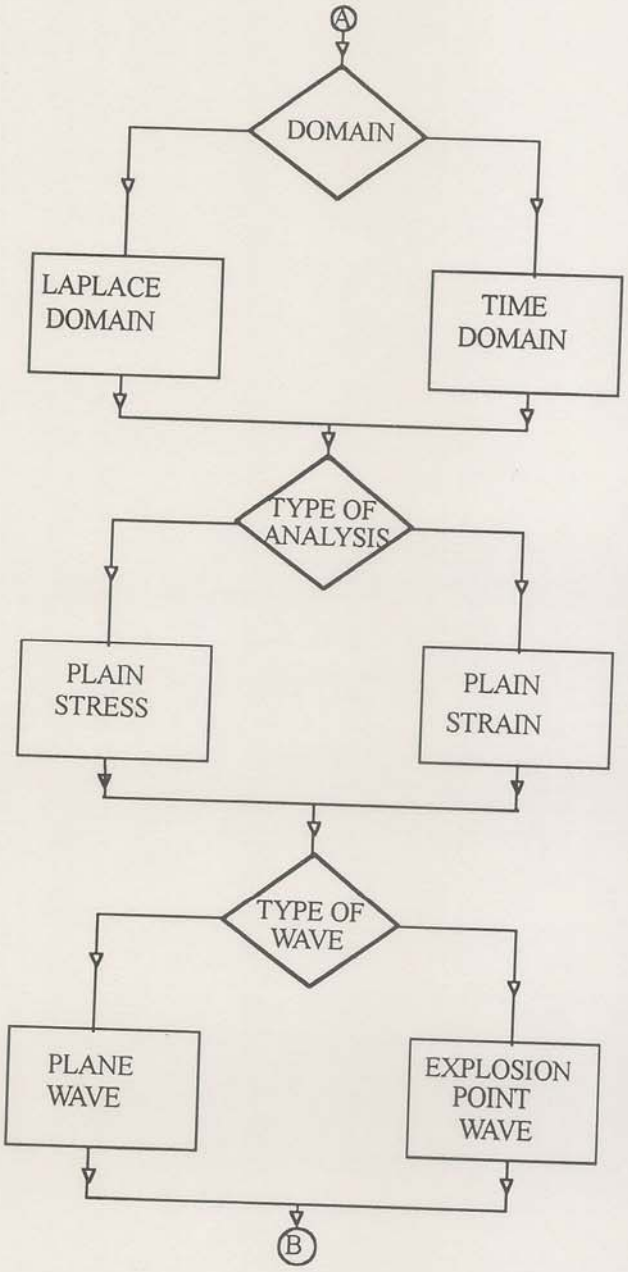


Figure 7.3 Preprocessing phase

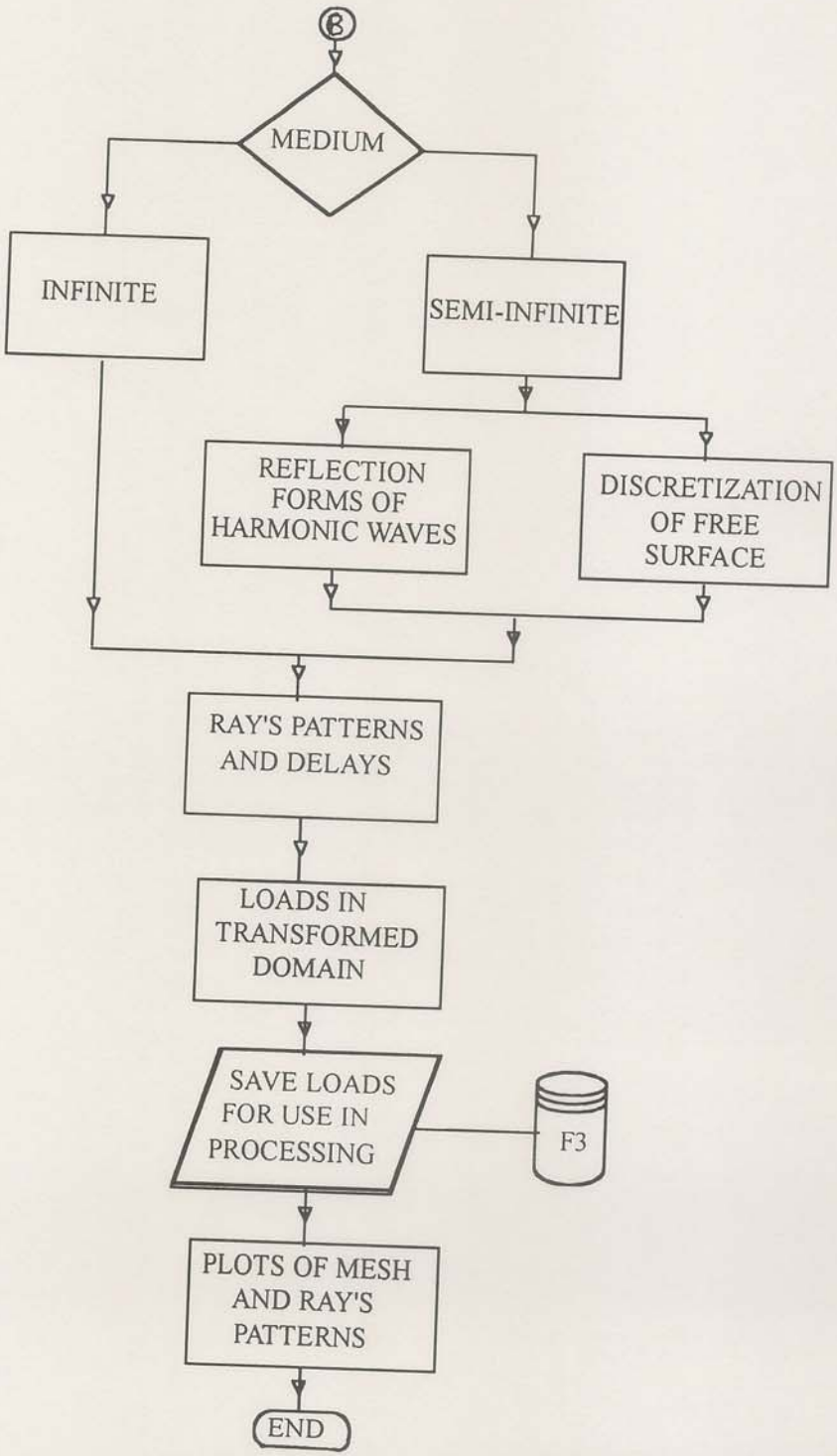


Figure 7.4 Preprocessing phase

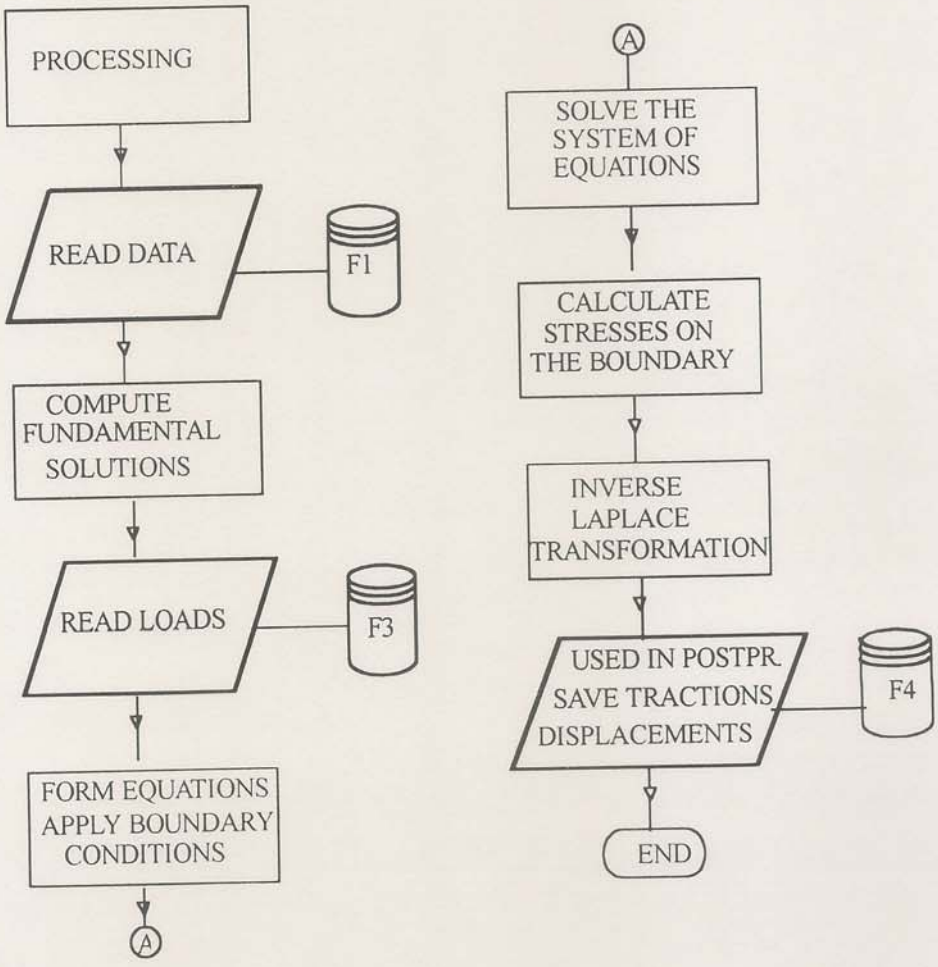


Figure 7.5 Processing phase of Laplace Domain program

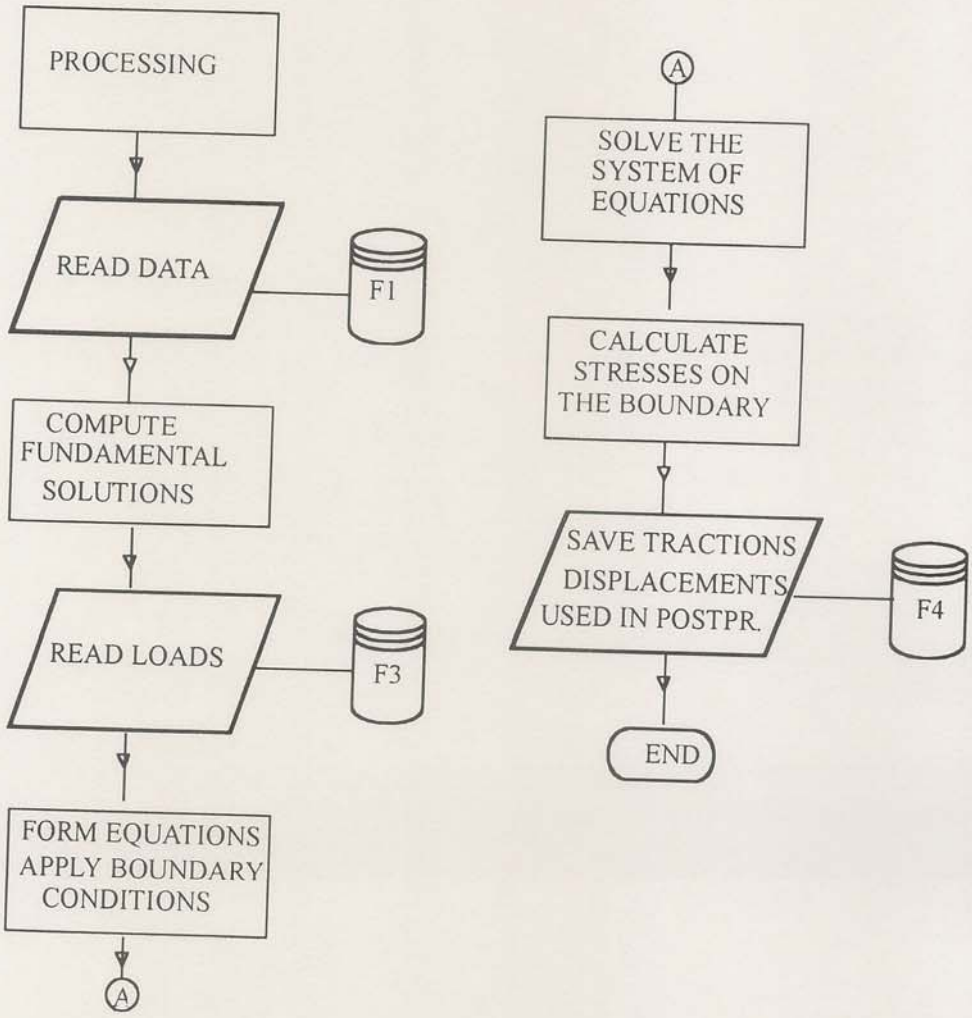


Figure 7.6 Processing phase of Time Domain program

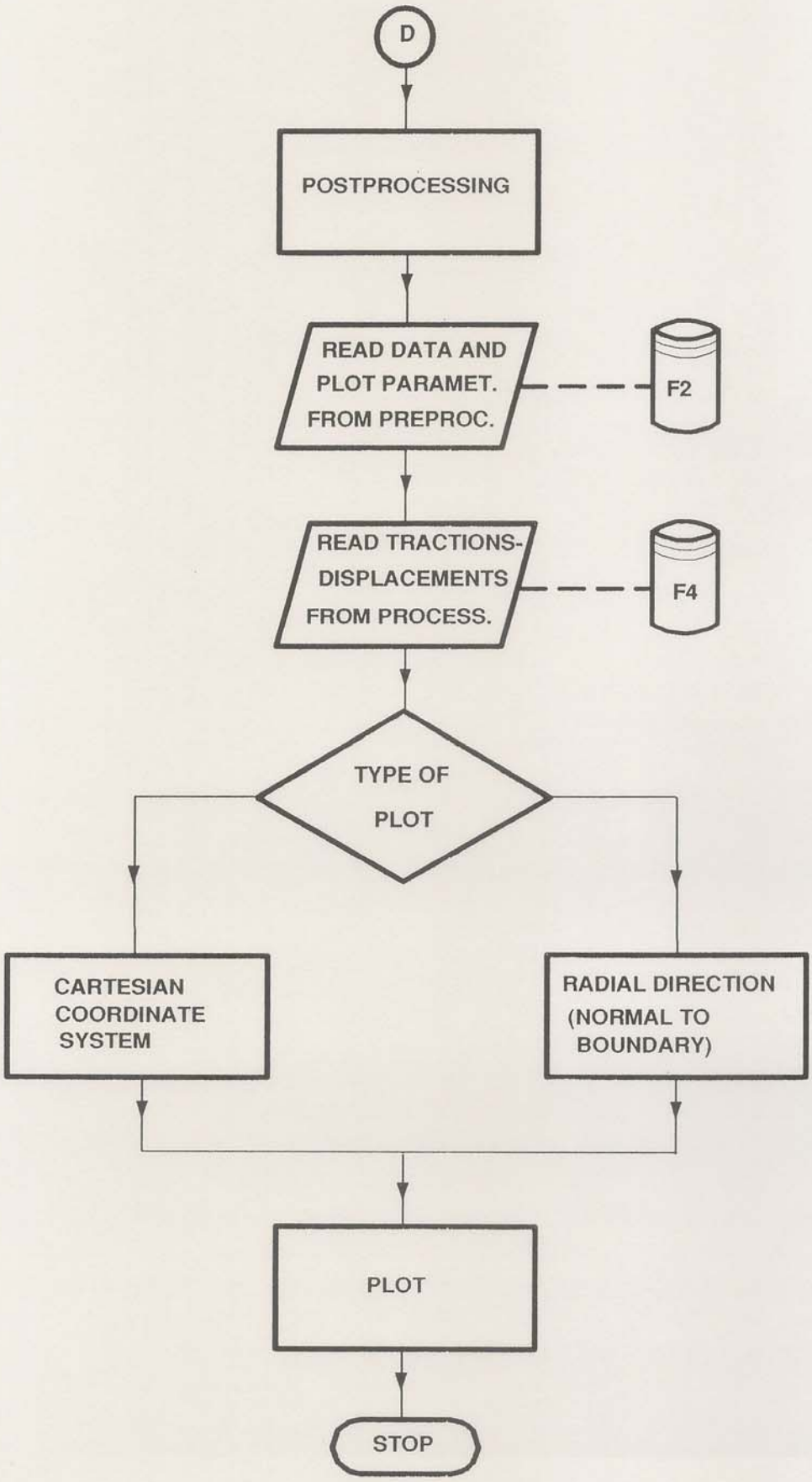
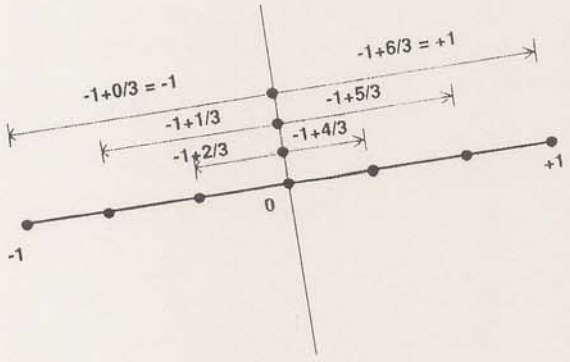
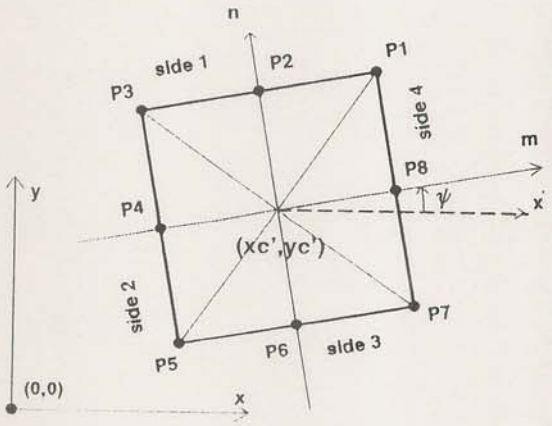


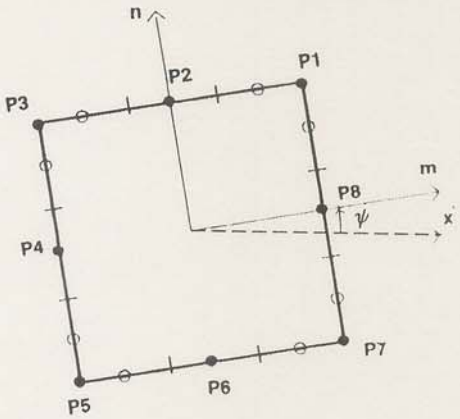
Figure 7.7 Postprocessing phase



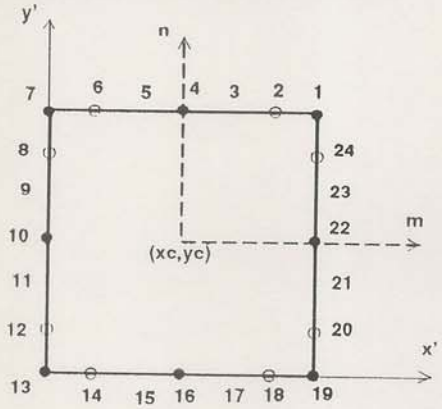
Interpolation of 5 nodes on a line



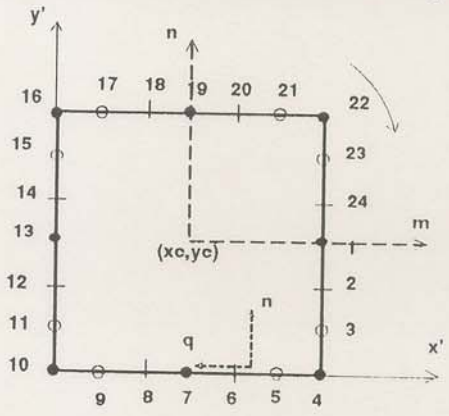
Square mesh with rotation and translation



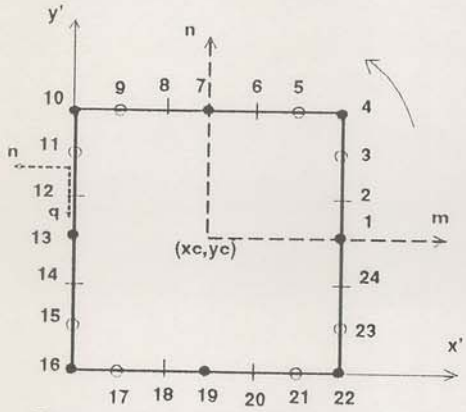
Interpolation of nodes on the sides of a square



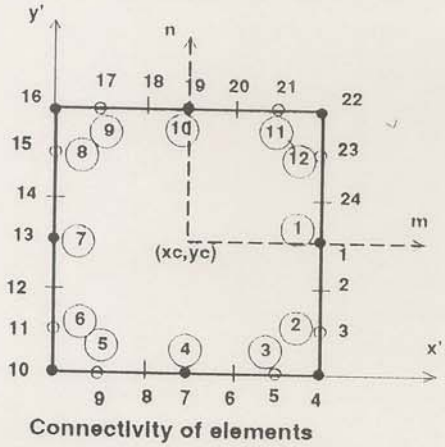
Initial numbering of the nodes



Square mesh for exterior problem (square cavity)



Square mesh for interior problem (solid)



Connectivity of elements

Figure 7.8 Mesh generation process for a square boundary with 12 elements

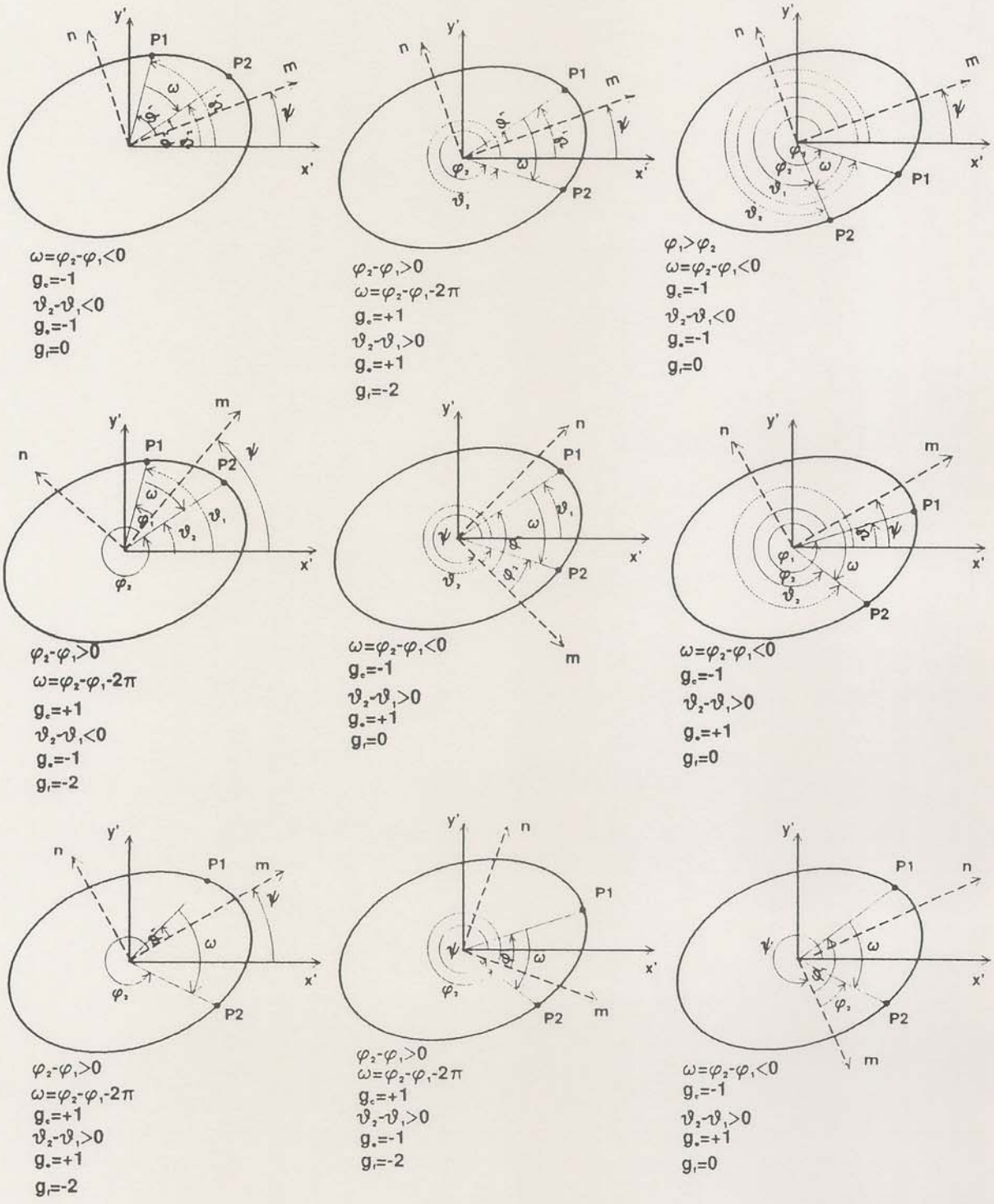


Figure 7.9 Elliptical mesh generation, exterior problem

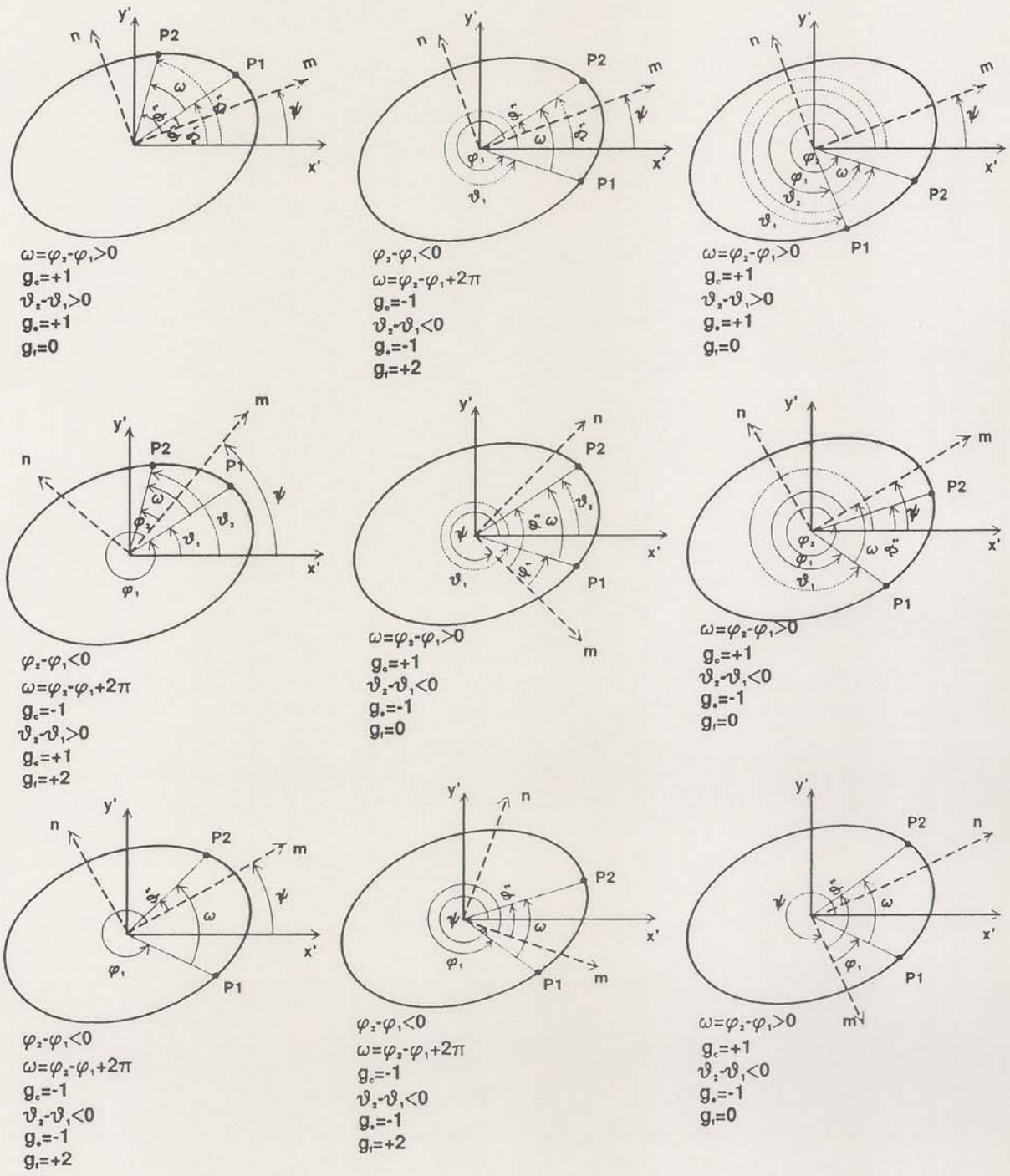


Figure 7.10 Elliptical mesh generation, interior problem

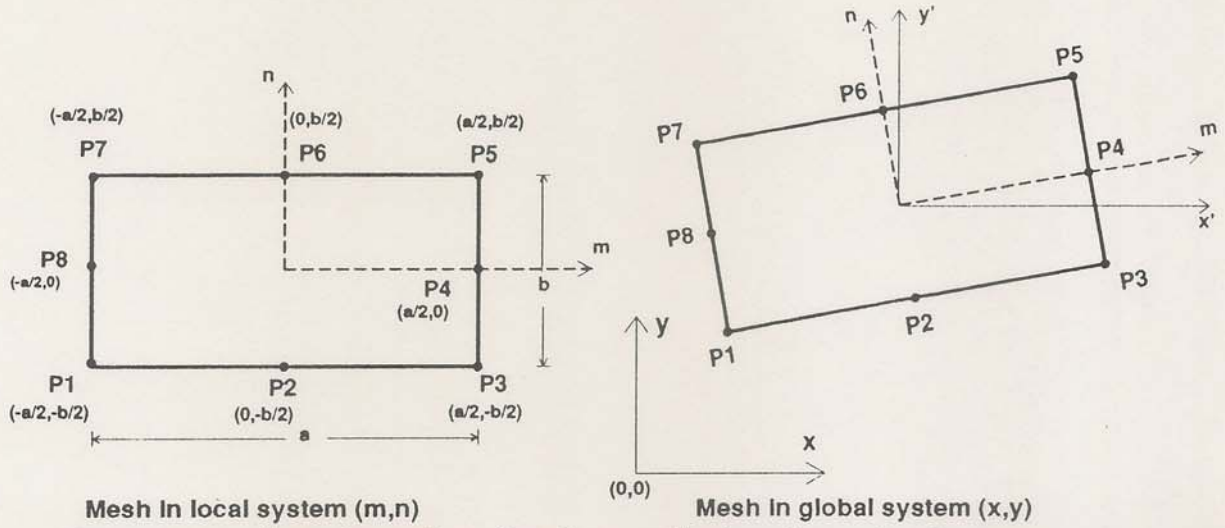


Figure 7.11 Transformation of node co-ordinates of a rectangle

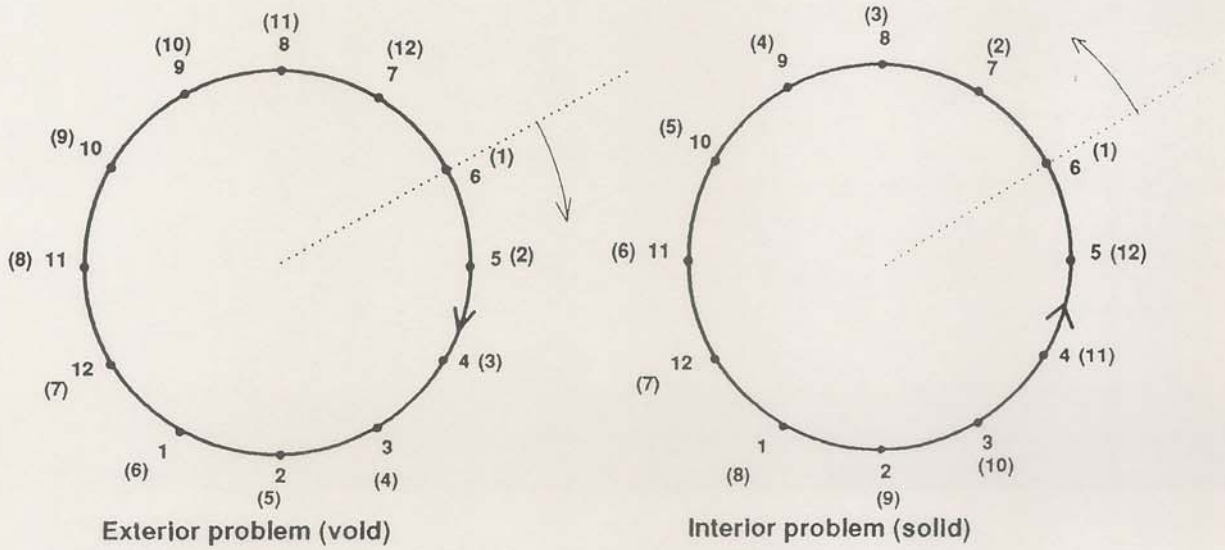


Figure 7.12 Node re-numbering for exterior and interior type problem

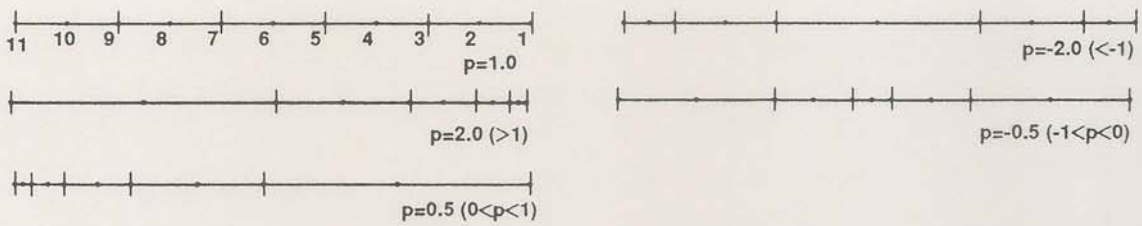


Figure 7.13 Types of quadratic elements on a line

APPENDIX A

FUNDAMENTAL SINGULAR SOLUTIONS FOR ELASTODYNAMICS

The boundary displacement and traction kernels in the transformed domain, for transient elastodynamic, are given by (Cruse and Rizzo,1968; Ahmad,1986)

$$\begin{aligned}
\overline{G}_{ij} &= \frac{1}{2\pi\mu}\left\{\psi\delta_{ij} - \varphi r_{,i} r_{,j}\right\} \\
\overline{F}_{ij} &= \rho\left\{\left(c_1^2 - 2c_2^2\right)\overline{G}_{ik,k} n_j + c_2^2\left(\overline{G}_{ij,k} + \overline{G}_{ik,j}\right) n_k\right\} \\
&= \frac{1}{2\pi}\left\{a\left(\delta_{ij} r_{,n} + r_{,j} n_i\right) + b\left(r_{,i} n_j - 2 r_{,i} r_{,j} r_{,n}\right)\right. \\
&\qquad \left. + c\left(r_{,i} r_{,j} r_{,n}\right) + d\left(r_{,i} n_j\right)\right\}
\end{aligned} \tag{A.1}$$

where

$$\begin{aligned}
\psi &= K_0(z_2) + \frac{1}{z_2}\left\{K_1(z_2) - \frac{c_2}{c_1} K_1(z_1)\right\} \\
\varphi &= K_2(z_2) - \frac{c_2^2}{c_1^2} K_2(z_1) \\
a &= \psi_{,r} - \frac{1}{r}\varphi \;, \qquad b = -\frac{2}{r}\varphi \\
c &= -2\varphi_{,r} \;, \qquad d = \left(\frac{c_1^2}{c_2^2} - 2\right)\left(\psi_{,r} - \varphi_{,r} - \frac{1}{r}\varphi\right)
\end{aligned} \tag{A.2}$$

and

$$
\begin{aligned}
r_{,n} &= \frac{\partial r}{\partial n} = \frac{\partial r}{\partial x_i}\frac{\partial x_i}{\partial n} = r_{,i}n_i \\
\psi_{,r} &= \frac{\partial \psi}{\partial r}\ , \qquad \varphi_{,r} = \frac{\partial \varphi}{\partial r} \\
z_1 &= \frac{sr}{c_1}\ , \qquad z_2 = \frac{sr}{c_2}
\end{aligned}
\tag{A.3}
$$

where n is the unit normal to the surface of the boundary vector.

K_0, K_1 and K_2 are the modified Bessel functions of the second kind and of the zero, first and second orders respectively having the following properties (Abramowitz, 1965):

$$
\begin{aligned}
K_2(z_i) &= K_0(z_i) + \frac{2}{z_i}K_1(z_i) \\
\frac{\partial}{\partial z_i}[K_0(z_i)] &= -K_1(z_i) \\
\frac{\partial}{\partial z_i}[K_1(z_i)] &= -K_0(z_i) - \frac{1}{z_i}k_1(z_i)
\end{aligned}
\tag{A.4}
$$

Using these properties and Equation (A.3) results in

$$
\begin{aligned}
\psi_{,r} = \frac{\partial \psi}{\partial r} = &-\frac{1}{r}K_0(z_2) - \left(\frac{s}{c_2} + \frac{2}{z_2 r}\right)K_1(z_2) \\
&+ \frac{c_2^2}{c_1^2 r}K_0(z_1) + \frac{2c_2^2}{sc_1 r^2}K_1(z_1)
\end{aligned}
\tag{A.5}
$$

and

$$
\begin{aligned}
\varphi_{,r} = \frac{\partial \varphi}{\partial r} = &-\frac{2}{r}K_0(z_2) - \left(\frac{z}{c_2} + \frac{4}{z_2 r}\right)K_1(z_2) \\
&+ \frac{2c_2^2}{c_1^2 r}K_0(z_1) + \left(\frac{c_2^2 s}{c_1^3} + \frac{4c_2^2}{sc_1 r^2}\right)K_1(z_1)
\end{aligned}
\tag{A.6}
$$

Modified Bessel functions

The modified Bessel functions of second kind are as following:

Ascending series For usual values of z_i ($10^{-8} < |z_i| \le 3.5$) are:

Zero order:

$$K_0(z_i) = -(\gamma + \ln(\zeta))I_0(z_i) + \frac{(\zeta^2)}{(1!)^2} + \left(1 + \frac{1}{2}\right)\frac{(\zeta^2)^2}{(2!)^2} + \left(1 + \frac{1}{2} + \frac{1}{3}\right)\frac{(\zeta^2)^3}{(3!)^2} + \ldots \quad \textbf{(A.7)}$$

where

$$\zeta = \frac{z_i}{2} \quad \textbf{(A.8)}$$

and γ is the Euler's constant given by

$$\gamma = \lim_{m\to\infty}\left[1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{m} - \ln(m)\right] \approx 0.5772156649 \quad \textbf{(A.9)}$$

I_0 is the modified Bessel function of the first kind, zero order, given by

$$I_0(z_i) = 1 + \frac{(\zeta^2)}{(1!)^2} + \frac{(\zeta^2)^2}{(2!)^2} + \frac{(\zeta^2)^3}{(3!)^2} + \ldots \quad \textbf{(A.10)}$$

First order:

$$K_1(z_i) = \frac{1}{z_i} + \ln\left(\frac{z_i}{2}\right)I_1(z_i) - \frac{z_i}{4}\sum_{k=0}^{\infty}\left\{(\psi(k+1) + \psi(k+2))\frac{(z_i^2/4)^k}{k!(k+1)!}\right\} \quad \textbf{(A.11)}$$

where

$$I_1(z_i) = \frac{z_i}{2}\sum_{k=0}^{\infty} \frac{(z_i^2/4)^k}{k!\,\Gamma(k+2)} \qquad \textbf{(A.12)}$$

and Γ is the gamma function of integer values and is

$$\Gamma(n+1) = 1\cdot 2\cdot 3 \ldots (n-1)\cdot n = n! \qquad \textbf{(A.13)}$$

and

$$\psi(1) = -\gamma\ , \qquad \psi(n) = -\gamma + \sum_{m=1}^{m-1} \frac{1}{n}\ , \quad (n \geq 2) \qquad \textbf{(A.14)}$$

Asymptotic expansions

Small argument expansion:

When z_i is very small ($0 \leq |z_i| \leq 10^{-8}$), then

$$\lim_{z_i \to 0} K_0(z_i) = -\ln(z_i)\ , \qquad \lim_{z_i \to 0} K_1(z_i) = \frac{1}{z_i} \qquad \textbf{(A.15)}$$

Large argument expansion:

When z_i is large ($3.5 < |z_i| \leq 100$), then

$$K_0(z_i) \simeq \sqrt{\frac{\pi}{2z_i}} e^{-z_i} \left\{ 1 - \frac{(1)^2}{8z_i} + \frac{(3)^2}{2!\,(8z_i)^2} - \frac{(3)^2(5)^2}{3!\,(8z_i)^3} + \frac{(3)^2(5)^2(7)^2}{4!\,(8z)^4} - \ldots \right\} \qquad \textbf{(A.16)}$$

$$K_1(z_i) \simeq \sqrt{\frac{\pi}{2z_i}} e^{-z_i} \left\{ 1 + \frac{(4-1^2)}{8z_i} + \frac{(4-1^2)(4-3^2)}{2!\,(8z_i)^2} + \frac{(4-1^2)(4-3^2)(4-5^2)}{3!\,(8z_i)^3} + \ldots \right\} \qquad \textbf{(A.17)}$$

For z_i even larger ($|z_i| \geq 100$) is

$$K_o(z_i) = K_1(z_i) \sim 0 \qquad \textbf{(A.18)}$$

Singular kernels

When the distance between the source node P and the receiver node Q approaches zero, the displacement and traction fundamental solutions become singular. Tullberg (1983) proved that the kernels tend to their static values when the parameters z_i satisfy the following conditions

$$\lim_{r \to 0} \left\{ Re[z_i] \simeq 0 \right\} \qquad \textbf{(A.19)}$$
$$\lim_{r \to 0} \left\{ Im[z_i] > 0 \right\}$$

APPENDIX B

FUNDAMENTAL SINGULAR SOLUTIONS FOR ELASTOSTATICS

The kernels in the elastostatic case are given by (Brebbia and Dominguez, 1989)

$$\begin{aligned}\Gamma_{i\varphi} &= \frac{1}{8\pi\mu(1-\nu)}\left\{-(3-4\nu)\lambda\nu(\rho)\delta_{i\varphi}+\rho_{,i}\rho_{,\varphi}\right\}\\ \Phi_{i\varphi} &= -\frac{1}{4\pi(1-\nu)\rho}\left\{\frac{\partial\rho}{\partial\nu}\left[(1-2\nu)\delta_{i\varphi}+2\rho_{,i}\rho_{,\varphi}\right]\right.\\ &\qquad \left.-(1-2\nu)\left(\rho_{,i}\nu_{\varphi}-\rho_{,\varphi}\nu_{i}\right)\right\}\end{aligned} \tag{B.1}$$

FUNDAMENTAL SINGULAR SOLUTIONS FOR TIME DOMAIN

The displacement and traction kernels for three dimensions, in the time domain, for transient elastodynamics, are given by (Eringen and Suhubi, 1975; Manolis, 1983) and is, for the displacement kernel

$$\begin{aligned}\Gamma_{i\varphi} = \frac{1}{4\pi\rho}\left\{\left(\frac{3\rho_{i}\rho_{\varphi}}{\rho^{3}}-\frac{\delta_{i\varphi}}{\rho}\right)\int_{1/\chi_1}^{1/\chi_2}\lambda\delta(\tau-\lambda\rho)\delta\lambda+\frac{\rho_{i}\rho_{\varphi}}{\rho^{3}\chi_1^{2}}\,\delta\left(\tau-\frac{\rho}{\chi_1}\right)\right.\\ \left.+\left(-\frac{\rho_{i}\rho_{\varphi}}{\rho^{3}}+\frac{\delta_{i\varphi}}{\rho}\right)\frac{1}{\chi_2^{2}}\,\delta\left(\tau-\frac{\rho}{\chi_2}\right)\right\}\end{aligned} \tag{B.2}$$

and for the traction kernel

$$\Phi_{\iota\kappa} = H_{\iota\varphi\kappa}\nu_{\varphi}$$

$$H_{\iota\varphi\kappa} = \frac{1}{4\pi}\left\{-6\chi_2^2\left[5\frac{\rho_\iota\rho_\varphi\rho_\kappa}{\rho^5} - \frac{\delta_{\iota\varphi}\rho_\kappa + \delta_{\iota\kappa}\rho_\varphi + \delta_{\varphi\kappa}\rho_\iota}{\rho^3}\right]\int_{1/\chi_1}^{1/\chi_2} \lambda\delta(\tau-\lambda\rho)\,\delta\lambda\right.$$

$$-2\left[6\frac{\rho_\iota\rho_\varphi\rho_\kappa}{\rho^5} - \frac{\delta_{\iota\varphi}\rho_\kappa+\delta_{\iota\kappa}\rho_\varphi+\delta_{\varphi\kappa}\rho_\iota}{\rho^3}\right]\frac{\chi_2^2}{\chi_1^2}\,\delta\left(\tau-\frac{\rho}{\chi_1}\right)$$

$$+\left[2\left(6\frac{\rho_\iota\rho_\varphi\rho_\kappa}{\rho^5} - \frac{\delta_{\iota\varphi}\rho_\kappa+\delta_{\iota\kappa}\rho_\varphi+\delta_{\varphi\kappa}\rho_\iota}{\rho^3}\right) - \frac{\delta_{\iota\kappa}\rho_\varphi+\delta_{\varphi\kappa}\rho_\iota}{\rho^3}\right]\delta\left(\tau-\frac{\rho}{\chi_2}\right)$$

$$+\left[-2\frac{\rho_\iota\rho_\varphi\rho_\kappa}{\rho^4\chi_2}\frac{\chi_2^3}{\chi_1^3} - \frac{\rho_\kappa\delta_{\iota\varphi}}{\rho^3}\left(1-2\frac{\chi_2^2}{\chi_1^2}\right)\frac{\rho}{\chi_1}\right]\frac{1}{\Delta\tau}\,\dot{\delta}\left(\tau-\frac{\rho}{\chi_1}\right)$$

$$\left.+\left[2\frac{\rho_\iota\rho_\varphi\rho_\kappa}{\rho^4\chi_2} - \frac{\delta_{\iota\kappa}\rho_\varphi+\delta_{\varphi\kappa}\rho_\iota}{\rho^3}\frac{\rho}{\chi_2}\right]\frac{1}{\Delta\tau}\,\dot{\delta}\left(\tau-\frac{\rho}{\chi_2}\right)\right\} \qquad \text{(B.3)}$$

APPENDIX C

ANALYTICAL INTEGRATION OF THE DYNAMIC KERNELS FOR CONSTANT TIME INTERPOLATION

Assuming constant time variation of the displacements and tractions during a time step α for the examining node i is

$$\upsilon_i(\tau) = u_i^{\alpha}\varphi_{\alpha}(\tau) \ , \qquad t_i(\tau) = t_i^{\alpha}\varphi_{\alpha}(\tau) \tag{C.1}$$

where u_i^{α} and t_i^{α} are the constant values of displacements and tractions during the α time step, taken as the mean of the respective values of the current and previous time steps which is

$$u_i^{\alpha} = \frac{1}{2}(u_i^{(\alpha-1)} + u_i^{(\alpha)}) \ , \qquad \alpha = 2,3,...,k-1 \tag{C.2}$$

and

$$t_i^{\alpha} = \frac{1}{2}(t_i^{(\alpha-1)} + t_i^{(\alpha)}) \ , \qquad \alpha = 2,3,...,k \tag{C.3}$$

the function $\varphi_{\alpha}$is

$$\varphi_{\alpha}(\tau) = H[\tau_{\alpha}] - H[\tau_{\alpha} - \Delta\tau] = H[\tau_{\alpha}] - H[\tau_{\alpha-1}] \tag{C.4}$$

where $\tau_{\alpha-1}=\tau_{\alpha}-\Delta\tau$, H is the Heaviside unit step function and φ_{α} takes value

$$\begin{aligned} &\varphi_{\alpha}(\tau) = 1 \qquad for \quad (\alpha-1)\Delta t \le \tau \le \alpha\Delta t \\ &\varphi_{\alpha}(\tau) = 0 \qquad for \quad \tau < (\alpha-1)\Delta t \quad or \quad \tau > \alpha\Delta t \end{aligned} \tag{C.5}$$

Integration of displacement kernel

The fundamental solutions of displacements G_{ij} in reference to Appendix A, are of the form where a_k^{ij} are functions of the distance r between the source and the receiver point P and Q

$$G_{ij} = a_1^{ij}(r)\int_{c_1^{-1}}^{c_2^{-1}} \delta(t-\lambda r)\lambda d\lambda + a_2^{ij}(r)\delta(t-r/c_1) + a_3^{ij}(r)\delta(t-r/c_2) \qquad \textbf{(C.6)}$$

respectively and c_1, c_2 are the dilatational and shear wave velocity. The first term of the second part of equation 3.6 is written

$$\begin{aligned} S_1 &= \sum_{\alpha=1}^{k} \int_{\tau_{\alpha-1}}^{\tau_\alpha} \left\{ \int_A (G_{ij} t_i) dA \right\} d\tau \\ &= \sum_{\alpha=1}^{k} \int_A \left\{ \int_{\tau_{\alpha-1}}^{\tau_\alpha} [G_{ij}(\tau_\alpha - \tau)]\, [t_i^\alpha \varphi_\alpha(\tau)] d\tau \right\} dA \end{aligned} \qquad \textbf{(C.7)}$$

where t_i^α is the traction for the i node at the α time step. Examining only the current time α, the term inside the summation, in view of equation (C.7), gives

$$\begin{aligned} S_{11} &= \int_A \left\{ \int_{\tau_{\alpha-1}}^{\tau_\alpha} [G_{ij}(\tau_\alpha - \tau)]\, [t_i^\alpha \varphi_\alpha(\tau)] d\tau \right\} dA \\ &= \int_A a_1^{ij}(r) t_i^\alpha \left\{ \int_{c_1^{-1}}^{c_2^{-1}} \left[\int_{\tau_{\alpha-1}}^{\tau_\alpha} \delta(\tau_\alpha - \lambda r - \tau)\, \varphi_\alpha(\tau)\, \lambda d\tau \right] d\lambda \right\} dA \\ &+ \int_A \left\{ a_2^{ij}(r) t_i^\alpha \int_{\tau_{\alpha-1}}^{\tau_\alpha} \delta(\tau_\alpha - r/c_1 - \tau)\ \varphi_\alpha(\tau)\, d\tau \right\} dA \end{aligned} \qquad \textbf{(C.8)}$$

Using the filtering property of Dirac Delta function (O'Neil,1991; Greenberg,1971)

$$\int_{-\infty}^{\infty} \delta(x-\xi) h(x) dx = h(\xi) \qquad \textbf{(C.9)}$$

equation (C.8) results in

$$
\begin{aligned}
S_{11} = & \int_A a_1^{ij}(r) t_i^{\alpha} \left\{ \int_{c_1^{-1}}^{c_2^{-1}} \lambda \varphi_{\alpha}(\tau_{\alpha} - \lambda r) d\lambda \right\} dA \\
& + \int_A \left\{ a_2^{ij}(r) t_i^{\alpha} \ \varphi_{\alpha}(\tau_{\alpha} - r/c_1) \right\} dA \\
& + \int_A \left\{ a_3^{ij}(r) t_i^{\alpha} \ \varphi_{\alpha}(\tau_{\alpha} - r/c_2) \right\} dA \\
& = \{t\}^{\alpha} [G]^{\alpha}
\end{aligned}
\qquad \textbf{(C.10)}
$$

the three terms of the right hand of equation (C.10) can be evaluated as follows:

a. <u>First term</u> of the right hand of equation (C.10)

$$
\begin{aligned}
A_1 = & \int_A a_1^{ij}(r) t_i^{\alpha} \left\{ \int_{c_1^{-1}}^{c_2^{-1}} \lambda \varphi_{\alpha}(\tau_{\alpha} - \lambda r) d\lambda \right\} dA \\
= & \int_A t_i^{\alpha} \ a_1^{ij}(r) \left[\frac{\lambda^2}{2} \varphi_{\alpha}(\tau_{\alpha} - \lambda r) \Big|_{\lambda = c_1^{-1}}^{c_2^{-1}} \right] dA
\end{aligned}
\qquad \textbf{(C.11)}
$$

It is

$$
\begin{aligned}
\varphi_{\alpha}(\tau_{\alpha} - r/c) &= H[\tau_{\alpha} - r/c] - H[\tau_{\alpha} - \Delta\tau - r/c] \\
&= H[\tau_{\alpha} - r/c] - H[\tau_{\alpha-1} - r/c]
\end{aligned}
\qquad \textbf{(C.12)}
$$

where $\tau_{\alpha-1}=\tau_{\alpha}-\Delta\tau$ and

$$
\begin{aligned}
\left[\frac{\lambda^2}{2} \varphi_{\alpha}(\tau_{\alpha} - \lambda r) \Big|_{\lambda = c_1^{-1}}^{c_2^{-1}} \right] &= \frac{\lambda^2}{2} H \left[\tau_{\alpha} - \lambda r \right] \Big|_{\lambda = c_1^{-1}}^{c_2^{-1}} - \frac{\lambda^2}{2} H \left[(\tau_{\alpha} - \Delta\tau) - \lambda r \right] \Big|_{\lambda = c_1^{-1}}^{c_2^{-1}} \\
&= I_{\alpha} - I_{\alpha-1}
\end{aligned}
\qquad \textbf{(C.13)}
$$

Applying the integration limits results in

$$I_\alpha = \frac{\lambda^2}{2} H\left[\tau_\alpha - \lambda r\right]\Big|_{\lambda=c_1^{-1}}^{c_2^{-1}} = \frac{1}{2c_2^2} H\left[\tau_\alpha - \frac{r}{c_2}\right] - \frac{1}{2c_1^2} H\left[\tau_\alpha - \frac{r}{c_1}\right]$$

$$= \begin{cases} 0 & \text{if} \quad \tau_\alpha < \dfrac{r}{c_1} \\ \dfrac{\tau_\alpha^2}{2r^2} - \dfrac{1}{2c_1^2} & \text{if} \quad \dfrac{r}{c_1} < \tau_\alpha < \dfrac{r}{c_2} \\ \dfrac{1}{2c_2^2} - \dfrac{1}{2c_1^2} & \text{if} \quad \tau_\alpha > \dfrac{r}{c_2} \end{cases} \qquad \textbf{(C.14)}$$

similarly

$$I_{\alpha-1} = \frac{\lambda^2}{2} H\left[(\tau_\alpha - \Delta\tau) - \lambda r\right]\Big|_{\lambda=c_1^{-1}}^{c_2^{-1}} = \frac{1}{2c_2^2} H\left[\tau_{\alpha-1} - \frac{r}{c_2}\right] - \frac{1}{2c_1^2} H\left[\tau_{\alpha-1} - \frac{r}{c_1}\right]$$

$$= \begin{cases} 0 & \text{if} \quad \tau_{\alpha-1} < \dfrac{r}{c_1} \\ \dfrac{\tau_{\alpha-1}^2}{2r^2} - \dfrac{1}{2c_1^2} & \text{if} \quad \dfrac{r}{c_1} < \tau_{\alpha-1} < \dfrac{r}{c_2} \\ \dfrac{1}{2c_2^2} - \dfrac{1}{2c_1^2} & \text{if} \quad \tau_{\alpha-1} > \dfrac{r}{c_2} \end{cases} \qquad \textbf{(C.15)}$$

So equation (C.11) is rewritten

$$A_1 = \int_A t_i^\alpha a_1^{ij}(r)\,(I_\alpha - I_{\alpha-1})\,dA \qquad \textbf{(C.16)}$$

b. <u>Second term</u> of the right hand of equation (C.10)

$$A_2 = \int_A a_2^{ij}(r) t_i^\alpha \ \varphi_\alpha(\tau_\alpha - r/c_1)\,dA$$

$$= \int_A t_i^\alpha \ a_2^{ij}(r)\left\{H\left[\tau_\alpha - r/c_1\right] - H\left[\tau_{\alpha-1} - r/c_1\right]\right\}dA \qquad \textbf{(C.17)}$$

c. Third term of the right hand of equation (C.10)

$$A_3 = \int_A a_3^{ij}(r) t_i^{\alpha}\ \varphi_{\alpha}(\tau_{\alpha} - r/c_2)\, dA$$

$$= \int_A t_i^{\alpha}\ a_3^{ij}(r) \left\{ H\left[\tau_{\alpha} - r/c_2\right] - H\left[\tau_{\alpha-1} - r/c_2\right] \right\} dA \qquad \textbf{(C.18)}$$

The constant values of the tractions t_i^a during a time interval, are calculated as the mean values of the tractions at the current and previous stations, as in equation (C.3). The contribution of the retarded time interval α-1, is zero for the two first time steps.

Instead of averaging the tractions between the time steps, constant multipliers β are introduced in the above equations with the following values:

(a). for the current time step, it is

$$\beta_1 = \begin{cases} 1.0 & for \quad \alpha=1 \\ 0.5 & for \quad \alpha>1 \end{cases} \qquad \textbf{(C.19)}$$

(b). for the retarded time step, it is

$$\beta_2 = \begin{cases} 0 & for \quad \alpha=1,2 \\ 0.5 & for \quad \alpha>2 \end{cases} \qquad \textbf{(C.20)}$$

Using these values, equation (C.18) is written

$$A_3 = \int_A t_i^{\alpha}\ a_3^{ij}(r) \left\{ \beta_1\ H\left[\tau_{\alpha} - r/c_2\right] - \beta_2\ H\left[\tau_{\alpha-1} - r/c_2\right] \right\} dA \qquad \textbf{(C.21)}$$

where now t_i^{α} represent the exact value of tractions at the examining time step α.

From the above analysis it is evident that because of the nature of the Heaviside function, the [G] matrix has the same value for all the time steps α=1,2,...,k. As a result matrix [G] is calculated only in the first time interval and maintains a constant value in the subsequent time steps.

Integration of traction kernel

The traction's fundamental solutions F_{ij} in reference to Appendix A, are of the form

$$F_{ij} = b_1^{ij}(r)\int_{c_1^{-1}}^{c_2^{-1}} \delta(t-\lambda r)\lambda d\lambda + b_2^{ij}(r)\delta(t-r/c_1) + \\ + b_3^{ij}(r)\delta(t-r/c_2)\, b_4^{ij}(r)\dot{\delta}(t-r/c_1) + b_5^{ij}(r)\dot{\delta}(t-r/c_2) \qquad \textbf{(C.22)}$$

where b_k^{ij} is function of the distance r between the source and the receiver point P and Q respectively. The second term of the second part of equation 3.6 is written as previously

$$S_2 = \sum_{\alpha=1}^{k} \int_{\tau_{\alpha-1}}^{\tau_\alpha} \left\{ \int_A \left(F_{ij} u_i\right) dA \right\} d\tau$$
$$= \sum_{\alpha=1}^{k} \int_A \left\{ \int_{\tau_{\alpha-1}}^{\tau_\alpha} \left[F_{ij}(\tau_\alpha - \tau)\right] \left[u_i^\alpha \varphi_\alpha(\tau)\right] d\tau \right\} dA \qquad \textbf{(C.23)}$$

where u_i^{α} is the displacement for the i node at the α time step. Examining only the current

$$
\begin{aligned}
S_{22} &= \int_{\tau_{\alpha-1}}^{\tau_\alpha} \left[\int_A (F_{ij} u_i) dA \right] d\tau = \int_A \left\{ \int_{\tau_{\alpha-1}}^{\tau_\alpha} [F_{ij}(\tau_\alpha - \tau)] \, [u_i^{\alpha} \varphi_\alpha(\tau)] d\tau \right\} dA \\
&= \int_A b_1^{ij}(r) u_i^{\alpha} \left\{ \int_{c_1^{-1}}^{c_2^{-1}} \left[\int_{\tau_{\alpha-1}}^{\tau_\alpha} \delta(\tau_\alpha - \lambda r - \tau) \varphi_\alpha(\tau) \lambda d\tau \right] d\lambda \right\} dA \\
&+ \int_A \left\{ b_2^{ij}(r) u_i^{\alpha} \int_{\tau_{\alpha-1}}^{\tau_\alpha} \delta(\tau_\alpha - r/c_1 - \tau) \;\; \varphi_\alpha(\tau) d\tau \right\} dA \\
&+ \int_A \left\{ b_3^{ij}(r) u_i^{\alpha} \int_{\tau_{\alpha-1}}^{\tau_\alpha} \delta(\tau_\alpha - r/c_2 - \tau) \;\; \varphi_\alpha(\tau) d\tau \right\} dA \\
&+ \int_A \left\{ b_4^{ij}(r) \int_{\tau_{\alpha-1}}^{\tau_\alpha} \dot{\delta}(\tau_\alpha - r/c_1 - \tau) \;\; u_i d\tau \right\} dA \\
&+ \int_A \left\{ b_5^{ij}(r) \int_{\tau_{\alpha-1}}^{\tau_\alpha} \delta(\tau_\alpha - r/c_2 - \tau) \;\; u_i d\tau \right\} dA \qquad \textbf{(C.24)} \\
&= \{u\}^{\alpha} [F]^{\alpha}
\end{aligned}
$$

interval α the term inside the summation, in view of equations (C.22) and (C.23), gives

The first three terms of the second part of equation (C.24) are given by expressions similar to equations (C.16),(C.17) and (C.18).

Using the following property for the time derivative of Dirac Delta function (Greenberg,1974)

$$
\int_{-\infty}^{\infty} \delta^{(j)}(x - \xi) h(x) dx = (-1)^j \, h^{(j)}(\xi) \qquad \textbf{(C.25)}
$$

the fourth term of equation (C.24) becomes

$$
\begin{aligned}
B_4 &= \int_A \left\{ b_4^{ij}(r) \int_{\tau_{\alpha-1}}^{\tau_\alpha} \dot{\delta}(\tau_\alpha - r/c_1 - \tau)\, u_j(\tau)\, d\tau \right\} dA \\
&= -\int_A \left\{ b_4^{ij}(r) \int_{\tau_{\alpha-1}}^{\tau_\alpha} \delta(\tau_\alpha - r/c_1 - \tau)\, \dot{u}_j(\tau)\, d\tau \right\} dA \\
&= -\int_A b_4^{ij}(r)\, \dot{u}_j(\tau - r/c_1)\, dA = -\int_A b_4^{ij}(r)\, \dot{u}_j(\tau)\, \varphi_a(\tau - r/c_1)\, dA \\
&= -\int_A b_4^{ij}(r) \left(\frac{u_i^{\alpha} - u_i^{\alpha-1}}{\Delta\tau} \right) \varphi_a(\tau - r/c_1)\, dA
\end{aligned}
\qquad \textbf{(C.26)}
$$

the same is valid for the 5th term, with only difference that the wave velocity c_1 is replaced by

$$
\begin{aligned}
S_{22} &= \int_A b_1^{ij}(r) u_i^{\alpha} \left\{ \int_{c_1^{-1}}^{c_2^{-1}} \lambda\, \varphi_\alpha(\tau_\alpha - \lambda r)\, d\lambda \right\} dA \\
&+ \int_A \left\{ b_2^{ij}(r) u_i^{\alpha}\ \varphi_\alpha(\tau_\alpha - r/c_1) \right\} dA + \int_A \left\{ b_3^{ij}(r) u_i^{\alpha}\ \varphi_\alpha(\tau_\alpha - r/c_2) \right\} dA \\
&- \int_A b_4^{ij}(r) \left(\frac{u_i^{\alpha} - u_i^{\alpha-1}}{\Delta\tau} \right) \varphi_a(\tau - r/c_1)\, dA - \int_A b_5^{ij}(r) \left(\frac{u_i^{\alpha} - u_i^{\alpha-1}}{\Delta\tau} \right) \varphi_a(\tau - r/c_2)\, dA
\end{aligned}
\qquad \textbf{(C.27)}
$$

c_2. Finally equation (C.24) becomes

From equation (C.27), it is clear that the [F] matrix at time interval α, keeps the same value between all the time steps α=1,2,...,k.

APPENDIX D

ANALYTICAL INTEGRATION OF THE DYNAMIC KERNELS FOR LINEAR TIME INTERPOLATION

Instead of constant time, linear time variation of the displacements and tractions during a time step α can be assumed. In this case is

$$\upsilon_i(\tau)=\left[\left(\frac{\tau_\alpha-\tau}{\Delta\tau}\right)u_i^{\alpha-1}+\left(\frac{\tau-\tau_{\alpha-1}}{\Delta\tau}\right)u_i^{\alpha}\right]\varphi_\alpha(\tau)$$

$$t_i(\tau)=\left[\left(\frac{\tau_\alpha-\tau}{\Delta\tau}\right)t_i^{\alpha-1}+\left(\frac{\tau-\tau_{\alpha-1}}{\Delta\tau}\right)t_i^{\alpha}\right]\varphi_\alpha(\tau) \qquad \textbf{(D.1)}$$

where u_i^{α},t_i^{α} and $u_i^{\alpha-1}$,$t_i^{\alpha-1}$ are the constant values of displacements and tractions of the current α and previous α-1 time steps respectively.

The matrices [G] and [F], keep the same value for all the time steps, as will be proved immediately.

Integration of displacement kernel

For the matrix [G]:

It is

$$S_1=\sum_{\alpha=1}^{k}\left\{\int_{\tau_{\alpha-1}}^{\tau_\alpha}\left[\int_A\left(G_{ij}t_i\right)dA\right]d\tau\right\}$$

$$=\sum_{\alpha=1}^{k}\int_A\left\{\int_{\tau_{\alpha-1}}^{\tau_\alpha}\left[G_{ij}(\tau_\alpha-\tau)\right]\left[\left(\frac{\tau_\alpha-\tau}{\Delta\tau}\right)t_i^{\alpha-1}+\left(\frac{\tau-\tau_{\alpha-1}}{\Delta\tau}\right)t_i^{\alpha}\right]\varphi_\alpha(\tau)\,d\tau\right\}dA \qquad \textbf{(D.2)}$$

replacing G_{ij} from equation (19), and examining only the current time step α, the term in the summation becomes

$$S_{11} = \int_A \left\{ \int_{\tau_{\alpha-1}}^{\tau_\alpha} \left[G_{ij}(\tau_\alpha - \tau) \right] \left[\left(\frac{\tau_\alpha - \tau}{\Delta\tau} \right) t_i^{\alpha-1} + \left(\frac{\tau - \tau_{\alpha-1}}{\Delta\tau} \right) t_i^{\alpha} \right] \varphi_\alpha(\tau)\, d\tau \right\} dA$$

$$= \int_A \left\{ \int_{\tau_{\alpha-1}}^{\tau_\alpha} a_1^{ij}(r) \left[\int_{c_1^{-1}}^{c_2^{-1}} \delta(\tau_\alpha - \lambda r - \tau)\lambda\, d\lambda \right] \left[\left(\frac{\tau_\alpha - \tau}{\Delta\tau} \right) t_i^{\alpha-1} + \left(\frac{\tau - \tau_{\alpha-1}}{\Delta\tau} \right) t_i^{\alpha} \right] \varphi_\alpha(\tau)\, d\tau \right\} dA$$

$$+ \int_A \left\{ \int_{\tau_{\alpha-1}}^{\tau_\alpha} a_2^{ij}(r)\, \delta(\tau_\alpha - r/c_1 - \tau) \left[\left(\frac{\tau_\alpha - \tau}{\Delta\tau} \right) t_i^{\alpha-1} + \left(\frac{\tau - \tau_{\alpha-1}}{\Delta\tau} \right) t_i^{\alpha} \right] \varphi_\alpha(\tau)\, d\tau \right\} dA$$

$$+ \int_A \left\{ \int_{\tau_{\alpha-1}}^{\tau_\alpha} a_3^{ij}(r)\, \delta(\tau_\alpha - r/c_2 - \tau) \left[\left(\frac{\tau_\alpha - \tau}{\Delta\tau} \right) t_i^{\alpha-1} + \left(\frac{\tau - \tau_{\alpha-1}}{\Delta\tau} \right) t_i^{\alpha} \right] \varphi_\alpha(\tau)\, d\tau \right\} dA \qquad \textbf{(D.3)}$$

$$= \{t\}^\alpha [F]^\alpha$$

a. First term of equation (D.3) can be written

$$A_1 = \int_A \left\{ \int_{\tau_{\alpha-1}}^{\tau_\alpha} a_1^{ij}(r) \left[\int_{c_1^{-1}}^{c_2^{-1}} \delta(\tau_\alpha - \lambda r - \tau)\lambda\, d\lambda \right] \left[\left(\frac{\tau_\alpha - \tau}{\Delta\tau} \right) t_i^{\alpha-1} + \left(\frac{\tau - \tau_{\alpha-1}}{\Delta\tau} \right) t_i^{\alpha} \right] \varphi_\alpha(\tau)\, d\tau \right\} dA$$

$$= \int_A \left\{ \left(\frac{\tau_\alpha t_i^{\alpha-1} - \tau_{\alpha-1} t_i^{\alpha}}{\Delta\tau} \right) a_1^{ij}(r) \left\{ \int_{c_1^{-1}}^{c_2^{-1}} \left[\int_{\tau_{\alpha-1}}^{\tau_\alpha} \delta(\tau_\alpha - \lambda r - \tau)\, \varphi_\alpha(\tau)\lambda\, d\tau \right] d\lambda \right\} dA \right.$$

$$\left. + \int_A \left(\frac{t_i^{\alpha} - t_i^{\alpha-1}}{\Delta\tau} \right) a_1^{ij}(r) \left\{ \int_{c_1^{-1}}^{c_2^{-1}} \left[\int_{\tau_{\alpha-1}}^{\tau_\alpha} \delta(\tau_\alpha - \lambda r - \tau)\, \tau\, \varphi_\alpha(\tau)\lambda\, d\tau \right] d\lambda \right\} dA \right. \qquad \textbf{(D.4)}$$

The first integral of equation (D.4) is calculated as in the case of constant time interpolation by using equations (D.16). The second integral is

$$A_{12} = \int_A \left(\frac{t_i^{\alpha} - t_i^{\alpha-1}}{\Delta\tau} \right) a_1^{ij}(r) \left\{ \int_{c_1^{-1}}^{c_2^{-1}} \left[\int_{\tau_{\alpha-1}}^{\tau_\alpha} \delta(\tau_\alpha - \lambda r - \tau)\, \tau\, \varphi_\alpha(\tau)\, \lambda\, d\tau \right] d\lambda \right\} dA$$

$$= \int_A \left(\frac{t_i^{\alpha} - t_i^{\alpha-1}}{\Delta\tau} \right) a_1^{ij}(r) \left\{ \int_{c_1^{-1}}^{c_2^{-1}} (\tau_\alpha - \lambda r)\, \varphi_\alpha(\tau_\alpha - \lambda r)\, \lambda\, d\lambda \right\} dA$$

$$= \int_A \left(\frac{t_i^{\alpha} - t_i^{\alpha-1}}{\Delta\tau} \right) a_1^{ij}(r) \left\{ \tau_\alpha \left[\frac{\lambda^2}{2} \varphi_\alpha(\tau_\alpha - \lambda r) \Big|_{\lambda=c_1^{-1}}^{c_2^{-1}} \right] - r \left[\frac{\lambda^3}{2} \varphi_\alpha(\tau_\alpha - \lambda r) \Big|_{\lambda=c_1^{-1}}^{c_2^{-1}} \right] \right\} \qquad \textbf{(D.5)}$$

the quantity

$$\left[\frac{\lambda^2}{2} \varphi_\alpha(\tau_\alpha - \lambda r) \Big|_{\lambda=c_1^{-1}}^{c_2^{-1}} \right] = I_\alpha - I_{\alpha-1} \qquad \textbf{(D.6)}$$

is evaluated from equations (D.14), (D.15)
similarly it is

$$\left[\frac{\lambda^3}{3} \varphi_\alpha(\tau_\alpha - \lambda r) \Big|_{\lambda=c_1^{-1}}^{c_2^{-1}} \right] = \frac{\lambda^3}{3} H\left[\tau_\alpha - \lambda r\right] \Big|_{\lambda=c_1^{-1}}^{c_2^{-1}} - \frac{\lambda^3}{3} H\left[(\tau_\alpha - \Delta\tau) - \lambda r\right] \Big|_{\lambda=c_1^{-1}}^{c_2^{-1}}$$

$$= J_\alpha - J_{\alpha-1} \qquad \textbf{(D.7)}$$

Applying the integration limits results in

$$J_\alpha = \frac{\lambda^3}{3} H\left[\tau_\alpha - \lambda r\right] \Big|_{\lambda=c_1^{-1}}^{c_2^{-1}} = \frac{1}{3c_2^3} H\left[\tau_\alpha - \frac{r}{c_2}\right] - \frac{1}{3c_1^3} H\left[\tau_\alpha - \frac{r}{c_1}\right]$$

$$= \begin{cases} 0 & \text{if} \quad \tau_\alpha < \dfrac{r}{c_1} \\ \dfrac{\tau_\alpha^3}{3r^3} - \dfrac{1}{3c_1^3} & \text{if} \quad \dfrac{r}{c_1} < \tau_\alpha < \dfrac{r}{c_2} \\ \dfrac{1}{3c_2^3} - \dfrac{1}{3c_1^3} & \text{if} \quad \tau_\alpha > \dfrac{r}{c_2} \end{cases} \qquad \textbf{(D.8)}$$

Similarly

$$J_{\alpha-1} = \frac{\lambda^3}{3} H\left[(\tau_\alpha - \Delta\tau) - \lambda r\right]\Big|_{\lambda=c_1^{-1}}^{c_2^{-1}} = \frac{1}{3c_2^3} H\left[\tau_{\alpha-1} - \frac{r}{c_2}\right] - \frac{1}{3c_1^3} H\left[\tau_{\alpha-1} - \frac{r}{c_1}\right]$$

$$= \begin{cases} 0 & \text{if} \quad \tau_{\alpha-1} < \dfrac{r}{c_1} \\ \dfrac{\tau_{\alpha-1}^3}{3r^3} - \dfrac{1}{3c_1^3} & \text{if} \quad \dfrac{r}{c_1} < \tau_{\alpha-1} < \dfrac{r}{c_2} \\ \dfrac{1}{3c_2^3} - \dfrac{1}{3c_1^3} & \text{if} \quad \tau_{\alpha-1} > \dfrac{r}{c_2} \end{cases} \qquad \textbf{(D.9)}$$

Equation (D.4) is rewritten

$$A_1 = \int_A \left(\frac{\tau_\alpha t_i^{\alpha-1} - \tau_{\alpha-1} t_i^{\alpha}}{\Delta\tau} \right) a_1^{ij}(r)\,(I_\alpha - I_{\alpha-1})\,dA$$

$$+ \int_A \left(\frac{t_i^{\alpha} - t_i^{\alpha-1}}{\Delta\tau} \right) a_1^{ij}(r) \left(\tau_\alpha (I_\alpha - I_{\alpha-1}) - r(J_\alpha - J_{\alpha-1}) \right) dA$$

$$= \int_A \left\{ \frac{(\tau_\alpha - \tau_{\alpha-1})(I_\alpha - I_{\alpha-1}) - r(J_\alpha - J_{\alpha-1})}{\Delta\tau} t_i^{\alpha} + \frac{r(J_\alpha - J_{\alpha-1})}{\Delta\tau} t_i^{\alpha-1} \right\} a_1^{ij}(r)\,dA$$

$$= \int_A \left\{ \left[(I_\alpha - I_{\alpha-1}) - \frac{r}{\Delta\tau}(J_\alpha - J_{\alpha-1}) \right] t_i^{\alpha} + \left[\frac{r}{\Delta\tau}(J_\alpha - J_{\alpha-1}) \right] t_i^{\alpha-1} \right\} a_1^{ij}(r)\,dA \qquad \textbf{(D.10)}$$

b. Second term of equation (D.3) is

$$A_2 = \int_A \left\{ \int_{\tau_{\alpha-1}}^{\tau_\alpha} a_2^{ij}(r)\,\delta(\tau_\alpha - r/c_1 - \tau) \left[\left(\frac{\tau_\alpha - \tau}{\Delta\tau} \right) t_i^{\alpha-1} + \left(\frac{\tau - \tau_{\alpha-1}}{\Delta\tau} \right) t_i^{\alpha} \right] \varphi_\alpha(\tau)\,d\tau \right\} dA$$

$$= \int_A \left(\frac{\tau_\alpha t_i^{\alpha-1} - \tau_{\alpha-1} t_i^{\alpha}}{\Delta\tau} \right) a_2^{ij}(r) \left[\int_{\tau_{\alpha-1}}^{\tau_\alpha} \delta(\tau_\alpha - r/c_1 - \tau)\,\varphi_\alpha(\tau)\,d\tau \right] dA$$

$$+ \int_A \left(\frac{t_i^{\alpha} - t_i^{\alpha-1}}{\Delta\tau} \right) a_2^{ij}(r) \left[\int_{\tau_{\alpha-1}}^{\tau_\alpha} \delta(\tau_\alpha - r/c_1 - \tau)\,\tau\,\varphi_\alpha(\tau)\,d\tau \right] dA \qquad \textbf{(D.11)}$$

in view of equations (C.17) and (D.5), equation (D.11) gives

$$A_2 = \int_A \left(\frac{\tau_\alpha t_i^{\alpha-1} - \tau_{\alpha-1} t_i^{\alpha}}{\Delta\tau} \right) a_2^{ij}(r) \left\{ H\left[\tau_\alpha - r/c_1\right] - H\left[\tau_{\alpha-1} - r/c_1\right] \right\} dA$$

$$+ \int_A \left(\frac{t_i^{\alpha} - t_i^{\alpha-1}}{\Delta\tau} \right) a_2^{ij}(r) (\tau_\alpha - r/c_1) \left\{ H\left[\tau_\alpha - r/c_1\right] - H\left[\tau_{\alpha-1} - r/c_1\right] \right\} dA$$

$$= \int_A \left(\frac{\tau_\alpha - \tau_{\alpha-1} - r/c_1}{\Delta\tau} t_i^{\alpha} + \frac{r/c_1}{\Delta\tau} t_i^{\alpha-1} \right) a_2^{ij}(r) \left\{ H[\tau_\alpha - r/c_1] - H[\tau_{\alpha-1} - r/c_1] \right\} dA$$

$$= \int_A \left\{ \left(1 - \frac{r}{c_1 \Delta\tau}\right) t_i^{\alpha} + \frac{r}{c_1 \Delta\tau} t_i^{\alpha-1} \right\} a_2^{ij}(r) \left\{ H\left[\tau_\alpha - r/c_1\right] - H\left[\tau_{\alpha-1} - r/c_1\right] \right\} dA \qquad \textbf{(D.12)}$$

c. Third term Finally the third term of equation (42) is given by a similar to equation (D.12) relation, with the wave velocity c_1 replaced by c_2.

Integration of traction kernel

For the matrix [F]:
It is

$$S_2 = \sum_{\alpha=1}^{k} \left\{ \int_{\tau_{\alpha-1}}^{\tau_\alpha} \left[\int_A \left(F_{ij} u_i \right) dA \right] d\tau \right\}$$

$$= \sum_{\alpha=1}^{k} \int_A \left\{ \int_{\tau_{\alpha-1}}^{\tau_\alpha} \left[F_{ij}(\tau_\alpha - \tau) \right] \left[\left(\frac{\tau_\alpha - \tau}{\Delta\tau} \right) u_i^{\alpha-1} + \left(\frac{\tau - \tau_{\alpha-1}}{\Delta\tau} \right) u_i^{\alpha} \right] \varphi_\alpha(\tau) d\tau \right\} dA \qquad \textbf{(D.13)}$$

replacing F_{ij} from equation (D.21), the term in the summation becomes

$$
\begin{aligned}
S_{22} &= \int_A \left\{ \int_{\tau_{\alpha-1}}^{\tau_\alpha} \left[F_{ij}(\tau_\alpha - \tau)\right] \left[\left(\frac{\tau_\alpha - \tau}{\Delta\tau} \right) u_i^{\alpha-1} + \left(\frac{\tau - \tau_{\alpha-1}}{\Delta\tau} \right) u_i^{\alpha} \right] \varphi_\alpha(\tau)\, d\tau \right\} dA \\
&= \int_A \left\{ \int_{\tau_{\alpha-1}}^{\tau_\alpha} b_1^{ij}(r) \left[\int_{c_1^{-1}}^{c_2^{-1}} \delta(\tau_\alpha - \lambda r - \tau)\lambda\, d\lambda \right] \left[\left(\frac{\tau_\alpha - \tau}{\Delta\tau} \right) u_i^{\alpha-1} + \left(\frac{\tau - \tau_{\alpha-1}}{\Delta\tau} \right) u_i^{\alpha} \right] \varphi_\alpha(\tau)\, d\tau \right\} dA \\
&\quad + \int_A \left\{ \int_{\tau_{\alpha-1}}^{\tau_\alpha} b_2^{ij}(r)\, \delta(\tau_\alpha - r/c_1 - \tau) \left[\left(\frac{\tau_\alpha - \tau}{\Delta\tau} \right) u_i^{\alpha-1} + \left(\frac{\tau - \tau_{\alpha-1}}{\Delta\tau} \right) u_i^{\alpha} \right] \varphi_\alpha(\tau)\, d\tau \right\} dA \\
&\quad + \int_A \left\{ \int_{\tau_{\alpha-1}}^{\tau_\alpha} b_3^{ij}(r)\, \delta(\tau_\alpha - r/c_2 - \tau) \left[\left(\frac{\tau_\alpha - \tau}{\Delta\tau} \right) u_i^{\alpha-1} + \left(\frac{\tau - \tau_{\alpha-1}}{\Delta\tau} \right) u_i^{\alpha} \right] \varphi_\alpha(\tau)\, d\tau \right\} dA \\
&\quad + \int_A \left\{ \int_{\tau_{\alpha-1}}^{\tau_\alpha} b_4^{ij}(r)\, \dot{\delta}(\tau_\alpha - r/c_1 - \tau)\, u_i(\tau)\, d\tau \right\} dA \\
&\quad + \int_A \left\{ \int_{\tau_{\alpha-1}}^{\tau_\alpha} b_5^{ij}(r)\, \dot{\delta}(\tau_\alpha - r/c_2 - \tau)\, u_i(\tau)\, d\tau \right\} dA
\end{aligned}
\qquad \textbf{(D.14)}
$$

The first three terms of the second part of equation (D.14) are given by expressions similar to the previous case of displacement kernel. The last two expressions, where the derivative of Delta function is involved, are evaluated as
in equation (C.26).

APPENDIX E

STRESS AND AMPLITUDE REFLECTION COEFFICIENTS

This Appendix deals with the mathematical analysis for the derivation of stress and amplitude reflection coefficients.

1. REFLECTION OF P WAVES

An incident P wave, striking on the free surface, is reflecting as P and SV wave (Figure 6.1). A local co-ordinate system is used with x axis parallel to the unit vector normal to the free surface line and direction to the right hand, travelling from the first defined node of the free line in the mesh, to the second. The y axis is tangent to the free surface line and at right angles to the x axis.

The angles of the normal unit vector with the x axis of the general co-ordinate system and the angle of incident P, reflected P and reflected SV waves are

$$\theta_{\nu}\ ,\quad \theta^{\prime}_{\Pi}\ ,\quad \theta^{\rho}_{\Pi}\ ,\quad \theta^{\rho}_{\Sigma\varsigma} \tag{E.1}$$

respectively.

The equations of a harmonic wave are used in order to extract the reflection stress coefficients. The equations in the local co-ordinate system are for the incident P wave:

$$
\begin{aligned}
u_1^{(0)} &= -A_0 \cos(\theta_P^i)\ \varphi_0 \\
u_2^{(0)} &= -A_0 \sin(\theta_P^i)\ \varphi_0 \\
\textit{where} \\
\varphi_0 &= \exp[ik_0(x_1\cos\theta_P^i + x_2\sin\theta_P^i - c_1 t)]
\end{aligned}
\qquad \textbf{(E.2)}
$$

for the reflected P wave:

$$
\begin{aligned}
u_1^{(1)} &= A_1 \cos(\theta_P^r)\ \varphi_1 \\
u_2^{(1)} &= A_1 \sin(\theta_P^r)\ \varphi_1 \\
\textit{where} \\
\varphi_1 &= \exp[ik_1(x_1\cos\theta_P^r + x_2\sin\theta_P^r - c_1 t)]
\end{aligned}
\qquad \textbf{(E.3)}
$$

and for the reflected SV wave:

$$
\begin{aligned}
u_1^{(2)} &= -A_2 \sin(\theta_S^r)\ \varphi_2 \\
u_2^{(2)} &= A_2 \cos(\theta_S^r)\ \varphi_2 \\
\textit{where} \\
\varphi_2 &= \exp[ik_2(x_1\cos\theta_S^r + x_2\sin\theta_S^r - c_2 t)]
\end{aligned}
\qquad \textbf{(E.4)}
$$

differentiating the above equations with respect to time results in for the incident P wave

$$
\begin{aligned}
u_{1,1}^{(0)} &= -A_0 \cos^2(\theta_P^i)\ ik_0\varphi_0 \\
u_{1,2}^{(0)} &= -A_0 \sin(\theta_P^i)\cos(\theta_P^i)\ ik_0\varphi_0 \\
u_{2,1}^{(0)} &= -A_0 \sin(\theta_P^i)\cos(\theta_P^i)\ ik_0\varphi_0 \\
u_{2,2}^{(0)} &= -A_0 \sin^2(\theta_P^i)\ ik_0\varphi_0
\end{aligned}
\qquad \textbf{(E.5)}
$$

for the reflected P wave

$$
\begin{aligned}
u_{1,1}^{(1)} &= A_1 \cos^2(\theta_P^r)\, ik_1\varphi_1 \\
u_{1,2}^{(1)} &= A_1 \sin(\theta_P^r)\cos(\theta_P^r)\, ik_1\varphi_1 \\
u_{2,1}^{(1)} &= A_1 \sin(\theta_P^r)\cos(\theta_P^r)\, ik_1\varphi_1 \\
u_{2,2}^{(1)} &= A_1 \sin^2(\theta_P^r)\, ik_1\varphi_1
\end{aligned}
\qquad \textbf{(E.6)}
$$

and for the reflected SV wave:

$$
\begin{aligned}
u_{1,1}^{(2)} &= -A_2 \sin(\theta_S^r)\cos(\theta_S^r)\, ik_2\varphi_2 \\
u_{1,2}^{(2)} &= -A_2 \sin^2(\theta_S^r)\, ik_2\varphi_2 \\
u_{2,1}^{(2)} &= A_2 \cos^2(\theta_S^r)\, ik_2\varphi_2 \\
u_{2,2}^{(2)} &= A_2 \sin(\theta_S^r)\cos(\theta_S^r)\, ik_2\varphi_2
\end{aligned}
\qquad \textbf{(E.7)}
$$

the constitutive equations are

$$
\begin{aligned}
\sigma_{11} &= (\lambda+2\mu)u_{1,1} + \lambda u_{2,2} \\
\sigma_{22} &= \lambda u_{1,1} + (\lambda+2\mu)u_{2,2} \\
\sigma_{12} &= \mu(u_{1,2} + u_{2,1})
\end{aligned}
\qquad \textbf{(E.8)}
$$

substituting from (E.5),(E.6) and (E.7) in (E.8) gives
for the incident P wave

$$
\begin{aligned}
\sigma_{11}^{(0)} &= -A_0(\lambda + 2\mu \cos^2(\theta_P^i))\, ik_0\varphi_0 \\
\sigma_{22}^{(0)} &= -A_0(\lambda + 2\mu \sin^2(\theta_P^i))\, ik_0\varphi_0 \\
\sigma_{12}^{(0)} &= -A_0 \mu \sin(2\theta_P^i)\, ik_0\varphi_0
\end{aligned}
\qquad \textbf{(E.9)}
$$

for the reflected P wave

$$\sigma_{11}^{(1)} = A_1(\lambda + 2\mu \cos^2(\theta_P^r))\, ik_1\varphi_1$$
$$\sigma_{22}^{(1)} = A_1(\lambda + 2\mu \sin^2(\theta_P^r))\, ik_1\varphi_1$$
$$\sigma_{12}^{(1)} = A_1\, \mu \sin(2\theta_P^r)\, ik_1\varphi_1 \qquad \textbf{(E.10)}$$

for the reflected SV wave

$$\sigma_{11}^{(2)} = -A_2\, \mu \sin(2\theta_S^r)\, ik_2\varphi_2$$
$$\sigma_{22}^{(2)} = A_2\, \mu \sin(2\theta_S^r)\, ik_2\varphi_2$$
$$\sigma_{12}^{(2)} = A_2\, \mu \cos(2\theta_S^r)\, ik_2\varphi_2 \qquad \textbf{(E.11)}$$

also the following relations are valid

$$k_1 = k_0\,, \qquad \theta_1 = \theta_0$$
$$k_0 c_1 = k_1 c_1 = k_2 c_2 = \omega \;\Rightarrow\; \frac{k_2}{k_0} = \frac{c_1}{c_2} = k$$
$$k = \frac{\sin(\theta_0)}{\sin(\theta_2)} = \frac{c_1}{c_2} = \sqrt{\frac{2(1-v)}{1-2v}} = \sqrt{\frac{\lambda+2\mu}{\mu}} \qquad \textbf{(E.12)}$$

$$\frac{\lambda}{\mu} + 2\cos^2(\theta_P^i) = k^2\cos(2\theta_S^r) \qquad \textbf{(E.13)}$$

and

incident P, reflected S wave:

*case*1: *case*2:

$$\theta_P^i = \pi - \theta_0 \qquad \theta_P^i = \pi + \theta_0$$

$$\theta_S^r = \pi + \theta_2 \qquad \theta_S^r = \pi - \theta_2$$

both cases:

$$k = \frac{\sin(\theta_0)}{\sin(\theta_2)} = -\frac{\sin(\theta_P^i)}{\sin(\theta_S^r)} \tag{E.14}$$

and

incident P, reflected P wave:

*case*1: *case*2:

$$\theta_P^i = \pi - \theta_0 \qquad \theta_P^i = \pi + \theta_0$$

$$\theta_P^r = \pi + \theta_2 \qquad \theta_P^r = \pi - \theta_2$$

both cases:

$$\cos(\theta_P^i) = \cos(\theta_P^r) = -\cos(\theta_0)$$

$$\sin^2(\theta_P^r) = \sin^2(\theta_P^i) = \sin^2(\theta_0)$$

$$\sin(\theta_P^r) = -\sin(\theta_P^i) \tag{E.15}$$

$$\sin(2\theta_P^i) = -\sin(2\theta_P^r)$$

Eor the ground free surface, the stresses in the normal direction to the surface line and the shear stresses in the local system must be zero, so

$$\sum \sigma_{11} = 0 , \qquad \sum \sigma_{12} = 0 \tag{E.16}$$

Consequently, by substituting the stresses from Equations (E.9), (E.10) and (E.11) into Equations (E.16) and taking into account Equations (E.12), (E.13), (E.14) and (E.15) results finally to the following expressions for the ratio of the amplitudes of the reflected P to the incident P wave :

$$\frac{A_1}{A_0} = \frac{k^2\cos^2(2\theta_S^r) + \sin(2\theta_P^i)\sin(2\theta_S^r)}{k^2\cos^2(2\theta_S^r) + \sin(2\theta_P^r)\sin(2\theta_S^r)} \frac{\varphi_0}{\varphi_1} \qquad \textbf{(E.17)}$$

and for the ratio of the amplitudes of the reflected SV to the incident P wave :

$$\frac{A_2}{A_0} = \frac{-2k\,\sin(2\theta_S^r)\cos(2\theta_P^r)}{k^2\cos^2(2\theta_S^r) + \sin(2\theta_P^r)\sin(2\theta_S^r)} \frac{\varphi_0}{\varphi_2} \qquad \textbf{(E.18)}$$

The following stress reflection coefficients result from Equations (9),(10) and (11) in view of Equations (E.12), (E.13), (E.14) and (E.15) giving :
for the normal to the free surface stress

$$\frac{\sigma_{11}^{(1)}}{\sigma_{11}^{(0)}} = -\frac{A_1}{A_0}\frac{\varphi_1}{\varphi_0}$$

$$\frac{\sigma_{11}^{(2)}}{\sigma_{11}^{(0)}} = \frac{k\mu\,\sin(2\theta_S^r)}{\lambda+2\mu\,\cos^2(\theta_P^i)}\frac{A_2}{A_0}\frac{\varphi_2}{\varphi_0} \qquad \textbf{(E.19)}$$

for the tangent to the free surface stress

$$\frac{\sigma_{22}^{(1)}}{\sigma_{22}^{(0)}} = -\frac{A_1}{A_0}\frac{\varphi_1}{\varphi_0}$$

$$\frac{\sigma_{22}^{(2)}}{\sigma_{22}^{(0)}} = \frac{-k\mu\,\sin(2\theta_S^r)}{\lambda+2\mu\,\cos^2(\theta_P^i)}\frac{A_2}{A_0}\frac{\varphi_2}{\varphi_0} \qquad \textbf{(E.20)}$$

and for the shear stress in the local system

$$\frac{\sigma_{12}^{(1)}}{\sigma_{12}^{(0)}} = \frac{-\sin(2\theta_P^r)}{\sin(2\theta_P^i)}\frac{A_1}{A_0}\frac{\varphi_1}{\varphi_0}$$

$$\frac{\sigma_{12}^{(2)}}{\sigma_{12}^{(0)}} = \frac{-k\,\cos(2\theta_S^r)}{\sin(2\theta_P^i)}\frac{A_2}{A_0}\frac{\varphi_2}{\varphi_0} \qquad \textbf{(E.21)}$$

2. REFLECTION OF SV WAVE

An incident SV wave, striking on the free surface of the ground, is reflecting as (Figure 6.2) either P or SV wave, depending on the magnitude of angle of incidence according to the critical angle (Figure 6.14).

The local co-ordinate system is used once more here.

The angles of the normal unit vector with the x axis of the general co-ordinate system and the angle of incident P, reflected P and reflected SV waves are

$$\theta_n \ , \quad \theta_S^i \ , \quad \theta_P^r \ , \quad \theta_{SV}^r \tag{E.22}$$

respectively.

The equations of a harmonic wave are used again in order to extract the reflection stress coefficients. The equations in the local co-ordinate system are
for the incident SV wave:

$$u_1^{(0)} = A_0 \sin(\theta_S^i)\ \varphi_0$$
$$u_2^{(0)} = -A_0 \cos(\theta_S^i)\ \varphi_0$$

where

$$\varphi_0 = \exp[ik_2(x_1\cos\theta_S^i + x_2\sin\theta_S^i - c_2 t)] \tag{E.23}$$

for the reflected P wave:

$$u_1^{(1)} = A_1 \cos(\theta_P^r)\ \varphi_1$$
$$u_2^{(1)} = A_1 \sin(\theta_P^r)\ \varphi_1$$

where

$$\varphi_1 = \exp[ik_1(x_1\cos\theta_P^r + x_2\sin\theta_P^r - c_1 t)] \tag{E.24}$$

and for the reflected SV wave:

$$u_1^{(2)} = -A_2 \sin(\theta_S^r)\ \varphi_2$$
$$u_2^{(2)} = A_2 \cos(\theta_S^r)\ \varphi_2 \qquad \textbf{(E.25)}$$

where

$$\varphi_2 = \exp[ik_2(x_1\cos\theta_S^r + x_2\sin\theta_S^r - c_2 t)]$$

differentiating the above equations with respect to time gives for the incident SV wave

$$u_{1,1}^{(0)} = A_0 \sin(\theta_S^i)\cos(\theta_S^i)\ ik_2\varphi_0$$
$$u_{1,2}^{(0)} = A_0 \sin^2(\theta_S^i)\ ik_2\varphi_0$$
$$u_{2,1}^{(0)} = -A_0 \cos^2(\theta_S^i)\ ik_2\varphi_0$$
$$u_{2,2}^{(0)} = -A_0 \sin(\theta_S^i)\cos(\theta_S^i)\ ik_2\varphi_0 \qquad \textbf{(E.26)}$$

for the reflected P wave

$$u_{1,1}^{(1)} = A_1 \cos^2(\theta_P^r)\ ik_1\varphi_1$$
$$u_{1,2}^{(1)} = A_1 \sin(\theta_P^r)\cos(\theta_P^r)\ ik_1\varphi_1$$
$$u_{2,1}^{(1)} = A_1 \sin(\theta_P^r)\cos(\theta_P^r)\ ik_1\varphi_1$$
$$u_{2,2}^{(1)} = A_1 \sin^2(\theta_P^r)\ ik_1\varphi_1 \qquad \textbf{(E.27)}$$

and for the reflected SV wave:

$$u_{1,1}^{(2)} = -A_2 \sin(\theta_S^r)\cos(\theta_S^r)\ ik_2\varphi_2$$
$$u_{1,2}^{(2)} = -A_2 \sin^2(\theta_S^r)\ ik_2\varphi_2$$
$$u_{2,1}^{(2)} = A_2 \cos^2(\theta_S^r)\ ik_2\varphi_2$$
$$u_{2,2}^{(2)} = A_2 \sin(\theta_S^r)\cos(\theta_S^r)\ ik_2\varphi_2 \qquad \textbf{(E.28)}$$

applying the constitutive equations and substituting from (E.26), (E.27) and (E.28) in (E.8) results

for the incident P wave

$$\sigma_{11}^{(0)} = A_0\mu \sin(2\theta_S^i))\ ik_2\varphi_0$$
$$\sigma_{22}^{(0)} = -A_0\mu \sin(2\theta_S^i)\ ik_2\varphi_0$$
$$\sigma_{12}^{(0)} = -A_0\mu \cos(2\theta_S^i)\ ik_2\varphi_0 \qquad \textbf{(E.29)}$$

for the reflected P wave

$$\sigma_{11}^{(1)} = A_1(\lambda+2\mu \cos^2(\theta_P^r))\ ik_1\varphi_1$$
$$\sigma_{22}^{(1)} = A_1(\lambda+2\mu \sin^2(\theta_P^r))\ ik_1\varphi_1$$
$$\sigma_{12}^{(1)} = A_1\mu \sin(2\theta_P^r)\ ik_1\varphi_1 \qquad \textbf{(E.30)}$$

for the reflected SV wave

$$u_{1,1}^{(2)} = -A_2 \sin(\theta_S^r)\cos(\theta_S^r)\ ik_2\varphi_2$$
$$u_{1,2}^{(2)} = -A_2 \sin^2(\theta_S^r)\ ik_2\varphi_2$$
$$u_{2,1}^{(2)} = A_2 \cos^2(\theta_S^r)\ ik_2\varphi_2$$
$$u_{2,2}^{(2)} = A_2 \sin(\theta_S^r)\cos(\theta_S^r)\ ik_2\varphi_2 \qquad \textbf{(E.31)}$$

also the following relations are valid

$$k_2 = k_0\ , \quad \theta_2 = \theta_0$$
$$k_0c_2 = k_1c_1 = k_2c_2 = \omega \Rightarrow \frac{k_0}{k_1} = \frac{c_1}{c_2} = k$$
$$k = \frac{\sin(\theta_1)}{\sin(\theta_0)} = \frac{c_1}{c_2} = \sqrt{\frac{2(1-v)}{1+v}} = \sqrt{\frac{\lambda+2\mu}{\mu}} \qquad \textbf{(E.32)}$$

$$\frac{\lambda}{\mu} + 2\cos^2(\theta_P^r) = k^2\cos(2\theta_S^r) \qquad \textbf{(E.33)}$$

also

incident S, reflected P wave:

*case*1: *case*2:

$$\theta_S^i = \pi - \theta_0 \qquad \theta_S^i = \pi + \theta_0$$

$$\theta_P^r = \pi + \theta_2 \qquad \theta_P^r = \pi - \theta_2$$

both cases:

$$k = \frac{\sin(\theta_1)}{\sin(\theta_0)} = -\frac{\sin(\theta_P^r)}{\sin(\theta_S^i)} \tag{E.34}$$

and

incident S reflected S wave:

*case*1: *case*2:

$$\theta_S^i = \pi - \theta_0 \qquad \theta_S^i = \pi + \theta_0$$

$$\theta_S^r = \pi + \theta_2 \qquad \theta_S^r = \pi - \theta_2$$

both cases:

$$\cos(\theta_S^i) = \cos(\theta_S^r) = -\cos(\theta_0)$$

$$\sin^2(\theta_S^r) = \sin^2(\theta_S^i) = \sin^2(\theta_0)$$

$$\sin(\theta_S^r) = -\sin(\theta_S^i) \tag{E.35}$$

$$\cos(2\theta_S^i) = \cos(2\theta_S^r)$$

Applying equations (E.16) for the free ground surface, the ratio of the amplitudes of the reflected and incident wave can be canculated as :

$$\frac{A_1}{A_0} = \frac{k\,\sin(4\theta_S^r)}{k^2\cos^2(2\theta_S^r) + \sin(2\theta_P^r)\sin(2\theta_S^r)}\,\frac{\varphi_0}{\varphi_1} \tag{E.36}$$

and the ratio of the amplitudes of the reflected SV to the incident P wave may be expressed as :

$$\frac{A_2}{A_0} = \frac{k^2\cos^2(2\theta_S^r) + \sin(2\theta_S^i)\sin(2\theta_P^r)}{k^2\cos^2(2\theta_S^r) + \sin(2\theta_P^r)\sin(2\theta_S^r)}\,\frac{\varphi_0}{\varphi_2} \tag{E.37}$$

the stress reflection coefficients are

for the normal to the free surface stress

$$\frac{\sigma_{11}^{(1)}}{\sigma_{11}^{(0)}} = \frac{\lambda+2\mu\ \cos^2(\theta_P^r)}{k\mu\ \sin(2\theta_S^i)}\frac{A_1}{A_0}\frac{\varphi_1}{\varphi_0}$$
$$\frac{\sigma_{11}^{(2)}}{\sigma_{11}^{(0)}} = \frac{A_2}{A_0}\frac{\varphi_2}{\varphi_0} \qquad \textbf{(E.38)}$$

for the tangent to the free surface stress

$$\frac{\sigma_{22}^{(1)}}{\sigma_{22}^{(0)}} = -\frac{\lambda+2\mu\ \sin^2(\theta_P^r)}{k\mu\ \sin)(2\theta_S^i)}\frac{A_1}{A_0}\frac{\varphi_1}{\varphi_0}$$
$$\frac{\sigma_{22}^{(2)}}{\sigma_{22}^{(0)}} = \frac{A_2}{A_0}\frac{\varphi_2}{\varphi_0} \qquad \textbf{(E.39)}$$

and for the shear stress in the local system

$$\frac{\sigma_{12}^{(1)}}{\sigma_{12}^{(0)}} = \frac{-\sin(2\theta_P^r)}{k\ \cos(2\theta_S^i)}\frac{A_1}{A_0}\frac{\varphi_1}{\varphi_0}$$
$$\frac{\sigma_{12}^{(2)}}{\sigma_{12}^{(0)}} = -\frac{A_2}{A_0}\frac{\varphi_2}{\varphi_0} \qquad \textbf{(E.40)}$$

APPENDIX F

USERS GUIDE

F.1 INTRODUCTION

Useful instructions are given here, for running the BEM package developed under the name "DYNAMICS".

The package consists of three main parts. The preprocessing, the processing and the postprocessing phase. These parts are independent programs that can be run either through a main guide program or independently. The main guide program starts by giving the command DYNAMICS and gives the ability to the user to select the desired phase from a menu on the screen.

The independent phases are the following :

a. Preprocessing phase : prebem
b. Processing phase for elastodynamic Laplace domain : bemlapl
c. Processing phase for elastodynamic Time domain : bemtim
d. Processing phase for multi-zonal elastodynamic Laplace domain : bemz
e. Processing phase for elastostatic case : elasim
f. Postprocessing phase : postbem

F.2 PREPROCESSING

Running this phase, the program asks for the

Date:
Time of run:
Name:
Input data filename:

The date, time and input data file are not obligatory.

The Input data file is necessary to be given. Its name may be any legal DOS file

name, of eight characters long. It is given without any extension, and the program automatically adds the extension .DAT. For instance for datafile given MYFILE, the program supposes that the data are held in the file MYFILE.DAT. This input file is an ASCII format file, where the data of the problem are kept. It should be created by any editor in ASCII format and saved in the current directory e.g. MYFILE.DAT. The way of creation of this file is explained in the following.

The program automatically appends to the given name, the following extensions

MYFILE.DAT : for the input file that contains the data

MYFILE.PRE : for the output file of preprocessing

MYFILE.PRO : for the output file of processing

MYFILE.POS : for the output file of postprocessing

During the running of the different phases, useful information and results for the user are placed in these files, which are essentially the printing files. Furthermore, the file MYFILE.PRE, will be the output printing file of preprocessing, that contains useful information of the input data file, error messages and results of the preprocessing phase. It is here that one may check first, if the program is not running properly.

In this phase other files are created also, that are needed for reading data in the processing and postprocessing phase. These files are overwritten in every run and are for internal use of the program :

for Laplace domain (elastodynamic case) the files are

BEMLTDAT : datafile

BEMLTTRA : data of incident tractions on the boundaries in x and y global co-ordinates (opposite sign of incoming) produced by subroutine DATALAP

BEMLTSTR : data of initial hoop stresses, used in subroutine HOOP of the BEM main program.

for time domain (elastodynamic case) the files are

BEMTDDAT : datafile

BEMTDTRA : data of incident tractions on the boundaries, in x and y global co-ordinates (opposite sign of incoming) produced by subroutine DATATIM

BEMTDSTR : data of initial hoop stresses, used in subroutine HOOP of

the BEM main program.

for Elastostatic case the files are

BEMELDAT : datafile

BEMELTRA : data of incident tractions on the boundaries, in x and y global co-ordinates (opposite sign of incoming) produced by subroutine DATATIM.

BEMELSTR : data of initial hoop stresses, used in subroutine HOOP of the BEM main program.

for all the cases

POSTDAT : file for writing data plotting data used in processing and postprocessing.

Furthermore, temporarily files are created for storing data, that are automatically erased at the end of the phase.

F.3 PROCESSING

The results of processing phase are kept in the file with the extension PRO.

Furthermore, in this phase another file is created for reading in the postprocessing phase, and is for internal only use of the program

for Laplace domain (elastodynamic case)

BEMLTHOO

for Time domain (elastodynamic case)

BEMTDHOO

for Elastostatic case

BEMELDAT

Also many temporarily files are created for storing data, that are being erased at the end of the phase.

F.4 POSTPROCESSING

In the postprocessing phase the results are being stored in the file with extension

POS.

If the plots for other nodes, or other kind of plots are needed in additional to those that have been given in the preprocessing data file, it is not necessary to run the processing phase again. Simply the node numbers or the kind of plots can be changed to the desired numbers, re-run the preprocessing and then run the postprocessing phase directly without passing through the processing phase again.

The same is valid for the case where the plots for more quantities is desired, e.g. if the user originally had declared in the data file the tangential stresses only (number 12 of PLOTS) and following the running of the processing and the postprocessing, it was decided to have the plots in the normal direction (number 11), and in the 3nd direction (number 14). In this case it is only necessary to change these plot numbers to the desired values in the data file with extension DAT and re-run the preprocessing immediately followed by running the postprocessing.

Finally, a file with the name THIRDGR is created, which contains the results for creating charts with a third package (e.g. Harvard Graphics, Excel, etc.).

F.5 RESTRICTIONS OF THE PROGRAM

Maximum number of boundaries: 20
Max number of segments in combined shapes: 10

F.6 INPUT DATA FILE

The following instructions are given for the creation of the input data file used in the preprocessing program. Every module of the input file, is introduced with an example, and then useful comments are following. For instance, a data file should look like this:

```
TITLE
2D L.D., Circle, Infinity, Plane Wave
TYPE.PROBLEM
DOMAIN ANALYSIS
  1       1
SECTION.TYPE
SHAPE ELEMENTS GEOMETRY
  1       8      5.0
MATERIAL.PROPERTIES
   E     NU   RO
20000.0 0.25  2.55
STEP.NUMBER
   20
TIME.TOTAL
  0.015



LOAD.STRESSES
PLANE.WAVE
LONGITUDINAL.STRESS TRANSVERSE.SHEAR ANGLE
 -10.0                 0.0         225.0
PLOTS
2 4 12 19 20 28 29 30
NODE.PLOTS
 15 13 11 9 7
END
```

The data are introduced in categories (modules) and sub-categories, e.g. in the module

```
LOAD.STRESSES
PLANE.WAVE
LONGITUDINAL.STRESS TRANSVERSE.SHEAR ANGLE
 -10.0                 0.0         225.0
```

where loading stresses are introduced, the main category (module) is LOAD.STRESSES, sub-category is PLANE.WAVE and finally the titles and the corresponding data are introduced

```
LONGITUDINAL.STRESS TRANSVERSE.SHEAR ANGLE
 -10.0                0.0         225.0
```

for simplifying the introduction of the titles and sub-titles, only the first four first letters are significant e.g. LONGITUDINAL.STRESS could introduced as LONG.

F.6.1. MODULE TITLE

Example

```
TITLE
Two-dimensional transformed Laplace domain
```

Explanations

In this module, the general title of the problem is declared by the given sub-titles, which are :

TITLE : A short title is provided up to 80 characters long.

Comments

This module must be the first in the data file

F.6.2. MODULE TYPE.PROBLEM

Example

```
TYPE.PROBLEM
 DOMAIN  ANALYSIS  UNITS
   1       1       1
```

Explanations

In this module, some general data of the problem are declared by the given sub-titles, which are :

DOMAIN : An integer value that represents :
- 1 Laplace transform domain (transient elastodynamic case),
- 2 time domain (transient elastodynamic case)
- 3 elastostatic case
- 4 Laplace transform domain (steady state elastodynamic case, (frequency domain))

ANALYSIS : integer value :
- 1 plain strain
- 2 plain stress

UNITS : This sub-module is not necessary. If it is omitted the default value is 1. An integer value is given, that represents
- 1: SI International system units
- 2: Imperial system units
- 3: Any combination of units, responsibility of user. This is not valid in case of explosion point wave.

Comments

The program does not depend on the units, except of the part that refers to the explosion point wave. It is responsibility of the user to use any units, that should be in

agreement with the length units in module SECTION.TYPE and the material properties in module MATERIAL.PROPERTIES.

In case that an explosion point wave is analyzed, the units must be 1 or 2 only and introduced exactly in the two above sub-module are prescribed.

It is suggested to use the units 1 or 2.

Independently of the units used, the final results are always in SI system (tractions in MPa, displacements in m), even if the data were given in the Imperial system.

If data and results are needed in the Imperial system then the following must be done :

(i). The data are given in the Imperial system and in addition the density ρ should be multiplied by 1000.

(ii). In the sub-module units the default case 1 (SI system) is used.

F.6.3. Module SECTION.TYPE

Example

```
SECTION.TYPE
BOUNDARIES
   1
SHAPE REGION ELEMENTS  GEOMETRY
 2     2      4      3.0
```

Explanations

In the present module the geometrical characteristics and the discretization into elements is introduced.

BOUNDARIES: the total number of closed or open boundaries that form the whole mesh (e.g. 1 circle+1 square+1 free surface =3)

SHAPE REGION ELEMENTS YELEMENTS GEOMETRY

A brief explanation of the sub-titles with indicating examples follows :

a. Simple shapes

SHAPE : integer number

1 ellipsis(or circle)

2 square

3 rectangle

4 straight line

-n combined shape, n is number of segments

Ellipse-circle: nshape=1.

For the circle case : are not all the data necessary . From the geometrical data, only radius is necessary. As ratio must be given 1.0 or if it is omitted, automatically the program gives ratio=1.

Straight line: it is always of region =4 or 3 (open surface free or not), for simple shapes

REGION : integer number. It is not necessary and if it is omitted, the program automatically accepts the default value 1 (opening, void)

1: closed surface, external problem (cavity)

2: closed surface, internal problem (inclusion, solid)

3: open surface. A straight line in simple shape, is always an open surface

4: open free traction surface (e.g. ground level), which can reflect the incident waves. The reflection is taking place only if the reflection conditions are fulfilled. This is checked automatically from the program.

5: enclosing elements surface, this surface is in anticlockwise direction (encloses the mesh, like region=2)

6: enclosing elements surface, this surface is in clockwise direction (encloses the mesh, like region=1). For combined shapes the number of region is given only once, and is valid for all the segments of the region.

In case that zones exist : all the boundaries are given as clock-wise and then the program automatically (using the data in module ZONES) arranges the rest.

ELEMENTS : integer number that declares the total number of elements of the boundaries.

a. ellipse : total number of elements

b. square : number of elements of only one side

c. rectangle : number of elements of x-side

d. segments : total number of elements

The number of elements must be greater then 0. Only in the case of free surface (region=4), it can be 0. (This is useful when somebody wants to find the reflection wave from the free surface (solution by the laws of reflection of harmonic waves), but in the BEM solution we do not require discretisation of the free surface.

YELEMENTS : integer number, given only for rectangle, which represents the number of elements in one of y sides.

GEOMETRY :

The numbers that are given under this common header, determine the geometrical data of the mesh. All of them must be introduced, without omitting any of these numbers, even if it is 0. They can be omitted only if the rest of the numbers to the end of the line are 0, or if all the values accept the default numbers.

e.g. for a square:

```
SHAPE ELEMENTS GEOMETRY
 2      2      5.0 0.0 0.0 0.0 1.0 2.0
```

the zeros and the number 1.0 for the type of the elements is given, even if it is a default value. If instead of the numbers 1.0 (type of elements) and 2.0 (type of discontinuity at corners) it was used the default values 1.0 and 1.0, then the data could stop at number 5.0.

The numbers represent in sequence :

SEMIAX: length of semiaxis (radius) of an ellipse (circle)

RATIO : ratio of the other axis to semiaxis = b/SEMIAX. For the case of circle, is not necessary to be given or is given equal to 1.

SIDE : length of one side of a square

XSIDE, YSIDE: length of sides of a rectangle.

DIS : it is given only for a square or a rectangle. It is not necessary, if omitted the default case is 1. The values it takes are :

1: no discontinuities at the corners. The solution at the corners is not quite accurate. At the corners, which are points of discontinuities of tractions (but not displacements), the loading at a node is taken by interpolating between the neighbourhood nodes. At the corner node, the interpolation for the loading is performed at an imaginary element, which has as start node the neighbouring node on the one side of the rectangle, middle node the current node and end node the neighbouring node on the other side of the rectangle. The solution suffers from an inaccuracy, even if the hoop stresses in subroutine HOOP the calculations are performed accurate, for every existing real element. The advantage of this method, is that at the corners no duplicated nodes are used. For the usual structures that do not have sharp corners, this is accuracy is acceptable.

2: discontinuities at the corners. The solution exactly at the corners is accurate. At the corners duplicate nodes are inserted, so the interpolation for the calculation of the loading at the node is performed for every real (existing) element per side. The disadvantage of this method is the increase of the number of nodes by 1 for every corner (total increase 4 nodes). The zero distance between the duplicate nodes does not cause problems, because the processing program has the capability of dealing with singular cases.

ATTENTION: Only nodes on the free surfaces are permitted to be duplicated. So in the case of two zones enclosed by square boundaries, only the corners on the inner free surface boundary are permitted to be duplicated. The corners at the outer boundary that forms the interface of the two zones, are not allowed to be duplicated.

XC,YC : co-ordinates of the centre of the shape, or the centre of an elliptical segment.

ROT : angle of rotation of the x local co-ordinate axis of the shape (or an elliptical segment) with the x global axis, measured in [degrees]

X0,Y0,XL,YL: co-ordinates of the start and end of the straight line segment

TYP : A real number that determines the type of straight line, used in simple shapes only, available for straight line, square and rectangle only. This sub-title is not necessary, since its absence, the default value 1 is used

(uniform size of elements in the line). It determines the size of elements in the line.

TYPX,TYPY : Real numbers, equivalent to TYP previous sub-title, which are used for a square or a rectangle for determining the type of lines of the 2 sides. (TYPX refers to the 1st and 3rd sides, TYPY to 2nd and 4th sides). If one of these numbers is 0, e.g. TYPY=0, then the value of introducing TYPY is calculated automatically from the program essentially about similar discretization near the corners for both sides that join at the corner. If both TYPX and TYPY are 0, then the default value 1 for both of them is accepted (equal elements).

Values of p=TYP:

1: uniform size of elements

```
|--.--|--.--|--.--|--.--|
 1   1  1  1  1  1  1  1
```

p>1: gradualy increasing elements, with ratio p

e.g. p=2.0:

```
|--.--|----.----|--------.--------|
 1  1   2   2      4         4
```

0<p<1: gradually decreasing elements, doubling

e.g. p=0.5:

```
|--------.--------|----.----|--.--|
    4         4      2     2   1  1
```

p<-1: gradually increasing-decreasing elements, with ratio p

e.g. p=-2.0:

```
|--.--|----.----|----.----|--.--|
 1  1   2   2     2    2   1  1
```

or

```
|--.--|----.----|--------.--------|----.----|--.--|
```

1 1 2 2 4 4 2 2 1 1

-1<p<0: gradualy decrising-incrising elements, with ratio p (p_1=1/p)

e.g. p=-0.5 (p_1=1/0.5=2):

|----.----|--.--|--.--|----.----|

2 2 1 1 1 1 2 2

or

|--------.--------|----.----|--.--|----.----|--------.--------|

4 4 2 2 1 1 2 2 4 4

also p=-0.8 (p_1=1/0.8=1.25):

|----.----|---.---|---.---|----.----|

1.25 1.25 1 1 1 1 1.25 1.25

and p=-0.2 (p_1=1/0.2=5):

|-----.-----|-.-|-.-|-----.-----|

5 5 1 1 1 1 5 5

TH1,TH2: angles of start and end of the elliptical (circular) segment with the x axis measured in [degrees]

b. Combined shapes

SHAPE : Here a negative integer number is given, which represents the total number of segments (individual shapes) that form the whole mesh. Up to 10 segments are allowed by the program.

REGION : the number of region, as in the simple case, common for all the segments. The rest of the given numbers are :

An integer (shape no.): This number is given without sub-title and determines the kind of shape of the segment. The following integer numbers are only accepted :

1 elliptical (or circular) segment

4 straight line segment

With these two types of segments, the combined shape is build.

ELEMENTS : integer number that declares the total number of elements of the segment.

GEOMETRY :

The numbers that are given under this common header, determine the geometrical data of the mesh. All of them must be introduced, without omitting any of these numbers, even if it is 0. They can be omitted only if the rest of the numbers to the end of the line are 0, or if all the values accept the default numbers.

Straight line case: four real numbers are given

X0,Y0,Xl,Yl,TYP : co-ordinates of the start and end of the line and TYP : type of straight line, as in simple shapes case.

Circular-elliptical segment case:

XC,YC,TH1,TH2,SEMIAX,RATIO,ROT :

XC,YC : Co-ordinates of segment's centre th1,th2: start and end angles of the segment given in [degrees]

SEMIAX : length of semiaxis (radius) of an ellipse (circle)

RATIO : ratio of the other axis to semiaxis=b/SEMIAX, for circle is not necessary or is given value 1.

ROT : angle of rotation of the x local axis of the shape measured in [degrees]

F.6.4. Module INTERNAL.POINTS

Example

```
INTERNAL.POINTS
 3
 XI     YI
15.0  -10.0
18.0  -8.0
20.0  -5.5
```

Explanations

In this module the geometrical data of the internal points (not on the boundary surfaces) are determined. By defining these points, the displacements will be calculated in the processing phase and will be plotted in the post-processing phase at the same points.

INTERNAL.POINTS : total number of internal points for calculation of tractions and displacements.

XI,YI : Co-ordinates of the internal points. The number of the set of co-ordinates should be equal to the number of points defined under INTERNAL.POINTS title.

F.6.5. Module MATERIAL.PROPERTIES.

Example

```
MATERIAL.PROPERTIES
  E     NU  RO  VISC MODE
2.0E7 0.25 2.7  23.96 1
```

Explanations

Here the properties of the uniform material are stated precisely.

E : Young's modulus

NU : Poison's ratio between 0.0 and 0.50

RO : mass density

VISC: Coefficient of viscosity of the material (n). Usually it is given as percentage of the shear modulus
e.g. $n=f \bullet \mu$, $f=0.005$, $\mu=4792$ (MPa) , $n=0.005 \bullet 4792=23.96$
so the value 23.96 is the data for VISC

MODE: Model for viscosity, only for Laplace domain
1: Kelvin solid
2: Maxwell fluid

If zones of different material exist, the properties of the surrounding (outer to the zones) material is introduced here. The material characteristics for the zones are introduced in module ZONES.

Comments

Special attention should be paid to the units used. For the International system the

following units should be used for the data :

SI International system units:

E [MPa]

RHO [Mg/m3]

Imperial system

E [lb/in^2] or (psi)

RHO [lb/ft^3]

e.g.

SI International system units:

E=20,000 [MPa]

NU=0.25, RHO=2.7 [Mg/m3]

Imperial system

E=8,992,500 lb/in^2 (psi)

NU=0.25, RHO=150.0 [lb/ft^3]

Transformations of the RHO (ρ) units:

The following trasnformations are performed automatically from the program, and are presented only for clarification.

In order to keep the units of mass density RHO, compatible with the other quantities, the program automatically proceeds to the following changes, which takes place in subroutine MATERMOD :

SI International system units:

E=20,000 [MPa]

NU=0.25 RHO=2.7/1000=0.0027 [MPa•s^2/m^2]

Imperial system

E=8,992,500 lb/in^2 (psi)

NU=0.25, RHO=150.0•0.0157 [Mg/m3]

=150.0•0.0157/1000.0 [MPa•s^2/m^2]

In case that RHO is extracted from the formula

$$\chi_2 = \sqrt{\frac{\mu}{\rho}} \quad \rightarrow \quad \rho = \frac{\mu}{\chi_2^2} \tag{Γ.1}$$

in Imperial system the units for RHO (ρ) are [psi•s^2/in^2] and should be

multiplied by $32.2 \bullet 12^4$ to convert them to [lb/ft^3].

The final results of the program are :

SI system :

After the processing phase has finished, the results are saved in a file with units:

tractions or stresses in [MPa]

displacements in [m]

In the postprocessing phase, the results are presented in the final graphs with units:

tractions or stresses in [MPa]

displacements in [mm]

Imperial system :

After the processing phase :

tractions or stresses in [psi]

displacements in [in]

In the postprocessing phase, the final graphs is presentent using :

tractions or stresses in [psi]

displacements in [in]

F.6.6. Module STEP.NUMBER

Example

Laplace domain:

STEP.NUMBER

20 2 2 5

Time domain:

STEP.NUMBER

20

Explanations

STEP.NUMBER :

An integer number that determines the number of the discrete time steps that the total time interval of module TIME.TOTAL is separated

In Laplace domain three more integers are optionally given, that are needed for the Fast Fourier Transformation (FFT), with the following characteristics:

N1 integer should have the value 2

If N1,N2,N3 are these 3 integers their product should be such that

N1xN2xN3 = STEP.NUMBER

e.g. for STEP.NUMBER=20 can be N1=2, N2=2, N3=5 which results in N1xN2xN3=20

If N1,N2,N3 are not given in the data then the program collects automatically these numbers N1xN2xN3=STEP.NUMBER as following :

2x1x2=4, 2x1x3=6, 2x2x2=8, 2x1x5=10 2x2x3=12,
2x2x4=16, 2x3x3=18, 2x2x5=20, 2x3x4=24, 2x3x5=30,
2x3x6=36, 2x4x5=40, 2x4x6=48, 2x5x5=50, 2x4x7=56,
2x5x6=60, 2x4x8=64, 2x5x7=70, 2x5x8=80, 2x5x9=90,
2x5x10=100, 2x6x9=108, 2x6x10=120, 2x7x9=126, 2x7x10=140,
2x8x10=160,2x9x10=180,2x10x10=200

these integer multipliers N1,N2 and N3 are equal to the values for the jp1 in subroutine TRIGO of the bem program e.g. for nstep=20 then jp1(1)=2, jp1(2)=2 and jp1(3)=5

F.6.7. Module TIME.TOTAL

Example

```
 TIME.TOTAL
0.015
```

Explanations

TIME.TOTAL : Real number, defines the total time interval that is being examined measured in [s]. This interval is separated into the time steps that are declared in module STEP.NUMBER.

F.6.8. Module HOOP.STRESSES

Example

```
HOOP.STRESSES
TOTAL.POINTS POSITION.ELEMENTS
2          1 7
```

Explanations

Here, the desired elements for calculating the hoop stresses are defined. This module is not obligatory and if it is not present, the program calculates the hoop stresses at the middle and edge nodes of all the elements.

TOTAL.POINTS :

Integer number, defining the total points where the hoop stresses are desired.

POSITION.ELEMENTS :

Integer number that declares the elements for calculating the hoop stresses. The number of these elements must be determined manually, by numbering the elements of the mesh, starting from the element that the positive x local axis cuts the mesh, and numbering in clockwise(anticlockwise) direction for exterior (interior) problem.

F.6.9. Module PRESCRIBED.CONDITIONS

Example

PRESCRIBED.CONDITIONS
TOTAL.NODES POSITION.NODE AXIS.NUMBER CONDITION.CODE
0

Explanations

The boundary conditions are introduced here.
This module is optional and in caseif it is not prescribed, the default case 0 of all-tractions prescribed conditions is accepted.

TOTAL.NODES :

Integer number taking values :

0 : all-tractions prescribed conditions. The given loads in module LOADS should be tractions on all the nodes of the mesh. The rest values in the sub-modulus are not needed.

N : integer number greater than 0, defining the total nodes that the prescribed tractions or displacement conditions are applied.

POSITION.NODE :

Integer number defining the number of the node where the traction or displacement prescribed condition is applied. The number of the node corresponding to the mesh numbering must be found manually, by numbering the nodes of the mesh, starting from the node that the positive x local axis cuts the mesh, and numbering in clockwise(anticlockwise) direction for exterior(interior) problem, (the middle nodes of quadratic elements must be numbered too). Alternatively the pre-processing phase can be run first without defining position nodes, examine the plot of the mesh on the screen and then execute again by giving in the present module the node numbers.

AXIS.NUMBER :

Integer number defining the global x or y axis to which the prescribe condition is applied :

1 for x global axis

2 for y global axis

CONDITION.CODE :

Integer number, determining the type of the prescribed condition, which should be traction or displacement.

This sub-module is optional and if is not present, the default value 0 of the prescribed traction condition is accepted.

The available values are :

0: prescribed traction

1: prescribed displacement

F.6.10. Module LOADS

In this module, the applied on the nodes loading is defined. This loading can be a transient loading of the following nature :

a. Plane wave

b. Harmonic wave

c. Inside a boundary pressure

d. Pressure on a surface of a boundary

e. Traction on nodes

f. Explosion point wave

Only one wave at a time can be applied.

F.6.10.1 PLANE WAVE

Example

```
LOADS
PLANE.WAVE
LONGITUDINAL.STRESS  VERTICAL.STRESS  TRANSVERSE.SHEAR
ANGLE  XDISTANCE  YDISTANCE
-1000.0          0.0          0.0        0.0
FUNCTION TRAISE
  1
```

Explanations

Here the characteristics of a stress transient shock wave, propagating in a plane wavefront are prescribed. This is for example the wave resulting from an underground explosion at a point at infinity.
It is applicable to the Laplace and time domain as well.

LONGITUDINAL.STRESS :

A real number equal to the magnitude of the incident P stress σ_{11} in the direction of propagation. In the usual case, this stress is compressive (negative) and because of the nature of the shock wave (there is not enough time to expand the material laterally), this stress generates normal to the direction of propagation compressional stress. This shear is equal to $\sigma_{22}'=\sigma_{11}\nu/(1-\nu)$, where ν is Poisson's ratio, and is added to the actual vertical stresses σ_{22} defined in the next sub-module VERTICAL.STRESS.

VERTICAL.STRESS :

A real number equal to the vertical to the direction of propagation incident P stress σ_{22}. In case that this stress is negative, it also generates an additional to the direction of propagation longitudinal stress as previously equal to $\sigma_{11}'=\sigma_{22}\nu/(1-\nu)$ that is added to the actual longitudinal stress σ_{11} defined previously in sub-module LONGITUDINAL.STRESS.

TRANSVERSE.SHEAR :

A real number equal to the normal incident SV shear stress σ_{12}.

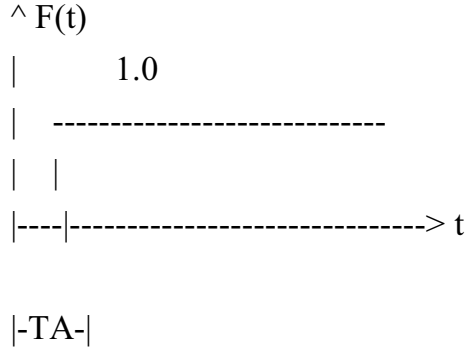
ANGLE :

The angle of attack of the axis of the incident wave (tale of the wave) with the x global co-ordinate axis, measured in degrees.

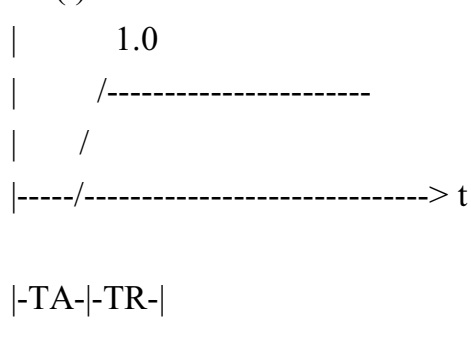
FUNCTION :

An integer number, determining the type of loading function. The following loading functions are available :

1 : HEAVISIDE STEP FUNCTION



2: FUNCTION OF LOADING



These loading functions are maintained constant during the total time period that is examined.

TRAISE :

Used only for FUNCTION=2. It is a real number equal to the raising time TR.

XDISTANCE, YDISTANCE :

The distance of the wavefront at time t=0, from the centre of the mesh is defined here.

This sub-module is optional, and if it is not present then the program automatically specifies a plane wavefront at the node of the mesh that it impinges (arrives) for time=0. This point is

for simple shapes :

circle : at a point on the periphery

ellipse: at a point on the periphery of a circle with radius the large axis square,

rectangle: at a point on the periphery of a circle with radius the diagonal.

for combined shapes:

at a point with co-ordinates PX,PY

If the sub-module XDISTANCE, YDISTANCE is specified then two real numbers are given which are the co-ordinates of the centre of wavefront at time t=0, corresponding the delays to the start measurement. If these are given, then the initial wavefront is at point (xdist,ydist) and is normal to the line that connects the centre of the mesh (xcen,ycen) with the point (xdist,ydist).

F.6.10.2. HARMONIC WAVE

Example

```
LOADS
HARMONIC.WAVE
FREQUENCY   LONGITUDINAL.STRESS   VERTICAL.STRESS
TRANSVERSE.SHEAR   ANGLE XDISTANCE YDISTANCE
 50.0     -1000.0        0.0        0.0        0.0      0.0     0.0
```

Explanations

This modules is used in Laplace domain only and refers to the harmonic wave

case.

FREQUENCY :

A real number equal to the frequency of the incident harmonic wave, measured in (Hz).

The rest of the sub-moduli are given as in the case of plane wave (section F.6.10.1).

F.6.10.3. INSIDE PRESSURE

Example

```
LOADS
INSIDE.PRESSURE
 2
BOUNDARY PRESSURE
     2          10.0
     3           8.0
```

Explanations

Here the pressure from inside a boundary is introduced. The pressure is applied and maintained constant.

INSIDE.PRESSURE:

The number of boundaries that internal pressure is applied.

BOUNDARY:

Number of boundary that the pressure applies. The number of boundaries is exactly as the one given in the SHAPE sub-modulus of modulus SECTION.TYPE (section F.6.3).

PRESSURE:

The pressure (MPa), that applies on the nodes, in the normal to the surface direction. It is positive for direction inward to the material normal direction e.g. pressure from inside a cavity (e.g. explosion from inside is positive).

F.6.10. 4 SURFACE PRESSURE

Example

```
LOADS
SURFACE.PRESSURE
     2
XPRESSURE YPRESSURE ELEMENTS
68.95          0.0        12 15
50.0          10.0         3  6
```

Explanations

The pressure on the surface of individual elements, is prescribed in this modulus.

SURFACE.PRESSURE:

An integer number that indicates the pressure cases. For every pressure case, an entry with the following sub-modulus must be given. By pressure cases, we do not mean the different pressures on the same element. On every element, only one pressure case is applied.

XPRESSURE:

The x co-ordinate of the pressure (MPa), that applies on the elements that follow (default value 0.0). The co-ordinate system used is referred to the general co-ordinate system.

YPRESSURE:

The y co-ordinate of the pressure (MPa), that applies on the elements that follow (default value 0.0).

ELEMENTS:

The elements that the pressure is applied. The numbering of the elements is the same with the element numbers of the mesh. A convenient way to find the elements numbering, is to run the preprocessing without the pressure modulus, determine the element numbers from the drawings and then execute again the preprocessing with the appropriate data for the present modulus.

F.6.10.5 TRACTION NODES

Example

```
TRACTION.NODES
     2
XPRESSURE YPRESSURE NODES
     10.0          5.0       11 12
     20.0          10.0       2  4
```

Explanations

In this modulus, the prescribed tractions on individual nodes is introduced.

TRACTION.NODES:

An integer number that indicates the traction cases. For every traction case, an entry with the following sub-modulus must be given. By traction cases, is not meant the different pressures on the same node. On every node, only one pressure case is applied.

XTRACTION:

The x co-ordinate of the traction (MPa), that applies on the nodes that follow (default value 0.0)

YTRACTION:

The y co-ordinate of the traction (MPa), that applies on the nodes that follow (default value 0.0)

NODES:

The nodes that the pressure is applied. The numbering of the nodes is the same with the node numbers of the mesh. A convenient way to find the numbering of the nodes, is to run the preprocessing without the pressure module, determine the node numbers from the drawings and then re-run the preprocessing with the appropriate data for the present modulus.

F.6.10.6 EXPLOSION-BLASTING

The incident loading of the nodes from an explosion at a point is introduced here.

Example

```
LOADS
EXPLOSION.POINT
FORMULA  XE   YE   W    B     F   N    N1  DECAY  SV    P
  1           5.5   0.0 250 160.0 1.0 2.75        1      1.0    0.0
```

Explanations

FORMULA:

An integer number expressing the type of formula used.

This is optional, and if is not found the default value 1 is accepted. The integer is :

1: cubic root formula

2: square root formula

3: other type formula

XE,YE:

Real numbers, are the co-ordinates of the centre of explosion in the global co-ordinate system measured in [m] or [ft]

W:

Real number, equivalent charge weight of the explosive, measured in [kgf] or [lbf] (force)

for bombs, this is the product of the gross weight, multiplied by the charge-to-weight ratio:

e.g. 500-lb GP bomb:

total gross weight : 520 lbf

charge-to-weight ratio : 51%

charge weight W = 520•0.51=265 lbf

B:

Real number, empirical coefficient of the formula. This is equal to the k_1 coefficient that is presented later.

This is optional, if is not given it is taken equal to

160 for FORMULA=1

2 for FORMULA=2

1.4 for FORMULA=3

F:

Real number, coupling factor.

This is optional. The default value given by the program is 1.0.

N:

Real number, is the attenuation coefficient.

This is optional. The default value given by the program is:

2.75 for FORMULA=1

1.57 for FORMULA=2

0.70 for FORMULA=3

N1:

Real number, coefficient needed only for FORMULA=3.

It optional. If in not given, then the following default value is accepted by the program :

1.50 for FORMULA=3

DECAY:

Integer, index for the decay of the stress wave with time.

This is optional. The default case given by the program is 1, with decay. The following values are available :

0: no decay with time, $P=P_0$

1: decay with time, $P=P_0 \bullet e^{-t/ta}$ where ta=arrival time of the wave.

SV:

Real number indicating that SV wave is generated from the explosion

(non-spherical charge). The number indicates also the ratio between SV and P wave e.g.

1.0 means that the SV wave is the same as the P wave

0.5 means that the SV wave is 0.5 times the P wave

This is optional and if is not given, then only P wave is taken (SV=0.0).

P:

Real number indicating that P wave is generated from the explosion (non-spherical charge). The number indicates also the ratio between SV and P wave e.g.

1.0 means that the P wave is the same as the P wave calculated from the formulae for cubic or square root

0.5 means that the P wave is 0.5 times the P wave calculated from the formulae for cubic or square root

If this is not given, then the default value is 1.0 P.

Comments

F.6.10.6.1 Explosion Formulae

There are three types of formulae, square root (usually for blasting), cubic root (usually for explosions) and other type of formulae. The square and cubic formulae, usually are used with the Imperial Units system. Here are examined the square and cubic root formulae that have original forms in the Imperial system, and other type of formulae that are given originally in the SI system.

The units system used, is determined in the module TYPE.PROBLEM, sub-module UNITS.

Important: the user has to give the coefficient $B=k_1$, in accordance to the following :

1. Cubic root formula

a. Given formula for the velocity in Imperial units :

$$\varsigma = \kappa\phi \left(\frac{P}{\Omega^{1/3}} \right)^{-\nu} \qquad (\Gamma.2)$$

where V[in/s] or [ft/s], R[ft], W[lbf] (force)

b. Extracted formula for the pressure

Imperial units :

$$\Pi = \kappa_1 \phi(\rho\chi) \left(\frac{P}{\Omega^{1/3}} \right)^{-\nu} \qquad (\Gamma.3)$$

where P[psi], ρc[pci/ft/s], R[ft], W[lbf] (force)

and k_1=k/12 in case that the formula used gives v[in/s]

=k in case that the formula used gives v[ft/s]

S.I. units :

$$\Pi = \kappa_3 \phi(\rho\chi) \left(\frac{P}{\Omega^{1/3}} \right)^{-\nu} \qquad (\Gamma.4)$$

where P[MPa], ρc[MPa•s/m], R[m], W[N] and

$$\kappa_3 = \frac{0.3048^{(\nu+1)}}{4.4482^{(\nu/3)}} \kappa_1 \qquad (\Gamma.5)$$

2. Square root formula

a. Given formula for the velocity in Imperial units :

$$\varsigma = \kappa\phi \left(\frac{P}{\Omega^{1/2}} \right)^{-\nu} \qquad (\Gamma.6)$$

where V[in/s] or [ft/s], R[ft], W[lbf] (force)

b. Extracted formula for the pressure

Imperial units :

$$\Pi = \kappa_1 \phi(\rho\chi) \left(\frac{P}{\Omega^{1/2}} \right)^{-\nu} \qquad (\Gamma.7)$$

where P[psi], ρc[pci/ft/s], R[ft], W[lbf] (force)

and k_1=k/12 in case that the formula used gives v[in/s]

=k in case that the formula used gives v[ft/s]

S.I. units :

$$\Pi = \kappa_3 \phi(\rho\chi) \left(\frac{P}{\Omega^{1/2}} \right)^{-\nu} \qquad (\Gamma.8)$$

where P[MPa], ρc[MPa•s/m], R[m], W[N] and

$$\kappa_3 = \frac{0.3048^{(\nu+1)}}{4.4482^{(\nu/2)}} \kappa_1 \qquad (\Gamma.9)$$

3. Other formulae

a. Given formula for the velocity in S.I. units

$$\varsigma = \kappa\phi \frac{\Omega^{\alpha}}{P^{\beta}} \qquad (\Gamma.10)$$

where v[mm/s] or [m/s], R[m], W[Kgf] (force)

b. Extracted formula for the pressure

S.I. units :

$$\Pi = \kappa_3 \phi(\rho\chi) \frac{\Omega^{\alpha}}{P^{\beta}} \qquad (\Gamma.11)$$

where P[MPa], ρc[MPa•s/m], R[m], W[N] and

$$\kappa_3 = 9.8066^{\alpha}\ \kappa_1 \qquad (\Gamma.12)$$

and k_1=1000/k in case that the formula used gives v[mm/s]

= 1/k in case that the formula used gives v[m/s]

In the above it is

f: coupling factor $0 \le f \le 1$

F.6.10.6.2 Particulars of the program

Assuming that one of the above formulae is used, and giving the type of formula used in FORMULA, and the type of units (given in module TYPE.PROBLEM, sub-module UNITS), the program automatically converts all the units in

P[MPa], pc[MPa•s/m], R[m], W[N]

by using the above square, cubic or other formula and the respective conversion factors for k_3.

It is the responsibility of the user to give the correct coefficients k_1, in sub-module $B=k_1$, as was explained above.

P: stress in radial direction, [MPa] or [psi]

Attention should be paid to the units used

SI international system:

XE,YE [m], W [Kgf (force)]

Imperial system :

XE,YE [ft], W [lbf (force)]

In the formulae given previously, R is the distance of the centre of explosion from the node of the mesh. This distance is calculated automatically from the program and is not given here.

More specifically

1. <u>Explosions from conventional weapons (FORMULA=1)</u>

The free-field stress wave generated by conventional weapons is

a. The formula for the velocity in Imperial units is Equation (F.2)

where V[ft/s], R[ft], W[lbf] (force)

b. Extracted formula for the pressure

Imperial units : Equation (F.3)

where P[psi], ρc[psi/ft/s], R[ft], W[lbf] (force)

and $B=k_1=k=160$ the default value, or any other value defined in sub-modulus B of modulus EXPLOSION.POINT.

S.I. units : Equation (F.4)

where P[Ma], ρc[MPa•s/m], R[m], W[N]

and k_3 is given by Equation (F.5)

2. Blasting (FORMULA=2)

B: This is the coefficient that the original formula uses for calculating the velocity. The default coefficient used is given by Dowding (1985): $v=25\bullet(R/W^{1/2})^{-n}$, where v[in/s]
so k=25 and $B=k_1=25/12=2.0$.

Other values of k usually used are :

for tunnels : k=2.7361 (n=0.5563, v[in/s]), so $B=k_1=2.7361/12=0.02$
for quarries : k=19.0976 (n=1.1967, v[in/s]), so $B=k_1=19.0976/12=1.5$
for pits:k=223.5916 (n=1.9078, v[in/s]),so $B=k_1=223.5916/12=19$

In the following, the calculation of the coefficients k_3 used in the above is demonstrated:

The velocity at a node by using the coefficient for tunnels is given by Equation (F.6)
where k=B for v[f/s], R[f], W[lbf] (force)
It is desirable v[m/s], R[m], W[kgf] and so another coefficient k_3 must be found, such that to satisfy Equation

$$\varsigma=\kappa_3\phi\left(\frac{P}{\Omega^{1/2}}\right)^{-\nu} \qquad (\Gamma.13)$$

where v[m/s], R[m], W[N].
So $[m/s]=K([m]/[N]^{1/2})^{-n}$
or $1/0.3048[in/s]=K\{(1/0.3048)^{-n}[f]/(1/4.4482)^{-n/2}[lbf]\}$
which results in

$K=4.4482^{n/2}/0.3048^{n+1}$

and finally the coefficient is

$k_3=k/K=k\bullet 0.3048^{n+1}/4.4482^{n/2}$

e.g. for k=B=2.7361, n=0.5563, this gives $k_3=0.0144$

3. Other type formulae (FORMULA=3)

B: For blasting, the default value used here, is taken
the Holberg and Persson (1980) for average Swidish bedrock

k=700, (a=0.70, b=1.50, v[mm/s]) so B=k_1=1000/700=1.4

This coefficient is transformed,as explained above, in the coefficient

$k_3=k_1 \bullet 9.8066^a=1.4 \bullet 9.8066^a$

where a, given in sub-module N

in order to perform all the calculations in the SI units.

N: coefficient for the power of W (a).

default value=0.70 (Holberg and Persson (1980) for average Swidish bedrock).

N1: coefficient for the power of R (b).

default value=1.50 (Holberg and Persson (1980) for average Swedish bedrock).

F.6.10.6.3 Acoustic impedance

The acoustic impedance ρc, is not given as input. It is calculated automatically from the program

ρ: mass density, is already given in module MATERIAL.PROPERTIES

c: seismic wave velocity. it is calculated automatically from the known values of ρ and E (given in module MATERIAL.PROPERTIES) by using the formulas:

$$\chi_1 = \sqrt{\frac{\lambda+2\mu}{\rho}}, \quad \chi_2 = \sqrt{\frac{\mu}{\rho}}$$
$$\mu = \frac{E}{2(1+\nu)}, \quad \lambda = \frac{\nu E}{(1+\nu)(1-2\nu)} \qquad (\Gamma.14)$$

where c_1 the dilatational and c_2 the shear wave velocity.

From the given values in module MATERIAL.PROPERTIES (in subroutine SECTMOD) the program calculates ρc as follows

S.I. units :

c [m/s], ρ [MPa•s^2/m^2], ρc [MPa•s/m]

Imperial units :

these units are converted to the S.I. units (in subroutine SECTMOD).

F.6.11 Module FREE.FIELD.STRESSES

Example

FREE.FIELD.STRESSES
1

Explanations

In this modulus the method of calculating the free field stresses in the case of existence of free surface of the ground, is prescribed.

An integer number is given as following:

0: default case. the incident field is applied on the surfaces of all the boundaries, including the free surface, and the solution is performed as in the case of the infinite regions (discretization of the free surface).

1: the incident field is reflected on the free surface, by the laws of reflection of harmonic waves, and the free field is calculated by adding the incident field and the reflected field

2: the free field is calculated by dividing the region in subregions and taking the equilibrium conditions on the surface of the subsurface

Comments

This sub-modulus is not necessary. If is not found, then the default value 0 is accepted.

F.6.12 Module ZONES

Example

```
ZONES
3
NUMBER BOU1 BOU2  E      NU  RO  VISC MODE
 1      1    2   60000  0.25 3.0 0.05  1
 2      3   -4   80000  0.30 5.0 0.05  1
 3      5    0   40000  0.15 2.0 0.05  1
```

Explanations

ZONES :

Integer number indicating the total number of zone boundaries (up to 5). In the case of zones existence, the mesh is considered as consisting of zones between boundaries. e.g. for a cavity with lining, one zone is the lining and another the surrounding material and the zones are lying one inside the other.
One zone can be consisted of up to five independent boundaries.

NUMBER :

Is the number of the zone 1,2,...

BOU1 :

Is the 1stt boundary that encloses the zone. This boundary is the one in contact with the outer surrounding material. In case that the zone extends to infinity this number is 0 (e.g. the outer surrounding to the lining material in case of a lined cavity)

BOU2 :

Is the 2nd boundary that encloses the zone. This boundary is the one in contact with the inner surrounding material. This is the inner surface of the lining, in case of one lining.

BOU3 :

Is the 3rd boundary that encloses the zone. This boundary is the one in contact with the inner surrounding material.

BOU4 :

Is the 4th boundary that encloses the zone. This boundary is the one in contact with the inner surrounding material.

BOU3 :

Is the 5th boundary that encloses the zone. This boundary is the one in contact with the inner surrounding material.

These values BOU1,... are not necessary, e.g. in case of a rigid inclusion (0 value). The number of the boundary but with negative value is given for the inner free surface of a zone (void inside).

E :

Young's modulus

NU :

Poison's ratio between 0.0 and 0.50

RO :

Mass density (e.g. kg/m3)

VISC:

Coefficient of viscosity expressed as fraction of shear modulus, taking values between 0.0 and 1.0. Only for Laplace domain

MODE :

Model for viscosity 1: Kelvin solid, 2: Maxwell fluid. Only for Laplace domain

Comments

This modulus is not necessary. It is used for zones (e.g. inclusions or linings). If the sub-modulus E, NU, RO, VISC, MODE are not given, then the program automatically accepts the same as the surrounding material, and calculates the tractions and displacements on the boundary of the inclusion. This is a convenient way of calculating the field at interior point.

F.6.13 Module GAUSSIAN.QUADRATURE

Example

```
GAUSSIAN.QUADRATURE
 8
```

Explanations

It controls the Gaussian quadrature for the numerical integration.
An integer number is given that expresses the order of Gaussian quadrature that is used for the numerical integration.
The default value is 6.

Comments

This modulus is not necessary. If it is omitted, the program automatically accepts the default values.

F.6.14. Module SUBELEMENTS

Example

```
SUBELEMENTS
SINGULAR
 6
NONSINGULAR
 7
 DMIN     DMAX      NSUB
 0.0      0.05      12
 0.05     0.10      9
 0.10     0.30      7
 0.30     0.70      5
 0.70     1.00      3
 1.00     2.00      2
 2.00     5.00      1
```

Explanations

This module controls the subdivision of the elements into sub-elements during the numerical integration for better accuracy.
Both values for singular and nonsingular cases must be given, even if one of them has the default values.

SINGULAR :

For the singular case an EVEN integer number is given. For the singular case, the source node is treated as identical with one of the nodes of the element.
The default value is 4.

NONSINGULAR :

Here the number of the cases that follow are determined. In the nonsingular case the source node is not identical with any of the nodes of the element.

DMIN DMAX NSUB :

In the nonsingular case, this subdivision depends on the minimum distance of the source node from the nodes of the examining element.

It is d=r/l, where r=min(r_1,r_2,r_3) and r_1,r_2,r_3 are the distances of the source node from the three nodes of the quadratic element and l is the length of the examining element.

For DMIN <= d < MAX the element is divided into NS subelements

The given numbers are up to 10 values (10 rows).

The default values are

DMIN	DMAX	NSUB	
0.0	1.0	3	(0.0 <= d < 1.0 then subelements =3)
1.0	2.0	2	(1.0 <= d < 2.0 then subelements =2)
2.0		1	(2.0 <= d then subelements =1)

In the nonsingular case, the program accepts the value of NSUB and automatically checks the following:

$$\left(\frac{\alpha}{4\rho}\right)^{2\kappa} \le \chi\alpha\ , \qquad 10^{-2} \le \chi' \le 10^{-4} \tag{Γ.15}$$

In this work c is taken equal to 10^{-4}.

From the relation (F.15) the Gaussian quadrature order k may be obtained.

Comments

This modulus is not necessary. If it is omitted, the program automatically accepts the default values.

F.6.15 Module PLOTS

Example

PLOTS
2 3

Explanations

This module controls the type of plots in the postprocessing phase. The graphs are either on the periphery of the boundaries (on the boundary) or in a Cartesian system (in a Cartesian system).

1: Hoop stress (on the boundary) in the normal direction.
2: Hoop stress (on the boundary) in the tangential direction.
3: Hoop stress (on the boundary) shear.
4: Hoop stress (on the boundary) in the third direction (only for plane strain).
5: Deformed shape of the mesh.
6: Plots at the internal points (in a Cartesian system).
11: Hoop stress (in a Cartesian system) in the normal direction.
12: Hoop stress (in a Cartesian system) in the tangential direction.
13: Hoop stress (in a Cartesian system) shear.
14: Hoop stress (in a Cartesian system) in the third direction (only for plane strain).
15: Stress x (not implemented).
16: Stress y (not implemented).
17: Shear xy (not implemented).
18: Stress in the third direction, only for plane strain (on the boundary).
19: Traction x (not implemented).
20: Traction y (not implemented).
21: Displacements x (in a Cartesian system).
22: Displacements y (in a Cartesian system).
26: For the stress plots on the periphery, write the values of stresses only at the nodes that are declared at NODE.PLOTS modulus. This is useful for

avoiding the congestion of the plot with numbers. If this number is given, but no nodes are declared at NODE.PLOTS, no values are written at all.

27: Plot the stresses on the periphery for every element, without taking the mean value between neighbourhing nodes (the default case is to take the mean value). This must not be used with plot numbers 11,12,13, in this case always the mean value between neighbourhing nodes is taken.

28: For the NODE.PLOTS, draw the graphs in the same page (one over the other).

29: For the NODE.PLOTS, draw the graphs in dimensionless x (time) axis as x values are taken the time steps numbers (1..NTINT).

30: Plots in the preprocessing phase are produced.

Comments

This modulus is not necessary. If not given, the default values are 1,2,3,4

F.6.16 Module DRAW.ZONES

Example

```
DRAW.ZONES
 1 2
```

Here the number of zones is determined, where belong the nodes and elements that follow.

In case that this modulus is not present, the plots for all the nodes are presented.

Explanations

This modulus is used to determine the number of zones where the prescribed in the following nodes and elements belong to and where the plots are needed. The zone numbers are tha same as the ones determined in modulus ZONES.

Comments

This sub-modulus is not necessary. If is not found, then the plots for all the nodes and elements that are determined in the following are presented.
A convenient way to find the number of the nodes and elements is by running first the preprocessing program without declaring modulus ZONE.PLOTS, watching the mesh on the screen and re-running again the preprocessing phase with the correct numbers.

F.6.17 Module NODE.PLOTS

Example
NODE.PLOTS
4 8 21

Explanations

This modulus is used to determine:

a. The nodes where plots in a Cartesian coordinate system are needed (plots: 11,12,13 of module PLOTS, section F.6.15). By default the program calculates the mean values between neighbourhing elements that the node connects e.g. left and right value at the node. If the number of the node is declared in the modulus ELEMENT.NODES (section F.6.17), then no mean values are calculated, and the values left and right to the node are plotted.

b. The nodes where the values of stresses will be written, for the case of plots: 26 (of module PLOTS, section F.6.15, plots on the periphery, with values of stresses at some nodes only).

Comments

For the corner nodes of a square or a rectangle where there is discontinuity of stress and traction, the program automatically duplicates these nodes by giving duplicate numbers. The numbering of these nodes is explained in the forthcoming, section F.7.

A convenient way to find the number of the nodes and elements is by running first the preprocessing program without declaring modulus NODE.PLOTS, watching the mesh on the screen and re-running again the preprocessing phase with the correct numbers. In case that the refinement of the mesh is such that is impossible to distinguish the node numbers on the screen, there is an alternative method of numbering manually : numbering the nodes of the mesh, starting from the node that the positive x local axis cuts the mesh, and numbering in clockwise (anticlockwise) direction for exterior(interior) problem, (the middle nodes of quadratic elements must be also numbered).

F.6.18 Module ELEMENT.NODES

Example

ELEMENT.NODES
25 50 25 51 26 51 26 52 35 70 35 71 36 71 36 72

Explanations

This modulus determines the plots for nodes that belong to an element.
The values at these nodes are the real values (without taking the mean value between neighbourhing elements as previously). This is very useful for nodes where there is discontinuity and the value of stress is needed to the left or right of the node e.g. node at the angle of two elements or node that joins more than 2 elements.
For square and rectangular shapes, the plots can bee produced by using the modulus NODE.PLOTS, without using the current modulus.
This modulus may be used in connection with the NODE.PLOTS modulus.
A pair of integer numbers is given for each entry:

a b, where a is the element that the node b belongs.

e.g. in the above example the following plots are need :

the plots for element 25, node 50 of this element
element 25, node 51 of this element
element 26, node 51 of this element
element 26, node 52 of this element
element 35, node 70 of this element
element 35, node 71 of this element
element 36, node 71 of this element
element 36, node 72 of this element

The drawings in the preprocessing phase are plotted on the screen always. But if a hardcopy is needed, plot number 30 must be given in modulus PLOTS (section F.6.15).

Comments

A convenient way to find the number of the nodes and elements is by running first the preprocessing program without declaring modulus ELEMENT.NODES, watching the mesh on the screen and re-running again the preprocessing phase with the correct numbers. In the case that the refinement of the mesh is such that is impossible to distinguish the node numbers on the screen, there is an alternative method of numbering manually : numbering the nodes of the mesh, starting from the node that the positive x local axis cuts the mesh, and numbering in clockwise (anticlockwise) direction for exterior (interior) problem, (the middle nodes of quadratic elements must be also numbered).

F.6.19 Module END

Example
END

Explanations

END :

This is the last modulus given, declaring the end of the data file.

F.7 POINTS OF DISCONTINUITIES

For the corner nodes of a square or a rectangle where there is discontinuity of stress and traction, the program automatically duplicates these nodes by giving there duplicate node numbers. The initial numbers of the corners belong to the elements that are encountered before the corner, travelling in the positive direction.

The duplicate numbers belong to the elements after the corner.

e.g. for a square with 40 elements (0 angle of rotation), the normal number of nodes is 80. The duplicate corner nodes take numbers:

(1). The left-upper corner, number 81 which belongs to the element 26 of the upper-horizontal side. (2). The left-lower corner takes number 82, which belongs to the element 16 of the left-vertical side. (3). The right-lower node takes number 83 that belongs to the element 6 of the lower-horizontal side. (4). The upper-right node takes number 84, that belongs to the element 36 of the right-vertical side.

If the plot at a corner node is needed, it must be declared exactly at which of these corner nodes the plot is wanted, by giving the respective number at the modulus NODE.PLOTS (section F.6.16). These duplicated nodes identical, with the same coordinates.

For discontinuities at corner nodes of other than rectangular shapes, the ELEMENT.NODES modulus must be used for the plots 'left' and 'right' of the node.

F.8 USEFUL COMMENTS ON THE DATA FILE

All the titles and subtitles must be in capital letters.

Comment lines can inserted by puting the first letter in the line (first column) the character '*'.

No coments allowed between the lines of numerical data inside a module

e.g. the following comment is not permited

```
 SHAPE REGION ELEMENTS GEOMETRY
  1      1       8      2.5
*      now combined shapes
 -1     4       6      6.0 5.0 -6.0 5.0
 or
       MATERIAL.PROPERTIES
*       in SI system
  E     MU    NU    RO
  20000  8000  0.25  2.55
```

If in the numerical data the letter o is found, it is converted to 0. Here it is supposed that a user's mistake ensountered.

For decimal points use the '.' e.g. 35.63, -46.1

The data must seperated by one or more blank characters (spaces).

All the lines of the data (titles, subtitles, numerical values), must have maximum length of 80 characters (from 1 to 80).

The titles, subtitles and the numerical values can be inserted free in the lines. Ensure only that the subtitles and the numerical values are given in the same order

e.g.

```
              MATERIAL.PROPERTIES
 E        MU     NU      RO            VISC
 2.0E7        8.0E6              0.25      2.7
```

here the subtitles E, MU, NU, RO, VISC have the numerical values 2.0E7, 8.0E6, 0.25, 2.7 respectively

F.9 HARDCOPY OF THE PLOTS

a. MSDOS Environment

For the MSDOS version of the program, the results are saved also in a file named THIRDGR in FORTRAN format. These data can be imported to any graphical package and can be processed.

e.g. to import the results to Harverd Graphics package :

Select : File,
Import,
ASCII delimeted character:
End of file delimiter
usual values: coma ,
space #32

b. UNIX Environment

The UNIRAS graphics package is used in connection with the program. For the creation of the plots subroutines of the UNIRAS library are called through the program, in FORTRAN language, and the plots are saved in a unipict file with name unipict.upi. In the Sunview workstation enviroment windows are selected. Hardcopies can be done from UNIEDIT. For this reason from a shell window of Sunview, enter UNIEDIT -d=msview. This opens the UNIEDIT. Collect EDIT CELL, give a name and then GET , UNIPICT FILE. Give the name unipict.upi for the unipict filename. Then collect SEGMENT, NUMBER .....(enter the number of plot screens you want to import).

Finally collect CELL, SAVE to save the cell file. This file can be retrieved and ploted by HARDCOPY.

F.10 COMPLETE EXAMPLE

A complete data file, should look like this:

```
TITLE
2D L.D., CIRCLE, INFINITY, PLANE WAVE
TYPE.PROBLEM
DOMAIN ANALYSIS
  1       1
SECTION.TYPE
SHAPE ELEMENTS GEOMETRY
  1       8      5.0
MATERIAL.PROPERTIES
   E     NU   RO
20000.0 0.25  2.55
STEP.NUMBER
   20
TIME.TOTAL
  0.015
LOAD.STRESSES
PLANE.WAVE
LONGITUDINAL.STRESS TRANSVERSE.SHEAR ANGLE
 -10.0                  0.0          225.0
PLOTS
2 4 12 19 20 28 29 30
NODE.PLOTS
 15 13 11 9 7
END
```

This file can be created by any a text editor, and a name is given by saving it, e.g. infpl1.exp.
Running the preprocessing, this is the name given, when the program asks for an input file.

APPENDIX G

SUBROUTINES OF THE PROGRAM

In the following sections, the main subroutines are presented, with a short explanation of their corresponding function.

G.1 PREPROCESSING PHASE

G.1.1 Reading of the input data

Before running the preprocessing phase, a data file must be prepared with all the necessary information, according to the specifications introduced in Appendix G. For the preparation of the data file any text editor with capabilities of saving in ASCII format may be used. This file is saved in ASCII format and its name is given as input at the prompt of the main preprocessing program.

The main program calls the basic subroutine MODULES, which manages the reading of the input data file. In order to perform the reading of the file, the following most important subroutines are called in sequence:

Subroutine TITLES : reads the main title of the problem (module "TITLE" section G.6.1 of Appendix G).

Subroutine TYPEMOD : reads the type of the problem (module "TYPE.PROBLEM" section G.6.2 of Appendix G).

Subroutine SECTMOD : reads the geometrical characteristics and information concerning discretization into elements (module "SECTION.TYPE" section G.6.3 of Appendix G).

Subroutine INTERMOD : reads the geometrical data of the internal points (not on the boundary surfaces) (module "INTERNAL.POINTS" section G.6.4 of Appendix G).

Subroutine MATERMOD : reads the properties of the uniform material (module "MATERIAL.PROPERTIES" section G.6.5 of Appendix G).

Subroutine STEPMOD : reads the number of the discrete time steps that the total examining time interval is devided (module "STEP.NUMBER" section G.6.5 of Appendix G).

Subroutine TIMEMOD : reads the number of the total time interval that is being examined (module "TIME.TOTAL" section G.6.7 of Appendix G).

Subroutine HOOPMOD : reads the desired elements for calculating the hoop stresses (module "HOOP.STRESSES" section G.6.8 of Appendix G).

Subroutine PRESMOD : reads the boundary conditions (module "PRESCRIBED.CONDITIONS" section G.6.9 of Appendix G).

Subroutine LOADMOD : reads the applied loading on the nodes (module "LOADS" section G.6.10 of Appendix G).

G.1.2 Mesh generation subroutines

Automatic mesh generation facilities have been introduced, that simplify considerably the introduction of the data. The basic subroutine GEOMETR controls the mesh generation by calling the following most important subroutines:

Subroutine GEOELL : generates an ellipsis and performs its discretization into elements and nodes.

Subroutine GEOBOX : generates a square or a rectangle and is responsible for its discretization into elements and nodes.

Subroutine GEOSTRLN : generates a straight line and introduces its discretization into elements and nodes.

Subroutine GEOCOMB : generates a compound geometrical shape consisting of straight lines and elliptical segments and it is also discretizing the shape into elements and nodes.

Subroutine NODECON : generates the topography of the mesh (connectivity of the nodes of each element).

G.1.2 Other subroutines

Subroutine FREESURF : keeps track of elements of special kind e.g. elements that belong to a free surface of the ground or elements that are lying on the inner or outer surface of a zone.

Subroutine WAVEFIR : determines the wavefront position, the equation of its line and calculates the distances of the nodes of the mesh from the wavefront, as specified in section 6.4.1.

Subroutine WRITEDAT : saves the data for their use in a later phase.

G.1.3 Subroutines for Incident and Reflected Stresses

a. Laplace Transform Domain

Subroutine DATALAP : This subroutine controls the creation of loading in Laplace transform domain. It also superimposes the incident and the reflected stresses of a plane or explosive wave.

Subroutine WINPRES : evaluates the incident stresses resulting from a pressure wave acting from the interior towards an opening.

Subroutine WINSURF : evaluates the incident stresses resulting from pressure on

the surface of the elements.

Subroutine WINCID : evaluates the incident stresses resulting from pressure on the surface of the elements, see section 6.4.

Subroutine WAVPLAN : evaluates the incident stresses resulting from a plane pressure wave, see section 6.4.2.

Subroutine WAVEXPL : evaluates the incident stresses resulting from an explosion at a point, see section 6.4.2.

Subroutine SURFLINE : the line of the free surface of the ground is calculated, see section 6.5.1.

Subroutine REFP : the reflection of P waves by the laws of harmonic waves is analyzed, see section 6.5.4.

Subroutine REFLCOND : the reflection conditions of P or SV wave on the free ground surface are examined and the incident wave is reflected only if they are fulfilled, see section 6.5.4.1.

Subroutine PATP : calculates numerically the ray patterns for P incident wave, see section 6.5.4.2.

Subroutine REFSV : the reflection of SV waves by the laws of harmonic waves is analyzed, see section 6.5.5.

Subroutine PATSV : calculates numerically the ray patterns for SV incident wave, see section 6.5.5.

Subroutine ZEXPLOSQ : Calculates the incident P pressures at a point generated by an explosion at a point, see section 6.6.1.a.

Subroutine DECEXPL : Calculates the transformed stresses in Laplace domain for a constant or decaying in time explosive wave, see section 6.6.1.a.

Subroutine EXPLMOR : Calculates the stress acting along the direction perpendicular to the radial one, see section 6.6.1.a.

Subroutine REFEXP : Finds the reflected P explosive wave, see section 6.7.2.

Subroutine REFCONEX : the reflection conditions of the P and SV explosive waves on the free ground surface are examined and the incident wave is reflected only if these conditions are fulfilled, see section 6.7.2.1.

Subroutines REFANGEX and TH34EX : Evaluate the angles of incident and reflected wave with the free ground surface, see section 6.7.2.1.b.

Subroutine PATEXP : calculates numerically the ray patterns for a P explosive incident wave, see section 6.7.2.1.b.

Subroutine REFEXSV : Finds the reflected SV explosive wave, see section 6.G.

Subroutine PATEXSV : calculates numerically the ray patterns for a SV explosive incident wave, see section 6.G.

b Time Domain

Subroutine DATATIM : This is the subroutine that controls the creation of loading in time domain. Furthermore, it also superimposes the incident and the reflected stresses of a plane or explosive wave.

Subroutine TWINPRES : evaluates the incident stresses resulting from a pressure wave acting from the interior towards an opening.

Subroutine TWINSURF : evaluates the incident stresses resulting from pressure on the surface of the elements.

Subroutine WINCIDIM : evaluates the incident stresses resulting from pressure on the surface of the elements, see section 6.4.

Subroutine TWAVPLAN : evaluates the incident stresses resulting from a plane pressure wave, see section 6.4.2.

Subroutine TWAVEXPL : evaluates the incident stresses resulting from an explosion at a point, see section 6.4.2.

Subroutine REFPTIM : the reflection of P waves by the laws of harmonic waves is analyzed, see section 6.5.4.

Subroutine TPATP : calculates numerically the ray patterns for P incident wave, see section 6.5.4.2.

Subroutine REFSVTIM : the reflection of SV waves by the laws of harmonic waves is analyzed, see section 6.5.5.

Subroutine TPATSV : calculates numerically the ray patterns for SV incident wave, see section 6.5.5.

Subroutine TWAVEXPL : for the Laplace domain case, the stresses are transformed in the Laplace transform domain, as in see section 6.4.2

Subroutine TEXPLOSQ : Calculates the incident P pressures at a point generated by an explosion at a point, see section 6.6.1.a.

Subroutine TDECEXPL : Calculates the stresses in time domain for a constant or decaying in time explosive wave, see section 6.6.1.a.

Subroutine TEXPLMOR : Calculates the stress acting along the direction perpendicular to the radial direction, see section 6.6.1.a.

Subroutine TREFEXP : Finds the reflected P explosive wave, see section 6.7.2.

Subroutine TPATEXP : calculates numerically the ray patterns for a P explosive incident wave, see section 6.7.2.1.b.

Subroutine TREFEXSV : Finds the reflected SV explosive wave, see section 6.G.

Subroutine TPATEXSV : calculates numerically the ray patterns for a SV explosive incident wave, see section 6.G.

G.1.4 Plotting Subroutines

Subroutine PLMESH : main subroutine controlling the plotting.

Subroutine PLOTDOM : plots the mesh.

Subroutine WAVFRONT : plots the wavefront line at t=0.

Subroutine RAYINCPL : plots the incident rays for a plane wave.

Subroutine RAYPLANE : plots the reflected rays for a plane wave.

Subroutine RAYINCEX : plots the incident rays for an explosion wave.

Subroutine RAYEXPL : plots the reflected rays for an explosion wave.

G.2. PROCESSING PHASE - LAPLACE DOMAIN

The multi-zonal case program in Laplace domain is presented next.

Subroutine DATAS : reads the data generated from the preprocessing phase.

Subroutine GAUSWE : Calculates the Gaussian quadrature points

Subroutine LAPIN : Performs the numerical inversion of Laplace transform.

Subroutine BEM : Control subroutine of the BEM.

Subroutine NMAIN : Controls the numerical integration subroutines.

Subroutine INTEGR : Performs the numerical integration.

Subroutine KERNEL2 : Calculates the 2-D Laplace domain kernels.

Subroutine VZISCOEL : Material characteristics for viscoelastic behaviour

Subroutine BESSEL : Bessel functions

Subroutine ZERNEL : Evaluates the 2-D elastostatic L.D. kernels

Subroutine CSHAPE : Shape functions for quadratic and linear elements, see Chapter 4, see section 4.4.2.

Subroutine PREGENR : Controls the generation of the matrix of tractions and displacements kernels.

Subroutine GENRF : Calculates the [F] part of the left hand side matrix [A] (Equation 2.31, Chapter 2) that refers to the zones boundaries only (Equations 5.10 and 5.15, Chapter 5).

Subroutine GENRF5 : Evaluates the [F] part of matrix [A] that refers to the non-zones boundaries.

Subroutine GENRGL : Calculates the [G] part of matrix [A] that multiplies the interface tractions.

Subroutine GENRG3 : Evaluates the [G] part of matrix [A] that refers to the non-zone boundaries.

Subroutine GENRGR1 : Calculates the [G] part of the right hand matrix (Equations 5.10 and 5.15, Chapter 5), that multiplies non-interface tractions.

Subroutine SOLVE : Reads the free field stresses (calculated and saved in a file in the preprocessing phase) and controls the solution of equations

Subroutine GENRONE : Calculates the [G] and [F] matrices for the non-zones case.

Subroutine BOUND : introduces the boundary condition (traction or displacement)

applied to the nodes.

Subroutine ZVPROD : generates the right hand matrix [B] (Equation 2.31, Chapter 2 and Equations 5.10 and 5.15, Chapter 5).

Subroutine ZMINV : Inverts the [A] matrix.

Subroutine ZAPROD : calculates the product of matrices $[A]^{-1}[B]$ (Equation 2.32, Chapter 2).

Subroutine HOOP : Finds the hoop stresses on the boundaries.

Subroutine VZISCOEL : material characteristics for viscoelastic behaviour.

Subroutine NMAININT : numerical integration for internal points.

Subroutine GENRINT : formulation and solution of Equations for the case of interior points (see section 2.2.2.4, Chapter 2).

Subroutine BOUNDINT : Applies the boundary conditions and solves for the interior displacements (Equation 2.35, Chapter 2).

Subroutine FASTINV1 : numerical inversion of Laplace transform for tractions.

Subroutine FASTINV2 : numerical inversion of Laplace transform for displacements.

Subroutine FFCT : performs Fast Fourier transform.

G.3 POSTPROCESSING PHASE

Subroutine DATAS : reads the data generated from the preprocessing and processing phase.

Subroutine PLMESH : main subroutine controlling the plotting.

Subroutine PLOTHOOP : plots the hoop stresses on the periphery of the opening for every time step.

Subroutine PLOTDEFOR : plots the deformed shape of the mesh for every time step.

Subroutine PLCARSTR : controls the plots in a Cartesian co-ordinate system.

Subroutine PLOTHOOPN : plots the hoop stresses versus time steps, in a Cartesian co-ordinate system.

Subroutine PLOTAXES : plots the x and y Cartesian axes .

Subroutine PLCARINT : plots the displacements versus time steps, in a Cartesian co-ordinate system, at interior points.

Subroutine PLOTDOM : plots the mesh.

Subroutine WAVFRONT : plots the wavefront line at t=0.

Subroutine MOREDAT : performs interpolation for the stresses, in order to have more data for smoother drawings.

Subroutine DISPDAT : performs interpolation for the displacements, in order to have more data for smoother drawings.

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